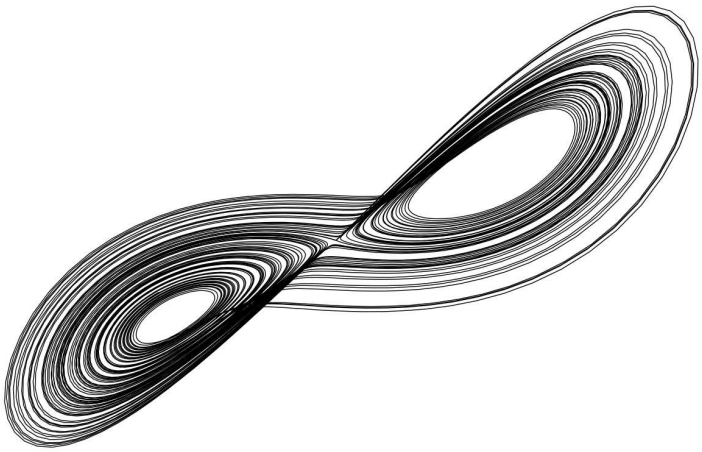


Modeling with partial differential equations

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Foreword

The objective of these introductory lectures is to present a few examples of models based on ordinary and partial differential equations. Mathematical models aim at describing the behavior of physical objects (predict), and are also useful to analyze data (understand) and to elaborate techniques to act on these objects (control). We will see that in many situations, conservation laws dictated by physics lead to partial differential equations, which thus play a crucial role in mathematical modelling.

Each of the four parts, is devoted to a specific type of equation: ordinary differential equations, the Poisson equation, the heat equation and equations modeling transport phenomena (advection and wave equations). For each topic, we provide examples of models, outline some mathematical properties of the equations and discuss numerical methods to approximate the solution. Through these examples, students will gain insight into the interplay between mathematical modeling, mathematical analysis, and simulation.

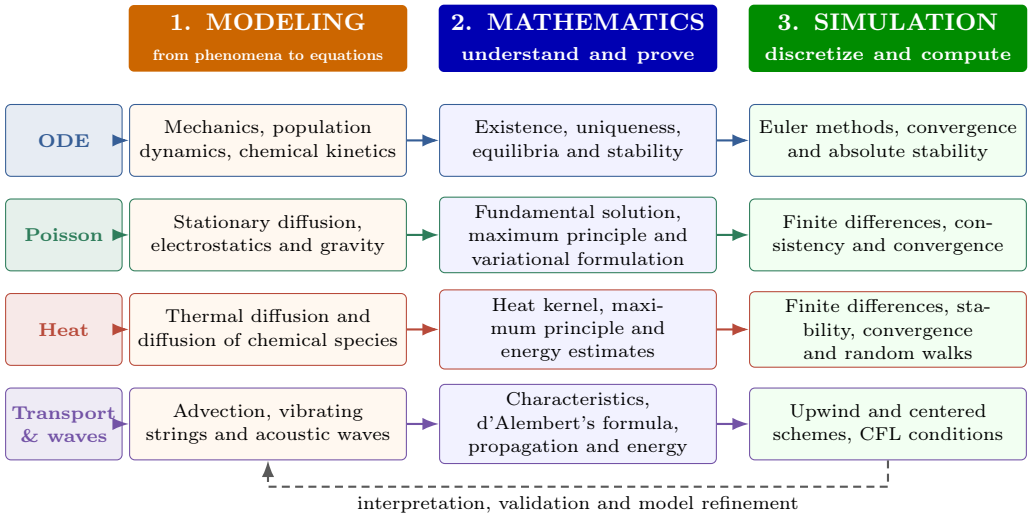


Figure 1: The common structure of the lectures: physical principles lead to mathematical models, analysis identifies their properties, and numerical methods make their simulation possible. Computations can in turn be used to assess and refine the models.

These lectures are non-standard in two ways. First, we deliberately rely on elementary mathematical tools, mostly already encountered in *Classes Préparatoires*. More advanced concepts from mathematical and numerical analysis will be introduced in other courses of the first year, and also during the second year for students in the IMI (*Ingénierie Mathématique et Informatique*) program. Second, we systematically illustrate the equations in practical sessions using numerical simulations with Python, with the support of artificial intelligence tools to help generate initial versions of the codes.

The lecture notes propose several exercises, with various difficulties (★: simple, ★★: intermediate, ★★★: hard). We expect all students to be able to solve the simple exercises.

> Some parts are highlighted with a bar in the margin: these are paragraphs which can be skipped on first reading. Students are not expected to master these sections to pass the exam.

In terms of bibliography, we heavily relied on excellent textbooks such as [6, 9, 12, 18].

These notes undoubtedly contain typos and errors—please do not hesitate to send any corrections or suggestions to tony.lelievre@enpc.fr.

Many thanks to the team in charge of the exercise sessions for this year: Nail Baloul, Vincent Boulard, Louis Carillo, Antonin Coatantiec, Théo Duez, Grégoire Elinck, and Jean-Baptiste Himbert.

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Part I

Ordinary differential equations

Chapter 1

Modeling using ordinary differential equations

1.1 Definition of ordinary differential equations

Ordinary differential equations are used to model the evolution of a state function $x(t)$ which typically depends on time $t \geq 0$. The idea is that in many situations, one can relate the time derivative $\dot{x}(t)$ of x at time t (the rate), with the current state $x(t)$. The state can be a scalar quantity ($x(t) \in \mathbb{R}$) or a vector ($x(t) \in \mathbb{R}^d$). This gives rise to differential equations of the form (if x is vector-valued, $\dot{x}(t)$ is simply the vector in $\mathbb{R}^d$ whose i-th component is $\dot{x}_i(t)$):

$$\dot{x}(t) = f(x(t))$$

for some function $f : \mathbb{R}^d \to \mathbb{R}^d$. For the function x to be uniquely defined, one necessarily needs an initial condition:

$$x(0) = x_0.$$

The objective of this part is to discuss solutions to this problem (called a Cauchy problem) in the following sense:

Definition 1. *Let $f : \mathbb{R}^d \to \mathbb{R}^d$ be a continuous function, and let us consider the initial value problem: find x such that*

$$\begin{cases} \dot{x}(t) = f(x(t)) \\ x(0) = x_0. \end{cases} \tag{1.1}$$

A solution to (1.1) *over the time interval $J \subset \mathbb{R}$ (such that $0 \in J$) is a $\mathcal{C}^1$ function $x : J \to \mathbb{R}^d$ which satisfies* (1.1)*, the first equation being satisfied for all $t \in J$.*

Notice that the solution requires to define both the function x and its domain of definition J. When $J = \mathbb{R}$, x is called a global solution. In many practical cases, we will be interested in solutions defined for positive time (this will be indicated in the problem by specifying $\dot{x}(t) = f(x(t))$ for $t \geq 0$), in which case $J = [0, a)$ for some $a > 0$.

We will be interested in three types of questions:

- Is the solution well defined, namely, does it exist and is it unique?
- What can be said about the stationary states, namely, constant (not depending on time) solutions?

- How to approximate numerically the solutions on a computer?

Exercise 1. ★*Assume that* $x : \mathbb{R} \to \mathbb{R}^d$ *is a global solution to* (1.1). *Let* $t_0 \in \mathbb{R}$. *Express the global solution to the problem*

$$\begin{cases} \dot{y}(t) = f(y(t)) \\ y(t_0) = x_0 \end{cases}$$

as a function of x.

1.2 Some models based on ordinary differential equations

Models based on ordinary differential equations are extremely common. Let us present a few examples from various scientific fields.

1.2.1 Mechanics and Hamiltonian systems

The evolution of a system of material points with position vector x is governed by Newton's law:

$$m\ddot{x} = F(x, \dot{x}),$$

where $\dot{x}$ denotes the first time derivative (velocity) and $\ddot{x}$ the second time derivative (acceleration). We assume here that the mass m is constant. More generally, m may be a tensor. The right-hand side $F(x, \dot{x})$ represents a force that depends on position and velocity.

Note that we can reduce this to a first-order system by introducing $v = \dot{x}$ and working in the phase space (x, v):

$$\begin{cases} \dot{x} = v, \\ \dot{v} = \frac{1}{m} F(x, \dot{x}). \end{cases} \tag{1.2}$$

Some examples of force fields that we will encounter later:

- For a pendulum under gravity, we have $x \in \mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$ and $F(x, \dot{x}) = -m\frac{g}{l}\sin x - k\dot{x}$, where g is gravitational acceleration, l is the length of the rod, and $k \geq 0$ is a viscous damping coefficient.
- The force may derive from a potential: $F(x) = -\nabla V(x)$. In that case, we have a Hamiltonian system. Setting $(q, p) = (x, mv)$ (where p is the momentum) and $H(q, p) = \frac{\|p\|^2}{2m} + V(q)$, the system can be rewritten as:

$$\begin{cases} \dot{q} = \dfrac{\partial H}{\partial p}(q, p), \\ \dot{p} = -\dfrac{\partial H}{\partial q}(q, p). \end{cases} \tag{1.3}$$

  Note that H is conserved along trajectories: $\frac{d}{dt} H(p(t), q(t)) = 0$ (prove it!★). In general, any quantity that remains constant along the trajectories is called a first integral of motion. The study of such systems is important in practice (e.g., in molecular dynamics or celestial mechanics).

1.2.2 Population dynamics

Ordinary differential equations are widely used to model the evolution of populations. An example is the Lotka-Volterra equations, a predator-prey model. Let P denote the density of predators and N the density of preys. The evolution of P and N satisfies:

$$\begin{cases} \dot{N} = N(a - bP), \\ \dot{P} = P(cN - d), \end{cases}$$

with a, b, c, d four strictly positive coefficients interpreted as follows: a is a birth rate for the prey, b is a predation rate, d is a mortality rate for the predators, and c is a reproduction rate for the predators.

Some solutions are shown in Figure 1.1. We observe that the solutions appear to be periodic, remain positive, and are confined along a curve in the (N, P) plane.

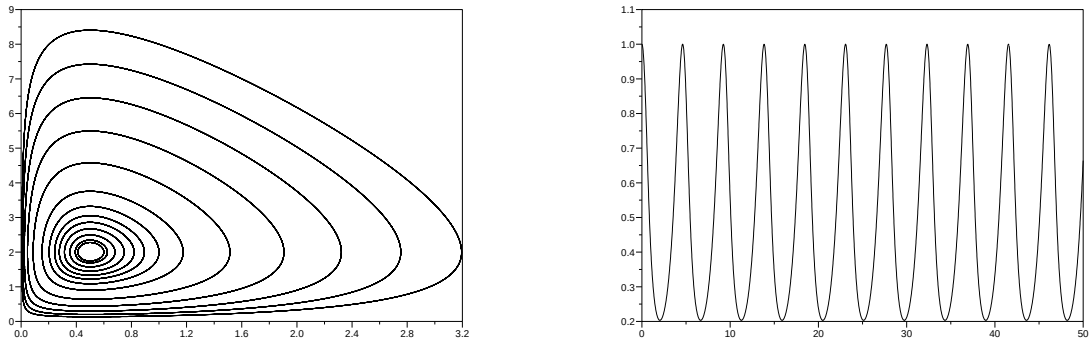


Figure 1.1: Solutions of the Lotka-Volterra equations. Left: several trajectories $(N(t), P(t))_{t\geq 0}$ for different initial conditions (horizontal: N, vertical: P). Right: one trajectory $t \mapsto N(t)$ (horizontal: t, vertical: N). Parameters: $a = 2$, $b = 1$, $c = 2$, $d = 1$.

Exercise 2. ★*Discuss the meaning of each term in the equations (sign, nonlinearity).*

Exercise 3. ★*Show that if (N, P) is a solution, then $\dot{N}(c - d/N) = \dot{P}(a/P - b)$. Deduce that $H(N, P) = cN - d \ln N + bP - a \ln P$ is conserved along trajectories.*

1.2.3 Meteorology

The equations governing the evolution of meteorological parameters (temperature, humidity, pressure, etc.) are very complex. A simplified system capturing some of their features was proposed by Lorenz in 1963. It is a 3-dimensional system of ordinary differential equations:

$$\begin{cases} \dot{x} = \sigma(y - x), \\ \dot{y} = rx - y - xz, \\ \dot{z} = xy - bz \end{cases}$$

Common values for the parameters are $\sigma = 10$, $r = 28$, and $b = 8/3$.

Figure 1.2 shows some solutions of these equations. We observe that trajectories diverge rapidly for very close initial conditions. It thus seems futile to attempt precise prediction based on specific initial conditions, as small variations lead to large differences in outcomes. A better approach might be to study the evolution of an *ensemble of initial conditions* (i.e., the propagation of a probability distribution through the dynamics), or the *average behavior along a trajectory* (trajectory averaging). This will not be discussed in these introductory lectures. Such analysis is at the heart of a branch of mathematics called dynamical systems theory [19].

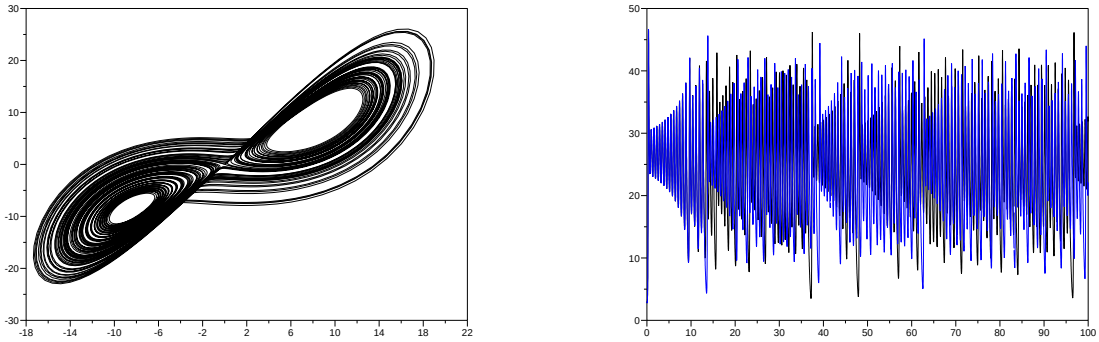
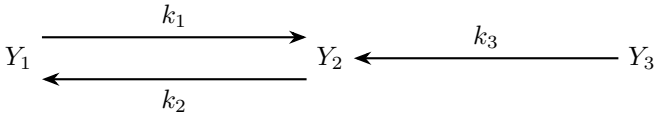


Figure 1.2: Solutions of the Lorenz equations. Left: a typical trajectory $(x(t), y(t))_{t\geq 0}$ (horizontal: x, vertical: y). Right: two trajectories $t \mapsto z(t)$ for two very close but different initial conditions ($(x, y, z)(0) = (1, 2, 3)$ and $(1.01, 2, 3)$). Horizontal: t, vertical: z. Parameters: $\sigma = 10$, $r = 28$, $b = 8/3$.

1.2.4 Chemical kinetics

Consider a chemical reaction involving three species:



The evolution of the concentrations $(y_i)_{1\leq i\leq 3}$ of the three species $(Y_i)_{1\leq i\leq 3}$ is ruled by the following system of ordinary differential equations:

$$\begin{cases} \dot{y_1} = -k_1 y_1 + k_2 y_2, \\ \dot{y_2} = k_1 y_1 - k_2 y_2 + k_3 y_3, \\ \dot{y_3} = -k_3 y_3. \end{cases}$$

In such problems, the coefficients k_i typically vary widely in magnitude. The dynamical system therefore contains very different time scales. It is important to consider this when discretizing the equations efficiently.

1.2.5 Some questions raised by these models

From these examples, we are naturally led to the following questions:

- Under what conditions is a model based on ordinary differential equations well-posed? This includes three aspects: (i) do the equations admit a solution? (ii) is the solution unique? (iii) does the solution depend continuously on the input data?
- How can we analyze the qualitative behavior of solutions? For example, are periodic solutions possible? What is the long-term behavior of the system? How sensitive are the solutions to initial conditions?
- How can we discretize these equations effectively to obtain accurate numerical approximations?

We will provide some answers to these questions in the following chapters.

Chapter 2

Existence, uniqueness and qualitative properties

We consider the problem introduced in Definition 1 (sometimes called a Cauchy problem): given a continuous function $f : \mathbb{R}^d \to \mathbb{R}^d$, find a $\mathcal{C}^1$ function $x : J \to \mathbb{R}^d$ such that

$$\begin{cases} \dot{x}(t) = f(x(t)) \text{ for } t \in J, \\ x(0) = x_0. \end{cases} \tag{2.1}$$

After having presented a few examples in Section 2.1, we will provide standard existence and uniqueness results in Sections 2.2 and 2.3 and finally discuss the longtime behaviors of solutions in the vicinity of stationary points in Section 2.4.

Remark 1. *We assume for simplicity that f does not depend on time (the problem is time-homogeneous). The existence and uniqueness results presented in this chapter, as well as the discretization schemes introduced in the next one can be generalized to the case of a time-dependent function f, namely, when $\dot{x}(t) = f(t, x(t))$ for a continuous function $f : \mathbb{R} \times \mathbb{R}^d \to \mathbb{R}^d$.*

2.1 A few examples

Let us first recall standard examples.

2.1.1 First order linear case

The solution to the problem

$$\begin{cases} \dot{x}(t) = ax(t) \text{ for } t \in \mathbb{R} \\ x(0) = x_0 \end{cases}$$

is: for all $t \in \mathbb{R}$, $x(t) = x_0 \exp(at)$.

Exercise 4. ★*Prove that the solution to the problem*

$$\begin{cases} \dot{x}(t) = a(t)x(t) \text{ for } t \in \mathbb{R} \\ x(0) = x_0 \end{cases}$$

for some continuous function $a : \mathbb{R} \to \mathbb{R}$ is $x(t) = x_0 \exp\left(\int_0^t a(s)\, ds\right)$.

From this, one can deduce the solution to the problem ($a \neq 0$)

$$\begin{cases} \dot{x}(t) = ax(t) + b \text{ for } t \in \mathbb{R} \\ x(0) = x_0 \end{cases}$$

by looking for a solution of the form $x(t) = \exp(at)y(t)$ for some y. Indeed, one has

$$\begin{aligned} \dot{y}(t) &= \exp(-at)(x(t) - ax(t)) \\ &= \exp(-at)b \end{aligned}$$

which yields $y(t) = y(0) + \int_0^t \exp(-as)b\,ds = x_0 + \frac{b}{a}(1 - \exp(-at))$. Thus, for all $t \in \mathbb{R}$,

$$x(t) = x_0 \exp(at) + \frac{b}{a}(\exp(at) - 1).$$

When $a < 0$, $\lim_{t\to\infty} x(t) = b/a$, while when $a > 0$, $\lim_{t\to\infty} x(t) = \infty$.

Exercise 5. ★★*Use the same technique to solve*

$$\begin{cases} \dot{x}(t) = ax(t) + b(t) \text{ for } t \in \mathbb{R} \\ x(0) = x_0 \end{cases}$$

for some continuous function $b : \mathbb{R} \to \mathbb{R}$.

Exercise 6. ★ *Assume that* $x(t)$ *is a* $\mathcal{C}^1$ *function such that, for all* $t \geq 0$, $\dot{x}(t) \leq ax(t)$. *Prove that for all* $t \geq 0$, $x(t) \leq x(0)\exp(at)$. Hint: consider again $y(t) = \exp(-at)x(t)$.

2.1.2 The vectorial case

One has a similar result in the vectorial case, namely, for the problem:

$$\begin{cases} \dot{x}(t) = Ax(t) \text{ for } t \in \mathbb{R} \\ x(0) = x_0 \end{cases} \tag{2.2}$$

where $x(t) \in \mathbb{R}^d$ is a vector and $A \in \mathbb{R}^{d\times d}$ is a matrix. In this case,

$$x(t) = \exp(tA)x_0.$$

We use here the matrix exponential which is defined by

$$e^{tA} = \sum_{n\geq 0} \frac{(tA)^n}{n!}.$$

We recall that $e^{tA} \in \mathbb{R}^{d\times d}$ is well defined since the series is absolutely convergent: for any submultiplicative [1] matrix norm $\|\cdot\|$, $\sum_{n\geq 0} \frac{(t\|A^n\|)}{n!} \leq \sum_{n\geq 0} \frac{(t\|A\|)^n}{n!} = \exp(t\|A\|) < \infty$.

Exercise 7. ★★*Show that the ensemble of functions satisfying* $\dot{x}(t) = Ax(t)$ *for* $A \in \mathbb{R}^{d\times d}$ *(for varying initial conditions) is a linear space of dimension* d.

---

[1] Let us recall that all norms are equivalent on a finite-dimensional space. A submultiplicative matrix norm is such that $\|AB\| \leq \|A\|\|B\|$ (and thus $\|A^n\| \leq \|A\|^n$), for all $A, B \in \mathbb{R}^{d\times d}$. For example, the Frobenius norm $\|A\|_F = \sqrt{\sum_{1\leq i,j\leq d}(A_{i,j})^2}$ is submultiplicative: this is just Cauchy-Schwarz, the Frobenius norm being the standard Euclidean norm applied to A seen as a d^2-dimensional vector. We will discuss below very useful submultiplicative norms, namely matrix norms which are induced by vector norms, see Section 6.2.2.

In the specific case when A can be diagonalized:

$$A = PDP^{-1}$$

for some diagonal matrix $D = \mathrm{diag}(\lambda_1, \dots, \lambda_d) \in \mathbb{R}^{d\times d}$, and some invertible matrix $P \in \mathbb{R}^{d\times d}$, then the solution can be easily obtained by rewriting the ordinary differential equations in the basis of eigenvectors. Remember indeed that the columns of $P = [p_1, \dots, p_d]$ are eigenvectors of A: for $i \in \{1, \dots, d\}$

$$Ap_i = \lambda_i p_i.$$

By introducing $y(t) = P^{-1}x(t)$, one gets

$$\begin{cases} \dot{y}(t) = Dy(t) \text{ for } t \in \mathbb{R} \\ y(0) = y_0 \end{cases}$$

where $y_0 = P^{-1}x_0$, whose solution is $y(t) = (y_1(t), \dots, y_d(t))$ with

$$y_i(t) = y_{0,i} \exp(\lambda_i t).$$

Exercise 8. ★*Another way to prove the previous result for diagonalizable matrix $A = PDP^{-1}$ is to use the fact that $\exp(tA) = P\exp(tD)P^{-1}$. Check this and complete the reasoning.*

Remark 2. *If matrices A and B commute then $\exp(A + B) = \exp(A)\exp(B)$, but this is not generally true.*

Exercise 9. ★★★*Let $A(t) = \begin{pmatrix} 1 & e^t \\ 0 & 0 \end{pmatrix}$, and let us consider the problem:*

$$\begin{cases} \dot{x}(t) = A(t)(x(t)) \text{ for } t \geq 0 \\ x(0) = x_0. \end{cases}$$

Compute $x(t)$ for general $x(0)$ and compare with $\exp\big(\int_0^t A(s)\,ds\big)x(0)$. Comment.

2.1.3 Second order linear case

Another standard example is the second order equation with constant coefficients ($a \neq 0$):

$$\begin{cases} a\ddot{x}(t) + b\dot{x}(t) + cx(t) = 0 \text{ for } t \in \mathbb{R} \\ (x(0), \dot{x}(0)) = (x_0, \dot{x}_0). \end{cases}$$

Let us first recall that this is indeed a problem of the form (2.1) by introducing $X(t) = (x(t), \dot{x}(t)) \in \mathbb{R}^2$, and actually of the form (2.2): we thus expect the ensemble of solutions for varying initial conditions to be to a linear space of dimension 2 (see Exercise 7).

By looking for solution of the form $x(t) = \exp(\lambda t)$, one gets the characteristic equation

$$a\lambda^2 + b\lambda + c = 0$$

and therefore, by introducing $\Delta = b^2 - 4ac$:

- if $\Delta > 0$, $x(t) = c_1 \exp(r_+ t) + c_2 \exp(r_- t)$, where $r_\pm = \frac{-b \pm \sqrt{\Delta}}{2a}$
- if $\Delta = 0$, $x(t) = (c_1 + c_2\, t)\exp(rt)$ where $r = \frac{-b}{2a}$
- if $\Delta < 0$, $x(t) = \exp(\alpha t)(c_1 \cos(\beta t) + c_2 \sin(\beta t))$ where $\alpha = \frac{-b}{2a}$ and $\beta = \frac{\sqrt{-\Delta}}{2a}$.

In all these cases, the two constants c_1 and c_2 are determined by the initial conditions $(x(0), \dot{x}(0)) = (x_0, \dot{x}_0)$.

Exercise 10. ★★★*Revisit this case by considering it as a 2 dimensional problem on $X(t) = (x(t), \dot{x}(t)) \in \mathbb{R}^2$, using the results of Section 2.1.2.*

2.1.4 Non existence, non uniqueness

Let us now give two examples to illustrate the fact that there may be no global solutions, or infinitely many solutions.

Let us consider the problem

$$\begin{cases} \dot{x}(t) = x^2(t) \text{ for } t \geq 0 \\ x(0) = x_0 \end{cases}$$

for some positive initial condition $x_0 > 0$.

If this problem admits a solution, then x is increasing, and thus $x(t) \geq x_0 > 0$. Therefore, one has $-\frac{d}{dt}\left(\frac{1}{x}\right) = 1$ which yields $-\frac{1}{x(t)} = -\frac{1}{x_0} + t$ and thus

$$x(t) = \frac{1}{1/x_0 - t}.$$

This solution exists on the time interval $[0, t_0)$ where $t_0 = 1/x_0$ is the so-called explosion time.

Exercise 11. ★*Check that if $x_0 > 0$, the solution is actually defined on $(-\infty, 1/x_0)$. What happens if $x_0 < 0$?*

Let us finally consider the problem

$$\begin{cases} \dot{x}(t) = \sqrt{|x(t)|} \text{ for } t \geq 0 \\ x(0) = 0. \end{cases}$$

This problem admits infinitely many solutions. Indeed, for any $t_0 \geq 0$, the function

$$x(t) = \begin{cases} 0 \text{ for } t \in [0, t_0] \\ \dfrac{(t - t_0)^2}{4} \text{ for } t > t_0 \end{cases}$$

is a solution (check it! ★).

2.2 The globally Lipschitz case

Here is a general existence and uniqueness result:

Theorem 2. *Let us assume that*

$$f : \mathbb{R}^d \to \mathbb{R}^d$$

is globally Lipschitz, i.e.

$$\exists L > 0, \forall x, y \in \mathbb{R}^d, \|f(x) - f(y)\| \leq L\|x - y\|.$$

Then, the problem (2.1) *admits a unique global solution (i.e. defined for all time $t \in \mathbb{R}$).*

Exercise 12. ★*Check that the two examples considered in Section 2.1.4 do not satisfy the assumptions of Theorem 2.*

2.2.1 Uniqueness

Let us prove the uniqueness part of Theorem 2. Let us assume that x_1 and x_2 are two solutions of (2.1). Let us define $e(t) = \|x_1(t) - x_2(t)\|^2$. Notice that $e(0) = 0$.

Then, for all $t \geq 0$,

$$\begin{aligned}\dot{e}(t) &= 2(x_1(t) - x_2(t)) \cdot (\dot{x_1}(t) - \dot{x_2}(t)) \\ &= 2(x_1(t) - x_2(t)) \cdot (f(x_1(t)) - f(x_2(t))) \\ &\leq 2\|x_1(t) - x_2(t)\|\|f(x_1(t)) - f(x_2(t))\| \\ &\leq 2L\|x_1(t) - x_2(t)\|^2 \\ &= 2Le(t).\end{aligned}$$

Therefore, by integration over the time interval $[0, t]$ (see Exercise 6), for all $t \geq 0$,

$$e(t) \leq e(0)\exp(2Lt) = 0.$$

This shows uniqueness of solutions on $\mathbb{R}_+$. The same argument can be used on $\mathbb{R}_-$ by considering $e(-t)$ for $t \geq 0$.

This concludes the proof of uniqueness.

2.2.2 Existence

This part can be skipped on first reading (another existence proof will be provided in Section 3.2.2).

The existence part of Theorem 2 can be proven using the Banach-Picard fixed point theorem and the fact that the space $\mathcal{C}^0([a, b], \mathbb{R}^d)$ (for $a < b$) endowed with the L^∞ norm is complete. The notion of completeness and these two facts will be proven in the lectures [4]. Let us state the result:

Theorem 3. *Let $a < b$ and let us consider $\mathcal{C}^0([a, b], \mathbb{R}^d)$ the space of continuous functions on the time interval $[a, b]$ with values in $\mathbb{R}^d$. Let us introduce the L^∞ norm defined by: for any $x \in \mathcal{C}^0([a, b], \mathbb{R}^d)$*

$$\|x\|_\infty = \max_{a \leq t \leq b} \|x(t)\|.$$

Let $T : \begin{cases} \mathcal{C}^0([a, b], \mathbb{R}^d) \to \mathcal{C}^0([a, b], \mathbb{R}^d) \\ x \mapsto T(x) \end{cases}$ be a contraction: $\exists k \in [0, 1)$, for all $x, y \in \mathcal{C}^0([a, b], \mathbb{R}^d)$,

$$\|T(x) - T(y)\|_\infty \leq k\|x - y\|_\infty.$$

Then, T admits a unique fixed point:

$$\exists! x \in \mathcal{C}^0([a, b], \mathbb{R}^d),\ T(x) = x.$$

Remark 3. *This theorem also holds in the simpler case where $\mathcal{C}^0([a, b], \mathbb{R}^d)$ is replaced by $\mathbb{R}^d$, in which case this is related to the celebrated Bolzano-Weierstrass theorem.*

In order to apply Theorem 3, let us introduce the integral form of (2.1). Let us consider the map

$$T : \begin{cases} \mathcal{C}^0([0, 1], \mathbb{R}^d) \to \mathcal{C}^0([0, 1], \mathbb{R}^d) \\ x \mapsto \left(t \mapsto x_0 + \displaystyle\int_0^t f(x(s))\, ds \right) \end{cases} \tag{2.3}$$

It is easy to check that x is a solution to (2.1) over the time interval $[0, 1]$ if and only if x is a fixed point of T (namely, $T(x) = x$). We work here on the time interval $[0, 1]$: the argument also holds on any time interval $[0, t_{max}]$ and we will show below that it allows to obtain finally a solution over the whole of $\mathbb{R}_+$.

The existence of a solution over the time interval $[0,1]$ is a consequence of the following lemma, and Theorem 3.

Lemma 4. *There exists $m > 1$ such that the map T^m is contraction, where T is defined by (2.3) and T^m denotes the map T composed with itself m times.*

Proof: For any x and y, one has, for all $t \geq 0$,

$$T(x)(t) - T(y)(t) = \int_0^t (f(x(s)) - f(y(s)))\, ds.$$

Therefore, for all $t \geq 0$,

$$\begin{aligned} \|T(x)(t) - T(y)(t)\| &= \left\| \int_0^t (f(x(s)) - f(y(s)))\, ds \right\| \\ &\leq \int_0^t \|f(x(s)) - f(y(s))\|\, ds \\ &\leq L \int_0^t \|x(s) - y(s)\|\, ds. \end{aligned}$$

By applying this inequality to $T(x)$ and $T(y)$, one gets for all $t \geq 0$,

$$\begin{aligned} \|T^2(x)(t) - T^2(y)(t)\| &\leq L \int_0^t \|Tx(s) - Ty(s)\|\, ds \\ &\leq L^2 \int_0^t \int_0^s \|x(r) - y(r)\|\, dr\, ds \end{aligned}$$

By induction, one thus gets that for any $m \geq 1$, for all $t \geq 0$,

$$\|T^m(x)(t) - T^m(y)(t)\| \leq L^m \int_0^t \int_0^{s_1} \int_0^{s_2} \dots \int_0^{s_{m-1}} \|x(r) - y(r)\|\, dr\, ds_{m-1} \dots ds_1.$$

This yields: for all $t \in [0,1]$,

$$\begin{aligned} \|T^m(x)(t) - T^m(y)(t)\| &\leq L^m \|x - y\|_{L^\infty} \int_0^t \int_0^{s_1} \int_0^{s_2} \dots \int_0^{s_{m-1}} dr\, ds_{m-1} \dots ds_1 \\ &= \frac{L^m t^m}{m!} \|x - y\|_\infty \\ &\leq \frac{L^m}{m!} \|x - y\|_\infty \end{aligned}$$

so that

$$\|T^m(x) - T^m(y)\|_\infty \leq \frac{L^m}{m!} \|x - y\|_\infty.$$

The proof is then complete by choosing m sufficiently large so that $\frac{L^m}{m!} < 1$. ♢

From this lemma and Theorem 3, we know that there exists a unique function $x \in \mathcal{C}^0([0,1], \mathbb{R}^d)$ such that $T^m(x) = x$. By applying the map T to this equality, one gets that $T^{m+1}(x) = T(x)$ so that $T(x)$ is also a fixed point of T^m, and thus (by uniqueness of the fixed point):

$$T(x) = x.$$

This result shows the existence of a solution over the time interval $[0,1]$. It is then easy to prove by induction the existence of a global solution over $\mathbb{R}$ by iterating the argument over $[1,2]$ (using as an initial condition the final value $x(1)$), and then over $[2,3]$, etc, and by finally completing the solution over $\mathbb{R}_-$ by noticing that $(x(-t))_{t\geq 0}$ also solves a similar ODE with again a Lipschitz right-hand side ($\frac{d}{dt}x(-t) = -f(x(-t))$).

Remark 4. *Notice that this proof actually provides also the uniqueness of the solution.*

2.2.3 Stability

Let us finally state a stability result.

Proposition 5 (Stability under perturbations)**.** *Let $f : \mathbb{R}^d \to \mathbb{R}^d$ be continuous and globally L-Lipschitz. Let $x : [0,T] \to \mathbb{R}^d$ be a $\mathcal{C}^1$ function such that*

$$\begin{cases} \dot{x} = f(x), \ \text{on } [0,T] \\ x(0) = x_0, \end{cases}$$

and let $y : [0,T] \to \mathbb{R}^d$ be a $\mathcal{C}^1$ function satisfying the perturbed problem

$$\begin{cases} \dot{y} = f(y) + \varepsilon, \ \text{on } [0,T] \\ y(0) = y_0, \end{cases}$$

where $\varepsilon : [0,T] \to \mathbb{R}^d$ is a continuous function. Then, for all $t \in [0,T]$,

$$\|x(t) - y(t)\| \leq e^{LT} \left(\|x_0 - y_0\| + \int_0^T \|\varepsilon(s)\| \, ds \right). \tag{2.4}$$

Note that the stability constant $\exp(LT)$ in (2.4) grows rapidly as T increases or when L is large. Therefore, this result is of limited use for understanding sensitivity to perturbations over long time horizons or for problems in which L is large. This issue will be addressed in Section 2.4.

The proof of this result is based on a very useful trick to analyze the growth of a solution to an ordinary differential equation: the Grönwall Lemma.

Lemma 6 (Grönwall's inequality)**.** *If $u \in \mathcal{C}^0([0,T],\mathbb{R})$ and for all $t \in [0,T]$,*

$$u(t) \leq \int_0^t a(s)\, u(s)\, ds + b(t)$$

with $a, b \in \mathcal{C}^0([0,T],\mathbb{R})$, $a \geq 0$, then for all $t \in [0,T]$,

$$u(t) \leq b(t) + \int_0^t b(s)\, a(s)\, \exp\left(\int_s^t a(r)\, dr \right) ds.$$

Proof: The key trick is to define $v(t) = \int_0^t a(s)\, u(s)\, \mathrm{d}s$. Then

$$\dot{v} = a\, u \leq a\, (v + b).$$

Hence

$$\frac{\mathrm{d}}{\mathrm{d}t} \left(e^{-\int_0^t a(r)\, \mathrm{d}r}\, v(t) \right) \ \leq \ e^{-\int_0^t a(r)\, \mathrm{d}r}\, a(t)\, b(t),$$

and integrating yields the desired bound. ♢

Exercise 13. ★★*Here is a discrete version of the Grönwall Lemma. Suppose a sequence* $(u_n)_{n\geq 0}$ *satisfies: for all* $n \geq 1$

$$u_n \leq \sum_{k=0}^{n-1} a_k\, u_k + b_n,$$

for some sequences $(a_n)_{n\geq 0}$ *and* $(b_n)_{n\geq 0}$ *with for all* $n \geq 0$, $a_n \geq 0$ *Show an upper bound on* u_n. Hint: Use induction.

Let us now prove Proposition 5 using similar ideas.

Proof: Let $w(t) = x(t) - y(t)$. One has $w(0) = x(0) - y(0)$ and $\dot{w}(t) = f(x(t)) - f(y(t)) - \varepsilon(t)$ and thus, for all $t \in [0, T]$,

$$w(t) = w(0) + \int_0^t f(x(s)) - f(y(s))\, ds - \int_0^t \varepsilon(s)\, ds.$$

By using the fact that f is L-Lipschitz, one gets: for all $t \in [0, T]$,

$$\|w(t)\| \leq \|w(0)\| + \int_0^t L\|w(s)\|\, ds + \int_0^t \|\varepsilon(s)\|\, ds.$$

Following the idea of the Grönwall Lemma, let us consider $\varphi(t) = \int_0^t L\|w(s)\|\, ds$. One has

$$\begin{aligned}
\dot{\varphi}(t) &= L\|w(t)\| \\
&\leq L\left(\|w(0\| + \varphi(t) + \int_0^t \|\varepsilon(s)\|\, ds\right) \\
&= L\varphi(t) + b(t)
\end{aligned}$$

where $b(t) = L\left(\|w(0)\| + \int_0^t \|\varepsilon(s)\|\, ds\right)$. Therefore $\frac{d}{dt}\left(\exp(-Lt)\varphi(t)\right) = \exp(-Lt)(\dot{\varphi}(t) - L\varphi(t)) \leq \exp(-Lt)b(t)$ which yields (since $\varphi(0) = 0$): for all $t \in [0, T]$,

$$\begin{aligned}
\varphi(t) &\leq \int_0^t \exp(L(t-s))b(s)\, ds \\
&= \left[-\frac{1}{L}\exp(L(t-s))b(s)\right]_0^t + \frac{1}{L}\int_0^t \exp(L(t-s))\dot{b}(s)\, ds \\
&= \frac{1}{L}\left(\exp(Lt)b(0) - b(t)\right) + \int_0^t \exp(L(t-s))\|\varepsilon(s)\|\, ds \\
&= \exp(Lt)\|w(0)\| - \frac{1}{L}b(t) + \int_0^t \exp(L(t-s))\|\varepsilon(s)\|\, ds.
\end{aligned}$$

Thus, for all $t \in [0, T]$,

$$\begin{aligned}
\|w(t)\| &\leq \|w(0)\| + \int_0^t L\|w(s)\|\, ds + \int_0^t \|\varepsilon(s)\|\, ds \\
&= \frac{1}{L}b(t) + \varphi(t) \\
&\leq \exp(Lt)\|w(0)\| + \int_0^t \exp(L(t-s))\|\varepsilon(s)\|\, ds \\
&\leq \exp(LT)\left(\|w(0)\| + \int_0^T \|\varepsilon(s)\|\, ds\right).
\end{aligned}$$

This concludes the proof. ♢

2.3 The locally Lipschitz case

It is actually possible to generalize the previous existence result to a locally Lipschitz function f. The function f can even by only defined on an open subset O of $\mathbb{R}^d$.

Theorem 7 (Cauchy–Lipschitz, Picard–Lindelöf)**.** *Let $O \subset \mathbb{R}^d$ be an open set, and assume $f : O \to \mathbb{R}^d$ is continuous and locally Lipschitz: for every $y \in O$, there exists a neighborhood[2] U of y where f is Lipschitz.*

Then, for any $y \in O$, there exists $\alpha > 0$ and a neighborhood U of y such that, for any initial condition $x_0 \in U$, the Cauchy problem

$$\begin{cases} \dot{x} = f(x) \\ x(0) = x_0 \end{cases} \tag{2.5}$$

admits a unique $\mathcal{C}^1$ solution $x : (-\alpha, +\alpha) \to O$.

This result is admitted, see [9, 10, 20] for a proof. Let us recall that any $\mathcal{C}^1$ function $f : O \to \mathbb{R}^d$ is locally Lipschitz (★prove it!).

Let us emphasize that the time span 2α is independent of $x_0 \in U$: this plays a crucial role in the proof of Theorem 9 below.

Of course, the same result holds by replacing $x(0) = x_0$ by $x(t_0) = x_0$ for any $t_0 \in \mathbb{R}$ (one can simply apply Theorem 7 to $t \to x(t_0 + t)$), in which case the solution is defined on the interval $(t_0 - \alpha, t_0 + \alpha)$.

Using this theorem, one can define a maximal solution as one that cannot be extended beyond its interval of definition (maximal solutions are well defined since if x and y are two solutions defined on $I \ni 0$ and $J \ni 0$ and such that $x(0) = y(0)$, then $x = y$ on $I \cap J$).

Corollary 8. *Under the assumptions of Theorem 7, there exists a unique maximal solution $x : J \to \mathbb{R}$, to the Cauchy problem (2.5) (with necessarily $0 \in J$), namely a solution defined on the interval J and which cannot by extended beyond J.*

Let us emphasize that this corollary does not give any information on the interval of definition J of the maximal solution. When this interval equals $\mathbb{R}$, the solution is said to be global. A key result to characterize global solution is the following.

Theorem 9. *Let $x : J \to O$ be a maximal solution of the Cauchy problem (2.5) on $J \subset \mathbb{R}$. If $\sup J < \infty$, then $x(t)$ leaves every compact subset of O as $t \to \sup J$: $\forall K$ compact set included in O, $\exists t^* \in J$ such that $\forall t \in (t^*, \sup J)$, $x(t) \notin K$.*

Likewise, if $\inf J > -\infty$, then $x(t)$ leaves every compact subset of O as $t \to \inf J$. If the maximal solution is not global, it therefore "blows up" on the boundary.

Proof: Let us consider the case $\sup J < \infty$ (the case $\inf J > -\infty$ is treated similarly). We argue by contradiction: otherwise, there exists an increasing sequence of times t_n converging to $\sup J$ such that $x(t_n) \in K$, and thus $x(t_n)$ converges to some $x^\infty \in K$ (possibly after extracting a subsequence). According to the Cauchy-Lipschitz theorem, there exists a solution to the Cauchy problem defined on an interval of width 2α, for any pair (t_0', x_0') in a neighborhood of $(\sup J, x^\infty)$ defining the initial condition. This therefore applies to $(t_m, x(t_m))$ if m is large enough, which then allows extending the solution over $(t_m, t_m + \alpha)$, and $t_m + \alpha$ will be greater than $\sup J$ for sufficiently large m, yielding a contradiction. ♢

[2] Let us recall that a neighborhood of y is an open set which contains y. In $\mathbb{R}^d$, one can think of an open ball centered at y, with radius $r > 0$.

For example, if f is locally Lipschitz on $O = \mathbb{R}^d$, and $\sup J < \infty$ (resp. $\inf J > -\infty$), then $\lim_{t\to\sup J} \|x(t)\| = \infty$ (resp. $\lim_{t\to\inf J} \|x(t)\| = \infty$).

As a corollary, if one can prove that a maximal solution remains bounded in a compact of O, then it is global in time. To obtain such boundedness one often uses a priori estimates such as the Grönwall's lemma 6. For example, if $f : \mathbb{R}^d \to \mathbb{R}^d$ grows at most linearly in x — i.e. there exist constants α, β such that

$$|f(t,x)| \leq \alpha\,|x| + \beta,$$

then any solution to the corresponding Cauchy problem is *global*.

Exercise 14. ★★*Prove this result.*

Exercise 15. ★★*Show that if the potential V is $\mathcal{C}^1$ and bounded below, then the Hamiltonian system*

$$\dot{q} = \partial_p H(q,p), \quad \dot{p} = -\partial_q H(q,p),$$

has a global solution. Hint: Energy ($\frac{1}{2}m\|v\|^2 + V(x)$) is conserved $\Longrightarrow$ velocity remains bounded $\Longrightarrow$ position stays bounded.

2.4 Study of the stability of equilibrium points

2.4.1 Phase portraits and notions of stability

Let us consider the Cauchy problem:

$$\begin{cases} \dot{x} = f(x), \\ x(0) = x_0, \end{cases} \tag{2.6}$$

where f is of class $\mathcal{C}^1$ on an open set $O \subset \mathbb{R}^d$. We know that for any initial condition $x_0 \in O$, there exist a maximal solution $x : J \to O$ defined on an interval $J \subset \mathbb{R}$ (notice that J may depend on x_0).

Definition 10. *Define the* flow $\phi(x_0,t)$ *as the solution $x(t)$ at time t of the Cauchy problem* (2.6) *with $x(0) = x_0$. The map*

$$\phi\colon (x_0,t) \mapsto x(t)$$

is continuous in (x_0,t) and is defined on an open set of $O \times \mathbb{R}$. It satisfies the flow property: for all x_0, for all s and t such that $s, s+t \in J$ where J is the time interval of the maximal solution,

$$\phi(\phi(x_0,s),t) = \phi(x_0,s+t). \tag{2.7}$$

The continuity on x_0 is a consequence of Proposition 5.

Remark 5. *If f is $\mathcal{C}^r$ ($r \geq 1$), it can be shown that $(x_0,t) \mapsto \phi(x_0,t)$ is also $\mathcal{C}^r$.*

Exercise 16. ★★

1. *Let us consider the linear case $\dot{x} = Ax$ for $A \in \mathbb{R}^{d\times d}$. Show that $\phi(x_0,t) = \exp(At)x_0$.*
2. *Let us now consider the problem:*

$$\begin{cases} \dot{x} = Ax + b(t) \text{ for } t \geq 0, \\ x(0) = x_0 \end{cases}$$

for some continuous function $b : \mathbb{R}_+ \to \mathbb{R}$. Show that

$$x(t) = \exp(tA)x_0 + \int_0^t \exp((t-s)A)b(s)\,ds.$$

This is the so-called Duhamel formula.

Exercise 17 (Expansion and contraction of volumes by a linear flow)**.** ★★*Let $A \in \mathbb{R}^{d\times d}$ and consider the linear vector field $f(x) = Ax$. Let ϕ be the flow associated with f, as introduced in Definition 10.*

1. *Recall that*
$$\phi(x_0, t) = \exp(tA)x_0.$$
Show that the Jacobian matrix
$$J(t) = \nabla_{x_0}\phi(x_0, t)$$
does not depend on x_0 and satisfies
$$J(t) = \exp(tA), \qquad \dot{J}(t) = AJ(t), \qquad J(0) = I.$$

2. *Set $j(t) = \det J(t)$. Using Jacobi's formula*
$$\frac{d}{dt}\det J(t) = \det J(t)\,\mathrm{tr}\big(J(t)^{-1}\dot{J}(t)\big),$$
show that
$$\frac{dj}{dt}(t) = \mathrm{tr}(A)j(t), \qquad j(0) = 1,$$
and deduce that
$$\det\big(\exp(tA)\big) = \exp\big(t\,\mathrm{tr}(A)\big).$$
Check that $\mathrm{div}\, f = \mathrm{tr}(A)$.

3. *Let B be a bounded measurable subset of $\mathbb{R}^d$. Using the change-of-variables formula, show that*
$$|\phi(B, t)| = \exp\big(t\mathrm{tr}(A)\big)|B|.$$
Discuss whether the flow preserves, expands, or contracts volumes according to the sign of $\mathrm{tr}(A)$.

4. *First consider $A = \lambda I$. Compute the flow and its Jacobian determinant directly, and check the preceding formula. Then consider, in dimension two,*
$$A = \begin{pmatrix} 1 & 0 \\ 0 & -2 \end{pmatrix}.$$
Show that this flow contracts areas although it expands distances in one direction. Explain why a negative trace describes contraction of volumes, but not necessarily contraction of all distances.

Exercise 18 (Expansion and contraction of volumes by a nonlinear flow)**.** ★★★*Let $f \in \mathcal{C}^1(O, \mathbb{R}^d)$ be a general, not necessarily linear, vector field, and let ϕ be the flow introduced in Definition 10. For $x_0 \in O$ and every time for which the flow is defined, set*
$$A(x_0, t) = \nabla_{x_0}\phi(x_0, t), \qquad j(x_0, t) = \det A(x_0, t).$$
The $d \times d$ matrix $A(x_0, t)$ is called the Jacobian of the function ϕ; its (i, j)-th component is $\frac{\partial \phi_i}{\partial x_{0,j}}(x_0, t)$. The quantity $j(x_0, t)$ measures the local change of volume induced by the flow (see Question 4 below).

1. *By differentiating the equation*

$$\partial_t \phi(x_0, t) = f(\phi(x_0, t))$$

with respect to x_0, show that

$$\partial_t A(x_0, t) = Df(\phi(x_0, t))A(x_0, t), \qquad A(x_0, 0) = I. \tag{2.8}$$

2. *The flow property* (2.7) *shows that $A(x_0, t)$ is invertible. Using Jacobi's formula*

$$\frac{d}{dt} \det M(t) = \det M(t) \operatorname{tr}\big(M(t)^{-1}\dot{M}(t)\big),$$

derive the following scalar ordinary differential equation:

$$\partial_t j(x_0, t) = \operatorname{div} f(\phi(x_0, t))j(x_0, t), \qquad j(x_0, 0) = 1. \tag{2.9}$$

3. *Deduce that*

$$j(x_0, t) = \exp\left(\int_0^t \operatorname{div} f(\phi(x_0, s))\, ds\right). \tag{2.10}$$

In particular, show that $j(x_0, t) > 0$.

4. *Let B be a bounded measurable subset of O on which the flow is defined up to time t. Using the change-of-variables formula, show that*

$$|\phi(B, t)| = \int_B j(x_0, t)\, dx_0.$$

Discuss the following three cases:

$$\operatorname{div} f = 0, \qquad \operatorname{div} f > 0, \qquad \operatorname{div} f < 0.$$

More precisely, what can be said if $\operatorname{div} f \leq -\alpha$ *or* $\operatorname{div} f \geq \alpha$ *for some $\alpha > 0$?*

5. *A collection of particles has initial mass density ρ_0. If the particles move according to the flow and mass is conserved, explain why their density ρ at time t satisfies*

$$\rho(t, \phi(x_0, t))j(x_0, t) = \rho_0(x_0).$$

Interpret this identity in the cases $\operatorname{div} f > 0$ *and* $\operatorname{div} f < 0$.

Definition 11. *We call the orbit of x_0 the set of $x(t)$ for t belonging to the time interval of the maximal solution:*

$$\mathcal{O}(x_0) = \{\phi(x_0, t),\, t \in J\}.$$

Note that by the Cauchy-Lipschitz theorem, since f is $\mathcal{C}^1$ on O, orbits cannot intersect. The set of orbits partitioning O is called the phase portrait. In practice, one draws some notable orbits in the phase portrait, along with the direction of the vector field f in certain regions. The phase portrait can be plotted without explicitly solving the equation, providing important qualitative insights into the behavior of solutions (see for example Section 2.4.5).

Exercise 19. ★★*Let us revisit the Lotka-Volterra equations from Section 1.2.2. Why does $N(t) > 0$ and $P(t) > 0$ for all $t \geq 0$ if $N(0) > 0$ and $P(0) > 0$? Using the fact that $H(N, P) = cN - d\ln N + bP - a\ln P$ is constant along trajectories, sketch the phase portrait for the Lotka-Volterra equation. Deduce that the solutions are time-periodic.*

Definition 12. *We call an equilibrium point, or fixed point, or stationary point, any point x^* such that $f(x^*) = 0$.*

Note that if x^* is an equilibrium point, then the orbit of x^* is reduced to the point itself: $\mathcal{O}(x^*) = \{x^*\}$.

In the linear case $f(x) = Ax$, the equilibrium points are the x such that $Ax = 0$ (so 0 is always an equilibrium point).

Definition 13. *A fixed point x^* is said to be stable if for every $\varepsilon > 0$, there exists $\delta > 0$ such that, for all x satisfying $\|x - x^*\| \leq \delta$ and all $t \geq 0$, we have $\|\phi(x,t) - x^*\| \leq \varepsilon$.*

In particular, if x^* is a stable fixed point and x is sufficiently close to x^*, the function $\phi(x,t)$ is defined for all $t \geq 0$ (as it remains in a compact set). This is sometimes called Lyapunov stability.

Exercise 20. ★*Discuss the stability of the equilibrium point $(0,0)$ for the following two systems:*
$\begin{cases} \dot{x} = -y \\ \dot{y} = x \end{cases}$ $\begin{cases} \dot{x} = -x \\ \dot{y} = y \end{cases}$

Definition 14. *A stationary point x^* is said to be asymptotically stable if it is stable and there exists a neighborhood of x^* such that for all x in this neighborhood, $\lim_{t\to\infty} \phi(x,t) = x^*$.*

Exercise 21 shows that the condition "there exists a neighborhood of x^* such that for all x in this neighborhood, $\lim_{t\to\infty} \phi(x,t) = x^*$" does not imply stability.

Note that in these notions of stability, the direction of time matters (we consider behavior for large positive times).

These notions of stability are different from the stability mentioned in Proposition 5: there are associated with a stationary point x^*.

Exercise 21. ★★★*Consider the problem:*

$$\begin{cases} \dot{x} = x - y - x(x^2+y^2) + \frac{xy}{\sqrt{x^2+y^2}}, \\ \dot{y} = x + y - y(x^2+y^2) - \frac{x}{\sqrt{x^2+y^2}}. \end{cases}$$

Show that the point $x^* = (1,0)$ is not stable, even though there exists a neighborhood of x^* such that for every x in this neighborhood, $\lim_{t\to\infty} \phi(x,t) = x^*$. (You may rewrite the system in polar coordinates as $\dot{r} = r(1-r^2)$, $\dot{\theta} = 2\sin^2(\theta/2)$.)

Exercise 22. ★★*The long-term qualitative behavior of solutions to an ordinary differential equation may change dramatically under small parameter variations (this is the subject of bifurcation theory). As an example, verify the following assertions:*

- *The problem*
$$\dot{x} = \mu x - x^3$$
has a unique stable stationary point for $\mu \leq 0$, which becomes unstable for $\mu > 0$, while two new stable equilibrium points appear for $\mu > 0$.

- *The problem*
$$\dot{x} = \mu + x^2$$
has two stationary points for $\mu < 0$, one stable and one unstable, which coalesce at $\mu = 0$ and disappear for $\mu > 0$.

It is very important in practice to study the stability of equilibrium points. Indeed, in many models, equilibrium points correspond to "operating points" of the system, and knowing whether the system stays close to these points under small perturbations is crucial for system reliability.

One method to study the stability of a stationary point is to linearize the ordinary differential equation around the stationary point, and then to analyze the associated linear problem. This is explained in the forthcoming two sections, by first considering the linear case in Section 2.4.2, and then present the linear stability analysis in Section 2.4.3).

2.4.2 The linear case

We consider the case $f(x) = Ax$, and study the stability of the stationary point $x^* = 0$.

Theorem 15. *The point $x^* = 0$ is asymptotically stable if and only if all the eigenvalues of A have strictly negative real parts.*

Proof: Let us recall that in the linear case, $x(t) = \exp(At)x_0$, and asymptotic stability is thus equivalent to $\lim_{t\to\infty} \exp(At) = 0$.

Let us first consider the case when A is diagonalizable in $\mathbb{C}$: $A = PDP^{-1}$ where $D = \mathrm{diag}(\lambda_1, \ldots, \lambda_d)$ is a diagonal matrix and $P \in \mathbb{C}^{d\times d}$ is invertible. Then, $\exp(tA) = P\exp(tD)P^{-1}$ and

$$\exp(tD) = \mathrm{diag}(\exp(t\lambda_1), \ldots, \exp(t\lambda_d)).$$

The result then follows from the fact that, for $\lambda \in \mathbb{C}$,

$$\lim_{t\to\infty} \exp(\lambda t) = 0 \iff \mathrm{Re}(\lambda) < 0.$$

The proof of the general case can be skipped on first reading. It relies on the so-called Jordan decomposition of a matrix. For any matrix $A \in \mathbb{C}^{d\times d}$, there exists an invertible matrix $P \in \mathbb{C}^{d\times d}$ such that

$$A = PJP^{-1},$$

where $J \in \mathbb{C}^{d\times d}$ is the co-called Jordan normal form of A, namely, a block-diagonal matrix consisting of Jordan blocks:

$$J = \begin{bmatrix} J_{m_1}(\lambda_1) & & 0 \\ & \ddots & \\ 0 & & J_{m_p}(\lambda_p) \end{bmatrix}, \quad J_{m_i}(\lambda_i) = \begin{bmatrix} \lambda_i & 1 & & 0 \\ & \lambda_i & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda_i \end{bmatrix}.$$

Each Jordan block $J_{m_i}(\lambda_i) \in \mathbb{C}^{m_i\times m_i}$ corresponds to an eigenvalue λ_i of A. All the Jordan blocks associated with an eigenvalue λ have size 1 if and only if $\mathrm{Ker}(A - \lambda\mathrm{Id}) = \mathrm{Ker}\left((A - \lambda\mathrm{Id})^m\right)$ where m if the algebraic multiplicity of λ (i.e. the exponent of $(x - \lambda)$ in the characteristic polynomial $\det(A - \lambda\mathrm{Id})$ of A): the eigenspace equals the generalized eigenspace.

Using the Jordan decomposition, we obtain $\exp(tA) = P\exp(tJ)P^{-1}$ and

$$\exp(tJ) = \begin{bmatrix} \exp(tJ_{k_1}(\lambda_1)) & & 0 \\ & \ddots & \\ 0 & & \exp(tJ_{k_p}(\lambda_p)) \end{bmatrix},$$

with

$$\exp\left(tJ_{k_i}(\lambda_i)\right)) = \exp(\lambda_i t) \begin{bmatrix} 1 & t & ... & \frac{t^{m_i-1}}{(m_i-1)!} \\ 0 & 1 & t & ... \\ 0 & ...0 & 1 & t \\ 0 & ... & 0 & 1 \end{bmatrix}.$$

The conclusion thus follows from the fact that for any integer k,

$$\lim_{t\to\infty} t^k \exp(\lambda t) = 0 \iff \mathrm{Re}(\lambda) < 0.$$

♢

Exercise 23. ★★★*Repeat this proof to show that the point $x^* = 0$ is stable if and only if the eigenvalues of A have real parts that are non-positive, and that for those with zero real part, the eigenspace equals the generalized eigenspace.*

Example 1 (Phase portraits in dimension 2)**.** *Consider the linear Cauchy problem*

$$\dot{x} = Ax$$

with x of dimension 2 and $A \in \mathbb{R}^{2\times 2}$. Depending on the spectrum of A, the critical point $(0,0)$ can be (see Figure 2.1):

- *A saddle: A has two real eigenvalues with opposite signs,*
- *A generic node (attractive ore repulsive): A has two distinct real eigenvalues with the same sign,*
- *A proper node or star (attractive ore repulsive): A has one real eigenvalue and is diagonalizable (it is thus proportional to* Id*),*
- *An improper node (attractive ore repulsive): A has one real eigenvalue and is not diagonalizable,*
- *A spiral or focus (attractive ore repulsive): A has two complex conjugate eigenvalues with non-zero real parts,*
- *A center : A has two purely imaginary complex conjugate eigenvalues.*

2.4.3 Linear stability analysis

Let us now consider the case of a non-linear function f, with regularity $\mathcal{C}^1$. Let x^* be a stationary point: $f(x^*) = 0$. By Taylor expansion, in the neighborhood of the equilibrium point x^*, the vector field f can be approximated as

$$f(x) = Df(x^*)(x - x^*) + o(\|x - x^*\|)$$

where $Df(x^*)$ is the Jacobian matrix defined by: for all $1 \le i, j \le d$,

$$[Df(x^*)]_{i,j} = \frac{\partial f_i}{\partial x_j}(x^*). \tag{2.11}$$

It is thus natural to expect that the stability of x^* can be analyzed in terms of the spectrum of $Df(x^*)$: this leads to the so-called linear stability analysis.

One has the following result.

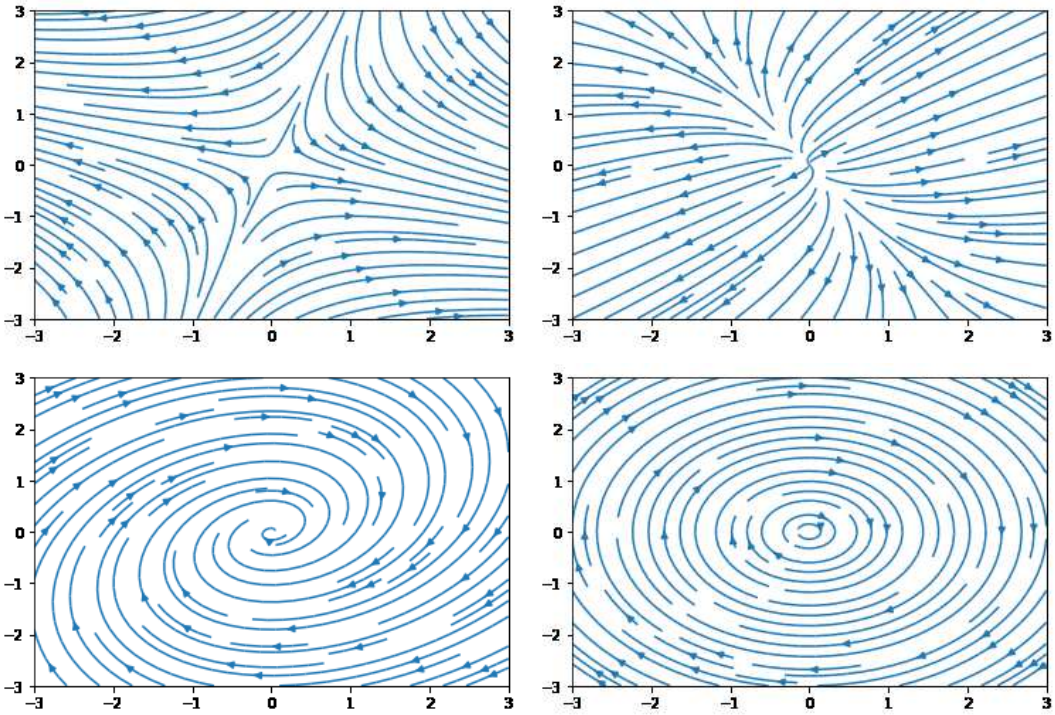


Figure 2.1: Phase portraits in dimension 2. From up to bottom and from left to right: Saddle point $A = \begin{pmatrix} 2 & -1 \\ -1 & -1 \end{pmatrix}$, Repulsive improper node $A = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}$, Attractive spiral $A = \begin{pmatrix} -1 & 2 \\ -2 & 0 \end{pmatrix}$, Center $A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

Theorem 16. *If all the eigenvalues of the matrix $Df(x^*)$ have strictly negative real parts, then the equilibrium point is asymptotically stable. Moreover, if one of the eigenvalues has a strictly positive real part, then the equilibrium point is unstable.*

The proof of this theorem is given in the next section.

Warning: If one of the eigenvalues has zero real part, the linearization method does not allow us to conclude about the stability of the stationary point. As an example, consider the stability of the stationary point $x^* = 0$ for the problem $\dot{x} = \lambda x^3$, depending on the sign of λ.

Remark 6. *This method of analyzing stability using the spectrum of the linearized problem is specific to finite dimensions: if we consider a problem of the type $\partial_t u = A(u)$ with A a nonlinear operator acting on a function u (u is a function of time and a spatial variable, and A is, for example, a differential operator), and u^* a steady state of the operator A ($A(u^*) = 0$), then the fact that the eigenvalues of the differential of the operator at u^* have strictly negative real parts does not, in general, imply that u^* is asymptotically stable. Results of this type in infinite dimensions typically rely on Lyapunov function methods, as presented in Section 2.4.4. Lyapunov functions are sometimes referred to as energy or entropy functions, as they often correspond to physical quantities.*

2.4.4 Proof of Theorem 16 by a Lyapunov function method

This section can be skipped on first reading.

To prove Theorem 16, one can use a Lyapunov function method.

Definition 17. *A Lyapunov function L for the ordinary differential equation (2.6) at an equilibrium point x^* is:*

- *a* continuous *function $L : V \to \mathbb{R}$, defined on a neighborhood V of x^*,*
- *such that x^* is a* strict minimum *of L on V (we may assume without loss of generality that the minimum value is 0): $L(x^*) = 0$ and $L(x) > 0$ if $x \neq x^*$,*
- *and such that for all $x \in V$ with $x \neq x^*$, the function $t \mapsto L(\phi(x,t))$ is* strictly decreasing *(on an open interval around 0 on which the flow is defined).*

To verify the third property, if L is of class $\mathcal{C}^1$, it suffices to check that $\nabla L(x) \cdot f(x) < 0$ for all $x \neq x^*$.

Theorem 18. *If the system admits a Lyapunov function in a neighborhood of the equilibrium point x^*, then this point is asymptotically stable.*

Proof: We can assume that V is relatively compact (i.e., its closure is compact), possibly after shrinking V. Let B be a ball centered at x^* contained in V, and let $\varepsilon > 0$ be the minimum of L on $V \setminus B$. Clearly, $L^{-1}[0, \varepsilon/2]$ is a compact set contained in B. (In other words, we can assume that the neighborhood V in Definition 17 is actually of the form $L^{-1}[0, \eta)$ for some $\eta > 0$.) Using this procedure, stability is easily proved (to be checked as an exercise). It remains to prove asymptotic stability.

Let $x \in L^{-1}[0, \varepsilon/2]$. Since we know that $t \mapsto L(\phi(x,t))$ is strictly decreasing, $\phi(x,t)$ remains in the compact set $L^{-1}[0, \varepsilon/2]$. In particular, the flow $\phi(x,t)$ is defined for all $t \in [0, \infty)$. Since $L(\phi(x,t)) \geq 0$, it then admits a limit a as $t \to \infty$. Moreover, by compactness,

we can construct a sequence $t_n \to \infty$ such that $\phi(x, t_n)$ converges to some x_∞. Clearly, $L(x_\infty) = a$.

Suppose $x_\infty \neq x^*$. Then for some $t > 0$, $L(\phi(x_\infty, t)) < a$. But $L(\phi(x_\infty, t)) = \lim_{n\to\infty} L(\phi(\phi(x, t_n), t)) = \lim_{n\to\infty} L(\phi(x, t + t_n,)) = a$. This is a contradiction. Therefore, $x_\infty = x^*$, and the point x^* is indeed asymptotically stable. ♢

Exercise 24. ★★*Let us consider a Lyapunov functions L such that for all $x \in V$ with $x \neq x^*$, the function $t \mapsto L(\phi(x, t))$ is decreasing (but not strictly decreasing) on an open interval around 0 where the function is defined. Revisit the proof to show that the equilibrium point is stable (but not asymptotically stable).*

We are now in position to prove Theorem 16.

Proof: Up to a change of variables $x \to x - x^*$, we may assume $x^* = 0$. The proof consists in exhibiting a Lyapunov function.

We begin by writing a Taylor expansion of f around $x^* = 0$:

$$f(x) = Df(0)x + |x|\epsilon(x),$$

where ϵ is a function defined on a neighborhood of 0 such that $\lim_{x\to 0} |\epsilon(x)| = 0$. In the sequel, we write

$$A = Df(0),$$

and $|\cdot|$ denotes the Euclidean norm for vectors, or the corresponding operator norm for matrices. We introduce the *linearized system*:

$$\dot{y} = Ay.$$

From the results in Section 2.4.2, we know that if all eigenvalues of the matrix $A = Df(0)$ have strictly negative real parts, then the equilibrium point 0 is asymptotically stable for the linearized system. Therefore, we would like to show that the nonlinear system behaves like the linearized one near the equilibrium point.

To this end, we introduce the Lyapunov function:

$$L(x) = \int_0^\infty |\exp(tA)x|^2 \, dt.$$

Note that L is well-defined because the eigenvalues of A are assumed to have strictly negative real parts, which implies that there exist constants $\alpha, c > 0$ such that

$$|\exp(tA)| \leq c \exp(-\alpha t).$$

It is clear that 0 is a strict minimum of L in a neighborhood of 0. We now check that $\nabla L \cdot f < 0$ outside $x^* = 0$:

$$\begin{aligned}
\nabla L(x) \cdot f(x) &= \int_0^\infty 2 \exp(tA)x \cdot (\exp(tA) f(x)) \, dt, \\
&= \int_0^\infty 2 \exp(tA)x \cdot (\exp(tA) Ax) \, dt + \int_0^\infty 2 \exp(tA)x \cdot (\exp(tA)|x|\epsilon(x)) \, dt, \\
&\leq \int_0^\infty \frac{d}{dt} |\exp(tA)x|^2 \, dt + \int_0^\infty 2 |\exp(tA)|^2 |x|^2 |\epsilon(x)| \, dt, \\
&\leq -|x|^2 + C|x|^2|\epsilon(x)|.
\end{aligned}$$

This shows that there exists a neighborhood of 0 such that $\nabla L \cdot f < 0$ outside of 0. Therefore, L is a Lyapunov function, and the point $x^* = 0$ is asymptotically stable by Theorem 18. $\diamondsuit$

Exercise 25. ★★★*Adapt the previous proof to verify that if one of the eigenvalues has strictly positive real part, then the equilibrium point is unstable.*

2.4.5 Example: the pendulum

Pendulum without Friction. We consider the frictionless pendulum problem:

$$\begin{cases} \dot{x} = v, \\ \dot{v} = -\sin x, \end{cases} \tag{2.12}$$

where $x \in 2\pi\mathbb{R}/\mathbb{Z}$ and $v \in \mathbb{R}$.

This system is a Hamiltonian system, and therefore admits a first integral:

$$E(x, v) = \frac{v^2}{2} - \cos x.$$

The fact that this quantity is conserved along trajectories implies that the solutions are bounded and thus defined for all times $t \in \mathbb{R}$.

To plot the phase portrait in the (x, v) plane, we start by drawing the level curves of the function E:

$$C_\alpha = \{(x, v) \mid E(x, v) = \alpha\}.$$

- For $\alpha > 1$, the level curves correspond to trajectories where the pendulum performs a complete rotation.
- For $-1 < \alpha < 1$, the level curves correspond to trajectories where the pendulum oscillates between two maximum angles.
- For $\alpha = -1$, the level curves reduce to the points $(2k\pi, 0)$.

The case $\alpha = 1$ is very particular. Note that if $(x, v) \in C_1 \cap \{v > 0\}$, then

$$v = \sqrt{2(1 + \cos(x))}.$$

From this, we deduce that for a trajectory such that $x(0) \in C_1 \cap \{v > 0\}$, we have:

$$t = \int_{x(0)}^{x(t)} \frac{1}{\sqrt{2(1 + \cos(u))}}\, du.$$

Indeed, for such a trajectory, $\frac{dx}{dt} = v(t) = \sqrt{2(1 + \cos(x(t)))}$. Letting $t \to +\infty$ in the equation above, we see that necessarily $\lim_{t\to\infty} x(t) = \pi$. Similarly, we can show that $\lim_{t\to-\infty} x(t) = -\pi$. This trajectory thus connects the points $(-\pi, 0)$ and $(\pi, 0)$ in infinite time. Such a trajectory that connects two different equilibrium points is called a heteroclinic trajectory.

Exercise 26. ★*Think about what kind of pendulum motion corresponds to this trajectory.*

By combining these results, we obtain the phase portrait shown in Figure 2.2.
The equilibrium points of the system are given by $x = k\pi$ ($k \in \mathbb{Z}$) and $v = 0$.

- The points $(x, v) = (2k\pi, 0)$ are stable. This can be verified by noting that trajectories remain in a neighborhood of $(2k\pi, 0)$, and that orbits do not intersect (it can also be proven using the Lyapunov function method from Exercise 24, using E minus its minimum as a Lyapounov function).

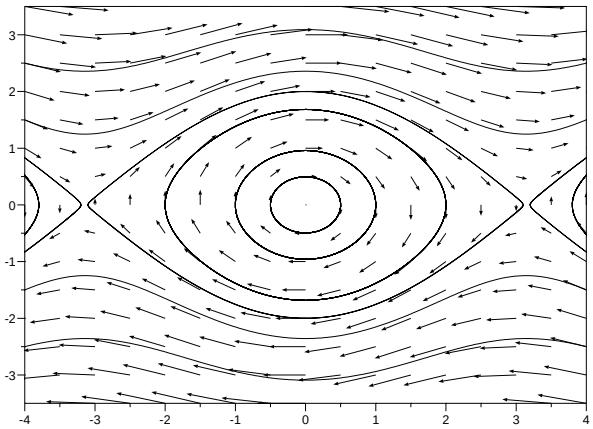


Figure 2.2: Phase portrait for the pendulum without friction. The horizontal axis represents the angle x, and the vertical axis represents the velocity v.

- The points $(x, v) = (\pi + 2k\pi, 0)$ are unstable. This can be seen in the phase portrait using the trajectories contained in C_1 analyzed above. These trajectories connect these equilibrium points, which implies instability.

We can also use a linear stability criterion. The Jacobian matrix of the system at (x, v) is:

$$\begin{bmatrix} 0 & 1 \\ -\cos x & 0 \end{bmatrix}.$$

At the point $(\pi + 2k\pi, 0)$, this becomes:

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix},$$

which has a strictly positive eigenvalue, indicating instability (see Exercise 25). The linear stability criterion cannot by used at the point $(x, v) = (2k\pi, 0)$ (since the Jacobian matrix has null eigenvalues).

Exercise 27. ★*Try applying the linear stability criterion to prove the stability of the points* $(x, v) = (2k\pi, 0)$. *Interpret the results in terms of the pendulum's physical behavior.*

Pendulum with Friction. We now consider the pendulum with friction:

$$\begin{cases} \dot{x} = v, \\ \dot{v} = -\sin x - \lambda v, \end{cases}$$

with $\lambda > 0$. Some typical trajectories in phase space are shown in Figure 2.3.

Exercise 28. ★*Interpret the trajectories shown in Figure 2.3 in terms of pendulum motion.*

The equilibrium points of the dynamics remain $(x, v) = (k\pi, 0)$ $(k \in \mathbb{Z})$.

The points $(2k\pi, 0)$ are now asymptotically stable. Indeed, along a trajectory, one has $\frac{d}{dt}E((x(t), v(t)) = -\lambda v^2(t)$, and since the times where $v = 0$ are isolated, this is a strictly decreasing function. This implies that if $(x(0), v(0))$ starts in a neighborhood of one of the points $(2k\pi, 0)$, it will converge to this point (see Theorem 18).

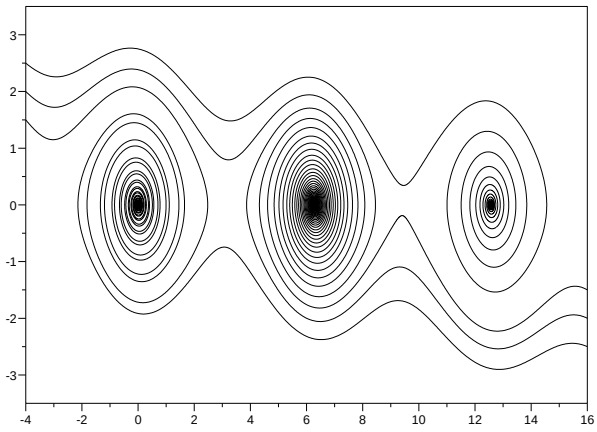


Figure 2.3: Some trajectories $(x(t), v(t))_{t\in\mathbb{R}}$ for the pendulum with friction.

The points $(\pi + 2k\pi, 0)$ remain unstable. This can be shown by examining the spectrum of the Jacobian at these points:

$$\begin{bmatrix} 0 & 1 \\ 1 & -\lambda \end{bmatrix},$$

which has a strictly positive eigenvalue, confirming instability.

Chapter 3

Numerical Approximations

We aim to compute an approximation of the solution x to the Cauchy problem

$$\begin{cases} \dot{x} = f(x), \\ x(0) = x_0. \end{cases}$$

where $f : \mathbb{R}^d \to \mathbb{R}^d$ is assumed to be L-Lipschitz continuous:

$$f \text{ is } L\text{-Lipschitz.} \tag{3.1}$$

In particular there exist a unique global solution $x : \mathbb{R} \to \mathbb{R}^d$. Our objective in Section 3.1 is to devise numerical methods to approximate this solution for $t \geq 0$. We consider a uniform time grid with time step Δt, we set $t_n = n\Delta t$ and we study numerical schemes that construct an approximation $x_n \approx x(t_n)$.

We aim to answer two types of questions:

- For fixed time horizon T, what can be said about the exact solution $(x(t))_{0 \leq t \leq T}$ versus the approximate solution $(x_n)_{0 \leq t \leq \lfloor T/\Delta t \rfloor}$ as $\Delta t \to 0$? This is a question of convergence, addressed in Section 3.2.
- For fixed Δt, has the approximate solution x_n as $n \to \infty$ a behaviour similar to the exact solution $x(t)$ as $t \to \infty$? This relates to the absolute stability of the scheme, discussed in Section 3.3.

3.1 Numerical schemes

We want to approximate the solution of the Cauchy problem on the time interval $[0, T]$, and we define $N(\Delta t) = T/\Delta t$ as the final time step. We assume Δt is such that $N(\Delta t) \in \mathbb{N} \setminus \{0\}$.

We restrict ourselves to one-step methods: x_{n+1} depends only on x_n:

$$x_{n+1} = x_n + \Delta t \, \Psi_{\Delta t}(x_n).$$

The increment function $\Psi_{\Delta t}$ of course depends on f. The function $\Phi_{\Delta t}(x) = x + \Delta t \, \Psi_{\Delta t}(x)$ is the "numerical flow" over a time interval of size Δt. It approximates the flow $\phi(\cdot, \Delta t)$ of the ordinary differential equation (see Definition 10).

Remark 7. *Remember that $\phi(\phi(x, \Delta t), \Delta t) = \phi(x, 2\Delta t)$. On the contrary, in general, $\Phi_{2\Delta t}(x) \neq \Phi_{\Delta t}(\Phi_{\Delta t}(x))$ (the difference actually gives information about the scheme's error).*

Here are a few examples:

- Explicit Euler method:
$$x_{n+1} = x_n + \Delta t\, f(x_n)$$
- Implicit Euler method:
$$x_{n+1} = x_n + \Delta t\, f(x_{n+1}) \tag{3.2}$$
- Trapezoidal method (also called Crank-Nicolson method):
$$x_{n+1} = x_n + \frac{\Delta t}{2}\left(f(x_n) + f(x_{n+1})\right)$$
- Midpoint method:
$$x_{n+1} = x_n + \Delta t\, f\left(\frac{x_n + x_{n+1}}{2}\right).$$
- Heun's method:
$$x_{n+1} = x_n + \frac{\Delta t}{2}\left(f(x_n) + f(x_n + \Delta t f(x_n))\right)$$

The idea to derive these scheme is to write the ordinary differential equation at a given time, and to approximate the derivative by a finite difference. For example, writing:

$$\frac{x(t_{n+1}) - x(t_n)}{\Delta t} \approx \dot{x}(t_n) = f(x(t_n))$$

one gets the explicit Euler method. Writing

$$\frac{x(t_{n+1}) - x(t_n)}{\Delta t} \approx \dot{x}(t_{n+1}) = f(x(t_{n+1}))$$

one gets the implicit Euler method. Writing

$$\frac{x(t_{n+1}) - x(t_n)}{\Delta t} \approx \dot{x}(t_{n+1/2}) = f(x(t_{n+1/2})) \approx \frac{f(x(t_n)) + f(x(t_{n+1}))}{2}$$

one gets the trapezoidal method. Writing

$$\frac{x(t_{n+1}) - x(t_n)}{\Delta t} \approx \dot{x}(t_{n+1/2}) = f(x(t_{n+1/2})) \approx f\left(\frac{x(t_n) + x(t_{n+1})}{2}\right)$$

one gets the midpoint method. Finally, the Heun's method is obtained from the trapezoidal method by replacing $x(t_{n+1})$ in the right-hand side by using the Euler explicit scheme.

The reader understands that there are infinitely many consistent discretizations. All these methods belong to a broader class of schemes called Runge-Kutta methods. See [9] and [11] for more details.

Definition 19. *A scheme is called explicit if, for any function f, the computation of x_{n+1} from x_n is of the form $x_{n+1} = T(x_n)$ with T an analytically known function of f. Conversely, a scheme is called implicit if computing x_{n+1} from x_n requires solving a (typically nonlinear) problem of the form $F(x_n, x_{n+1}) = 0$.*

For implicit schemes, the function $\Psi_{\Delta t}$ generally lacks a simple analytical expression. However, we assume it is well-defined, i.e., x_{n+1} is well-defined in terms of x_n. This is not trivial in general. For example, in our setting (3.1), if $\Delta t L < 1$, one can prove that there exists a unique solution to the Implicit Euler scheme (this can be done using Banach-Picard fixed-point argument, as discussed in Section 2.2.2). In practice, implicit problems are solved using iterative methods such

as fixed point iterations, considering the limit $k \to \infty$ (if $\Delta t L < 1$, it can be shown that, for any initial condition $x_n^{(0)}$, $\lim_{k\to\infty} x_{n+1}^{(k)} = x_{n+1}$ solution to (3.2)) of

$$x_{n+1}^{(k+1)} = x_n + \Delta t f(x_{n+1}^{(k)})$$

or the Newton method (these questions will be discussed in particular in the lectures [14]). Therefore, from a practical viewpoint, the difference between implicit and explicit methods is that an explicit method requires only a finite number of evaluations of f to compute the increment $\Psi_{\Delta t}$, whereas an implicit method requires *a priori* an infinite number of evaluations of f. Of course, in practice, only a finite number of iterations of these algorithms are used (until a numerical convergence criterion is met), but it should be noted that a step of an implicit method typically requires many more evaluations of f than a step of an explicit method.

In practice, the CPU time of a numerical method for solving an ordinary differential equation is considered proportional to the number of evaluations of the function f: all other operations are negligible. Thus, the computational cost of a method's step is evaluated by the number of evaluations of the function f. An implicit method is therefore typically much more expensive than an explicit method. Anticipating what follows, note that the interest of implicit methods is related to stability issues as discussed in Section 3.3.

Remark 8. *The explicit (resp. implicit) Euler method is sometimes called the forward (resp. backward) Euler method.*

Exercise 29. ★*In the examples above, distinguish between explicit and implicit schemes.*

3.2 Convergence

A natural question at this stage is: is x_n a good approximation of $x(t_n)$? To answer this question, we need to analyze two phenomena:

- What is the error made at each time step? This error is called the consistency error.
- How do these errors interact over the iterations? In other words, can error amplification occur? This analysis is related to the notion of stability.

3.2.1 Convergence of the explicit Euler scheme

Let us take the example of the explicit Euler scheme. Over a time step $[t_n, t_{n+1}]$, we have for the approximate solution:

$$x_{n+1} = x_n + \Delta t\, f(x_n),$$

and for the exact solution:

$$x(t_{n+1}) = x(t_n) + \int_{t_n}^{t_{n+1}} f(x(s))\, ds.$$

Let $e_n = x(t_n) - x_n$ be the error at time t_n, then:

$$\begin{aligned} e_{n+1} &= e_n + \int_{t_n}^{t_{n+1}} f(x(s))\, ds - \Delta t\, f(x_n) \\ &= e_n + \int_{t_n}^{t_{n+1}} [f(x(s)) - f(x(t_n))]\, ds + \Delta t[f(x(t_n)) - f(x_n)]. \end{aligned}$$

Let $\varepsilon_n = \int_{t_n}^{t_{n+1}} [f(x(s)) - f(x(t_n))]\, ds$ be the error made by the scheme over the time step $[t_n, t_{n+1}]$, starting from the point $x(t_n)$ at time t_n: this is the consistency error. Using the fact that f is Lipschitz, we get:

$$|e_{n+1}| \leq |e_n|(1 + L\Delta t) + |\varepsilon_n|.$$

Applying a discrete Grönwall lemma (see Exercise 13), we obtain (since $1 + L\Delta t \leq \exp(L\Delta t)$):

$$\begin{aligned} |e_n| &\leq |e_0|(1 + L\Delta t)^n + \sum_{k=0}^{n-1} (1 + L\Delta t)^{n-1-k} |\varepsilon_k| \\ &\leq |e_0| \exp(Lt_n) + \sum_{k=0}^{n-1} \exp(L(t_n - t_{k+1}))|\varepsilon_k|. \end{aligned}$$

This allows us to analyze how the errors accumulate over time. To complete the convergence proof, we need to estimate the consistency error ε_k: this will depend on the regularity of the solution x to the original continuous problem.

Assume that $f \in \mathcal{C}^1$, and hence $x \in \mathcal{C}^2$. We write:

$$\begin{aligned} |\varepsilon_n| &= \left| \int_{t_n}^{t_{n+1}} [f(x(s)) - f(x(t_n))]\, ds \right| \\ &\leq \int_{t_n}^{t_{n+1}} |\dot{x}(s) - \dot{x}(t_n)|\, ds \\ &\leq \int_{t_n}^{t_{n+1}} \int_{t_n}^{s} |\ddot{x}(r)|\, dr\, ds \\ &\leq \Delta t \int_{t_n}^{t_{n+1}} |\ddot{x}(r)|\, dr. \end{aligned}$$

We then obtain the estimate: for all $n \in \{0, \ldots, N(\Delta t)\}$,

$$|e_n| \leq |e_0| \exp(Lt_n) + \Delta t \int_0^{t_n} \exp(L(t_n - s))|\ddot{x}(s)|\, ds.$$

This yields convergence of order $O(\Delta t)$: we say the scheme converges with order 1.

We have introduced in this simple example the essential ingredients of the convergence proof for a numerical scheme approximating an ordinary differential equation. This will be generalized to more complex schemes in Section 3.2.3. Before going to that, we show in the next section how one can actually build a solution to the original continuous in time problem by using the explicit Euler numerical scheme.

Exercise 30. ★★★*Let us assume that f is simply Lipschitz, so that x is only $\mathcal{C}^1$. Prove that*

$$|\varepsilon_n| \leq \Delta t \sup_{|t-s| \leq \Delta t} |\dot{x}(t) - \dot{x}(s)|$$

and deduce that for all $n \in \{0, \ldots, N(\Delta t)\}$,

$$|e_n| \leq |e_0| \exp(Lt_n) + \sup_{|t-s| \leq \Delta t} |\dot{x}(t) - \dot{x}(s)| \frac{\exp(Lt_n) - 1}{L}.$$

Prove that the modulus of continuity

$$\sup_{|t-s| \leq \Delta t} |\dot{x}(t) - \dot{x}(s)|$$

of the solution $\dot{x}$ tends to 0 as $\Delta t \to 0$

This shows the convergence of the numerical scheme, but without any rate of convergence.

3.2.2 Existence of solution to an ordinary differential equation through a numerical scheme

The construction of an approximation scheme typically yields existence results. The objective of this section is to illustrate this fact for ordinary differential equations, by proving the existence of a solution by using the explicit Euler scheme[1].

The proof relies on a fundamental theorem of analysis which identifies the compact subsets of the ensemble of continuous functions equipped with the L^∞ norm (namely, the topology of the uniform convergence): this is called the Arzelà-Ascoli theorem.

Theorem 20 (Arzelà-Ascoli)**.** *Let $\left(x^{(n)} : [0,T] \to \mathbb{R}^d\right)_{n\geq 0}$ by a sequence of continuous functions such that*

- *Uniform boundedness:*
$$\sup_n \|x^{(n)}\|_{L^\infty([0,T])} < \infty$$
where $\|x^{(n)}\|_{L^\infty([0,T])} = \max_{t\in[0,T]} \|x^{(n)}(t)\|$;
- *Uniform equicontinuity: for any $\varepsilon > 0$, there exists $\delta > 0$, such that for any n, for any $s,t \in [0,T]$,*
$$|t-s| < \delta \Longrightarrow \|x^{(n)}(t) - x^{(n)}(s)\| < \varepsilon. \tag{3.3}$$

Then, there exists a subsequence $(x^{(n_k)})_{k\geq 0}$ and a continuous function $x : [0,T] \to \mathbb{R}^d$ such that $(x^{(n_k)})_{k\geq 0}$ converges uniformly to x:

$$\lim_{k\to\infty} \|x^{(n_k)} - x\|_{L^\infty([0,T])} = 0.$$

Proof:

The proof is based on a very useful trick called the diagonal extraction.

Let us first introduce $D = (t_m)_{m\geq 0}$ a dense countable subset of $[0,T]$ (one can for example build D using rational numbers). At the pont t_1, the sequence $(x^{(n)}(t_1))_{n\geq 0}$ is bounded, and thus one can extract a subsequence $(x^{(n)}(t_1))_{n\in N_1}$ such that $(x^{(n)}(t_1))_{n\in N_1}$ converges. Then, at point t_2, one can consider the sequence $(x^{(n)}(t_2))_{n\in N_1}$ which is again bounded, and thus extract a subsequence $(x^{(n)}(t_2))_{n\in N_2}$ which converges, with $N_2 \subset N_1$. This can be pursued by induction, to define a nested sequence $N_{k+1} \subset N_k$.

The diagonal extraction consists then in considering the sequence $x^{(n_k)}$ where n_k is the k-th element of N_k. It is easy to check that for any m, $x^{(n_k)}(t_m)$ converges: let us denote by $x(t_m)$ the limit, where $x : D \to \mathbb{R}^d$ is for the moment only defined on D. The objective is now to define x everywhere on $[0,T]$.

Let us prove that for any $\varepsilon > 0$, there exists k_0 such that for any $t \in [0,T]$, for any $k,l \geq k_0$,
$$\|x^{(n_k)}(t) - x^{(n_l)}(t)\| < \varepsilon. \tag{3.4}$$
(This means that the sequence $x^{(n)}$ satisfies uniformly in t a Cauchy criterium.) Indeed, by compactness, there exists a finite ensemble $\{s_1,\dots,s_I\} \subset D$ such that for any $t \in [0,T]$, there exists $i \in \{1,\dots,I\}$, such that $|t-s_i| < \delta$, where δ is the equicontinuity parameter associated with $\varepsilon/3$. The ensemble $\{s_1,\dots,s_I\}$ is called a δ-net. Moreover, there exists k_0 such that for any $k,l \geq k_0$ and for any $i \in \{1,\dots,I\}$, $\|x^{(n_k)}(s_i) - x^{(n_l)}(s_i)\| < \varepsilon/3$ (since the sequences are $(x^{(n_k)}(s_i))_{m\geq 0}$ convergent). Therefore, for any $t \in [0,T]$, for any $k,l \geq k_0$, by

[1] In the original lectures given by Augustin-Louis Cauchy in 1824, the first existence proof was actually based on this idea, using polygonal (Euler) approximations.

choosing s_i such that $|t - s_i| < \delta$,

$$\|x^{(n_k)}(t) - x^{(n_l)}(t)\| \leq \|x^{(n_k)}(t) - x^{(n_k)}(s_i)\| + \|x^{(n_k)}(s_i) - x^{(n_l)}(s_i)\| + \|x^{(n_l)}(s_i) - x^{(n_l)}(t)\| < \varepsilon.$$

Now, for any $t \in [0,T] \setminus D$, the sequence $(x^{(n_k)}(t))_{k\geq 0}$ is bounded and thus converges up to the extraction of a subsequence. From the property (3.4), it only admits a unique accumulation point, and thus the whole sequence $(x^{(n_k)}(t))_{k\geq 0}$ converges to a well defined limit denoted by $x(t)$.

The fact that the limit $x(t)$ is continuous (actually equicontinuous) is a direct consequence of the uniform equicontinuity property, by sending n to ∞ in (3.3). In particular, by using again the same δ as above, one has: for any $s,t \in [0,T]$, $|s-t| < \delta \Longrightarrow \|x(s) - x(t)\| < \varepsilon/3$.

The fact that $(x^{(n_k)})_{k\geq 0}$ uniformly converges to x is then a consequence of the equicontinuity properties. Indeed, for any $\varepsilon > 0$, using the δ-net introduced above, by choosing k_1 sufficiently large so that for any $i \in \{1,\dots,I\}$, for all $k \geq k_1$, $\|x^{(n_k)}(s_i) - x(s_i)\| < \varepsilon/3$, one has: for any $k \geq k_1$ and for any $s \in [0,T]$

$$\|x^{(n_k)}(s) - x(s)\| \leq \|x^{(n_k)}(s) - x^{(n_k)}(s_i)\| + \|x^{(n_k)}(s_i) - x(s_i)\| + \|x(s_i) - x(s)\| < \varepsilon$$

where s_i is chosen such that $|s_i - s| < \delta$.

♢

We are now in position to state the result of this section.

Theorem 21. *Let us consider the Cauchy problem*

$$\dot{x} = f(x(t))$$

with initial condition $x(0) = x_0$. Let us assume that

$$f : \mathbb{R}^d \to \mathbb{R} \text{ is continuous.} \tag{3.5}$$

For $\beta > 0$, let us define $M := \sup_{\|x-x_0\|\leq\beta} \|f(x)\|$. Notice that $M < \infty$ since f is continuous, and thus $f(\{x, \|x - x_0\| \leq \beta\})$ is compact.

Then, there exists a solution to the Cauchy problem defined on $[0, \beta/M]$.

Exercise 31. ★★*Let us assume that f is globally Lipschitz with constant L. (i) Show that one can choose, for any $\beta > 0$, $M = \|f(x_0)\| + L\beta$. (ii) Prove that for any x_0, one can choose β such that $\beta/M > \frac{1}{2L}$. (iii) Conclude to the existence of a unique global solution, using the result of Section 2.2.1.*

Proof: Let us define $T = \beta/M$, $\Delta t_N = T/N$, and $t_k^N = k\Delta t_N$ for $k \in \{0,\dots,N\}$. The solution to the Euler scheme on the grid $(t_k^N)_{k\in\{0,\dots,N\}}$ is defined by: $x_0^{(N)} = x_0$ and, for $k \in \{0,\dots,N-1\}$,

$$x_{k+1}^{(N)} = x_k^{(N)} + \Delta t_N f(x_k^{(N)}).$$

The linear interpolant $x^{(N)}(t)$ is defined by: for all $k \in \{0,\dots,N-1\}$, $x^{(N)}(t_k) = x_k^{(N)}$ and

$$\frac{dx^{(N)}(t)}{dt} = f(x_k^{(N)}) \text{ for } t \in (t_k, t_{k+1}).$$

Notice that $x^{(N)} : [0,T] \to \mathbb{R}^d$ is a continuous function, and that

$$x^{(N)}(t) = x_0 + \int_0^t f(x^{(N)}(\lfloor s \rfloor_N))ds$$

where $\lfloor s \rfloor_N = \lfloor s/\Delta t_N \rfloor \Delta t_N$ denotes the largest gridpoint in $(t_k^N)_{k\in\{0,\dots,N\}}$ smaller than s.

For any N, and any $t \in [0,T]$, one has $\|x^{(N)}(t) - x_0\| \leq \beta$. Indeed, for a fixed n, let $t_0 = \inf\{t \in [0,T], \|x^{(N)}(t) - x_0\| > \beta\}$. Necessarily, by continuity, $t_0 > 0$ and $\|x^{(N)}(t_0) - x_0\| = \beta$ so that

$$\beta = \|x^{(N)}(t_0) - x_0\| = \left\| \int_0^{t_0} f(x^{(N)}(\lfloor s \rfloor_N)\, ds \right\| \leq \int_0^{t_0} \|f(x^{(N)}(\lfloor s \rfloor_N)\|\, ds \leq M t_0$$

which shows that $t_0 \geq \beta/M = T$.

As a consequence, the family of functions $x^{(N)}$ is uniformly bounded, and uniformly Lipschitz (and thus equicontinuous). By the Arzelà-Ascoli theorem, up to the extraction of a subsequence, $x^{(N)}$ converges uniformly to a continuous function $x : [0,T] \to \mathbb{R}^d$:

$$\lim_{N\to\infty} \|x^{(N)} - x\|_{L^\infty([0,T])} = 0.$$

Now, one has, for any N and any $t \in [0,T]$,

$$x^{(N)}(t) = x_0 + \int_0^t g^{(N)}(s) ds$$

where $g^{(N)}(s) = f(x^{(N)}(\lfloor s \rfloor_N))$ is such that $\forall s \in [0,t]$ (i) $\lim_{N\to\infty} g^{(N)}(s) = f(x(s))$ and (ii) for all N, $\|g^{(N)}(s)\| \leq M$. Therefore, by the dominated convergence theorem, by passing to the limit $N \to \infty$, one gets for any $t \in [0,T]$,

$$x(t) = x_0 + \int_0^t f(x(s)) ds.$$

♢

3.2.3 Convergence analysis of one-step methods

The analysis requires to study the consistency of the scheme, then the stability of the scheme, to finally the convergence.

Consistency. We first define the consistency error.

Definition 22. *The consistency error on a time interval of size Δt and for an initial condition x_0 is the quantity*

$$\varepsilon_{\Delta t}(x_0) = x(\Delta t) - x_1$$

where x is the solution of the Cauchy problem:

$$\begin{cases} \dot{x} = f(x), \\ x(0) = x_0, \end{cases}$$

and x_1 is obtained by the numerical scheme

$$x_1 = x_0 + \Delta t\, \Psi_{\Delta t}(x).$$

In terms of flow, we have $\varepsilon_{\Delta t}(x_0) = \phi(x_0, \Delta t) - \Phi_{\Delta t}(x_0)$.

The numerical scheme is said to be consistent[2] if, for any continuous Lipschitz function f, and for any x_0, it holds $\varepsilon_{\Delta t}(x_0) = o(\Delta t)$ as $\Delta t \to 0$. In other words,

$$\lim_{\Delta t \to 0} \frac{\varepsilon_{\Delta t}(x_0)}{\Delta t} = 0.$$

[2] with the ordinary differential equation $\dot{x} = f(x)$

For an integer $p \geq 1$, the scheme is said to be consistent of order p if, for any function f of class $\mathcal{C}^p$, and for any x_0,

$$\varepsilon_{\Delta t}(x_0) = \mathcal{O}(\Delta t^{p+1}) \quad \text{as } \Delta t \to 0.$$

That is, there exist $\Delta t^* > 0$ and $K > 0$ such that for all $\Delta t \leq \Delta t^*$,

$$\left| \frac{\varepsilon_{\Delta t}(x_0)}{\Delta t^{p+1}} \right| \leq K.$$

In short, the consistency error is the difference between the exact and the numerical flow over one time step, for a given initial condition. The consistency error is typically obtained by Taylor expansions of the error, assuming as much regularity as needed on f (or equivalently, on the exact solution).

Let us take the example of the explicit Euler scheme. If x is the exact solution such that $x(0) = x_0$, and $x_1 = x_0 + \Delta t f(x_0)$, then there exists $\theta \in [0,1]$ such that:

$$\begin{aligned} x(\Delta t) &= x(0) + \Delta t \dot{x}(0) + \ddot{x}(\theta \Delta t)\frac{\Delta t^2}{2}, \\ &= x_0 + \Delta t f(x_0) + \ddot{x}(\theta \Delta t)\frac{\Delta t^2}{2}. \end{aligned}$$

We deduce that:

$$\begin{aligned} |\varepsilon_{\Delta t}(x_0)| &= |x(\Delta t) - x_1|, \\ &\leq \sup_{\theta \in [0,1]} |\ddot{x}(\theta \Delta t)| \frac{\Delta t^2}{2}. \end{aligned}$$

Now, we have $\ddot{x}(t) = Df(x(t))f(x(t))$ (where we recall Df is the Jacobian matrix of f, see Equation (2.11)), and thus, assuming $f \in \mathcal{C}^1$, we have $\sup_{\theta \in [0,1]} |\ddot{x}(\theta \Delta t)| < \infty$. This shows that the scheme is consistent of order 1.

Going back to the numerical schemes introduced above, one can show the following (do it★★★!): explicit and implicit Euler methods are of order 1, trapezoidal, Heun's and midpoint methods are of order 2.

Stability. To prove convergence, we need to analyze how errors accumulate over iterations. This requires some stability of the numerical scheme, which will be a consequence of the fact that $\Phi_{\Delta t}(\cdot)$ is $(1 + C\Delta t)$-Lipschitz.

We have the following stability property of the numerical flow $\Phi_{\Delta t}$ (compare with Proposition 5).

Proposition 23. *Assume that the increment function $\Psi_{\Delta t}$ is Lipschitz, with a Lipschitz constant $\overline{L}$ independent of Δt. Then the numerical scheme is stable in the following sense: if (x_n) satisfies*

$$x_{n+1} = x_n + \Delta t \Psi_{\Delta t}(x_n),$$

and (y_n) satisfies

$$y_{n+1} = y_n + \Delta t \Psi_{\Delta t}(y_n) + \varepsilon_n,$$

then

$$\max_{0 \leq n \leq N(\Delta t)} |x_n - y_n| \leq \exp(\overline{L}T) \left(|x_0 - y_0| + \sum_{k=0}^{N(\Delta t)-1} |\varepsilon_k| \right).$$

Proof: Let $e_n = x_n - y_n$. Then

$$e_{n+1} = e_n + \Delta t(\Psi_{\Delta t}(x_n) - \Psi_{\Delta t}(y_n)) - \varepsilon_n.$$

Hence,

$$|e_{n+1}| \leq (1 + \overline{L}\Delta t)|e_n| + |\varepsilon_n|.$$

Using a discrete Grönwall lemma (see Exercise 13),

$$|e_n| \leq (1 + \overline{L}\Delta t)^n |e_0| + \sum_{k=0}^{n-1} (1 + \overline{L}\Delta t)^{n-1-k} |\varepsilon_k|,$$

which yields the result using $(1 + \overline{L}\Delta t) \leq \exp(\overline{L}\Delta t)$. ♢

Using this stability, we prove the following convergence result:

Proposition 24. *Assume that f satisfies (3.1), and that the scheme is consistent of order p along the solution $(x(t))_{0\leq t\leq T}$ in the following sense: there exists $K > 0$ such that for all $t \in [0, T]$, for all Δt,*

$$|\varepsilon_{\Delta t}(x(t))| \leq K\Delta t^{p+1}.$$

Assume that the increment function $\Psi_{\Delta t}(x)$ is Lipschitz continuous with respect to x, with Lipschitz constant $\overline{L}$ independent of Δt. Also assume $x(0) = x_0$. Then the scheme is convergent: there exists $C > 0$ such that for all $\Delta t > 0$,

$$\sup_{0\leq n\leq N(\Delta t)} |x(t_n) - x_n| \leq C\Delta t^p. \tag{3.6}$$

We can take $C = \frac{K}{\overline{L}}(\exp(\overline{L}T) - 1)$.

Proof: We write

$$\begin{aligned}
|x(t_{n+1}) - x_{n+1}| &\leq |x(t_{n+1}) - \Phi_{\Delta t}(x(t_n))| + |\Phi_{\Delta t}(x(t_n)) - x_{n+1}| \\
&\leq |\phi(x(t_n), \Delta t) - \Phi_{\Delta t}(x(t_n))| + |\Phi_{\Delta t}(x(t_n)) - \Phi_{\Delta t}(x_n)| \\
&\leq |\varepsilon_{\Delta t}(x(t_n))| + (1 + \overline{L}\Delta t)|x(t_n) - x_n| \\
&\leq K\Delta t^{p+1} + (1 + \overline{L}\Delta t)|x(t_n) - x_n|.
\end{aligned}$$

Using the same argument as in the stability result above, we deduce:

$$\begin{aligned}
|x(t_n) - x_n| &\leq (1 + \overline{L}\Delta t)^n |x(0) - x_0| + K\Delta t^{p+1} \sum_{k=0}^{n-1} (1 + \overline{L}\Delta t)^{n-1-k} \\
&\leq K\Delta t^{p+1} \frac{(1 + \overline{L}\Delta t)^n - 1}{\overline{L}\Delta t},
\end{aligned}$$

which yields the result using $(1 + \overline{L}\Delta t) \leq \exp(\overline{L}\Delta t)$. ♢

We now understand why the consistency definition uses Δt^{p+1}: the accumulation of errors cancels one Δt, resulting in convergence of order Δt^p.

From the previous results, we can retain the so-called Lax principle: consistency and stability imply convergence.

Remark 9. *For consistent one-step numerical schemes, the fact that $\Psi_{\Delta t}$ is Lipschitz typically follows from assumption (3.1) on f. In the case of implicit schemes, the function $\Psi_{\Delta t}$ may only be defined for $\Delta t \leq \Delta t^*$, in which case $\Psi_{\Delta t}$ is Lipschitz and the scheme is stable only for sufficiently small time steps. One can check that the convergence proof also applies if the consistency and Lipschitz properties of $\Psi_{\Delta t}$ are only true for sufficiently small time steps. In any case, in these convergence theorems, we are only interested in the asymptotic behavior as $\Delta t \to 0$.*

Remark 10. *Looking closely at the previous proof, we understand that the constant multiplying Δt^p (for a method of order p) in the error estimates (3.7) and (3.6) actually depends on the regularity of the solution, and thus on the regularity of f. For a scheme of order p, the constant typically depends on L^∞ norms of derivatives of order $p+1$ of the exact solution, or equivalently, of order p of f. If the solution is not sufficiently regular, we cannot achieve convergence of order p, even if the scheme is consistent of order p. Similarly, the larger the norms of the derivatives of f up to order p, the larger the consistency error will be. It follows that it is useless to use high-order schemes if we know the solution is irregular. To obtain a better approximation in this case, it is better to reduce the time step δt. One can even use an estimate of the consistency error to derive an adaptive time-stepping method: this is called an* a posteriori *error estimate, which refers to an estimate of the error computed from the approximate result (see for example [9]).*

3.2.4 An alternative convergence proof of one-step methods

This section can be skipped on first reading.

In the proof of Proposition 24, we estimated the error $|x(t_{n+1}) - x_{n+1}|$ at time t_{n+1} in terms of the error at time t_n by introducing $\Phi_{\Delta t}(x(t_n))$. We then used a stability property of the numerical flow $\Phi_{\Delta t}$. We can also perform a convergence proof by introducing $\phi(x_n, \Delta t)$ and using a stability property of the continuous flow ϕ. Recall that the flow $\phi(\cdot, \Delta t)$ is $\exp(L\Delta t)$-Lipschitz, since f is globally L-Lipschitz (see (3.1)), which reflects the stability of the continuous problem (see Proposition 5).

Proposition 25. *Assume that f satisfies (3.1), and that the scheme is consistent of order p in the following sense: let $B = \{(t, x(t)), t \in [0,T]\}$ be the trajectory of the exact solution, and let $\mathcal{V}(B, \eta)$ be an η-neighborhood of B. Assume there exist $\eta > 0$ and $K > 0$ such that for all $(t, x) \in \mathcal{V}(B, \eta)$,*

$$|\varepsilon_{\Delta t}(x)| \leq K \Delta t^{p+1}.$$

Assume also that $x(0) = x_0$. Then the scheme is convergent: there exist constants $C > 0$ and $\Delta t^* > 0$ such that for all $\Delta t < \Delta t^*$,

$$\sup_{0 \leq n \leq N(\Delta t)} |x(t_n) - x_n| \leq C \Delta t^p. \tag{3.7}$$

We can take $C = \frac{K}{L}(\exp(LT) - 1)$ and $\Delta t^* = \left(\frac{\eta L}{K(\exp(LT)-1)}\right)^{1/p}$.

Remark 11. *For a scheme of order p, the constant in the consistency error estimate $O(\Delta t^{p+1})$ depends on derivatives of f up to order p. In particular, if these derivatives are bounded by a constant K, we can take $\varepsilon = \infty$ and $\Delta t^* = \infty$.*

Proof: The proof follows the same structure as Proposition 24, but this time we introduce the point $\phi(x_n, \Delta t)$, and use the fact that $x \mapsto \phi(x, \Delta t)$ is $\exp(L\Delta t)$-Lipschitz.

We write

$$\begin{aligned} |x(t_{n+1}) - x_{n+1}| &\leq |x(t_{n+1}) - \phi(x_n, \Delta t)| + |\phi(x_n, \Delta t) - x_{n+1}| \\ &\leq |\phi(x(t_n), \Delta t) - \phi(x_n, \Delta t)| + |\phi(x_n, \Delta t) - \Phi_{\Delta t}(x_n)| \\ &\leq \exp(L\Delta t)|x(t_n) - x_n| + |\varepsilon_{\Delta t}(x_n)|. \end{aligned}$$

We used Proposition 5 to estimate the first term.

Now consider the recurrence hypothesis:

$$\mathrm{H}(n): \quad |x(t_n) - x_n| \leq \frac{K}{L}(\exp(Ln\Delta t) - 1)\Delta t^p.$$

Clearly H(0) is true. Suppose H(n) holds. Since

$$|x(t_n) - x_n| \leq \frac{K}{L}(\exp(LT) - 1)\Delta t^p,$$

if $\Delta t < \Delta t^*$, then $|x(t_n) - x_n| < \varepsilon$ and hence $(t_n, x_n) \in \mathcal{V}(B, \eta)$, so that

$$|\varepsilon_{\Delta t}(x_n)| \leq K\Delta t^{p+1}.$$

Therefore,

$$\begin{aligned}
|x(t_{n+1}) - x_{n+1}| &\leq \exp(L\Delta t)|x(t_n) - x_n| + K\Delta t^{p+1} \\
&\leq \exp(L\Delta t)\left(\frac{K}{L}(\exp(Ln\Delta t) - 1)\Delta t^p\right) + K\Delta t^{p+1} \\
&\leq \frac{K}{L}\exp(L(n+1)\Delta t)\Delta t^p - \frac{K}{L}\Delta t^p(\exp(L\Delta t) - L\Delta t) \\
&\leq \frac{K}{L}(\exp(L(n+1)\Delta t) - 1)\Delta t^p,
\end{aligned}$$

since $\exp(x) \geq 1 + x$. This proves H($n+1$). ♢

In summary, propositions 24 and 25 yield convergence using two different stability properties (C denoting a positive constant):

- Proposition 24 relies on the stability of the numerical flow, which is a consequence of the fact that $\Phi_{\Delta t}$ is $(1 + C\Delta t)$-Lipschitz.
- Proposition 25 relies on the stability of the continuous flow, which is a consequence of the fact that $\phi(\cdot, \Delta t)$ is $C^{\Delta t}$-Lipschitz,

We refer to Figure 3.1 for a graphical illustration of the two error analysis methods.

Remark 12 (Rounding Errors)**.** *The notion of stability introduced in Proposition 23 is fundamental: it shows that the numerical scheme defines a* well-posed *problem in the sense that a small perturbation in the data implies a small perturbation in the result. This is important in practice because floating-point computation on computers inevitably introduces rounding errors. Proposition 23 allows us to control how these errors propagate [17, p. 479].*

3.3 Absolute stability

The convergence analyses presented in Section 3.2 may be impractical because the constants C in Propositions 24 and 25 can be very large, resulting in poor estimates. In particular, these estimates allow comparison between the exact and approximate solutions over a bounded time interval, but they say nothing about long-time behavior since the constant C depends exponentially on the integration time T.

Let us illustrate these difficulties with a simple test problem. Consider the solution of $\dot{x} = -\mu x$ with $\mu > 0$, which yields $x(t) = x(0)\exp(-\mu t)$. Using the explicit Euler scheme to discretize the equation:

$$x_{n+1} = x_n - \mu\Delta t x_n,$$

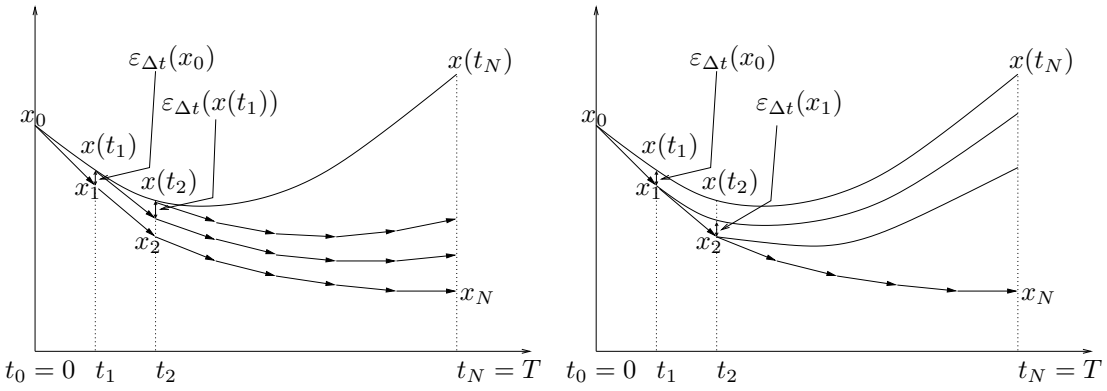


Figure 3.1: The two methods of error analysis. A solid line indicates a solution obtained by the exact flow. An arrowed line indicates a solution obtained by applying the numerical scheme. For Proposition 24 (left diagram), we introduce approximate solutions starting from the points $(t_i, x(t_i))$ and observe how the consistency errors $\varepsilon_{\Delta t}(x(t_i))$ are transported by the numerical scheme. For Proposition 25 (right diagram), we introduce exact solutions starting from the points (t_i, x_i) and observe how the consistency errors $\varepsilon_{\Delta t}(x_i)$ are transported by the exact flow.

we obtain $x_n = (1 - \mu\Delta t)^n x_0$. In particular, if $\Delta t > 2/\mu$, the sequence (x_n) becomes unbounded, while the exact solution tends to zero as $t \to \infty$.

In such cases, analyzing long-time behavior is crucial (even more so than consistency analysis) to obtain reasonable approximations. This leads to the study of *absolute stability* (or A-stability) of the scheme (not to be confused with the stability introduced earlier in Proposition 23). The notion of absolute stability is closely related to the asymptotic stability of stationary points introduced in Section 2.4 for continuous-time problems.

For this analysis, we fix Δt and let T and thus $N(\Delta t) = T/\Delta t$ tend to infinity. To understand what happens, we consider a one-dimensional test problem:

$$\dot{x} = \lambda x, \tag{3.8}$$

with $\lambda \in \mathbb{C}$.

To justify the importance of this model problem, note that at each time step, after linearizing the right-hand side and diagonalizing the Jacobian (assuming it is diagonalizable), we essentially solve problems of this type.

For the continuous problem, it is clear that $x(t)$ tends to zero as $t \to \infty$ for any initial condition as soon as $\mathrm{Re}\,(\lambda) < 0$. A numerical method is said to be absolutely stable if it satisfies the same property:

Definition 26. *A numerical method is said to be absolutely stable if, when applied to the test problem (3.8) with* $\mathrm{Re}\,(\lambda) < 0$*, it satisfies: for any initial condition* x_0*,*

$$\lim_{n\to\infty} |x_n| = 0.$$

Remark 13. *The idea of designing numerical schemes such that the approximations satisfy properties of the continuous problem is widespread in numerical analysis. Here, we focus on long-time behavior. One may also seek to preserve invariants of the continuous dynamics at the discrete level. For Hamiltonian systems, good schemes for this purpose are symplectic schemes (see [11]).*

The one-step schemes considered above, when applied to the test problem (3.8), take the form:

$$x_{n+1} = R(\lambda\Delta t)x_n,$$

where $R : \mathbb{C} \to \mathbb{C}$ is called the *stability function* of the method.

Remark 14. *For homogeneity reasons, it is natural that the parameters λ and Δt appear in the numerical schemes only through the product $\lambda\Delta t$.*

Clearly, for a numerical method to be absolutely stable, we must have that if $\mathrm{Re}\,(\lambda) < 0$ (equivalent to $\mathrm{Re}\,(z) < 0$ for $z = \lambda\Delta t$), then $|R(z)| < 1$.

Definition 27. *The set of $z \in \mathbb{C}$ such that $|R(z)| < 1$ is called the* region of absolute stability *of the numerical method:*

$$\mathcal{A} = \{z \in \mathbb{C} \mid |R(z)| < 1\}.$$

Proposition 28. *A numerical scheme is absolutely stable for all parameters λ and Δt if and only if $\{z \in \mathbb{C} \mid \mathrm{Re}\,(z) < 0\} \subset \mathcal{A}$. Such a scheme is called (unconditionally) absolutely stable.*

In practice, one can use the following result, which follows from the maximum principle for analytic functions (see [10, p.43]):

Proposition 29. *A numerical scheme is absolutely stable if and only if $|R(i\lambda)| < 1$ for all $\lambda \in \mathbb{R}$, and R is analytic on the complex half-plane $\{z \in \mathbb{C} \mid \mathrm{Re}\,(z) < 0\}$.*

Remark 15. *For some authors (see e.g. [10, 19]), the region of absolute stability is defined as $\{z \in \mathbb{C} \mid |R(z)| \leq 1\}$, and a method is (unconditionally) absolutely stable if $\{z \in \mathbb{C} \mid \mathrm{Re}\,(z) \leq 0\} \subset \{z \in \mathbb{C} \mid |R(z)| \leq 1\}$. In this case, the method is absolutely stable if, when applied to the test problem (3.8) with $\mathrm{Re}\,(\lambda) \leq 0$, it satisfies: for any initial condition x_0, there exists $C > 0$ such that for all $n > 0$, $|x_n| < C$. Compare this with the two definitions of stability for fixed points: Definition 13 and Definition 14.*

Let us look at some examples. For the explicit Euler scheme, we have $R(z) = 1 + z$. The region of stability is the disk centered at -1 with radius 1. The scheme is therefore stable under the condition $|1 + z| < 1$, which imposes a restriction on the time step. This scheme is said to be conditionally absolutely stable.

For the implicit Euler scheme, $R(z) = (1 - z)^{-1}$, so the region of stability is the set of z such that $|1 - z| > 1$ (the complement of the disk centered at 1 with radius 1). The scheme is unconditionally absolutely stable.

Exercise 32. ★★*Study the absolute stability of the numerical schemes introduced earlier. Under what condition on $\theta \in [0, 1]$ are the θ-schemes:*

$$x_{n+1} = x_n + \Delta t f\left(t_n + \theta\Delta t, (1 - \theta)x_n + \theta x_{n+1}\right)$$

or

$$x_{n+1} = x_n + \Delta t\left((1 - \theta)f(x_n) + \theta f(x_{n+1})\right)$$

A-stable?

Solution*: For the trapezoidal method (Crank-Nicolson) or the midpoint method (which are the same when applied to a linear problem), we obtain $R(z) = \frac{1+z/2}{1-z/2}$. Thus, $\mathcal{A} = \{z \in \mathbb{C} \mid |z + 2| < |z - 2|\} = \{z \in \mathbb{C} \mid \mathrm{Re}\,(z) < 0\}$, and the scheme is A-stable. More generally, for the θ-schemes, we obtain $R(z) = \frac{1+(1-\theta)z}{1-\theta z}$, and we verify that the scheme is A-stable if and only if $\theta \in [0.5, 1]$. Thus, there exist implicit methods that are not A-stable.*

For Heun's method, we obtain $R(z) = 1 + z + z^2/2$, and the scheme is not A-stable because if $z \in \mathbb{R}$ tends to $-\infty$, then $|R(z)|$ tends to $+\infty$.

Part II

The Poisson equation

Chapter 4

Models using the Laplace operator

4.1 Definition of the Laplace operator and of the Laplace and Poisson problems

The Laplace operator is a typical example of a linear differential operator, namely a mapping which takes a function and returns a linear combination of partial derivatives of the function. More precisely, one has: for any twice differentiable function $u : \mathbb{R}^d \to \mathbb{R}$,

$$\Delta u = \sum_{i=1}^{d} \frac{\partial^2 u}{\partial x_i^2}$$

where u is a function of $x = (x_1, \ldots, x_d) \in \mathbb{R}^d$.

One useful rewriting of the Laplace operator involves two other operators: the gradient and the divergence. For any differentiable function $u : \mathbb{R}^d \to \mathbb{R}$, the gradient operator returns of d-dimensional vector ∇u whose i-th component is:

$$[\nabla u]_i = \frac{\partial u}{\partial x_i}.$$

The divergence operator is defined on a vector-valued function (a.k.a. a vector field): for a smooth function $F : \mathbb{R}^d \to \mathbb{R}^d$ the divergence of F is defined by:

$$\text{div } (F) = \sum_{i=1}^{d} \frac{\partial F_i}{\partial x_i}$$

where, as above x_i denotes the i-th component of the underlying space variable x, and, likewise, F_i is the i-th component of the vector-valued function F. Now, one can check (do it!★) the following fundamental formula: for any twice differentiable function $u : \mathbb{R}^d \to \mathbb{R}$,

$$\Delta u = \text{div } (\nabla u).$$

The Laplace and Poisson problems appear in a variety of physical contexts. The Laplace problem is written as: find a function $u : \mathbb{R}^d \to \mathbb{R}$ such that

$$\Delta u = 0. \tag{4.1}$$

A function u satisfying this equation is called a harmonic function. Constant functions of course satisfy (4.1) but also harmonic polynomials such as $u(x) = x_1^2 - x_2^2$ for example (for $d \geq 2$).

Exercise 33. *Let u be a harmonic function, and $O \in \mathbb{R}^{d\times d}$ an orthogonal matrix. Let $v(x) = u(Ox)$. Prove that v is harmonic.*

The Poisson problem consists in, for a given function $f : \mathbb{R}^d \to \mathbb{R}$ finding a function $u : \mathbb{R}^d \to \mathbb{R}$ such that

$$-\Delta u = f. \tag{4.2}$$

The Laplace problem is thus a specific example of the Poisson problem where $f = 0$.

Thanks to the linearity of the operator, we observe that if $-\Delta u_1 = f_1$ and $-\Delta u_2 = f_2$ then for any $\lambda \in \mathbb{R}$, one solution to the problem $-\Delta u = f_1 + \lambda f_2$ is simply $u = u_1 + \lambda u_2$. This is sometimes called the superposition principle.

We will only be interested in so-called classical solutions, namely solutions such that u is twice differentiable. As will become clear below, for these equations to be well posed (namely admits a unique classical solution), one needs to make precise the behavior of u when $x \to \infty$, as well as assumptions on the function f.

Exercise 34. ★*Identify all the solution to the Laplace equation in dimension* 1*. Check that if u is bounded, then u is constant.*

We will also consider versions of this problem posed on a subset Ω of $\mathbb{R}^d$, in which case we will need boundary conditions to ensure the uniqueness of the solution.

Remark 16. *The Laplace and Poisson equations are prototypical examples of so-called elliptic linear problems, which are generally of the form:*

$$-\text{div}\,(D\nabla u) = f$$

for $D : \mathbb{R}^d \to \mathbb{R}^{d\times d}$ a space-dependent symmetric matrix such that $D \geq \alpha\text{Id}$ (i.e. $D(x) - \alpha\text{Id}$ is symmetric non negative for all x) for some constant $\alpha > 0$.

4.2 General derivation of the Laplace and Poisson equations

The Laplace equation (4.1) naturally appears to model the stationary state of a scalar valued function $u : \mathbb{R}^d \to \mathbb{R}$ which represents the density of some quantity (think of a concentration, or a potential energy), which depends on $x \in \mathbb{R}^d$, typically the space variable (so that $d \in \{1, 2, 3\}$).

Here is the general context where the Laplace equation appears. Let $F : \mathbb{R}^d \to \mathbb{R}^d$ be the flux associated with the quantity u: for any infinitesimal $(d-1)$-dimensional plane, with normal n, $F \cdot n$ represent the flow (per unit surface) of the quantity u going through the plane in the direction of n. At stationary state, for any $\Omega \subset \mathbb{R}^d$, one has (what comes in somewhere through the boundary of Ω should come out somewhere else through the boundary of Ω):

$$\int_{\partial\Omega} F(x) \cdot n(x)\sigma(dx) = 0 \tag{4.3}$$

where σ is the surface measure on $\partial\Omega$, and $n(x)$ is the outgoing unit normal to Ω at point $x \in \partial\Omega$.

Exercise 35. ★★*Let $\Omega \subset \mathbb{R}^d$ be defined by $\Omega = \{x \in \mathbb{R}^d, \phi(x) < 0\}$ for a smooth function ϕ such that $\nabla\phi(x) \neq 0$ for all $x \in \partial\Omega = \{x \in \mathbb{R}^d, \phi(x) = 0\}$. Check that the unit outward normal at point $x \in \partial\Omega$ is given by: $n(x) = \frac{\nabla\phi(x)}{|\nabla\phi(x)|}$.*

In order to proceed, we need a fundamental theorem of integration called the (Gauss) divergence formula (also called Stokes formula).

Proposition 30. *Let $\Omega \subset \mathbb{R}^d$ be a $\mathcal{C}^1$ domain*[1]*. Let $\Psi : \overline{\Omega} \to \mathbb{R}^d$ be a vector field such that $\Psi \in \mathcal{C}^1(\overline{\Omega}, \mathbb{R}^d)$. Then,*

$$\int_{\partial\Omega} \Psi(x) \cdot n(x)\sigma(dx) = \int_{\Omega} \text{div } (\Psi)(x)\, dx.$$

Notice that in Proposition 30, we assume that Ψ is continuous up to the boundary of Ω. This implies in particular that ψ is well defined on $\partial\Omega$.

From this proposition, one gets

$$\int_{\partial\Omega} F(x) \cdot n(x)\sigma(dx) = \int_{\Omega} \text{div } (F)(x)dx$$

and thus, using (4.3) and the fact that Ω is arbitrary, we obtain

$$\text{div } F = 0. \tag{4.4}$$

Exercise 36. ★*Let Ω be a $\mathcal{C}^1$ domain. Prove that for any function $u \in \mathcal{C}^1(\overline{\Omega}, \mathbb{R})$,*

$$\int_{\partial\Omega} u\, n_i\, d\sigma = \int_{\Omega} \partial_{x_i} u$$

where n_i denotes the i-th component of the unit outward normal n to Ω.

Exercise 37. ★★*Let Ω be a $\mathcal{C}^1$ domain. Let $u : \overline{\Omega} \to \mathbb{R}$ and $v : \overline{\Omega} \to \mathbb{R}$ be $\mathcal{C}^2$ functions. For any $x \in \partial\Omega$, we denote*

$$\partial_n u(x) = \nabla u(x) \cdot n(x)$$

the so-called normal derivative of u on $\partial\Omega$ (and likewise for v of course).

By using Proposition 30, prove the following three fundamental Green's identities[2]*:*

$$\int_{\Omega} \Delta u = \int_{\partial\Omega} \partial_n u\, d\sigma$$

$$\int_{\Omega} \nabla u \cdot \nabla v = -\int_{\Omega} v\Delta u + \int_{\partial\Omega} v\partial_n u\, d\sigma$$

$$\int_{\Omega} (v\Delta u - u\Delta v) = \int_{\partial\Omega} (v\partial_n u - u\partial_n v)\, d\sigma.$$

The last step to get the Laplace equation (4.1) requires a modelling assumption to relate the flux F to the physical quantity u. In many contexts (see below for some specific examples), one assumes that the flux F is proportional to the gradient of u but points in the opposite direction (flux implies flows from regions with large u to regions with small u, in order to locally equilibrate u): for all x,

$$F(x) = -a\nabla u(x)$$

where $a > 0$ is here assumed to be constant. By using now equation (4.4) one finally gets $0 = \text{div } (F) = -\text{div } (a\nabla u)$ which yields the Laplace equation:

$$\Delta u = 0.$$

---

[1]The precise definition of a $\mathcal{C}^1$ domain is out of the scope of these lectures. A simple example of a $\mathcal{C}^1$ domain is when $\Omega = \{x \in \mathbb{R}^d, \phi(x) < 0\}$ for some $\mathcal{C}^1$ function $\phi : \mathbb{R}^d \to \mathbb{R}$ such that $\nabla\phi(x) \neq 0$ for all $x \in \partial\Omega = \{x \in \mathbb{R}^d, \phi(x) = 0\}$. The unit outward normal of a $\mathcal{C}^1$ domain is defined everywhere on $\partial\Omega$. The surface measure σ of a $\mathcal{C}^1$ domain can also be rigorously defined.

[2]Here and in the following, depending on whether we would like to make explicit the integration variable or not, we either use the notation $\int_\Omega u$ or $\int_\Omega u(x)\, dx$ to denote the integral of u on Ω. Likewise $\int_{\partial\Omega} u\, d\sigma = \int_{\partial\Omega} u(x)\, \sigma(dx)$.

From the previous reasoning, one understands that, roughly speaking, the Laplace operator performs a "local averaging" of u.

In the previous derivation, we assumed that the system was stationary, and that there was no creation or consumption of the physical quantity u. When this is not the case, (4.3) is replaced by:

$$\int_{\partial\Omega} F(x) \cdot n(x)\sigma(dx) = \int_{\Omega} f(x)dx \tag{4.5}$$

where f represents the source term: creation (resp. consumption) of the physical quantity at point x if $f(x) > 0$ (resp. if $f(x) < 0$) per unit volume. By following the same reasoning (check it!★), one gets the Poisson equation:

$$-a\Delta u = f.$$

Exercise 38. ★*Assume that u satisfies the Poisson equation $-a\Delta u(x) = f(x)$ for all $x \in \Omega$ where $\Omega \subset \mathbb{R}^d$. Prove that necessarily, one has:*

$$-a \int_{\partial\Omega} \partial_n u \, d\sigma = \int_{\Omega} f.$$

How can this be interpreted?

The results of the following exercise will be useful below.

Exercise 39. ★★*Let $f : \mathbb{R}^d \to \mathbb{R}$ be a continuous function, integrable over $\mathbb{R}^d$. The transformation to polar coordinates implies:*

$$\int_{\mathbb{R}^d} f(x)\, dx = \int_0^\infty \left(\int_{\partial B(0,r)} f d\sigma \right) dr.$$

Deduce from that the following:

$$\frac{d}{dr} \int_{B(0,r)} f(x)\, dx = \int_{\partial B(0,r)} f d\sigma.$$

Let $\omega_d = |B(0,1)|$ denotes the volume of the unit ball in dimension d. Check that:

$$|B(0,r)| := \int_{B(0,r)} 1\, dx = \omega_d\, r^d$$

$$|\partial B(0,r)| := \int_{\partial B(0,r)} 1\, d\sigma = d\, \omega_d\, r^{d-1}.$$

Exercise 40. ★★★*We aim at proving that $\omega_d = \frac{\pi^{d/2}}{\Gamma(d/2+1)}$ where $\Gamma(z) = \int_0^\infty t^{z-1} \exp(-t)\, dt$ is the so-called gamma function. Let us recall that $\int_{\mathbb{R}} \exp(-t^2)\, dt = \sqrt{\pi}$. Prove on the one hand that $\int_{\mathbb{R}^d} \exp(-|x|^2)dx = \pi^{d/2}$. Prove on the other hand that $\int_{\mathbb{R}^d} \exp(-|x|^2)dx = \omega_d\, d \int_0^\infty \exp(-r^2) r^{d-1}\, dr$ and that $\int_0^\infty \exp(-r^2) r^{d-1} dr = (1/2)\Gamma(d/2)$. Conclude.*

4.3 Boundary conditions

In many practical contexts, we will actually consider the Laplace or Poisson equation only in a bounded domain Ω:

$$-\Delta u = f \text{ in } \Omega$$

either because it only holds in this domain, or because we already know u outside Ω and we therefore only needs to determine u in Ω.

In this case, for the problem to have a unique solution, it will be necessary to add additional conditions on the boundary of Ω. Two typical conditions are used in practice.

The Dirichlet boundary conditions consist in imposing the value of the quantity u on $\partial\Omega$:

$$u = g \text{ on } \partial\Omega.$$

The Neumann boundary conditions consists in imposing the value of the outgoing flux $F \cdot n$ on $\partial\Omega$:

$$\partial_n u = g \text{ on } \partial\Omega.$$

Notice that to be well defined, these boundary conditions require some regularity on u in order to define its boundary values. For example, for a continuous function $g : \partial\Omega \to \mathbb{R}$, one could consider $u \in \mathcal{C}(\overline{\Omega})$ for Dirichlet, and $u \in \mathcal{C}^1(\overline{\Omega})$ for Neumann.

4.4 Stationary thermal diffusion

Let $T(x) \in \mathbb{R}$ denotes the temperature at point x, and $Q(x) \in \mathbb{R}^d$ the heat flux at point x. Let $f(x)$ denotes the associated source term: if $f(x) > 0$ heat is created at point x and if $f(x) < 0$ heat is consumed at point x.

Then the relation $Q(x) = -k\nabla T(x)$ is called the Fourier law of heat conduction (k being the so-called heat conductivity) and the Poisson equation

$$-k\Delta T = f$$

is the stationary heat equation.

4.5 Stationary diffusion of chemical species

Let $u(x) \in \mathbb{R}$ denote the concentration of a chemical species at point x (think for example of a pollutant). Let $f(x)$ denotes the associated source term: if $f(x) > 0$, chemical species are created at point x (think for example at a factory producing the pollutant) while if $f(x) < 0$, chemical species are absorbed at point x (think for example at places where chemical reactions consume the pollutant).

Then the relation $F(x) = -D\nabla u(x)$ is called the Fick law of diffusion (D being the so-called diffusion coefficient or diffusivity) and the Poisson equation

$$-D\Delta u = f$$

is the stationary diffusion equation.

4.6 Electrostatics

Let us consider a static electric field $E : \mathbb{R}^3 \to \mathbb{R}^3$. By the Maxwell-Faraday equation, $\text{curl}(E) = 0$ (this is called a curl-free or irrotational vector field) which implies that $E = -\nabla V$ for some electric potential V. Then, from Gauss Maxwell-Coulomb (or Gauss) law, $\text{div}\,(D) = \rho$, where $\rho : \mathbb{R}^3 \to \mathbb{R}$ is the charge density, and D is the electric flux density. For a linear, isotropic, homogeneous material, $D = \epsilon E$ where ϵ is the permittivity. One thus obtains the Poisson equation, which is:

$$-\Delta V = \frac{\rho}{\epsilon}.$$

Exercise 41. ★★*Let us assume that* $E : \mathbb{R}^3 \to \mathbb{R}^3$ *is* $\mathcal{C}^1$ *and satisfies* $\mathrm{curl}(E) = 0$. *Let us recall that* $\mathrm{curl}(E) : \mathbb{R}^3 \to \mathbb{R}^3$ *and:*

$$\mathrm{curl}(E) = \begin{bmatrix} \partial_{x_2} E_3 - \partial_{x_3} E_2 \\ \partial_{x_3} E_1 - \partial_{x_1} E_3 \\ \partial_{x_1} E_2 - \partial_{x_2} E_1 \end{bmatrix}.$$

In short, $\mathrm{curl}E = \nabla \times E$ *where* $\times$ *denotes the cross product. Let us now consider* $V(x) = \int_0^1 E(tx) \cdot x \, dt$. *Show that* $\nabla V = E$.

It can actually be shown that for any simply connected domain Ω *(roughly speaking, a domain without holes) and* $\mathcal{C}^1$ *vector field* $E : \Omega \to \mathbb{R}^d$

$$\mathrm{curl}E = 0 \text{ in } \Omega \Longrightarrow \exists V : \Omega \to \mathbb{R}, E = \nabla V.$$

However, for non simply connected domains, this is wrong as can be checked with the following example.

Let us consider, in dimension 2, the domain $\Omega = \mathbb{R}^2 \setminus (0,0)$ *and the vector field (smooth on* Ω*):*

$$E(x_1, x_2) = \left(-\frac{x_2}{x_1^2 + x_2^2}, \frac{x_1}{x_1^2 + x_2^2} \right).$$

Check that $\mathrm{curl}(E) = 0$ *(namely* $\partial_{x_2} E_1 = \partial_{x_1} E_2$*), and* $\int_0^{2\pi} E(x(\theta)) \cdot \dot{x}(\theta) d\theta = 2\pi \neq 0$, *where* $x(\theta) = (\cos(\theta), \sin(\theta))$. *Show that the former relation implies that* E *cannot be written as* $E = -\nabla V$ *for some* V. *Is it possible to write* $E = -\nabla V$ *for some* V *on the domain* $\mathbb{R}^2 \setminus \{(x, 0), x \geq 0\}$*?*

4.7 Newtonian gravity

The gravitational force experienced by a point particle of mass m is equal to mg, where $g : \mathbb{R}^3 \to \mathbb{R}^3$ is the so-called gravitational field. The Gauss law for gravity states that $\mathrm{div}\,(g) = -4\pi G\rho$ where ρ is the mass density and G is the universal gravitational constant.

Moreover, the gravitational field is irrotational ($\mathrm{curl}g = 0$) which implies that there exists a gravitational potential V such that $g = -\nabla V$. By using the Gauss law for gravity, one then obtains the Poisson equation:

$$-\Delta V = -4\pi G\rho.$$

Chapter 5

Existence, uniqueness and qualitative properties of solutions

We present an integral representation approach to classical solutions to the Laplace and Poisson equations (Equations (4.1) and (4.2)) in the spirit of [6, Chapter 2]. We start by considering these problems in the whole of $\mathbb{R}^d$. The fundamental solution to the Laplacian is introduced in Section 5.1, and from this, a solution to the Poisson equation is presented in Section 5.2. To get uniqueness results, we discuss properties of harmonic functions in Section 5.3. Finally, we extend the previous analysis to the Dirichlet problem on bounded domains in Section 5.4, and discuss an alternative approach in Section 5.5. In this section, we consider problems in dimension $d \geq 2$.

5.1 Fundamental solution of Laplace equation

In this section, we will identify a family of solutions to the Laplace equation (4.1):

$$\Delta u = 0.$$

Since the equation is invariant under rotation (see Exercise 33), it is natural to look for a solution of the form $u(x) = v(|x|)$ (where $|x| = \sqrt{\sum_{i=1}^d x_i^2}$ denotes the Euclidean norm of x).

Let us thus assume that $u(x) = v(|x|)$. Notice that (check it!★)

$$\nabla u(x) = v'(|x|)\frac{x}{|x|}$$

and thus

$$\text{div } \nabla u(x) = v''(|x|) + v'(|x|)\text{div } \left(\frac{x}{|x|}\right)$$

where $\text{div } \left(\frac{x}{|x|}\right) = \sum_{i=1}^d \partial_{x_i}\left(\frac{x_i}{|x|}\right) = \frac{d}{|x|} - \sum_{i=1}^d \frac{x_i^2}{|x|^3} = \frac{d-1}{|x|}$. This shows that

$$\Delta u(x) = v''(|x|) + \frac{d-1}{|x|}v'(|x|).$$

We are thus led to find a function $v : \mathbb{R}_+ \to \mathbb{R}$ such that

$$v''(r) + \frac{d-1}{r}v'(r) = 0.$$

Assuming that $v'(r) \neq 0$, one thus gets $(\ln v')'(r) = \frac{1-d}{r}$, and thus $v'(r) = \frac{c}{r^{d-1}}$ for some constant c_0. This finely yields the following solution $v : \mathbb{R}_+^* \to \mathbb{R}$:

$$v(r) = \begin{cases} c_0 \ln(r) + c_1 \text{ if } d = 2 \\ \dfrac{c_0}{r^{d-2}} + c_1 \text{ if } d \geq 3 \end{cases}$$

for some constants c_0 and c_1.

We can now introduce the function of interest in the following, namely the fundamental solution of $-\Delta$, also known as the Newtonian potential.

Definition 31. *The fundamental solution of $-\Delta$ in $\mathbb{R}^d$ is the function $\Phi : \mathbb{R}^d \setminus \{0\} \to \mathbb{R}$ defined by:*

- *For $d = 2$,*
$$\Phi(x) = -\frac{1}{2\pi} \ln |x|.$$
- *For $d \geq 3$,*
$$\Phi(x) = \frac{1}{d(d-2)\omega_d} \frac{1}{|x|^{d-2}},$$
where ω_d is the volume of the unit ball in $\mathbb{R}^d$.

The choice for the constants in the definition of Φ will be justified below (see Remark 17).

Notice that, by construction $\Delta\Phi = 0$ in $\mathbb{R}^d \setminus \{0\}$. However Φ is singular at 0. What can be shown is that, in the sense of distribution,

$$-\Delta\Phi = \delta_0 \tag{5.1}$$

where δ_0 is the Dirac distribution at the origin. The notion of distribution will be introduced in the lectures [4]. The precise meaning of this equality is given in the next proposition. The idea is to "multiply" the left and right-hand sides of (5.1) by a generic smooth test function φ (with compact support) and to "integrate" over $\mathbb{R}^d$. This yields $\varphi(0)$ in the right-hand side. In the left-hand side, notice that in (5.2), all the derivatives are transferred from Φ to φ, in order to have a well defined integral ; $\int_{\mathbb{R}^d} \Delta\Phi\varphi$ is not well defined because of the singularity of Φ at 0, whereas $\int_{\mathbb{R}^d} \Phi\Delta\varphi$, which is formally obtained from the former integral by integration by parts, is indeed well defined.

Let us introduce a notation that will be useful here and in the following. For any $k \geq 0$, for any open set $U \subset \mathbb{R}^d$ (one can possibly take $U = \mathbb{R}^d$), the set of $\mathcal{C}^k(U)$ functions with compact support is:

$$\mathcal{C}_c^k(U) = \{\varphi : U \to \mathbb{R}, \text{ such that } \varphi \in \mathcal{C}^k \text{ and } \mathrm{supp}(\varphi) \text{ is compact}\}$$

where $\mathrm{supp}(\varphi) = \overline{\{x, \varphi(x) \neq 0\}}$ is the support of φ.

Exercise 42. *Functions which are $C^\infty(\mathbb{R}^d)$ with compact support play a crucial role in analysis in order to build smooth approximations of non-smooth functions (see for example the mollification procedure in the proof of Theorem 39) or to introduce a smooth restriction of a function to a compact (see for example the localization procedure in the proof of Theorem 67). We check in this exercise that such functions do exist, and illustrate how they can be used.*

1. ★*Show that the function $\eta : \mathbb{R} \to \mathbb{R}$ defined by*
$$\eta(x) = \begin{cases} \exp\left(-\dfrac{1}{x^2-1}\right) \text{ for } x \in [-1,1] \\ 0 \text{ for } x \in (-\infty,-1) \cup (1,+\infty) \end{cases}$$
is in $C_c^\infty(\mathbb{R})$.

2. ★*Give an example of a function in* $C^\infty(\mathbb{R}^d)$ *with support* $[-1,1]^d$*, and one with support in* $B(0,1)$.

3. ★★*Let* $g:\mathbb{R}^d\to\mathbb{R}$ *be a continuous function. For* $x\in\mathbb{R}^d$*, let* $\eta_\varepsilon(x)=\frac{C}{\varepsilon^d}\eta\left(\frac{|x|}{\varepsilon}\right)$ *where* $C=\left(\int_{\mathbb{R}^d}\eta(|x|)\,dx\right)^{-1}$*. Check that* $\int_{\mathbb{R}^d}\eta_\varepsilon=1$*. Let* $g_\varepsilon(x)=\int_{\mathbb{R}^d}\eta_\varepsilon(x-y)g(y)\,dy$*. Prove that (i)* $g_\varepsilon\in C^\infty(\mathbb{R}^d)$ *(ii) for any* $x\in\mathbb{R}^d$*,* $\lim_{\varepsilon\to0}g_\varepsilon(x)=g(x)$ *and (iii) for any compact* $K\subset\mathbb{R}^d$*,* $\lim_{\varepsilon\to0}\|g_\varepsilon-g\|_{L^\infty(K)}=0$.

4. ★★*Let* $g:\mathbb{R}^d\to\mathbb{R}$ *be a* C^∞ *function. Show that it is possible to build a* C^∞ *function* $f:\mathbb{R}^d\to\mathbb{R}$ *such that* $f=g$ *on* $B(0,1/2)$ *and* $\mathrm{supp}(f)=B(0,1)$.

Proposition 32. *For any* $\varphi\in\mathcal{C}^2_c(\mathbb{R}^d)$*,*

$$-\int_{\mathbb{R}^d}\Phi\Delta\varphi=\varphi(0). \tag{5.2}$$

Proof: Let us first notice that $\int_{\mathbb{R}^d}\Phi\Delta\varphi$ is well defined since $\ln(|x|)$ (resp. $|x|^{2-d}$) is integrable around 0 in dimension $d=2$ (resp. $d>2$).

Step 1: Preliminary computations. Fix $\varphi\in C^\infty_c(\mathbb{R}^d)$. Choose $R>0$ so that $\mathrm{supp}\,\varphi\subset B_R$, where in this proof, $B_R=B(0,R)\subset\mathbb{R}^d$ denotes the ball of radius R. For $0<\varepsilon<R$, define the annulus $\Omega_{\varepsilon,R}=B_R\setminus\overline{B_\varepsilon}$. Divergence formula on $\Omega_{\varepsilon,R}$ gives (see Exercise 37)

$$\int_{\Omega_{\varepsilon,R}}\Phi(\Delta\varphi)\,dx=\int_{\partial\Omega_{\varepsilon,R}}\left(\Phi\,\partial_n\varphi-\varphi\,\partial_n\Phi\right)d\sigma \tag{5.3}$$

because $\Delta\Phi=0$ on $\Omega_{\varepsilon,R}$. On ∂B_R we have $\varphi\equiv0$ (for R large enough), so that boundary contribution vanishes. Therefore one obtains:

$$\int_{\Omega_{\varepsilon,R}}\Phi(\Delta\varphi)\,dx=\int_{|x|=\varepsilon}\Phi\,\partial_n\varphi\,d\sigma\;-\;\int_{|x|=\varepsilon}\varphi\,\partial_n\Phi\,d\sigma.$$

Let us emphasize that in the surface integrals on $\{x,|x|=\varepsilon\}$, n denotes the unit outward normal to $\partial\Omega_{\varepsilon,R}$ so that $n(x)=-\frac{x}{|x|}=-\frac{x}{\varepsilon}$. We let $\varepsilon\to0$ and estimate the two terms in the right-hand side.

Step 2: The first term tends to 0. For $d\geq3$, on $|x|=\varepsilon$ one has $\Phi(x)=C\,\varepsilon^{2-d}$ for some constant C, $|\partial_n\varphi|=|\nabla\varphi\cdot n|\leq\|\nabla\varphi\|_\infty$, and $|\partial B_\varepsilon|=|\partial B_1|\,\varepsilon^{d-1}$. Hence

$$\Big|\int_{|x|=\varepsilon}\Phi\,\partial_n\varphi\,d\sigma\Big|\leq C\,\|\nabla\varphi\|_\infty\,\varepsilon^{2-d}\,|\partial B_1|\,\varepsilon^{d-1}=C\,|\partial B_1|\,\|\nabla\varphi\|_\infty\,\varepsilon\to0.$$

Likewise, for $d=2$, on $|x|=\varepsilon$, $|\Phi(\varepsilon)|=C|\ln\varepsilon|$ and $|\partial B_\varepsilon|=2\pi\varepsilon$, so the product is $O(\varepsilon|\log\varepsilon|)\to0$.

Step 3: The second term tends to $-\varphi(0)$. Since φ is $\mathcal{C}^1$ at 0, $\varphi(x)=\varphi(0)+o(1)$ uniformly for x on $\{x,|x|=\varepsilon\}$. For $d\geq3$, for x such that $|x|=\varepsilon$,

$$\begin{aligned}\partial_n\Phi(x)&=\nabla\Phi(x)\cdot n(x)\\&=\frac{-1}{d\omega_d}|x|^{1-d}\frac{x}{|x|}\cdot\frac{-x}{|x|}\\&=\frac{|x|^{1-d}}{d\omega_d}=\frac{\varepsilon^{1-d}}{d\omega_d}\end{aligned} \tag{5.4}$$

so (using the fact that $|\partial B_\varepsilon|=d\omega_d\varepsilon^{d-1}$, see Exercise 39)

$$\int_{|x|=\varepsilon}\varphi\,\partial_n\Phi\,d\sigma=\varphi(0)\int_{|x|=\varepsilon}\partial_n\Phi\,d\sigma+o(1)=\varphi(0)\frac{\varepsilon^{1-d}}{d\omega_d}d\omega_d\varepsilon^{d-1}+o(1)=\varphi(0)+o(1).$$

Likewise, for $d = 2$, $\partial_n \Phi = \frac{1}{2\pi} \frac{1}{\varepsilon}$ on $\{x, |x| = \varepsilon\}$ and $|\partial B_\varepsilon| = 2\pi\varepsilon$, hence the same limit $\varphi(0)$.
Step 4: Conclusion. Letting $\varepsilon \to 0$ in (5.3) we obtain (by dominated convergence for the left-hand side)

$$\int_{B_R} \Phi(\Delta\varphi)\, dx = -\varphi(0).$$

Because φ vanishes outside B_R, the left-hand side equals $\int_{\mathbb{R}^n} \Phi(\Delta\varphi)\, dx$. This concludes the proof of Proposition 32.

♢

Remark 17. *The constants in the definition of Φ have been chosen in order to get exactly $\varphi(0)$ in the right-hand side in Proposition 32 (and not a multiple of $\varphi(0)$).*

Remark 18. *From a physical viewpoint, if one thinks of the electrostatics interpretation of the Poisson equation (see Section 4.6), in view of* (5.1)*, Φ can be understood as the electric potential created by a point charge at* 0*.*

5.2 A solution to the Poisson equation in $\mathbb{R}^d$

Let us first proceed formally. As seen above (see (5.1), $-\Delta\Phi = \delta_0$, and thus, for any $y \in \mathbb{R}^d$, one has $-\Delta_x \Phi(x - y) = \delta_y(dx)$.

Let us consider a continuous function $f : \mathbb{R}^d \to \mathbb{R}$ with compact support. Let $h > 0$ be a space grid and $(y_k^h)_{k \geq 0}$ be a numbering of the points in $\mathrm{supp}(f) \cap h\mathbb{Z}^d$. Using the linearity of the equation, one thus expects that a solution to

$$-\Delta_x u = h^d \sum_{k=1}^{K} f(y_k) \delta_{y_k}(dx)$$

is

$$u(x) = h^d \sum_{k=1}^{K} f(y_k) \Phi(x - y_k).$$

By sending h to 0, it is thus natural to think that a solution to the Poisson equation

$$-\Delta u = f$$

could be given by the convolution:

$$u(x) = \int_{\mathbb{R}^d} f(y) \Phi(x - y) dy. \tag{5.5}$$

Definition 33. *For two functions $f : \mathbb{R}^d \to \mathbb{R}$ and $g : \mathbb{R}^d \to \mathbb{R}$ the convolution of f and g is the function $f * g : \mathbb{R}^d \to \mathbb{R}$ defined by, for all $x \in \mathbb{R}^d$, such that $\int_{\mathbb{R}^d} |f(y)g(x - y)|\, dy < \infty$,*

$$f * g(x) = \int_{\mathbb{R}^d} f(y) g(x - y)\, dy.$$

The convolution is well defined on $\mathbb{R}^d$ for example if f is integrable ($\int_{\mathbb{R}^d} |f| < \infty$) and g is bounded.

Exercise 43. ★*Assume that f is integrable and g is bounded. Show that for all $x \in \mathbb{R}^d$, $f * g(x) = \int_{\mathbb{R}^d} f(y) g(x - y)\, dy$ is well defined and that $f * g(x) = \int_{\mathbb{R}^d} f(x - y) g(y)\, dy$. Check that*

$$\int_{\mathbb{R}^d} |f * g| \leq \left(\int_{\mathbb{R}^d} |f| \right) \left(\int_{\mathbb{R}^d} |g| \right).$$

From the result of the previous exercise, it is natural to expect that $f * g$ can be defined as soon as $\int_{\mathbb{R}^d} |f| < \infty$ and $\int_{\mathbb{R}^d} |g| < \infty$. This is indeed the case, but this resquires to introduce Lebesgue integrals, which is a generalization of Riemann integrals. This will be explained in the lectures [4].

The next Theorem shows that the function u defined by (5.5) is indeed a solution to the Poisson equation, under appropriate assumptions on f.

Theorem 34. *Let $f \in \mathcal{C}_c^2(\mathbb{R}^d)$, and let us define u by (5.5). Then $u \in \mathcal{C}^2(\mathbb{R}^d)$ and $-\Delta u = f$ in $\mathbb{R}^d$.*

Proof: Since f is continuous with compact support (thus bounded) and $y \mapsto \Phi(x-y)$ is integrable, $u(x)$ is well defined, for any $x \in \mathbb{R}^d$. Moreover, one has $u(x) = \int_{\mathbb{R}^d} f(y)\Phi(x-y)dy = \int_{\mathbb{R}^d} f(x-y)\Phi(y)dy$. By dominated convergence, since f, $\partial_{x_i} f$ and $\partial_{x_i,x_j} f$ are bounded functions and Φ is integrable, one can check that $u \in \mathcal{C}^2(\mathbb{R}^d)$ and

$$-\Delta_x u(x) = \int_{\mathbb{R}^d} (-\Delta f)(x-y)\Phi(y)dy.$$

Let us fix $x \in \mathbb{R}^d$ and introduce $g_x : y \in \mathbb{R}^d \mapsto f(x-y) \in \mathbb{R}$. One can check that $\Delta f(x-y) = \Delta_y g_x(y)$. As a consequence, for any $x \in \mathbb{R}^d$, using Proposition 32,

$$\begin{aligned} -\Delta_x u(x) &= -\int_{\mathbb{R}^d} \Delta_y g_x(y)\Phi(y)dy \\ &= g_x(0) = f(x) \end{aligned}$$

which concludes the proof. ♢

We thus have constructed one solution to the Poisson equation. The question is now to understand the uniqueness. Using the linearity of the equation, if u and v are C^2 functions such that $-\Delta u = -\Delta v = f$, then $-\Delta(u-v) = 0$. Therefore, understanding the uniqueness of solutions to the Poisson equation amounts to understanding the uniqueness of solutions to the Laplace equation, namely harmonic functions. This requires to study specifically some properties of harmonic functions (in particular, uniqueness will be a consequence of so-called maximum principles).

Remark 19. *In Theorem 34, we assume $f \in \mathcal{C}_c^2(\mathbb{R}^d)$. The result actually holds under weaker conditions on f (typically f Hölder-continuous is sufficient).*

5.3 Some properties of harmonic functions

5.3.1 The mean-value formula

Here and in the following, we use the notation: for any domain $U \subset \mathbb{R}^d$ such that $|U| < \infty$ and for any integrable function $f : U \to \mathbb{R}$,

$$\fint_U f = \frac{1}{|U|}\int_U f$$

where, we recall, $|U| = \int_{\mathbb{R}^d} 1_U(x)\,dx$ denotes the volume of U. Likewise, assuming $|\partial U| < \infty$,

$$\fint_{\partial U} f = \frac{1}{|\partial U|}\int_{\partial U} f d\sigma$$

where, $|\partial U| = \int_{\partial U} d\sigma$ denotes the surface of ∂U.

Theorem 35. *Let U be an open set of $\mathbb{R}^d$ and let us assume that $u \in \mathcal{C}^2(U)$ satisfies $-\Delta u = 0$ in U. Then for any $x \in U$ and any $r > 0$ such that $\overline{B(x,r)} \subset U$,*

$$u(x) = \fint_{\partial B(x,r)} u \, d\sigma = \fint_{B(x,r)} u.$$

Proof: Fix $x \in U$ and $r > 0$ with $B(x,r) \subset U$. For $\rho \in (0,r)$ set

$$S_\rho := \partial B(x,\rho), \qquad I(\rho) := \int_{S_\rho} u \, d\sigma, \qquad m(\rho) := \fint_{S_\rho} u \, d\sigma = \frac{I(\rho)}{|S_\rho|}.$$

Using the change of variables $y = x + \rho\theta$ with $\theta \in S_1$ (so $d\sigma_y = \rho^{d-1} \, d\sigma_\theta$), we get

$$I(\rho) = \rho^{d-1} \int_{S_1} u(x + \rho\theta) \, d\sigma_\theta$$

so that, since $|S_\rho| = \rho^{d-1}|S_1|$,

$$m(\rho) = \fint_{S_1} u(x + \rho\theta) \, d\sigma_\theta.$$

Differentiating with respect to ρ yields

$$\begin{aligned} m'(\rho) &= \fint_{S_1} \nabla u(x + \rho\theta) \cdot \theta \, d\sigma_\theta \\ &= |S_1|^{-1} \rho^{d-1} \int_{S_\rho} \nabla u(y) \cdot \frac{y - x}{\rho} \, d\sigma_y \\ &= \fint_{S_\rho} \partial_n u \, d\sigma \end{aligned}$$

where n denotes the unit outward normal to $B(0,\rho)$. By the divergence formula,

$$\int_{S_\rho} \partial_n u \, d\sigma = \int_{B(x,\rho)} \Delta u = 0,$$

because u is harmonic. Hence $m'(\rho) = 0$ and $m(\rho)$ is constant in ρ. Taking the limit $\rho \to 0$ and using continuity of u gives

$$m(\rho) = \lim_{\rho \to 0} m(\rho) = u(x),$$

so

$$u(x) = \fint_{S_\rho} u \, d\sigma \qquad \text{for all } \rho \in (0,r),$$

and in particular for $\rho = r$.

To obtain the mean value over the ball, write (see Exercise 39)

$$\int_{B(x,r)} u = \int_0^r \left(\int_{S_\rho} u \, d\sigma \right) d\rho = \int_0^r |S_\rho| \, m(\rho) \, d\rho = \int_0^r |S_\rho| \, u(x) \, d\rho = u(x) \, |B(x,r)|.$$

Dividing by $|B(x,r)|$ yields

$$u(x) = \fint_{B(x,r)} u.$$

This completes the proof. ♢

Exercise 44. ★★★*Prove the converse statement: if $u \in \mathcal{C}^2(U)$ is such that for all $x \in U$, for all $r > 0$ such that $B(x,r) \subset U$, $u(x) = \fint_{\partial B(x,r)} u \, d\sigma$, then u is harmonic in U.*

The mean-value formula has two crucial consequences that are presented in the next two sections: the Liouville theorem, which will provide a uniqueness result for the Poisson equation in $\mathbb{R}^d$ (see Section 5.3.2), and the maximum principle, which will provide a uniqueness result for the Dirichlet problem on a bounded domain (see Section 5.3.3 and Section 5.4).

5.3.2 Liouville theorem

We will prove in this section a uniqueness result for harmonic functions.

Theorem 36 (Liouville theorem)**.** *Assume that $u \in \mathcal{C}^2(\mathbb{R}^d)$ is harmonic and with sublinear growth at infinity (sometimes written $|u(x)| = o(|x|)$ when $x \to \infty$) in the following sense:*

$$\forall \varepsilon > 0,\ \exists M > 0,\ \forall x \in \mathbb{R}^d, |x| > M \Rightarrow |u(x)| \leq \varepsilon |x|. \tag{5.6}$$

Then u is constant.

The standard Liouville theorem states that a harmonic bounded function is constant. Theorem 36 is a stronger version (indeed, if u is bounded, (5.6) holds with $M = \|u\|_{L^\infty}/\varepsilon$).

Proof: Notice that for any $i \in \{1, \dots, d\}$, $\partial_{x_i} u$ is harmonic (differentiate the Laplace equation with respect to x_i). We use here the regularity result from Theorem 39 which shows that $\partial_{x_i} u \in \mathcal{C}^2(\mathbb{R}^d)$. Let $x_0 \in \mathbb{R}^d$ and $r > 0$. Then, by the mean-value property and divergence formula (in particular Exercise 36),

$$\begin{aligned}
|\partial_{x_i} u(x_0)| &= \left| \fint_{B(x_0,r)} \partial_{x_i} u \right| \\
&= \frac{1}{r^d \omega_d} \left| \int_{B(x_0,r)} \partial_{x_i} u \right| \\
&= \frac{1}{r^d \omega_d} \left| \int_{\partial B(x_0,r)} u\, n_i d\sigma \right| \\
&\leq \frac{1}{r^d \omega_d} \int_{\partial B(x_0,r)} |u| d\sigma \\
&\leq \frac{1}{r^d \omega_d} |\partial B(x_0, r)| \|u\|_{L^\infty(\partial B(x_0,r))} \\
&= \frac{d}{r} \|u\|_{L^\infty(\partial B(x_0,r))}.
\end{aligned} \tag{5.7}$$

By the sublinear growth assumption, (5.7) goes to 0 when r goes to infinity. Indeed, fix $\varepsilon > 0$ and consider the associated M in (5.6). For any $x \in \partial B(x_0, r)$, if $r > M + |x_0|$, then $|x| > M$ and thus $|u(x)| \leq \varepsilon |x|$. Therefore, for $r > M + |x_0|$,

$$|\partial_{x_i} u(x_0)| \leq \frac{d}{r} \|u\|_{L^\infty(\partial B(x_0,r))} \leq \frac{d}{r} \varepsilon \sup_{\partial B(x_0,r)} |x| \leq \frac{d}{r} \varepsilon (r + |x_0|) = d\varepsilon \left(1 + \frac{|x_0|}{r}\right)$$

which can be made smaller than $(d+1)\varepsilon$ by choosing r large enough. Thus $|\partial_{x_i} u(x_0)| = 0$

Since $x_0 \in \mathbb{R}^d$ and $i \in \{1, \dots, d\}$ are arbitrary, this shows that $\nabla u = 0$, and thus u is constant. ♢

Remark 20. *It can actually be shown that if u is harmonic on $\mathbb{R}^d$ and bounded from below (or from above), then u is constant (see for example [7, Corollary 3.12]).*

Remark 21. *There exists infinitely many harmonic functions which do not satisfy the sublinear growth assumption. Think for example of linear functions $u(x) = a \cdot x$ (for $a \in \mathbb{R}^d$) or harmonic polynomials such as $u(x) = x_1^2 - x_2^2$ (for $d \geq 2$).*

As a direct consequence of Liouville theorem, one gets the following existence and uniqueness (up to an additive constant) result of solution to the Poisson equation in $\mathbb{R}^d$.

Corollary 37 (Existence and uniqueness of solutions to the Poisson equation in $\mathbb{R}^d$)**.** *Let $f \in \mathcal{C}_c^2(\mathbb{R}^d)$. Let u be a solution of $-\Delta u = f$ in $\mathbb{R}^d$ with sublinear growth at infinity (in the sense of (5.6)). Then, there exists a constant $C \in \mathbb{R}$ such that for all $x \in \mathbb{R}^d$*

$$u(x) = \int_{\mathbb{R}^d} \Phi(x-y) f(y)\, dy + C. \tag{5.8}$$

Proof: In dimension $d \geq 3$, $\lim_{|z|\to\infty} \Phi(z) = 0$. Therefore, the function $\tilde{u}(x) = \int_{\mathbb{R}^d} \Phi(x-y) f(y)\, dy$ (which is a solution to $-\Delta u = f$, see Theorem 34) is bounded and thus satisfies (5.6). Therefore, for any other solution u to $-\Delta u = f$, the function $(u - \tilde{u})$ is harmonic and bounded, and thus constant.

In dimension $d = 2$, let $\tilde{u}(x) = -\frac{1}{2\pi} \int_{\mathbb{R}^2} \ln(|x-y|) f(y)\, dy$. One has $-\Delta u = f$ in $\mathbb{R}^2$, and as $|x| \to \infty$

$$\tilde{u}(x) = -\frac{1}{2\pi} \left(\int_{\mathbb{R}^2} f \right) \ln|x| + O\left(\frac{1}{|x|} \right).$$

Thus, if u is such that $-\Delta u = f$, then $v = u - \tilde{u}$ is harmonic and has sublinear growth at infinity and thus satisfies (5.6). This implies that v is constant. ♢

Remark 22. *The sublinear growth assumption in Corollary 37 is necessary to select, in some sense, the "right solutions" to the Poisson equation on $\mathbb{R}^d$, namely the ones defined by convolution (see Equation (5.8)). This is a typical example where mathematics meet physics: mathematicians give a proper meaning to the notion of solutions to a partial differential equation derived from physical principles, in order to obtain a uniquely well defined solution (here, up to an additive constant) which matches the one physicists are referring to. There may be many appropriate functional settings: in these lectures, we stick to classical solutions, but in forthcoming lectures, much more powerful approaches will be introduced [4, 11].*

The reader may retain the following simple observation: the stricter the notion of solution (i.e. the smaller the ensemble of functions where the solution is looked for), the more difficult it is to prove existence but the easier it is to get uniqueness.

5.3.3 Maximum principles on a bounded domain

Theorem 38. *Assume that $U \subset \mathbb{R}^d$ is open and bounded. Assume that $u \in \mathcal{C}^2(U) \cap \mathcal{C}(\overline{U})$ is harmonic in U. The weak maximum principle states that*

$$\max_{\overline{U}} u = \max_{\partial U} u.$$

The strong maximum principle states that, if in addition U is connected and there exists $x_0 \in U$ such that $u(x_0) = \max_{\overline{U}} u$, then u is constant in U.

Proof: Notice that since u is continuous on the compact set $\overline{U}$, the maximum is well defined: there exists $x_* \in \overline{U}$ such that $u(x_*) = \max_{\overline{U}} u$.

Let us start by proving the strong maximum principle. Let us thus assume that U is connected and there exists $x_0 \in U$ such that $u(x_0) = \max_{\overline{U}} u$. Choose $r > 0$ so that $\overline{B(x_0, r)} \subset U$ (such a $r > 0$ exists since $x_0 \in U$ and U is open). By the mean-value property,

$$u(x_0) = \fint_{B(x_0,r)} u.$$

Since $u \leq u(x_0)$ on $\overline{U}$, in particular on $B(x_0, r)$, the average above can equal $u(x_0)$ only if $u \equiv u(x_0)$ on $B(x_0, r)$. This shows that the set

$$\Omega := \{x \in U : u(x) = u(x_0)\}$$

is open. It is also obviously non-empty and relatively closed in U since u is continuous. Therefore, $\Omega = U$ since U is connected.

Let us now prove the first assertion. Let us recall that $x_* \in \overline{U}$ is such that $u(x_*) = \max_{\overline{U}} u$. If $x_* \in \partial U$, we are done. Suppose instead that $x_* \in U$. The strong maximum principle shows that then, u is constant on the connected component V of U which contains x_*. By continuity of u, this implies that $\max_{\overline{U}} u = u(x_*) = \max_{\partial V} u = \max_{\partial U} u$. ♢

Of course, by changing u in $-u$, one easily obtains similar statements as in Theorem 38 (write them!★) involving $\min_{\overline{U}} u$ and $\min_U u$.

Exercise 45. ★*Show that the weak maximum principle is equivalent to the following statement: let $u \in \mathcal{C}^2(U) \cap \mathcal{C}(\overline{U})$ be a harmonic function on an open bounded set $U \subset \mathbb{R}^d$. Then,*

$$u \leq 0 \text{ on } \partial U \Longrightarrow u \leq 0 \text{ on } U.$$

Again, by replacing u by $-u$, a similar statement can be made with $u \geq 0$ on $\partial U \Longrightarrow u \geq 0$ on U.

Exercise 46. ★*Give an example of a harmonic function u an on open bounded set U which is not constant.*

Exercise 47. ★*Let $U \subset \mathbb{R}^d$ be an open and bounded set. Let $u \in \mathcal{C}^2(U) \cap \mathcal{C}(\overline{U})$ be a harmonic function. Assume that $u = 0$ on ∂U. Prove that $u = 0$ on U.*

Exercise 48. ★*Prove the weak and strong maximum principle (the results of Theorem 38) in dimension $d = 1$ by an elementary argument.* Hint: what are the solutions to $u'' = 0$?

5.3.4 Additional results: regularity and Harnack inequality for harmonic functions

Let us state two additional results whose proof are beyond the scope of these lectures. Let us start with a so-called regularity result.

Theorem 39 (Regularity of harmonic functions)**.** *Let U be an open set of $\mathbb{R}^d$. Let $u : U \to \mathbb{R}$ be a $\mathcal{C}^2$ function such that $-\Delta u = 0$ on U. Then u is C^∞ on U.*

This result may seem surprising: the differential operator $-\Delta$ only involves derivatives of the form $-\partial_{x_i,x_i}$, but this theorem shows that for harmonic functions, all the derivatives are well defined and continuous functions.

Proof: The proof is divided into three steps.

Step 1: Introduction of a mollifier. Let $\eta : \mathbb{R}^d \to \mathbb{R}$ be defined by

$$\eta(x) = C \exp\left(\frac{1}{|x|^2 - 1}\right) 1_{|x|<1}$$

$C > 0$ being chosen such that $\int_{\mathbb{R}^d} \eta = 1$. Notice that $\eta : \mathbb{R}^d \to \mathbb{R} \in \mathcal{C}^\infty(\mathbb{R}^d)$. For $\varepsilon > 0$, let $\eta_\varepsilon \in \mathcal{C}^\infty(\mathbb{R}^d)$ be defined by

$$\eta_\varepsilon(x) = \frac{1}{\varepsilon^d} \eta\left(\frac{x}{\varepsilon}\right).$$

Notice that $\text{supp}(\eta_\varepsilon) \subset B(0, \varepsilon)$ and $\int_{\mathbb{R}^d} \eta_\varepsilon = 1$.

Step 2: Definition and properties of the mollification. Let us now introduce the open set $U_\varepsilon \subset U$:

$$U_\varepsilon = \{x \in U, \mathrm{dist}(x, \partial U) > \varepsilon\}$$

and let us define

$$\forall x \in \varepsilon \in U_\varepsilon,\ u_\varepsilon(x) = \int_{\mathbb{R}^d} \eta_\varepsilon(x-y)u(y)\,dy.$$

The function u_ε is called a mollification of u and η_ε is a mollifier. Notice that $u_\varepsilon(x) = \int_{B(0,\varepsilon)} \eta_\varepsilon(y)u(x-y)\,dy$. One can check the fundamental properties on u_ε (these properties actually holds assuming only that $u \in \mathcal{C}(U)$, see Exercise 42):

(i) $u_\varepsilon \in \mathcal{C}^\infty(U_\varepsilon)$

(ii) for any $x \in U$, $\lim_{\varepsilon\to 0} u_\varepsilon(x) = u(x)$

(iii) for any compact $K \subset U$, $\lim_{\varepsilon\to 0} \|u_\varepsilon - u\|_{L^\infty(K)} = 0$

The function u_ε is thus a smooth approximation of u. We will only used the first property (i) to conclude the proof.

Step 3: Conclusion of the proof. Let us now show that $u = u_\varepsilon$ on $U = U_\varepsilon$, which will imply that $u \in \mathcal{C}^\infty(U_\varepsilon)$ for all $\varepsilon > 0$ and thus concludes the proof.

For $r \in \mathbb{R}$, let us denote $\zeta(r) = C\exp\left(\frac{1}{r^2-1}\right)1_{|x|<1}$ so that for $x \in \mathbb{R}^d$, $\eta(x) = \zeta(|x|)$. For any $x \in U_\varepsilon$, using Exercise 39 and the mean-value formula (see Theorem 35),

$$\begin{aligned}
u_\varepsilon(x) &= \int_{B(0,\varepsilon)} \eta_\varepsilon(y)u(x-y)\,dy \\
&= \varepsilon^{-d}\int_{B(0,\varepsilon)} \eta(y/\varepsilon)u(x-y)\,dy \\
&= \varepsilon^{-d}\int_0^\varepsilon \int_{\partial B(0,r)} \eta(y/\varepsilon)u(x-y)\sigma(dy)\,dr \\
&= \varepsilon^{-d}\int_0^\varepsilon \zeta(r/\varepsilon)\int_{\partial B(x,r)} u(z)\sigma(dz)\,dr \\
&= u(x)\varepsilon^{-d}\int_0^\varepsilon \zeta(r/\varepsilon)r^{n-1}|\partial B(0,1)|dr \\
&= u(x)\varepsilon^{-d}\int_0^\varepsilon \int_{\partial B(0,r)} \eta(y/\varepsilon)\sigma(dy)\,dr \\
&= u(x)\int_{B(0,\varepsilon)} \eta_\varepsilon = u(x).
\end{aligned}$$

This concludes the proof. ♢

Theorem 40 (Harnack's inequality)**.** *Let U be an open subset of $\mathbb{R}^d$. Let V be an open bounded connected subset of U such that $\overline{V} \subset U$. Then, there exists a positive constant V (which only depends on V) and such that, for all harmonic functions u on U with $u \geq 0$,*

$$\sup_V u \leq C \inf_V u.$$

As a consequence, for all $x, y \in V$,

$$C^{-1}u(y) \leq u(x) \leq Cu(y).$$

In some sense, Harnack's inequality states that the values of u within V are all comparable. Intuitively, since the Laplace operator "locally averages" the values of u, this holds since V is at positive distance from ∂U.

Proof: Let $r = dist(V, \partial U)/4 > 0$. Let $x, y \in V$ be such that $|x - y| \leq r$. Then, by the mean-value property, since $B(y, r) \subset B(x, 2r)$

$$\begin{aligned} u(x) &= \fint_{B(x,2r)} u = |B(x,2r)|^{-1} \int_{B(x,2r)} u \\ &\geq |B(x,2r)|^{-1} \int_{B(y,r)} u \\ &= 2^{-d} \fint_{B(y,r)} u = 2^{-d} u(y). \end{aligned}$$

This implies that, for all $x, y \in V$ such that $|x - y| \leq r$,

$$2^{-d}u(y) \leq u(x) \leq 2^d u(y).$$

Since V is connected and $\overline{V}$ is compact, one can cover V by a finite number of balls $B(x_i, r)$: $V \subset U_{i=1}^{I} B(x_i, r)$. Necessarily, for any $i \in \{1, \ldots, I\}$, there exists $j \in \{1, \ldots, I\}$ with $j \neq i$ and such that $B(x_i, r) \cap B(x_j, r) \neq \emptyset$. For any two points $x, y \in V$, one can thus connect x and y by a chain of points separated by a distance at most r, which yields:

$$u(x) \geq 2^{-dI} u(y).$$

This concludes the proof. ♢

5.4 Dirichlet problem on a bounded domain

As already mentioned above, in many practical cases, we will be interested in solutions to the following Dirichlet problem. Let $\Omega \subset \mathbb{R}^d$ be a $\mathcal{C}^1$ bounded open set. Our objective is to consider solutions to

$$\begin{cases} -\Delta u = f \text{ in } \Omega, \\ \quad u = g \text{ on } \partial\Omega. \end{cases} \tag{5.9}$$

In all this section, we assume that

$$f \in \mathcal{C}(\Omega) \text{ and } g \in \mathcal{C}(\partial\Omega) \tag{5.10}$$

and we consider solutions with at least the following regularity: $u \in \mathcal{C}^2(\Omega) \cap \mathcal{C}(\overline{\Omega})$ (in particular, the values of u on $\partial\Omega$ are well defined).

5.4.1 Uniqueness of solutions

The uniqueness of solutions is a direct consequence of the weak maximum principle stated in Theorem 38.

Proposition 41. *There exists at most one solution of* (5.9) *with regularity* $\mathcal{C}^2(\Omega) \cap \mathcal{C}(\overline{\Omega})$.

Proof: Let $u \in \mathcal{C}^2(\Omega) \cap \mathcal{C}(\overline{\Omega})$ and $v \in \mathcal{C}^2(\Omega) \cap \mathcal{C}(\overline{\Omega})$ be two functions satisfying (5.9). Let $w = u - v$. Then w is harmonic in Ω and such that $w = 0$ on $\partial\Omega$. Thus, by the weak maximum principle of Theorem 38, $w \leq 0$ on Ω. By applying the same reasoning to $-w$, one gets that $w \geq 0$ and thus $w = 0$. ♢

5.4.2 Representation of the solution using Green's function

Our objective in this section is to obtain an integral representation of the solution to (5.9), assuming that it is in $\mathcal{C}^2(\overline{\Omega})$.

Let $u \in \mathcal{C}^2(\overline{\Omega})$. Let $x \in \Omega$ and $\varepsilon > 0$ such that $B(x, \varepsilon) \subset \Omega$. Let us introduce

$$V_\varepsilon = \Omega \setminus B(x, \varepsilon).$$

Notice that $\partial V_\varepsilon = \partial\Omega \cup \partial B(x, \varepsilon)$.

From divergence formula (see Exercise 37) and the fact that Ω is bounded, one gets

$$\int_{V_\varepsilon} (u(y)\Delta\Phi(x-y) - \Phi(x-y)\Delta u(y))\, dy = \int_{\partial V_\varepsilon} (u(y)\partial_n\Phi(x-y) - \Phi(x-y)\partial_n u(y))\, \sigma(dy)$$

where, we recall, n denotes the unit outward normal to V_ε. Recall that $\Delta\Phi(x-y) = 0$ in V_ε. Moreover, using (5.4),

$$\int_{\partial B(x,\varepsilon)} u(y)\partial_n\Phi(x-y)\, \sigma(dy) = \frac{\varepsilon^{1-d}}{d\omega_d} |\partial B(x,\varepsilon)| \fint_{\partial B(x,\varepsilon)} u(y)\, \sigma(dy)$$

so that, by the continuity of u, $\lim_{\varepsilon\to 0} \int_{\partial B(x,\varepsilon)} u(y)\partial_n\Phi(x-y)\, \sigma(dy) = u(x)$. Besides,

$$\int_{\partial B(x,\varepsilon)} \Phi(x-y)\partial_n u(y)\, \sigma(dy) \leq \|\nabla u\|_{L^\infty(\Omega)} \|\Phi\|_{L^\infty(\partial B(0,\varepsilon))} |\partial B(0,1)| \varepsilon^{d-1}$$

and thus $\lim_{\varepsilon\to 0} \int_{\partial B(x,\varepsilon)} \Phi(x-y)\partial_n u(y)\, \sigma(dy) = 0$. Therefore, by sending $\varepsilon \to 0$ one obtains:

$$u(x) = -\int_\Omega \Phi(x-y)\Delta u(y)\, dy + \int_{\partial\Omega} (\Phi(x-y)\partial_n u(y) - u(y)\partial_n\Phi(x-y))\, \sigma(dy). \tag{5.11}$$

This formula does not yield $u(x)$ in terms of f and g since it involves $\partial_n u$ which is unknown.

For any $x \in \Omega$, let us now introduce the so-called corrector function $h_x : \Omega \to \mathbb{R}^d$ such that

$$\begin{cases} -\Delta_y h_x(y) = 0 \text{ for all } y \in \Omega, \\ h_x(y) = \Phi(y-x) \text{ for all } y \in \partial\Omega. \end{cases} \tag{5.12}$$

Notice that even though $y \mapsto \Phi(y-x)$ is singular at point x, we only need in (5.12) this function on $\partial\Omega$ where it is smooth, since $x \in \Omega$. We will admit the following result:

Proposition 42. *For smooth domain Ω, there exists a unique solution $h_x \in \mathcal{C}^2(\Omega) \cap \mathcal{C}^0(\overline{\Omega})$ to* (5.12).

Definition 43. *The Green's function for the Dirichlet problem on Ω is defined by: for all $x, y \in \Omega$, with $x \neq y$,*

$$G(x,y) = \Phi(y-x) - h_x(y).$$

Theorem 44 (A representation formula for solutions to the Dirichlet problem)**.** *Let $u \in \mathcal{C}^2(\overline{\Omega})$ be a solution to* (5.9)*. Then*

$$u(x) = \int_\Omega G(x,y) f(y)\, dy - \int_{\partial\Omega} \partial_n G(x,y) g(y) \sigma(dy) \tag{5.13}$$

where $\partial_n G(x,y) = \nabla_y G(x,y) \cdot n(y)$ is the outward normal derivative of G with respect to the y variable.

Proof: From divergence formula, using (5.12), one has:

$$\begin{aligned}\int_\Omega (-h_x(y)\Delta u(y))\ dy &= \int_{\partial\Omega} (u(y)\partial_n h_x(y) - h_x(y)\partial_n u(y))\ \sigma(dy)\\ &= \int_{\partial\Omega} (u(y)\partial_n h_x(y) - \Phi(x-y)\partial_n u(y))\ \sigma(dy).\end{aligned}$$

Therefore, by adding this equality to (5.11), one gets

$$u(x) - \int_\Omega h_x(y)\Delta u(y)\,dy = -\int_\Omega \Phi(x-y)\Delta u(y)\,dy + \int_{\partial\Omega} (u(y)\partial_n h_x(y) - u(y)\partial_n \Phi(x-y))\ \sigma(dy).$$

Thus, if $u \in \mathcal{C}^2(\overline{\Omega})$ satisfies (5.9), then $u(x) = \int_\Omega G(x,y)f(y)\,dy - \int_{\partial\Omega} \partial_n G(x,y)g(y)\sigma(dy)$. ♢

Let us insist on the fact that Theorem 44 provides a representation formula for a $\mathcal{C}^2(\overline{\Omega})$ solution: it does not tell us what are the assumptions on f and g such that the right-hand side of (5.13) indeed defines a $\mathcal{C}^2(\overline{\Omega})$ solution.

In order to build solutions to the problem (5.9), one strategy then consists in showing the existence of a corrector function h_x, and then to check that u defined by (5.13) indeed satisfies the Poisson equation (5.9). This has to be done case by case, for each domain Ω. This calls for other more robust approaches: in Section 5.5, we quickly present another mathematical approach based on so-called variational formulations of the problem and which will be more throughly developed in forthcoming mathematical lectures ; in Chapter 6, we will explain how to approximate solutions to (5.9) (assuming there exists a solution) without using Green functions.

Exercise 49. *Let us consider the* $1d$*-Dirichlet problem on the interval* $I = (0,1)$*:*

$$\begin{cases} -u'' = f \text{ on } (0,1) \\ u(0) = u(1) = 0 \end{cases}$$

Show that the solution is

$$u(x) = \int_0^L G(x,y)f(y)\,dy$$

where $G(x,y) = x(1-y)1_{x<y} + y(1-x)1_{y<x}$.

Remark 23. *A practical way to represent the Green function for a partial differential equation is to look for a probabilistic representation of the solution. For Poisson equations, this can be done by using the Brownian motion, a stochastic process that will be introduced in the lectures [3].*

Remark 24. *In terms of distribution, the Green function* $G(x,y)$ *satisfies:*

$$\begin{cases} -\Delta_y G(x,y) = \delta_x(dy) \text{ in } \Omega, \\ G(x,y) = 0 \text{ on } \partial\Omega. \end{cases}$$

5.5 Variational formulation of the Dirichlet problem

As mentioned above, looking for explicit representations has some limitations. A much more fruitful viewpoint is to use co-called variational formulations of the partial differential equations. This requires new tools in functional analysis[1], which will be the subject of forthcoming mathematical lectures [4, 11]. Let us present a result which points in that direction: u is the solution to the Dirichlet problem (5.9) if and only if it minimizes a so-called energy functional.

[1] Functional analysis is a branch of mathematical analysis to build appropriate topologies on vector spaces of functions. The main difference between such vector spaces and $\mathbb{R}^d$ is that these are infinite dimensional spaces, which imply many new phenomena.

Theorem 45 (Dirichlet's principle)**.** *Let Ω be a $\mathcal{C}^1$, bounded open set of $\mathbb{R}^d$. Let*

$$\mathcal{F} = \{w \in \mathcal{C}^2(\overline{\Omega}),\, w = g \text{ on } \partial\Omega\}.$$

Let us define the energy functional: for all $w \in \mathcal{F}$,

$$I(w) = \int_\Omega \left(\frac{1}{2}|\nabla w|^2 - fw\right).$$

Then $u \in \mathcal{C}^2(\overline{\Omega})$ solves the Dirichlet problem (5.9) if and only if $I(u) = \min_{w\in\mathcal{F}} I(w)$.

Proof: Let us perform the proof in two steps.

From (5.9) to the minimization problem. Let $u \in \mathcal{C}^2(\overline{\Omega})$ be a solution to the Dirichlet problem (5.9). Our objective is to show that for any function $v \in \mathcal{F}$, $I(v) \geq I(u)$ which is equivalent to show that for any $v \in \mathcal{F}_0$,

$$I(u+v) \geq I(u)$$

where

$$\mathcal{F}_0 = \{w \in \mathcal{C}^2(\overline{\Omega}),\, w = 0 \text{ on } \partial\Omega\}.$$

Now, let us compute

$$\begin{aligned} I(u+v) - I(u) &= \int_\Omega \frac{1}{2}\left(|\nabla(u+v)|^2 - |\nabla u|^2\right) - fv \\ &= \int_\Omega \frac{1}{2}\nabla(2u+v)\cdot\nabla v - fv \\ &= \frac{1}{2}\int_\Omega |\nabla v|^2 + \int_\Omega (\nabla u\cdot\nabla v - fv). \end{aligned} \tag{5.14}$$

Now, by multiplying the equation satisfied by u by the function v and integrating over Ω, one gets after integration by parts (using the fact that $v = 0$ on $\partial\Omega$):

$$\int_\Omega \nabla u\cdot\nabla v = \int_\Omega fv.$$

Thus, one obtains:

$$I(u+v) - I(u) = \frac{1}{2}\int_\Omega |\nabla v|^2 \geq 0.$$

From the minimization problem to (5.9). Conversely, let us assume that $u \in \mathcal{F}$ satisfies $I(u) = \min_{w\in\mathcal{F}} I(w)$. Let us consider a (non zero) function $v \in \mathcal{C}_c^2(\Omega)$. Notice that $u + v \in \mathcal{F}$ (since $v = 0$ on $\partial\Omega$) and therefore, for any $t \in \mathbb{R}$

$$I(u+tv) - I(u) \geq 0.$$

By using (5.14), one thus gets: for any $t \in \mathbb{R}$,

$$\frac{t^2}{2}\int_\Omega |\nabla v|^2 + t\int_\Omega (\nabla u\cdot\nabla v - fv) \geq 0$$

which implies that (this is a second order polynomial with roots 0 and $2\int_\Omega (\nabla u\cdot\nabla v - fv)\left(\int_\Omega |\nabla v|^2\right)^{-1}$: the second root has to be zero for this polynomial to be non negative):

$$\int_\Omega \nabla u\cdot\nabla v = \int_\Omega fv.$$

After integration by parts, one thus gets: for any $v \in \mathcal{C}^2_c(\Omega)$,

$$\int_\Omega (-\Delta u - f)v = 0$$

which implies that u solves (5.9). ♢

This result opens a new route to the analysis of partial differential equations, by looking for a solution as the minimizer of the energy functional I, or equivalently a solution to the so-called variational formulation: find $u \in \mathcal{F}$ such that:

$$\int_\Omega \nabla u \cdot \nabla v = \int_\Omega fv, \quad \forall v \in \mathcal{F}_0.$$

This so-called variational approach will be explored in the forthcoming lectures [11], and will require to introduce new functional spaces $\mathcal{F}$ and $\mathcal{F}_0$ (the so-called Sobolev spaces). It also opens the route to very efficient numerical methods (so-called Galekin approaches, using finite elements or spectral basis, see [14, 5]).

Exercise 50. ★*Let $u, v \in \mathcal{C}^2(\overline{\Omega})$ be two solutions to the Dirichlet problem* (5.9)*. Prove that $\int_\Omega |\nabla(u-v)|^2 = 0$. Deduce that $u = v$.*

Chapter 6

Discretization with the finite difference method

The finite difference method is a discretization technique for partial differential equations which consists in replacing the differential operators by Taylor expansions on a grid. It is relatively flexible, even though it is restricted to simple geometries (typically tensor products of segments, namely $\Omega = (a_1, b_1) \times \ldots \times (a_d, b_d)$), and structured grids (tensor products of one-dimensional grids). More flexible approaches will be introduced in forthcoming lectures (in particular finite element methods, which require variational formulations of partial differential equations).

We are interested in this Chapter in introducing finite difference discretization of the Dirichlet problem: find $u \in \mathcal{C}^2(\Omega) \cap \mathcal{C}^0(\overline{\Omega})$ such that

$$\begin{cases} -\Delta u = f \text{ in } \Omega \\ \quad u = g \text{ on } \partial\Omega \end{cases} \tag{6.1}$$

for some continuous functions $f : \Omega \to \mathbb{R}$ and $g : \partial\Omega \to \mathbb{R}$. We assume that the solution exists (this has been discussed in the previous chapter), and we would like to find a numerical method to approximate it on a computer.

The principle of approximations of differential operators by finite differences is presented in Section 6.1. Convergence analysis for a one-dimensional Dirichlet problem and a two-dimensional Dirichlet problem are then presented in Sections 6.2 and 6.3.

The sections devoted to the two-dimensional case can be skipped on first reading.

6.1 Finite difference approximations

6.1.1 The 1d case

Let us consider a function $u : \mathbb{R} \to \mathbb{R}$ $\mathcal{C}^2$ in a neighborhood of a point $x \in \mathbb{R}$. Then, the Taylor expansion yields: for any $h > 0$, there exists $h_1 \in (0, h)$ such that

$$u(x+h) = u(x) + hu'(x) + \frac{h^2}{2}u''(x+h_1).$$

One thus has: for $h_0 > 0$, for all $h \in (0, h_0)$,

$$\left| u'(x) - \frac{u(x+h) - u(x)}{h} \right| \leq Ch$$

where $C = \frac{1}{2}\sup_{[x,x+h_0]}|u''|$. The finite difference $\frac{u(x+h)-u(x)}{h}$ is a consistent approximation of $u'(x)$, with an error of order $O(h)$.

Let us introduce the general definition of an approximation of order p.

Definition 46. *A finite difference approximation L_h is an operator which to a function u and a point x associates a real number $L_hu(x)$ built using values of u on the grid $(u(x+kh))_{k\in\{-K,\dots,K\}}$, for some K.*

A finite difference approximation L_h is a consistent approximation at order p of a differential operator L if, for any function u sufficiently regular in a neighborhood of a point x, for any $h_0 > 0$, there exists C such that for all $h \in (0, h_0)$,

$$|Lu(x) - L_hu(x)| \leq Ch^p.$$

We have shown that, the forward difference

$$D_h^+u(x) = \frac{u(x+h)-u(x)}{h}$$

yields an approximation of order 1 of $u'(x)$. Likewise the backward difference

$$D_h^-u(x) = \frac{u(x)-u(x-h)}{h}$$

is an approximation of order 1 of $u'(x)$. Finally, the centered difference

$$D_hu(x) = \frac{u(x+h)-u(x-h)}{2h}$$

is an approximation of order 2 of $u'(x)$, assuming that u is $\mathcal{C}^3$ in a neighborhood of a point $x \in \mathbb{R}$ (check it★).

Let us now turn to approximations of the second order derivative. Let us assume that u is $\mathcal{C}^4$ in a neighborhood of a point $x \in \mathbb{R}$. Then the three point approximation

$$D_h^2u(x) = \frac{u(x+h)-2u(x)+u(x-h)}{h^2}$$

is an approximation of order 2 of $u''(x)$. Indeed: for $h > 0$, there exists $h_+ \in (0, h)$ and $h_- \in (0, h)$ such that:

$$u(x+h) = u(x) + hu'(x) + \frac{h^2}{2}u''(x) + \frac{h^3}{6}u^{(3)}(x) + \frac{h^4}{24}u^{(4)}(x+h_+)$$
$$u(x-h) = u(x) - hu'(x) + \frac{h^2}{2}u''(x) - \frac{h^3}{6}u^{(3)}(x) + \frac{h^4}{24}u^{(4)}(x-h_-).$$

By adding these two equalities, one gets, using the intermediate value theorem:

$$u''(x) = D_h^2u(x) + \frac{h^2}{12}u^{(4)}(x+h_\pm)$$

for some $h_\pm \in (-h, h)$. One thus gets: for $h_0 > 0$, for any $h \in (0, h_0)$,

$$|D_h^2u(x) - u''(x)| \leq Ch^2 \tag{6.2}$$

where $C = \frac{1}{12}\sup_{[x-h_0,x+h_0]}u^{(4)}$.

Remark 25. *When proving that a finite difference is an approximation at order p of a differential operator of order k at point x, one needs to assume that the function u is $\mathcal{C}^{k+p}$ in a neighborhood of x. If u is less regular, this limits the order of convergence. For example, if u is only $\mathcal{C}^3$ in a neighborhood of x, one only has $|D_h^2u(x) - u''(x)| \leq Ch$.*

6.1.2 The 2d case

Let us now consider a function $u : \mathbb{R}^2 \to \mathbb{R}$ (it is easy to generalize what follows to any dimension $d > 2$ – see however Remark 36).

If u is $\mathcal{C}^2$ in a neighborhood of a point $x = (x_1, x_2) \in \mathbb{R}^2$, the Taylor expansion yields: for any $\delta x = (\delta x_1, \delta x_2) \in \mathbb{R}^2$,

$$u(x + \delta x) = u(x) + \nabla u(x) \cdot \delta x + \delta x^T \nabla^2 u(x) \delta x + \dots$$

and more generally, If u is $\mathcal{C}^{k+1}$ in a neighborhood of x,

$$u(x + \delta x) = \sum_{k=0}^{K} \sum_{\alpha \in \mathbb{N}^2, |\alpha| = k} \frac{(\delta x)^\alpha}{\alpha!} D^\alpha u(x) + O(|\delta x|^{k+1})$$

where $\alpha = (\alpha_1, \alpha_2) \in \mathbb{N}^2$ is a so-called multi-index with length $|\alpha| = \alpha_1 + \alpha_2$ and we used the notations:

$$(\delta x)^\alpha = \delta x_1^{\alpha_1} \delta x_2^{\alpha_2}, \quad \alpha! = \alpha_1! \alpha_2!, \quad \text{and} \quad D^\alpha u = \frac{\partial^{|\alpha|} u}{\partial x_1^{\alpha_1} \partial x_2^{\alpha_2}}$$

(by convention $0! = 1$ and $D^{(0,0)} u = u$).

By writing order four expansions on $u(x + (h, 0))$, $u(x - (h, 0))$, $u(x + (0, h))$ and $u(x - (0, h))$, one can check that if u is $\mathcal{C}^{(4)}$ in a neighborhood of x, there exists $h_0 > 0$ such that for any $h \in (0, h_0)$,

$$|\Delta u(x) - \Delta_h u(x)| \leq C h^2 \tag{6.3}$$

where $\Delta_h u$ is the so-called 5-point approximation of the Laplacian:

$$(\Delta_h u)(x_1, x_2) = \frac{u(x_1 + h, x_2) + u(x_1 - h, x_2) + u(x_1, x_2 + h) + u(x_1, x_2 - h) - 4u(x_1, x_2)}{h^2}.$$

Remark 26. *In some sense $(\Delta_h u)(x_1, x_2)$ compares the local average of u on the neighboring points $\frac{1}{4}(u(x_1 + h, x_2) + u(x_1 - h, x_2) + u(x_1, x_2 + h) + u(x_1, x_2 - h))$ with the value $u(x_1, x_2)$: it is zero is u is "locally equilibrated".*

6.2 The finite difference method for the Dirichlet problem in dimension 1

Let us consider the problem in dimension 1, for $\Omega = (0, 1)$,

$$\begin{cases} -u''(x) + c(x)u(x) = f(x) \text{ for } x \in (0, 1) \\ u(0) = a \text{ and } u(1) = b \end{cases} \tag{6.4}$$

Compared to (6.1), we introduce a term $+c(x)u(x)$ so that the solution cannot be simply expressed using integrals. We assume that c is continuous and non negative (the two exercises 52 and 53 illustrate why this is a natural hypothesis to get uniqueness of solutions). Then, it can be shown that there exists a $\mathcal{C}^2(\Omega) \cap \mathcal{C}(\overline{\Omega})$ solution to (6.4).

Exercise 51. ★*Write an explicit solution to* (6.4) *when $c = 0$, in terms of a double integral of f.*

Exercise 52. ★★*Using the technique introduced in Exercise* (50)*, prove that if $c(x) \geq 0$,* (6.4) *admits a unique solution in $\mathcal{C}^2(\Omega) \cap \mathcal{C}(\overline{\Omega})$.*

Exercise 53. ★★★*Let us consider the case $c(x) = c$ is a constant. Identify the eigenfunctions $u \neq 0$ and eigenvalues c solution to*

$$\begin{cases} -u''(x) + cu = 0 \text{ for } x \in (0,1) \\ u(0) = u(1) = 0. \end{cases}$$

Check that $c < 0$. For such a constant c, show that the solution to (6.4) *admits infinitely many solutions.*

6.2.1 The finite difference problem

The first step is to introduce a discretization of the interval $(0,1)$: $(x_i)_{0 \leq i \leq I}$ where $x_i = \frac{i}{I}$. Let us denote by $h = \frac{1}{I}$ the mesh size. The second step is to translate the problem in terms of values of u on the grid $(x_i)_{0 \leq i \leq I}$. At the extreme points x_0 and x_I, one has:

$$u(x_0) = a \text{ and } u(x_I) = b.$$

At the interior points of Ω, we would like to find u satisfying the partial differential equation: for all $i \in \{1, \dots, I-1\}$,

$$-u''(x_i) + c(x_i)u(x_i) = f(x_i).$$

Let us now introduce the finite approximation $D_h^2 u(x_i)$ of $u''(x_i)$ in order to rewrite this equations only in terms of values of u on the mesh: for all $1 \leq i \leq I-1$,

$$-\frac{u(x_{i+1}) - 2u(x_i) + u(x_{i-1})}{h^2} + c(x_i)u(x_i) \simeq f(x_i).$$

Notice that we use here finite difference approximations of order 2.

We are now ready to introduce the finite difference discretization of (6.4), by replacing in the previous equation $u(x_i)$ by an unknown U_i, for $i \in \{1, \dots, I-1\}$ (the extreme values $u(x_0) = a$ and $u(x_I) = b$ are known, and one thus only needs to determine $(U_i)_{1 \leq i \leq I-1}$). The problem is the following: find $U \in \mathbb{R}^{I-1}$ such that:

$$\forall i \in \{1, \dots, I-1\}, \qquad \sum_{j=1}^{I-1} A_{i,j} U_j = C_i + F_i \tag{6.5}$$

where $A \in \mathbb{R}^{(I-1)\times(I-1)}$, $C \in \mathbb{R}^{I-1}$ and $F \in \mathbb{R}^{I-1}$ are defined by:

$$A = A_0 + A_1$$

with

$$A_0 = \frac{1}{h^2} \begin{bmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & -1 & 2 & -1 \\ 0 & \cdots & 0 & -1 & 2 \end{bmatrix} \text{ and } A_1 = \begin{bmatrix} c(x_1) & 0 & 0 & \cdots & 0 \\ 0 & c(x_2) & 0 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & 0 & c(x_{I-2}) & 0 \\ 0 & \cdots & 0 & 0 & c(x_{I-1}) \end{bmatrix} \tag{6.6}$$

$$C = \frac{1}{h^2} \begin{pmatrix} a \\ 0 \\ \vdots \\ 0 \\ b \end{pmatrix} \text{ and } F = \begin{pmatrix} f(x_1) \\ f(x_2) \\ \vdots \\ f(x_{I-2}) \\ f(x_{I-1}) \end{pmatrix}.$$

The matrix C is obtained by using the boundary conditions on u. We thus obtain a system of linear equations.

Proposition 47. *The matrix A is symmetric positive definite, and thus invertible. In particular, there exists a solution U to* (6.5).

Proof: The matrix A is clearly symmetric.
For any vector $V = (V_1, \dots, V_{I-1}) \in \mathbb{R}^{(I-1)}$, one has

$$V^T A V = V^T A_0 V + V^T A_1 V$$

and one obviously has $V^T A_1 V \geq 0$ since $c \geq 0$. Moreover, setting by convention $V_0 = V_I = 0$, one has:

$$\begin{aligned}
-h^2 V^T A_0 V &= \sum_{i=1}^{I-1} V_i (V_{i+1} - 2V_i + V_{i-1}) \\
&= \sum_{i=1}^{I-1} V_i \left((V_{i+1} - V_i) - (V_i - V_{i-1}) \right) \\
&= \sum_{i=1}^{I-1} V_i (V_{i+1} - V_i) - \sum_{i=1}^{I-1} V_i (V_i - V_{i-1}) \\
&= \sum_{i=1}^{I-1} V_i (V_{i+1} - V_i) - \sum_{i=0}^{I-2} V_{i+1} (V_{i+1} - V_i) \text{ [by a change of index in the second sum]} \\
&= \sum_{i=0}^{I-1} V_i (V_{i+1} - V_i) - \sum_{i=0}^{I-1} V_{i+1} (V_{i+1} - V_i) \text{ [since } V_0 = V_I = 0\text{]} \\
&= -\sum_{i=0}^{I-1} (V_{i+1} - V_i)^2 \leq 0.
\end{aligned}$$

Since A_0 is a finite difference approximation of $-\partial_{x,x}$, the relation $-V^T A_0 V = -\sum_{i=0}^{I-1} \left(\frac{V_{i+1} - V_i}{h} \right)^2$ can be seen as a discrete version of the integration by parts formula $\int_0^1 v'' v = -\int_0^1 (v')^2$ for any function $v \in \mathcal{C}^2((0,1)) \cap C^0([0,1])$ with $v(0) = v(1) = 0$. This computation implies that $V^T A V \geq 0$: the matrix A is positive.

Besides, if $V^T A V = 0$, then, necessarily, $V^T A_0 V = 0$ which implies thanks to the previous computation that $\forall i \in \{0, \dots, I-1\}$, $V_{i+1} - V_i = 0$ which means that all the components of V are the same. But since $V_0 = V_I = 0$, this implies that $V = 0$. The matrix A is thus symmetric positive definite. ♢

Let us insist on the fact that, even though this is not indicated explicitly, the quantities introduced here (A, C, F, and then U solution to (6.5)) depend on $h = \frac{1}{I}$. In particular, when $h \to 0$, namely when $I \to \infty$, the size of U increases.

6.2.2 Convergence

In order to analyze the convergence, we will proceed in two steps by first studying the consistency of the numerical scheme and then its stability.

Let us introduce the linear operator $\pi : \mathcal{C}((0,1)) \to \mathbb{R}^{I-1}$ (which implicitly depends on $h = 1/I$) defined by, for any continuous fonction $v \in \mathcal{C}((0,1))$,

$$\forall j \in \{1, \dots, I-1\}, \ (\pi(v))_j = v(x_j).$$

Our objective is to analyse the error

$$E = \pi(u) - U$$

where U is the solution to the discretization problem (6.5) and u is the solution to the continuous problem (6.4).

In the following, we assume that u is a $\mathcal{C}^4([0,1])$ solution to (6.4).

Consistency. A finite difference discretization of a partial differential equation is said to be consistent if, when plugging the solution to the original (continuous in space) problem in the finite difference equation, the remaining error ε goes to zero when $h \to 0$. Typically, we will use the l^∞ norm on the vector ε to define and quantify the consistency (or truncation) error. Let us illustrate this on our example.

On the specific $1d$ example considered above, if one sets $U = \pi(u)$ in the finite difference problem (6.5), one gets

$$A\pi(u) = C + F + \varepsilon$$

where for any $i \in \{1, \dots, I-1\}$,

$$\varepsilon_i = u''(x_i) - D_h^2 u(x_i).$$

Indeed, for any $i \in \{1, \dots, I-1\}$,

$$\begin{aligned} -D_h^2 u(x_i) &= -u''(x_i) + \varepsilon_i \\ &= -c(x_i)u(x_i) + f(x_i) + \varepsilon_i. \end{aligned}$$

Now, using the assumption that $u \in \mathcal{C}^4([0,1])$, we have that the exists C and $h_0 > 0$ such that, for all $h \in (0, h_0)$

$$\|\varepsilon\|_\infty \leq Ch^2 \tag{6.7}$$

since we know that $D_h^2 u(x)$ is a finite difference approximation of order 2 of $u''(x)$, see Equation (6.5). Here and in the following

$$\|\varepsilon\|_\infty = \sup_{1 \leq i \leq I-1} |\varepsilon_i| \tag{6.8}$$

denotes the l^∞ norm of a vector.

Stability. We have on the one hand

$$AU = C + F$$

and on the other hand

$$A\pi(u) = C + F + \varepsilon.$$

Since ε is small, we expect $\pi(u)$ to be close to U. The quantification of this is related to the so-called stability of the numerical scheme: how does a small perturbation of the data (here the right hand-side of the linear system (6.5) affect the result?

Let us recall that $E = \pi(u) - U$ is the error vector associated to the numerical scheme. By linearity, one obviously has:

$$AE = \varepsilon.$$

Therefore

$$E = A^{-1}\varepsilon \tag{6.9}$$

and our objective is to estimate $\|E\|$ for some norm, knowing that $\|\varepsilon\|_\infty \leq Ch^2$. We will typically use two types of norms: the l^∞ norm already introduced above, and the l^2 norm defined by: for any vector $V \in \mathbb{R}^{I-1}$,

$$\|V\|_2 = \sqrt{\frac{1}{I-1} \sum_{i=1}^{I-1} V_i^2}. \tag{6.10}$$

The l^2 norm is thus a normalized Euclidean norm, see Exercise 54 for a justification of this normalization. Notice that for any vector $V \in \mathbb{R}^{I-1}$,

$$\|V\|_2 \leq \|V\|_\infty.$$

Exercise 54. ★*Check that for any function $v \in \mathcal{C}_c(0,1)$, $\lim_{h\to 0} \|\pi_h(v)\|_2^2 = \int_0^1 v^2$.*

Exercise 55. ★*In finite dimension, all norms are equivalent. Show in particular that for any vector $V \in \mathbb{R}^{I-1}$,*

$$\|V\|_\infty \leq \sqrt{I-1}\|V\|_2.$$

From Equation (6.9), we know that $\|E\|_\infty \leq K\|\varepsilon\|_\infty$ and $\|E\|_2 \leq K\|\varepsilon\|_2 \leq K\|\varepsilon\|_\infty$ for some constant K, since $\varepsilon \mapsto A^{-1}\varepsilon$ is a linear continuous map. The problem is now to identify the optimal K, and to check that it does not depend catastrophically on the discretization parameter h. The optimal K is defined by matrix norms introduced below.

A brief aside on matrix norms.

Definition 48. *Let $\|\cdot\|$ be a vectorial norm on $\mathbb{R}^n$ and let $A \in \mathbb{R}^{n\times n}$ be a square matrix. Then the matrix norm induced by $\|\cdot\|$ (a.k.a. the operator norm or the triple norm[1]) is defined by:*

$$\|A\| = \sup_{x\in\mathbb{R}^n, \|x\|\leq 1} \|Ax\|.$$

The quantity $\|A\|$ is well defined as the supremum of a continuous function over a compact set. One can check the following standard facts (do it!★★):

$$\|A\| = \sup_{x\in\mathbb{R}^n, \|x\|\leq 1} \|Ax\| = \max_{x\in\mathbb{R}^n, \|x\|\leq 1} \|Ax\| = \max_{x\in\mathbb{R}^n, \|x\|= 1} \|Ax\|$$

and the norm properties: $\|A\| = 0 \Longrightarrow A = 0$, and

$$\forall \lambda \in \mathbb{R}, \forall A \in \mathbb{R}^{n\times n},\ \|\lambda A\| = |\lambda|\|A\|,$$

$$\forall A, B \in \mathbb{R}^{n\times n}, \|A + B\| \leq \|A\| + \|B\|.$$

Moreover, one has the so-called sub-multiplicative properties:

$$\forall A \in \mathbb{R}^{n\times n}, \forall x \in \mathbb{R}^n,\ \|Ax\| \leq \|A\|\|x\|,$$

$$\forall A, B \in \mathbb{R}^{n\times n}, \|AB\| \leq \|A\|\|B\|.$$

In the following, with a slight abuse of notation, we will use $\|\cdot\|_2$ (resp. $\|\cdot\|_\infty$) to denote the operator norm associated with the l^2 (resp. l^∞) norm on $\mathbb{R}^n$.

Exercise 56. ★ *Using the result of Exercise 55, show that for any $A \in \mathbb{R}^{n\times n}$,*

$$\|A\|_\infty \leq \sqrt{n}\|A\|_2$$

where, we recall the l^∞ and l^2 norms on $\mathbb{R}^n$ are defined by $\|U\|_\infty = \sup_{1\leq i\leq n} |U_i|$ and $\|U\|_2^2 = \frac{1}{n}\sum_{i=1}^n U_i^2$.

Exercise 57. ★★*Let $A \in \mathbb{R}^{n\times n}$.*

[1] In some textbooks, the triple norm is denoted $|||\cdot|||$: we prefer to stick to the simplest notation $\|\cdot\|$, with a slight abuse of notation since we thus use the same symbol to denote the vectorial norm and the associated matrix norm.

- *Show that*
$$\|A\|_2 = \sqrt{\rho(A^T A)}$$
where ρ is the spectral radius, namely the largest eigenvalue (in absolute value).

- *Show that*
$$\|A\|_\infty = \max_{1\le i\le n} \sum_{j=1}^{n} |A_{i,j}|.$$

Stability and matrix norms. From (6.7) and (6.9), one has that the error vector $E = \pi(u) - U$ satisfies: for small enough h,

$$\|E\|_\infty \le \|A^{-1}\|_\infty \|\varepsilon\|_\infty \le C\|A^{-1}\|_\infty h^2$$

and

$$\|E\|_2 \le \|A^{-1}\|_2 \|\varepsilon\|_2 \le \|A^{-1}\|_2 \|\varepsilon\|_\infty \le C\|A^{-1}\|_2 h^2$$

and it thus remains to estimate $\|A^{-1}\|_\infty$ (resp. $\|A^{-1}\|_2\|$) to get a convergence result in l^∞ (resp. in l^2). More precisely, we would like to check whether these matrix norms can be bounded from above by a constant independent of h, in which case we will get convergence with the same scaling as for the consistency error, namely, $O(h^2)$.

Stability in the l^2 norm. Let us start with a stability analysis in the l^2-norm. From Exercise 57, we know that (since A is symmetric positive) $\|A^{-1}\|_2 = \sqrt{\rho(A^{-T}A^{-1})} = \sqrt{\rho(A^{-2})} = \rho(A^{-1})$ and thus

$$\|A^{-1}\|_2 = \max \operatorname{spec}(A^{-1}) = (\min \operatorname{spec}(A))^{-1}$$

where $\operatorname{spec}(A)$ denotes the spectrum of A, namely the ensemble of eigenvalues of A. Besides, since $c \ge 0$,

$$\min \operatorname{spec}(A) \ge \min \operatorname{spec}(A_0).$$

The following lemma provides the spectrum of $\tilde{A}_0 = h^2 A_0$.

Lemma 49. *Let us consider the matrix $\tilde{A}_0 \in \mathbb{R}^{(I-1)\times(I-1)}$ defined by:*

$$\tilde{A}_0 = \begin{bmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & -1 & 2 & -1 \\ 0 & \cdots & 0 & -1 & 2 \end{bmatrix}$$

Let us define, for $i \in \{1, \dots, I-1\}$,

$$\lambda_i = 4\sin^2\left(\frac{i\pi}{2I}\right)$$

and $v_i \in \mathbb{R}^{I-1}$ with

$$\forall j \in \{1, \dots, I-1\}, [v_i]_j = \sin\left(\frac{ij\pi}{I}\right).$$

Then for $i \in \{1, \dots, I-1\}$,

$$\tilde{A}_0 v_i = \lambda_i v_i.$$

Proof: Fix $i \in \{1, \dots, I-1\}$. One can compute for any $j \in \{1, \dots, I-1\}$ (using the fact that $\sin(0) = \sin(\pi) = 0$ and trigonometric formulas),

$$\begin{aligned}[\tilde{A}_0 v_i]_j &= -\sin\left(\frac{i(j-1)\pi}{I}\right) + 2\sin\left(\frac{ij\pi}{I}\right) - \sin\left(\frac{i(j+1)\pi}{I}\right) \\ &= 2\sin\left(\frac{ij\pi}{I}\right)\left(1 - \cos\left(\frac{i\pi}{I}\right)\right) \\ &= \lambda_i [v_i]_j\end{aligned}$$

which concludes the proof. ♢

From this lemma, and the fact that $A_0 = \frac{1}{h^2}\tilde{A}_0$, one gets that the eigenvalues of A_0 are $\frac{\lambda_1}{h^2} < \ldots < \frac{\lambda_{I-1}}{h^2}$. Therefore,

$$\min \operatorname{spec}(A_0) = \frac{4}{h^2}\sin^2\left(\frac{\pi}{2I}\right) = \frac{4}{h^2}\sin^2\left(\frac{\pi h}{2}\right)$$

and thus, using the fact that there exists $C > 0$ such that $\sin(x) \geq Cx$ for $x \in [0, \pi/2]$, one concludes that there exists $C > 0$ such that, for all $h > 0$,

$$\min \operatorname{spec}(A_0) \geq C.$$

We thus have shown the following stability result:

Proposition 50. *There exists $C > 0$ such that, for all $h > 0$,*

$$\|A^{-1}\|_2 \leq C$$

where A is defined in (6.6).

Remark 27. *Notice that* $\min \operatorname{spec}(A_0)$ *converges when $h \to 0$ to π^2 which is nothing but the first eigenvalue of the operator $Lu = -u''$ on $(0,1)$ with Dirichlet boundary conditions (see Exercise 53).*

Stability in the l^∞ norm. Let us now turn to the estimate of the matrix norm $\|A^{-1}\|_\infty$. This relies on two ingredients: (i) a discrete maximum principle and (ii) a so-called barrier vector.

Proposition 51. *The numerical scheme* (6.5) *satisfies a discrete maximum principle in the following sense. Let $F \in \mathbb{R}^{I-1}$ be such that $F \geq 0$ (which means $F_i \geq 0$ for any $i \in \{1, \dots, I-1\}$). Then the solution $V \in \mathbb{R}^{I-1}$ to $AV = F$ also satisfies $V \geq 0$.*

Proof: Let us assume that $AV = F$ with $F \geq 0$ et let us consider $m = \min_{1 \leq j \leq I-1} V_j$. Let us argue by contradiction, and assume that $m < 0$. Let us moreover extend the vector V by introducing $V_0 = V_I = 0$.

Let us consider $k = \max\{j, V_j = m\}$. We have $k \in \{1, \dots, I-1\}$, $V_k = m$, $V_{k+1} > V_k$ and $V_{k-1} \geq V_k$ which yields

$$V_{k+1} - 2V_k + V_{k-1} > 0.$$

Besides, one has $-\frac{V_{k+1} - 2V_k + V_{k-1}}{h^2} + c(x_k)V_k = F_k$ and thus

$$V_{k+1} - 2V_k + V_{k-1} = h^2(c(x_k)V_k - F_k) \leq 0$$

since $V_k = m < 0$, $c(x_k) \geq 0$ and $F_k \geq 0$.

This leads to a contradiction, and thus $V \geq 0$. ♢

Remark 28. *Proposition 51 can be seen as a discrete equivalent to the weak maximum principle for the Laplace operator stated in Theorem 38 (see also Exercise* (45)*).*

Exercise 58. ★★★*Let $A \in \mathbb{R}^{n\times n}$ be a matrix such that:*

$$\forall i,j \in \{1,\dots,n\} \text{ with } i \neq j, a_{i,j} \leq 0$$

$$\forall i \in \{1,\dots,n\}, a_{i,i} \geq \sum_{j\in\{1\dots n\}\setminus\{i\}} |a_{i,j}|$$

with at least one i such that the inequality is strict. Prove that A satisfies the weak maximum principle: $AV \geq 0$ implies $V \geq 0$.

Lemma 52. *Let us consider the vector $\varphi \in \mathbb{R}^{I-1}$ defined by: for all $j \in \{1,\dots,I-1\}$,*

$$\varphi_j = \frac{x_j(1-x_j)}{2}.$$

Then

$$A\varphi \geq 1$$

(which means $[A\varphi]_j \geq 1$ for any $j \in \{1,\dots,I-1\}$).

Proof: For any $j \in \{1,\dots,I-1\}$ (by setting be convention $\varphi_0 = \varphi_{I+1} = 0$),

$$\begin{aligned}
[A\varphi]_j &= -\frac{\varphi_{j+1} - 2\varphi_j + \varphi_{j-1}}{h^2} + c(x_j)\varphi_j \\
&\geq -\frac{1}{h^2}(\varphi_{j+1} - 2\varphi_j + \varphi_{j-1}) \text{ [since } \varphi \geq 0 \text{ and } c \geq 0] \\
&= -\frac{1}{2h^2}\left((j+1)h - (j+1)^2h^2 - 2jh + 2(jh)^2 + (j-1)h - (j-1)^2h^2\right) \\
&= -\frac{1}{2}(-(j+1)^2 + 2j^2 - (j-1)^2) \\
&= 1
\end{aligned}$$

which concludes the proof. ♢

Remark 29. *The intuition to build this barrier vector comes from the continuous problem. Indeed, let us consider $\varphi(x) = \frac{x(1-x)}{2}$. One has $\varphi(0) = \varphi(1) = 0$ and $-\varphi''(x) = 1$ for all $x \in [0,1]$. Since $\varphi \geq 0$, this implies that for all $x \in [0,1]$,*

$$-\varphi''(x) + c(x)\varphi(x) \geq 1$$

for all $x \in [0,1]$. Lemma 52 is nothing but a discrete counterpart of that.

Proposition 53. *For all $h > 0$,*

$$\|A^{-1}\|_\infty \leq \frac{1}{8}$$

where A is defined in (6.6)*.*

Proof: Let $U \in \mathbb{R}^{I-1}$ and $F \in \mathbb{R}^{I-1}$ satisfy $AU = F$. We would like to prove that

$$\|U\|_\infty \leq \frac{1}{8}\|F\|_\infty.$$

Let us consider the barrier vector φ defined in Lemma 52, and let us introduce $V = \|F\|_\infty\varphi$. From Lemma 52, one has $AV \leq \|F\|_\infty$ and thus

$$A(V-U) = AV - F \leq \|F\|_\infty - F \geq 0.$$

This implies, thanks to Proposition , that $V - U \geq 0$ and thus

$$U \leq \|F\|_\infty \varphi.$$

Likewise,

$$A(V + U) = AV + F \leq \|F\|_\infty + F \geq 0,$$

and this implies that $V + U \geq 0$ and thus

$$-U \leq \|F\|_\infty \varphi.$$

Therefore,

$$\|U\|_\infty \leq \|\varphi\|_\infty \|F\|_\infty = \frac{1}{8}\|F\|_\infty$$

since $\max_{x\in[0,1]} \frac{x(1-x)}{2} = \frac{1}{8}$. ♢

Summary and convergence results. Let us summarize the convergence results we have obtained in this section, by combining the consistency result (6.7), the equation (6.9) satisfied by the error E, and the stability result stated in Proposition 53.

Theorem 54. *Let us consider the one-dimensional Dirichlet problem* (6.4) *with solution* $u : (0,1) \to \mathbb{R}$ *and the associated finite difference approximation* (6.5) *with solution* $U \in \mathbb{R}^{I-1}$. *Let us assume that* $u \in \mathcal{C}^4([0,1])$. *Then, there exists* $C > 0$ *such that, for any* $h > 0$,

$$\|U - \pi(u)\|_2 \leq \|U - \pi(u)\|_\infty \leq Ch^2.$$

Let us conclude this section with a series of important remarks.

Remark 30. *From Exercise 56 and Proposition 50, we know that*

$$\|A^{-1}\|_\infty \leq \sqrt{I-1}\|A^{-1}\|_2 \leq Ch^{-1/2}$$

for h *sufficiently small. This estimate is not optimal (compare with what we have obtained in Proposition 53) and would only yield a convergence of order* $O(h^{3/2})$ *instead of the optimal rate* $O(h^2)$.

Remark 31. *The reader may be surprised that we performed both a stability analysis in* l^∞ *and in* l^2 *norm, since the* l^∞ *convergence result of course implies the* l^2 *convergence result. This is in preparation of the forthcoming part on the heat equation: we will encounter in Section 9.2.2 numerical schemes which are stable in* l^2*-norm but not in* l^∞*-norm.*

Remark 32. *The reasoning performed in this section is yet another illustration of the so-called Lax principle already encountered in Section 3.2: consistency and stability imply convergence.*

Remark 33. *If* u *is not* $\mathcal{C}^4([0,1])$, *this will have an impact on the truncation error and thus on the convergence rate. For example, if* u *is only in* $\mathcal{C}^3([0,1])$ *and not*[2] *in* $\mathcal{C}^4([0,1])$, *then the consistency error is only of order* $O(h)$ *(see Remark 25) and thus the error is also only of order* $O(h)$.

In summary, the take home message is the following: what dictate the rate of convergence of a numerical method is the combination of two factors: (i) the order of the numerical scheme, and (ii) the regularity of the solution.

[2]From a practical viewpoint, having a very large L^∞ norm for $u^{(4)}$ (which would imply the constant C in (6.7) to be very large) is essentially equivalent to not being in $\mathcal{C}^4([0,1])$...

Remark 34. *The convergence analysis of finite difference methods is quite cumbersome, in particular the stability analysis which essentially has to be done for any new operator, and new finite difference discretization (for example to find suitable barrier vectors). Moreover, finite difference methods are limited to very specific grids, and thus to specific domains. There exists much more flexible techniques, such as spectral methods, and finite element methods (introduced for example in the course [5]), which rely on the variational formulation of the partial differential equations, and for which the stability (in the so-called energy norm, which would correspond to the l^2-norm here) comes for free.*

6.3 The finite difference method for the Dirichlet problem in dimension 2

Let us consider the following Poisson problem with Dirichlet boundary conditions in dimension two, on the square $\Omega = (0,1)^2$:

$$\begin{cases} -\Delta u = f & \text{in } \Omega, \\ u = g & \text{on } \partial\Omega, \end{cases} \tag{6.11}$$

where $f \in \mathcal{C}(\Omega)$ and $g \in \mathcal{C}(\partial\Omega)$. It can be shown that there exists a $\mathcal{C}^2(\Omega) \cap \mathcal{C}(\overline{\Omega})$.

Our goal is to approximate u using finite difference methods on a uniform grid, assuming that $u \in \mathcal{C}^{(4)}(\overline{\Omega})$.

6.3.1 The finite difference problem

Let $I \in \mathbb{N} \setminus \{0\}$, $h = 1/I$ and let us define the grid $(x_{i,j})_{0 \le i,j \le I}$ where

$$x_{i,j} = (ih, jh).$$

We write the partial differential equation $-\Delta u(x) = f(x)$ on all the interior points of this grid, and we approximate the Laplacian by the 5-point approximation operator Δ_h. We thus obtain the following:

$$\begin{cases} -\dfrac{u(x_{i+1,j}) + u(x_{i-1,j}) + u(x_{i,j+1}) + u(x_{i,j-1}) - 4u(x_{i,j})}{h^2} = f(x_{i,j}), \text{ for } x_{i,j} \in \Omega, \\ u(x_{i,j}) = g(x_{i,j}) \text{ for } x_{i,j} \in \partial\Omega. \end{cases} \tag{6.12}$$

Notice that, contrary to the one dimensional case, the grid points are here labelled by two indices. There are $(I+1)^2$ grid points in $\overline{\Omega}$: $(I-1)^2$ interior points in Ω, and $4I$ points on $\partial\Omega$. We will denote the set of indices corresponding to interior and boundary points by

$$\Omega_h = \{(i,j),\, x_{(i,j)} \in \Omega\} \text{ and } \partial\Omega_h = \{(i,j),\, x_{(i,j)} \in \partial\Omega\}.$$

In order to easily manipulate the vector of unknowns, one has to choose a single index labelling of the interior points, for example (with a slight abuse of notation, we keep the notation $x_\cdot$ for this new labelling): for $1 \le i,j \le I-1$,

$$x_{i,j} = x_{i+(j-1)(I-1)}.$$

We will denote by $k = i + (j-1)(I-1) \in \{1, \dots, (I-1)^2\}$ the new index and $K = (I-1)^2$ the total number of unknowns. The interior grid points are numbered from left to right, and then from the bottom to the top.

The finite difference approximation is (compare with the 1d case): find $U \in \mathbb{R}^K$ such that

$$AU = C + F \tag{6.13}$$

where $A \in \mathbb{R}^{K \times K}$ is is the block tridiagonal matrix defined as

$$A = \frac{1}{h^2} \begin{bmatrix} L & -\mathrm{Id} & 0 & \cdots & 0 \\ -\mathrm{Id} & L & -\mathrm{Id} & \ddots & \vdots \\ 0 & -\mathrm{Id} & L & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -\mathrm{Id} \\ 0 & \cdots & 0 & -\mathrm{Id} & L \end{bmatrix}, \text{ where } L = \begin{bmatrix} 4 & -1 & 0 & \cdots & 0 \\ -1 & 4 & -1 & \ddots & \vdots \\ 0 & -1 & 4 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -1 \\ 0 & \cdots & 0 & -1 & 4 \end{bmatrix} \in \mathbb{R}^{(I-1)\times(I-1)}. \tag{6.14}$$

The zeros in A correspond to the $(I-1)\times(I-1)$ null matrix and Id denotes the $(I-1)\times(I-1)$ identity matrix. The vectors $C \in \mathbb{R}^K$ and $F \in \mathbb{R}^K$ are defined by:

$$F = \begin{pmatrix} f(x_1) \\ f(x_2) \\ \vdots \\ f(x_{I-1}) \\ f(x_{1+(I-1)}) \\ f(x_{2+(I-1)}) \\ \vdots \\ f(x_{2(I-1)}) \\ f(x_{1+2(I-1)}) \\ f(x_{2+2(I-1)}) \\ \vdots \end{pmatrix}, \text{ and } C = \frac{1}{h^2} \begin{pmatrix} \big(g(x_{1,0}) + g(x_{0,1})\big) \\ g(x_{2,0}) \\ \vdots \\ \big(g(x_{I-1,0}) + g(x_{I,1})\big) \\ g(x_{0,2}) \\ 0 \\ \vdots \\ g(x_{I,2}) \\ g(x_{0,3}) \\ 0 \\ \vdots \end{pmatrix},$$

As in the 1d case, one can check that A is symmetric positive definite and thus invertible, so that U is well defined.

The fundamental argument to check the positive-definiteness is that, for any $V \in \mathbb{R}^K$, using the labelling with the indices (i,j), and completing the vector V by zeros on the boundary:

$$V^T A V = \sum_{0 \le i,j \le I-1} (V_{(i+1,j)} - V_{(i,j)})^2 + (V_{(i,j+1)} - V_{(i,j)})^2.$$

6.3.2 Convergence

The convergence proof can be done following exactly the same reasoning as in the 1d case: consistency, stability and then convergence. We will not provide all the details: it is a good exercise to fill the gaps!

As above, let us define the error $E = \pi(u) - U$ where for all $1 \le i,j \le I-1$, $\pi(u)_{(i,j)} = u(x_{(i,j)})$ and u is the solution to the original problem (6.11).

Consistency. As in the one-dimensional case, one can introduce the consistency vector $\epsilon \in \mathbb{R}^K$ defined by:

$$A\pi(u) = C + F + \varepsilon.$$

Using the consistency analysis of the Laplacian by the 5-point approximation (see (6.3)), one can show that

$$\|\varepsilon\|_\infty \leq Ch^2.$$

Stability. From $AU = C + F$ and $A\pi(u) = C + F + \varepsilon$, we get the equation on the error $E = \pi(u) - U$:

$$AE = \varepsilon$$

and it thus remains to analyze the stability of the scheme to get an estimate of the error on E.

Let us concentrate on the l^∞ stability (the l^2 stability can also be proven following the same reasoning as in 1d). On can prove that the numerical scheme satisfies a discrete maximum principle, using the same reasoning as in 1d:

$$AV \geq 0 \Longrightarrow V \geq 0$$

Actually, one can show the discrete strong maximum principle:

Proposition 55. *Let $(V_{(i,j)})_{0\leq i,j\leq I}$ be a vector defined on the whole grid, and let us assume that $\Delta_h V \geq 0$ on the interior points[a]. Then*

$$\max_{(i,j)\in\Omega_h} V_{(i,j)} \leq \max_{(i,j)\in\partial\Omega_h} V_{(i,j)}.$$

Moreover, one has equality if and only if V is constant.

Proof: Assume by contradiction that $\max_{(i,j)\in\Omega_h} V_{(i,j)} > \max_{(i,j)\in\partial\Omega_h} V_{(i,j)}$, and let (i_0, j_0) be an interior point where the maximum is achieved. The four neighbors of (i_0, j_0) are $(i_0 + \eta, j_0 + \zeta)_{\eta,\zeta\in\{-1,+1\}}$. One has

$$4V_{(i_0,j_0)} = \sum_{\eta,\zeta\in\{-1,+1\}} V_{(i_0+\eta,j_0+\zeta)} - h^2(\Delta_h V)_{(i_0,j_0)} \leq \sum_{\eta,\zeta\in\{-1,+1\}} V_{(i_0+\eta,j_0+\zeta)} \leq 4V_{(i_0,j_0)}$$

which shows than on the neighboring points, $V_{(i,j)} = V_{(i_0,j_0)}$. This argument propagates up to the boundary, and we get that V is constant, which leads to a contradiction. The second statement is obtained by a similar reasoning. ♢

Exercise 59. ★*Show that Proposition 55 implies the weak maximum principle $AV \geq 0 \Longrightarrow V \geq 0$.*

Let us now introduce the barrier vector $(\varphi_{(i,j)})_{0\leq i,j\leq I}$ defined by

$$\varphi_{(i,j)} = \varphi(x_{i,j})$$

with $\varphi(x) = \frac{1}{4}\left(\left(x - \frac{1}{2}\right)^2 + \left(y - \frac{1}{2}\right)^2\right)$. Notice that φ is not zero on $\partial\Omega$. We have:

$$\Delta_h\varphi = 1 \text{ on the interior points}$$

and

$$0 \leq \varphi \leq \frac{1}{8}.$$

Using the strong maximum principle and the barrier vector $(\varphi_{(i,j)})_{0\leq i,j\leq I}$ one can prove the following:

Proposition 56. *For all $h > 0$,*

$$\|A^{-1}\|_\infty \leq \frac{1}{8}$$

where A is defined by (6.14)*.*

Proof: Let $U \in \mathbb{R}^K$ and $F \in \mathbb{R}^K$ be such that $AU = F$, and let us extend U up to the boundary points by 0, so that $U = (U_{(i,j)})$ is defined for all $0 \leq i, j \leq I$. One has $\Delta_h U = -AU = -F$ on the interior points and thus, on the interior points,

$$\Delta_h(\|F\|_\infty \varphi - U) = \|F\|_\infty + F \geq 0.$$

Therefore, by Proposition 55, using the fact that $0 \leq \varphi \leq \frac{1}{8}$,

$$\max_{\Omega_h}(-U) \leq \max_{\Omega_h}(-U + \|F\|_\infty \varphi) \leq \max_{\partial\Omega_h}(-U + \|F\|_\infty \varphi) = \frac{1}{8}\|F\|_\infty.$$

Likewise, $\Delta_h(\|F\|_\infty \varphi + U) = \|F\|_\infty - F \geq 0$ and thus $\max_{\Omega_h} U \leq \frac{1}{8}\|F\|_\infty$ which finally yields

$$\|U\|_\infty \leq \frac{1}{8}\|F\|_\infty.$$

♢

Remark 35. *Compared to the 1d case, we used a stronger maximum principle since the barrier function we use is not zero on the boundary.*

Convergence. By combining the consistency and the stability analysis, we finally get the following convergence result:

Theorem 57. *Let us consider the two-dimensional Dirichlet problem* (6.11) *with solution $u : (0,1) \to \mathbb{R}$ and the associated finite difference approximation* (6.13) *with solution $U \in \mathbb{R}^K$. Let us assume that $u \in \mathcal{C}^4(\overline{\Omega})$. Then, there exists $h_0 > 0$ and $C > 0$ such that, for any $h \in (0, h_0)$,*

$$\|U - \pi(u)\|_2 \leq \|U - \pi(u)\|_\infty \leq Ch^2.$$

Remark 36. *Let us conclude this chapter with a remark concerning the computational complexity of finite difference methods. In dimension $d = 1$, to get an error of order $\varepsilon = h^2$, we have to solve a linear system with dimension of order ε^{-1} (the matrix in* (6.5) *is of size $(I-1)^2 \sim 1/h^2$). In dimension $d = 2$, to get an error of order $\varepsilon = h^2$, we have to solve a linear system with dimension ε^{-2} (the matrix in* (6.13) *is of size $K^2 = (I-1)^4 \sim 1/h^4$). More generally, finite difference schemes in dimension d lead to linear systems whose dimensions scale exponentially with d. In practice, it is thus very difficult to use them in dimension $d > 4$. This is very much related to the curse of dimensionality which essentially states that approximating all functions $u : [0,1]^d \to \mathbb{R}$ within a given regularity class (for example Lipschitz)*

with an error of order ε require a number of evaluations of $u(x)$ for more than $\left(\frac{1}{\varepsilon}\right)^d$ values of x. To break the curse of dimensionality, one needs to rely on some a priori knowledge on u which enables to use adapted (less computationally demanding) discretizations.

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[a] With a slight abuse of notation, for a vector $V = V_{(i,j)}$ defined on $\Omega_h \cup \partial\Omega_h$, we denote by $\Delta_h V$ the vector defined on Ω_h by $[\Delta_h V]_{(i,j)} = \frac{V_{(i+1,j)}+V_{(i-1,j)}+V_{(i,j+1)}+V_{(i,j-1)}-4V_{(i,j)}}{h^2}$.

Part III

The heat equation

Chapter 7

Models based on the heat equation

The equation we will be interested in in this part (the heat equation) combines aspects of the first two parts: it will involve a derivative in time and a Laplace operator on space variables. The heat equation models diffusion phenomena: evolution of temperature, chemical species. Let us first present how to derive this equation, and give examples of physical phenomena it models.

7.1 Derivation of the heat equation

The heat equation is one example of a conservation law described generally by so-called continuity or transport equations. Heat equation describes the evolution of a scalar quantity u through a physical phenomenon called diffusion.

Let us derive the heat equation in three steps (this is very close to the reasoning we used to derive the Poisson equation in Chapter 4). First we will carry out a balance of gains and losses in a domain Ω, taking into account a bulk "source and sink" term f (per unit of time and volume), and a flux term $F \cdot n$ (per unit of time and surface) on the boundary $\partial\Omega$, where n denotes the unit out-going normal to Ω. This will lead to the integral form of the conservation law. Then we will derive the differential form of the conservation law. Finally, we will assume a linear relation between the flux F and the gradient of the quantity u to get the heat equation.

Let us consider a domain $\Omega \subset \mathbb{R}^d$. On the time interval $(t, t+\Delta t)$, the evolution $\int_\Omega u(t+\Delta t, x)\,dx - \int_\Omega u(t,x)\,dx$ of the quantity u within Ω is ruled by two kinds of phenomena. First, within the domain Ω, some volume "source or sink" phenomenon modifies the quantity u by $\int_t^{t+\Delta t}\int_\Omega f(s,x)\,dx\,ds$. Second, some fluxes F on the boundaries will also make the quantity u evolve by $-\int_t^{t+\Delta t}\int_{\partial\Omega} F(s,x)\cdot n(x)\sigma(dx)\,ds$ where σ denotes the surface measure. By convention, a positive flux $F(t,x)\cdot n(x)$ represents a loss: this explains the minus sign. One thus obtains

$$\int_\Omega u(t+\Delta t,x)\,dx - \int_\Omega u(t,x)\,dx = \int_t^{t+\Delta t}\left(\int_\Omega f(s,x)\,dx - \int_{\partial\Omega} F(s,x)\cdot n(x)\sigma(dx)\right)ds.$$

By dividing by Δt and sending Δt to 0, one obtains formally the integral form of the conservation law: for any $t>0$, for any domain Ω,

$$\int_\Omega \frac{\partial u}{\partial t}(t,x)\,dx = \int_\Omega f(t,x)\,dx - \int_{\partial\Omega} F(t,x)\cdot n(x)\sigma(dx). \tag{7.1}$$

This concludes the first step.

To perform the second step, we rely on the divergence formula (see Proposition 30): $\int_{\partial\Omega} F\cdot n d\sigma = \int_\Omega \operatorname{div} F$. Using this formula and the integral form of the conservation law (7.1), one gets:

for any $t > 0$, for any domain Ω

$$\int_\Omega \left(\frac{\partial u}{\partial t} - f + \text{div } (F) \right) (t,x)\, dx = 0.$$

For this equation to be true for any Ω, u and F have to obey the differential form of the conservation law: for any $t > 0$, for any x,

$$\frac{\partial u}{\partial t}(t,x) + \text{div } (F)(t,x) = f(t,x). \tag{7.2}$$

This concludes the second step.

The last step requires a modeling assumption: in order to obtain a closed equation on u, one needs a formula (sometimes called a constitutive law) relating F and u. It is natural to assume that $F(t,x)$ is proportional to $-\nabla u(t,x)$ (the flux is oriented such that, for two neighboring regions A and B, if u in A is larger that u in B, then the flux induces a flow from A to B, so that the value of u equilibrates locally[1]):

$$F(t,x) = -D\nabla u(t,x)$$

where $D > 0$ is a modeling parameter. By using this relation in (7.2), one finally obtains the prototypical heat equation:

$$\frac{\partial u}{\partial t}(t,x) - D\Delta u(t,x) = f(t,x). \tag{7.3}$$

As for an ordinary differential equation, this will be supplemented with an initial condition at time $t = 0$:

$$u(0,x) = u_0(x).$$

Moreover, as for the Poisson equation, we will be either interested in this problem in the whole space $x \in \mathbb{R}^d$, or in an open bounded domain $x \in \Omega \subset \mathbb{R}^d$, in which case one adds boundary conditions for the problem to be well posed. These could be either Dirichlet boundary conditions (the value of u is given on $\partial\Omega$):

$$u = g \text{ on } \partial\Omega.$$

or Neumann boundary conditions (the outgoing flux $F \cdot n$ is given on $\partial\Omega$):

$$\partial_n u = g \text{ on } \partial\Omega.$$

Notice that, contrary to the stationary case in Section 4.3, the boundary data g may depend on time. We will stick to homogeneous in time Dirichlet boundary conditions in these introductory lectures.

Remark 37. *From a modeling viewpoint, for the heat equation over $(0,T] \times \Omega$ to be well posed ($T > 0$ and $\Omega \subset \mathbb{R}^d$ a bounded domain), it is natural to impose initial and boundary conditions on*

$$\Gamma_T = \{0\} \times \Omega \cup [0,T] \times \partial\Omega.$$

This set is called the parabolic boundary of $(0,T] \times \Omega$, and will play an important role below, see Theorem 62.

Remark 38. *The heat equation is the prototypical example of so-called parabolic linear problems, which are generally of the form:*

$$\partial_t u - \text{div } (D\nabla u) = f$$

for $D : \mathbb{R}^d \to \mathbb{R}^{d\times d}$ a space-dependent symmetric matrix such that $D \geq \alpha \text{Id}$ (i.e. $D(x) - \alpha\text{Id}$ is symmetric non negative for all x) for some $\alpha > 0$.

[1] Remember that the gradient of u points in the direction of increasing values of u.

Remark 39. *When one looks for a stationary solution to* (7.3)*, namely a function u which does not depend on time (this requires f not depending on time either), one obtains the Poisson equation $-D\Delta u = f$ that we studied in Part II.*

7.2 Evolution of the temperature in a domain

Similarly to Section 4.4, let us consider $T(t,x) \in \mathbb{R}$ the temperature at time t and point x of a material. In some materials, the associated heat flux Q satisfies the Fourier law: $Q(t,x) = -k\nabla T(t,x)$, k being the so-called thermal diffusivity. One thus obtains the heat equation:

$$\partial_t T - k\Delta T = f$$

where the Laplace operator acts on the space variable, and f denotes a source term. For bounded domains Ω, Dirichlet boundary conditions consist in imposing the value of the temperature on $\partial\Omega$, whereas Neumann boundary conditions consist in imposing the heat flux on $\partial\Omega$ (for example, a perfectly insulating boundary would be modeled using a zero flux boundary condition).

7.3 Diffusion of chemical species

As in Section 4.5, let $u(t,x) \in \mathbb{R}$ denote the concentration of a chemical species at time t and point x. In some models, one states that the associated flux F is related to the concentration gradient by the Fick law $F(t,x) = -D\nabla u(t,x)$, which leads to the following diffusion equation:

$$\partial_t u - D\Delta u = f$$

where f again denotes an associated source term.

There is a deep connection between such diffusion equations and probabilistic models for diffusions based on Brownian motions. This will be explored below at the discrete level, see Section 9.3.

Remark 40 (Generative models and diffusion equations)**.** *In generative models, neural network architectures are used in order to produce new data starting from a large pool of examples. The idea is to start with an initial condition $u(0,x)$ which is essentially the uniform law on the examples (this is a probability distribution) and to apply as a forward process a variant of the heat equation, which converges to a Gaussian distribution. The reverse process then learns a neural approximation to the reverse-time heat equation, transforming a newly drawn Gaussian sample back into structured data. Diffusion models are deeply tied to partial differential equations: they generate data by numerically solving a learned reverse heat flow.*

Chapter 8

Mathematical analysis of the heat equation

We derive an integral representation formula to classical solutions to the heat equation, relying on the fundamental solution of the heat equation in $\mathbb{R}_+ \times \mathbb{R}^d$: we are looking for solutions $u : (t, x) \in \mathbb{R}_+ \times \mathbb{R}^d \mapsto u(t, x) \in \mathbb{R}$ to the equation[1]:

$$\begin{cases} \partial_t u - \Delta u = f \text{ for } t > 0, x \in \mathbb{R}^d, \\ u(0, x) = u_0(x) \text{ for } x \in \mathbb{R}^d. \end{cases} \tag{8.1}$$

We will essentially stick to the problem posed in the whole of $\mathbb{R}^d$ here. In Section 8.1, we derive the fundamental solution for the heat equation, a.k.a. the heat kernel. A solution to (8.1) is then obtained as an integral (convolution) against the heat kernel in Section 8.2. Section 8.3 is then devoted to uniqueness results, as well as important mathematical properties of solutions to the heat equation. Finally, Section 8.4 can be seen as a prologue to variational methods which will be developed in forthcoming mathematical lectures [4, 11].

8.1 Fundamental solution for the heat equation

The following derivation is informal. Rigorous mathematical statements will come afterwards. We would like to identify a solution to the heat equation (8.1), for a specific initial condition u_0 concentrated at $x = 0$. It is natural to look for a radially symmetric solution in space $u(t, x) = v(t, |x|)$. Moreover, we observe that if $u(t, x)$ satisfies $\partial_t - \Delta u = 0$, so does $u(\lambda^2 t, \lambda x)$ for $\lambda > 0$ (check it!★): this naturally leads us to look for a solution which depends on x through $|x|/\sqrt{t}$. Finally, one also notes that $\int_{\mathbb{R}^d} u$ is conserved in time. This leads us to look for a solution of the form[2]:

$$u(t, x) = t^{-d/2} v\left(\frac{|x|}{\sqrt{t}}\right). \tag{8.2}$$

Exercise 60. ★*Assume that $v : \mathbb{R}^d \to \mathbb{R}^d$ is such that $\int_{\mathbb{R}^d} |v| < \infty$. Check that $\int_{\mathbb{R}^d} t^{-d/2} v\left(\frac{|x|}{\sqrt{t}}\right) dx$ is independent of $t > 0$.* Hint: use a change of variable.

[1] Here and in the following, the differential operators div , ∇, Δ are applied to the space variable x. We will indicate it explicitly when needed.

[2] Such solutions are called self similar: the solution at a given positive time determines the solution for all time $t > 0$.

If u is of the form (8.2), then (using the fact that $\mathrm{div}(x/|x|) = (d-1)/|x|$):

$$\partial_t u(t,x) = -\frac{d}{2} t^{-d/2-1} v\left(\frac{|x|}{\sqrt{t}}\right) - \frac{1}{2} t^{-d/2-3/2} |x| v'\left(\frac{|x|}{\sqrt{t}}\right)$$

$$\nabla u(t,x) = t^{-d/2-1/2} \frac{x}{|x|} v'\left(\frac{|x|}{\sqrt{t}}\right)$$

$$\mathrm{div}(\nabla u)(t,x) = t^{-d/2-1/2} \frac{d-1}{|x|} v'\left(\frac{|x|}{\sqrt{t}}\right) + t^{-d/2-1} v''\left(\frac{|x|}{\sqrt{t}}\right).$$

Thus, if u satisfies $\partial_t u - \Delta u = 0$, then v satisfies (we introduce $r = |x|/\sqrt{t}$):

$$-\frac{d}{2} v(r) - \frac{1}{2} r v'(r) - \frac{d-1}{r} v'(r) - v''(r) = 0$$

which leads to

$$-\frac{r^{1-d}}{2} (r^d v(r))' - r^{1-d} (r^{d-1} v')' = 0.$$

Thus, assuming that v and v' go to 0 when $r \to \infty$, one gets $-\frac{1}{2} r^d v(r) - r^{d-1} v'(r) = 0$ and thus

$$v'(r) = -v(r) \frac{r}{2}.$$

Therefore, for some constant C,

$$v(r) = C \exp\left(-\frac{r^2}{4}\right).$$

This leads us to introduce the heat kernel:

$$\Phi(t,x) = \begin{cases} \dfrac{1}{(4\pi t)^{d/2}} \exp\left(-\dfrac{|x|^2}{4t}\right) \text{ for } t > 0, x \in \mathbb{R}^d, \\ 0 \text{ for } t \leq 0, x \in \mathbb{R}^d. \end{cases} \tag{8.3}$$

The normalizing constant is chosen so that the following proposition holds.

Proposition 58. *The heat kernel satisfies: for all $t > 0$, $\int_{\mathbb{R}^d} \Phi(t,x)\, dx = 1$.*

Proof: By the change of variable $z = x/\sqrt{4t}$, one has

$$\frac{1}{(4\pi t)^{d/2}} \int_{\mathbb{R}^d} \exp\left(-\frac{|x|^2}{4t}\right) dx = \frac{1}{\pi^{d/2}} \int_{\mathbb{R}^d} \exp\left(-|z|^2\right) dz.$$

By Fubini, one gets

$$\int_{\mathbb{R}^d} \exp\left(-|z|^2\right) dz = \left(\int_{-\infty}^{+\infty} \exp(-y^2)\, dy\right)^d$$

and the result follows form the fact that $\int_{-\infty}^{+\infty} \exp(-y^2)\, dy = \sqrt{\pi}$. ♢

Exercise 61 (Gauss integral)**.** *We would like to show that $\int_{-\infty}^{+\infty} \exp(-y^2)\, dy = \sqrt{\pi}$. Let $I = \int_{-\infty}^{+\infty} \exp(-y^2)\, dy$. Prove that $I^2 = \int_{\mathbb{R}^2} \exp(-|x|^2)\, dx$. By using a change of variable to polar coordinates, show that $I^2 = \int_0^\infty \int_0^{2\pi} \exp(-r^2) r\, d\theta\, dr$. Conclude.*

Remark 41. *A more direct route to derive the heat kernel is to use Fourier transform: this will be explained in the lectures [4, 11].*

Notice first that the function Φ is $\mathcal{C}^\infty(\mathbb{R}\times\mathbb{R}^d\setminus(0,0))$, and that, by construction,

$$\partial_t\Phi(t,x)-\Delta\Phi(t,x)=0 \text{ for } (t,x)\in\mathbb{R}\times\mathbb{R}^d\setminus(0,0).$$

The function Φ is singular at $(t,x)=(0,0)$. Actually, what can be shown is that, in the sense of distribution (distributions will be introduced in the lectures [4], compare with (5.1)):

$$\partial_t\Phi-\Delta\Phi=\delta_{(0,0)}.$$

As already explained above for the fundamental solution of $-\Delta$ (see the discussion before Proposition 32), an equality in distribution holds when multiplied by a smooth and compactly supported test functions φ, integrated over time and space, and with all the derivatives transferred to the test function by integration by parts. This is made precise in the next proposition. We denote by $C_c^{1,2}(\mathbb{R}\times\mathbb{R}^d)$ the functions which are $\mathcal{C}^1$ in time and $\mathcal{C}^2$ in space, and with compact support in $\mathbb{R}\times\mathbb{R}^d$.

Proposition 59. *For any $\varphi\in\mathcal{C}_c^{1,2}(\mathbb{R}\times\mathbb{R}^d)$,*

$$\int_{\mathbb{R}}\int_{\mathbb{R}^d}\Phi(t,x)\left(-\partial_t\varphi-\Delta\varphi\right)(t,x)\,dt\,dx=\varphi(0,0).$$

Proof: Notice first that $\Phi(t,x)$ is integrable at $(0,0)$. Indeed, for any $t_0>0$ and $r>0$, using the change of variable $y=x/\sqrt{4t}$,

$$\begin{aligned}\int_{-t_0}^{t_0}\int_{B(0,r)}\Phi(t,x)\,dx\,dt&=\int_0^{t_0}\frac{1}{(4\pi t)^{d/2}}\int_{B(0,r)}\exp\left(-\frac{|x|^2}{4t}\right)dx\,dt\\&=\frac{1}{\pi^{d/2}}\int_0^{t_0}\int_{B(0,r/\sqrt{4t})}\exp\left(-|y|^2\right)dy\,dt\\&\le\frac{1}{\pi^{d/2}}\int_0^{t_0}\int_{\mathbb{R}^d}\exp\left(-|y|^2\right)dy\,dt=t_0<\infty.\end{aligned}$$

In particular, since $-\partial_t\varphi-\Delta\varphi$ is continuous with compact support,

$$I=\int_{\mathbb{R}}\int_{\mathbb{R}^d}\Phi(t,x)\left(-\partial_t\varphi-\Delta\varphi\right)(t,x)\,dt\,dx$$

is indeed well defined.

By the Lebesgue dominated convergence theorem, one has therefore:

$$I=\lim_{\varepsilon\to0^+}I_\varepsilon$$

where $\varepsilon\to0^+$ means that the limit is taken for $\varepsilon>0$ going to zero, and, for $\varepsilon>0$,

$$I_\varepsilon=\int_{|t|>\varepsilon}\int_{\mathbb{R}^d}\Phi(t,x)\left(-\partial_t\varphi-\Delta\varphi\right)(t,x)\,dx\,dt=\int_\varepsilon^\infty\int_{\mathbb{R}^d}\Phi(t,x)\left(-\partial_t\varphi-\Delta\varphi\right)(t,x)\,dx\,dt,$$

since $\Phi=0$ for $t<0$. By integration by parts (notice that all the functions are regular on $[\varepsilon,\infty)\times\mathbb{R}^d$),

$$\begin{aligned}I_\varepsilon&=\int_\varepsilon^\infty\int_{\mathbb{R}^d}\left(\partial_t\Phi-\Delta\Phi\right)(t,x)\varphi(t,x)\,dx\,dt+\int_{\mathbb{R}^d}\Phi(\varepsilon,x)\varphi(\varepsilon,x)\,dx\\&=\int_{\mathbb{R}^d}\frac{1}{(4\pi\varepsilon)^{d/2}}\exp\left(-\frac{|x|^2}{4\varepsilon}\right)\varphi(\varepsilon,x)\,dx.\end{aligned}$$

By the change of variable $y = \frac{x}{\sqrt{4\varepsilon}}$, one thus gets:

$$I_\varepsilon = \int_{\mathbb{R}^d} \frac{1}{\pi^{d/2}} \exp\left(-|y|^2\right) \varphi(\varepsilon, \sqrt{4\varepsilon}y)\, dy.$$

By the dominated convergence theorem, one obtains:

$$\lim_{\varepsilon \to 0^+} I_\varepsilon = \varphi(0,0),$$

which concludes the proof. ♢

Exercise 62. ★*Let $\varphi : \mathbb{R}^d \to \mathbb{R}$ be a continuous compactly supported function. Check that $\int_{\mathbb{R}^d} \Phi(t, x-y)\varphi(y)\, dy$ is well defined for all $t > 0$. Prove that*

$$\lim_{t \to 0^+} \int_{\mathbb{R}^d} \Phi(t, x-y)\varphi(y)\, dy = \varphi(x).$$

8.2 Integral representation of the solution in $\mathbb{R}^d$

Using the heat kernel (8.3), one can exhibit some solutions to (8.1).

8.2.1 The initial-value problem

We would like to exhibit a solution to the heat equation (8.1) with $f = 0$, using the fundamental solution Φ defined by (8.3). Let us notice that for any $t > 0$, for any $y > 0$, one has $\partial_t \Phi(t, x-y) - \Delta_x \Phi(t, x-y) = 0$ and from Exercise 62, $\lim_{t \to 0^+} \Phi(t, x-y) = \delta_x(dy)$. On thus expects that the convolution

$$u(t,x) = \int_{\mathbb{R}^d} \Phi(t, x-y) u_0(y)\, dy \tag{8.4}$$

solves (8.1) with $f = 0$.

Theorem 60. *Let u_0 be a bounded continuous function in $\mathbb{R}^d$. The function $(t,x) \in \mathbb{R}_+^* \times \mathbb{R}^d \mapsto u(t,x)$ defined by (8.4) solves the heat equation (8.1) with $f = 0$ in the following sense:*

- *$u \in \mathcal{C}^\infty(\mathbb{R}_+^* \times \mathbb{R}^d)$;*
- *for all $t > 0$ and $x \in \mathbb{R}^d$, $\partial_t u(t,x) - \Delta u(t,x) = 0$;*
- *for all $x \in \mathbb{R}^d$, $\lim_{t \to 0^+} u(t,x) = u_0(x)$.*

In particular, the last item in the theorem shows that u can be extended by continuity to a continuous function $\mathcal{C}(\mathbb{R}_+ \times \mathbb{R}^d)$.

Proof: Let us first notice that the integral defining u in (8.4) makes sense for all $t > 0$ and $x \in \mathbb{R}^d$ since $\Phi(t, \cdot)$ is integrable over $\mathbb{R}^d$ and u_0 is bounded.

For $t > 0$ and $x \in \mathbb{R}^d$, by using the dominated convergence theorem, one can prove that

$$\partial_t u(t,x) = \int_{\mathbb{R}^d} \partial_t \Phi(t, x-y) u_0(y)\, dy$$

and

$$\Delta_x u(t,x) = \int_{\mathbb{R}^d} \Delta_x \Phi(t, x-y) u_0(y)\, dy.$$

Let us for example write the details for the latter case. For $t_0 > 0$, and $x_0 \in \mathbb{R}^d$, for $\delta > 0$, $x \in B(x_0, \delta) \mapsto \Delta_x \Phi(t, x-y) u_0(y)$ is $\mathcal{C}^2$ and is bounded from above by $C \exp(-\alpha |y|^2)$ for C and

α well chosen, uniformly in $x \in B(x_0, \delta)$. The dominated convergence theorem then shows that one can interchange Δ_x and $\int_{\mathbb{R}^d}$ at $x = x_0$. The same argument shows that u is $\mathcal{C}^\infty$ on $\mathbb{R}_+^* \times \mathbb{R}^d$.

In particular, one has:

$$\partial_t u(t,x) - \Delta_x u(t,x) = \int_{\mathbb{R}^d} \left(\partial_t \Phi(t, x-y) - \Delta_x \Phi(t, x-y)\right) u_0(y)\, dy = 0$$

since $\partial_t \Phi - \Delta \Phi = 0$.

Finally, one gets

$$\begin{aligned}
\int_{\mathbb{R}^d} \Phi(t, x-y) u_0(y)\, dy &= \int_{\mathbb{R}^d} \frac{1}{(4\pi t)^{d/2}} \exp\left(-\frac{|x-y|^2}{4t}\right) u_0(y)\, dy \\
&= \int_{\mathbb{R}^d} \frac{1}{(4\pi t)^{d/2}} \exp\left(-\frac{|y|^2}{4t}\right) u_0(x-y)\, dy \\
&= \int_{\mathbb{R}^d} \frac{1}{\pi^{d/2}} \exp\left(-|z|^2\right) u_0(x - \sqrt{4t} z))\, dy
\end{aligned}$$

which converges to $u_0(x)$ when $t \to 0$, again by dominated convergence. ♢

Remark 42. *Notice a surprising phenomenon: when u_0 is non negative with compact support (and non-zero) and $f = 0$, then the solution u to the heat equation* (8.1) *defined by* (8.4) *is such that $u(t,x) > 0$ for all $x \in \mathbb{R}^d$ and $t > 0$ (even for infinitesimally small time t). In other words, a compactly supported initial distribution of temperature instantly spreads to the whole of $\mathbb{R}^d$. Thus, the heat equation propagates perturbation (energy) at infinite speed, which may seem unphysical. However, the validity of the heat equation as a model of temperature diffusion is still extremely good for classical physics and engineering applications.*

Remark 43. *The solution u defined by* (8.4) *admits a probabilistic representation (check it when $d = 1$★):*

$$u(t,x) = \mathbb{E}\left(u_0(x + \sqrt{2t} G)\right)$$

where $G \sim \mathcal{N}(0, \mathrm{Id})$ is a centered Gaussian vector with identity covariance matrix (i.e. $G = (G_1, \ldots, G_d)$ with G_i independent and identically distributed Gaussian random variables with zero mean and variance one). Indeed, the heat kernel (8.3) *is, for $t > 0$, the density of a centered Gaussian vector with covariance $2t\mathrm{Id}$. There is a deep connection between the heat equation (more generally parabolic equations) and the Brownian motion (more generally diffusion processes) which gives rise to a variety of mathematical developments and numerical methods.*

8.2.2 The non-homogeneous problem

Let us now consider the heat equation (8.1) with a general right-hand side f which depends on $t > 0$ and $x \in \mathbb{R}^d$. The idea is to apply the Duhamel principle (see Exercise 16) to guess a good candidate for the solution:

$$u(t,x) = \int_{\mathbb{R}^d} \Phi(t, x-y) u_0(y)\, dy + \int_0^t \int_{\mathbb{R}^d} \Phi(t-s, x-y) f(s,y)\, dy\, ds. \tag{8.5}$$

Theorem 61. *Let u_0 be a bounded continuous function on $\mathbb{R}^d$ and $f \in \mathcal{C}^{1,2}(\mathbb{R}_+ \times \mathbb{R}^d)$ with compact support*[3] *in $\mathbb{R}_+ \times \mathbb{R}^d$. The function $(t,x) \in \mathbb{R}_+^* \times \mathbb{R}^d \mapsto u(t,x)$ defined by* (8.4) *solves the heat equation* (8.1) *in the following sense:*

[3]Notice that a function with compact support in $\mathbb{R}_+ \times \mathbb{R}^d$ can be non zero on $t = 0$ since $[0, t_0]$ is a compact subset of $\mathbb{R}_+$.

- $u \in \mathcal{C}^{1,2}(\mathbb{R}_+^* \times \mathbb{R}^d)$;
- *for all* $t > 0$ *and* $x \in \mathbb{R}^d$, $\partial_t u(t,x) - \Delta u(t,x) = f(t,x)$;
- *for all* $x \in \mathbb{R}^d$, $\lim_{t \to 0^+} u(t,x) = u_0(x)$.

Proof:

This result has already been proven in Theorem 60 in the case $f = 0$ and $u_0 \neq 0$. By the superposition principle, it is thus enough to prove Theorem 61 for $f \neq 0$ and $u_0 = 0$ (check it!★). Let us thus consider $f \in \mathcal{C}^{1,2}(\mathbb{R}_+ \times \mathbb{R}^d)$ with compact support in $\mathbb{R}_+ \times \mathbb{R}^d$ and u defined for $t > 0$ and $x \in \mathbb{R}^d$ by:

$$u(t,x) = \int_0^t \int_{\mathbb{R}^d} \Phi(t-s, x-y) f(s,y)\, dy\, ds.$$

This integral is well defined since f has compact support, Φ is $\mathcal{C}^\infty(\mathbb{R} \times \mathbb{R}^d \setminus (0,0))$ and is integrable at $(t,x) = (0,0)$. Moreover, one has

$$u(t,x) = \int_0^t \int_{\mathbb{R}^d} \Phi(s,y) f(t-s, x-y)\, dy\, ds.$$

By dominated convergence theorem, using the fact that $\partial_t f$ and $\frac{\partial^2 f}{\partial x_i \partial x_j}$ are bounded continuous functions, one gets that $u \in C^{1,2}(\mathbb{R}_+^* \times \mathbb{R}^d)$ with, for all $t > 0$ and $x \in \mathbb{R}^d$,

$$\partial_t u(t,x) = \int_{\mathbb{R}^d} \Phi(t,y) f(0, x-y)\, dy + \int_0^t \int_{\mathbb{R}^d} \Phi(s,y) \partial_t f(t-s, x-y)\, dy\, ds$$

and

$$\frac{\partial^2 u}{\partial x_i \partial x_j}(t,x) = \int_0^t \int_{\mathbb{R}^d} \Phi(s,y) \frac{\partial^2 f}{\partial x_i \partial x_j}(t-s, x-y)\, dy\, ds.$$

Therefore, for $\varepsilon > 0$

$$\begin{aligned}(\partial_t u - \Delta_x u)(t,x) &= \int_{\mathbb{R}^d} \Phi(t,y) f(0,x-y)\, dy + \int_0^t \int_{\mathbb{R}^d} \Phi(s,y)(\partial_t - \Delta_x) f(t-s, x-y)\, dy\, ds \\ &= I_\varepsilon + J_\varepsilon \end{aligned} \tag{8.6}$$

where

$$I_\varepsilon = \int_{\mathbb{R}^d} \Phi(t,y) f(0,x-y)\, dy + \int_\varepsilon^t \int_{\mathbb{R}^d} \Phi(s,y)(\partial_t - \Delta_x) f(t-s, x-y)\, dy\, ds$$

and

$$J_\varepsilon = \int_0^\varepsilon \int_{\mathbb{R}^d} \Phi(s,y)(\partial_t - \Delta_x) f(t-s, x-y)\, dy\, ds.$$

Notice that $|J_\varepsilon| \leq \|(\partial_t - \Delta_x) f\|_{L^\infty} \int_0^\varepsilon \int_{\mathbb{R}^d} \Phi(s,y)\, dy\, ds$ and thus

$$\lim_{\varepsilon \to 0} J_\varepsilon = 0$$

since Φ is integrable. Moreover, using the fact that $\partial_t f(t-s, x-y) = -\partial_s f(t-s, x-y)$, $\Delta_x f(t-s, x-y) = \Delta_y f(t-s, x-y)$ and an integration by parts (all the functions are regular

for $s \geq \varepsilon > 0$),

$$\begin{aligned} I_\varepsilon &= \int_{\mathbb{R}^d} \Phi(t,y) f(0,x-y)\,dy + \int_\varepsilon^t \int_{\mathbb{R}^d} \Phi(s,y)(-\partial_s - \Delta_y) f(t-s,x-y)\,dy\,ds \\ &= \int_{\mathbb{R}^d} \Phi(t,y) f(0,x-y)\,dy + \int_\varepsilon^t \int_{\mathbb{R}^d} (\partial_s - \Delta_y)\Phi(s,y) f(t-s,x-y)\,dy\,ds \\ &\quad - \int_{\mathbb{R}^d} \Phi(t,y) f(0,x-y)\,dy + \int_{\mathbb{R}^d} \Phi(\varepsilon,y) f(t-\varepsilon,x-y)\,dy \\ &= \int_{\mathbb{R}^d} \Phi(\varepsilon,y) f(t-\varepsilon,x-y)\,dy \end{aligned}$$

since $(\partial_s - \Delta_y)\Phi(s,y) = 0$ for $s \geq \epsilon > 0$ and $y \in \mathbb{R}^d$. Thus, by the change of variable $z = y/\sqrt{4\varepsilon}$,

$$\begin{aligned} I_\varepsilon &= \int_{\mathbb{R}^d} \frac{1}{(4\pi\varepsilon)^{d/2}} \exp\left(-\frac{|y|^2}{4\varepsilon}\right) f(t-\varepsilon,x-y)\,dy \\ &= \int_{\mathbb{R}^d} \frac{1}{\pi^{d/2}} \exp\left(-|z|^2\right) f(t-\varepsilon,x-\sqrt{4\varepsilon}z)\,dz \end{aligned}$$

which converges to $f(t,x)$ by dominated convergence.

This concludes the proof, by letting $\varepsilon \to 0$ in (8.6).

♢

We have constructed one solution to the heat equation. We need now to address the question of uniqueness. Thanks to the linearity of the equation, this amounts to understand the uniqueness of solution to the heat equation (8.1) for $f = u_0 = 0$. This will be addressed in the forthcoming sections by proving maximum principles, together with some fundamental properties of the heat equation.

8.3 Mathematical properties of the heat equation

The solution to the heat equation we built previously on $\mathbb{R}_+ \times \mathbb{R}^d$ by convolution (see (8.5) and Theorem 8.5), exhibits remarkable properties. First, if $u_0 \geq 0$ and $f \geq 0$, then $u \geq 0$. Second, in the homogeneous case ($f = 0$), even if u_0 is only bounded continuous, the function u defined by convolution is actually $\mathcal{C}^\infty$ for any time $t > 0$ and any x in the domain. We will show in this section that these two properties (maximum principle, and instantaneous smoothing) are actually satisfied for any solutions to the heat equation in an appropriate ensemble of functions. As for the Poisson equation, maximum principles will in turn provide uniqueness results.

8.3.1 Maximum principle on a bounded subset of $\mathbb{R}^d$

Let us start with a simple exercise which will be useful to understand the proof of the weak maximum principle.

Exercise 63. ★*Let $u : \mathbb{R} \to \mathbb{R}$ be a C^2 function. If $\max_{[-1,1]} u(x) = u(0)$ prove that $u'(0) = 0$ and $u''(0) \leq 0$. If $\max_{[-1,1]} u(x) = u(1)$ prove that $u'(1) \geq 0$.*

Theorem 62 (Weak maximum principle for the heat equation on a bounded subset of $\mathbb{R}^d$)**.** *Let $T > 0$ and $\Omega \subset \mathbb{R}^d$ be an open bounded subset of $\mathbb{R}^d$, and let $u \in C^{1,2}((0,T] \times \Omega) \cap C([0,T] \times \overline{\Omega})$ satisfy*

$$\partial_t u - \Delta u \leq 0 \quad \text{in } (0,T] \times \Omega.$$

Let

$$\Gamma_T = \{0\} \times \Omega \cup [0,T] \times \partial\Omega$$

denote the so-called parabolic boundary of $(0,T] \times \Omega$. *Then*

$$\max_{[0,T]\times\overline{\Omega}} u = \max_{\Gamma_T} u.$$

Notice that the parabolic boundary is not the boundary of the cylinder $(0,T) \times \Omega$ (it only contains the sides and bottom; it does not include the top $\{T\} \times \Omega$).

Proof: Since $[0,T] \times \overline{\Omega}$ is compact, $\max_{[0,T]\times\overline{\Omega}} u$ is well defined.

If $\partial_t u - \Delta u < 0$ in $(0,T] \times \Omega$, one can argue by contradiction. Indeed, if there exists $(t_0, x_0) \in [0,T] \times \overline{\Omega} \setminus \Gamma_T$ such that $u(t_0,x_0) = \max_{[0,T]\times\overline{\Omega}} u$, then[4]:

$$\partial_t u(t_0,x_0) \geq 0 \text{ and } \Delta u(t_0,x_0) \leq 0$$

and thus $(\partial_t u - \Delta u)(t_0,x_0) \geq 0$ which contradicts the hypothesis.

In order to extend the argument to the hypothesis $\partial_t u - \Delta u \leq 0$ in $(0,T] \times \Omega$, let us consider, for $\varepsilon > 0$,

$$v_\varepsilon(t,x) = u(t,x) - \varepsilon t.$$

Then, $\partial_t v_\varepsilon - \Delta v_\varepsilon \leq -\varepsilon < 0$ in $(0,T] \times \Omega$ and thus,

$$\max_{[0,T]\times\overline{\Omega}} v_\varepsilon = \max_{\Gamma_T} v_\varepsilon.$$

Notice that $v_\varepsilon(t,x) \leq u(t,x) \leq v_\varepsilon(t,x) + \varepsilon T$ so that $\lim_{\varepsilon\to 0} \max_{[0,T]\times\overline{\Omega}} v_\varepsilon = \max_{[0,T]\times\overline{\Omega}} u$. Besides, let $(t_\varepsilon, x_\varepsilon) \in \Gamma_T$ be a point such that $\max_{[0,T]\times\overline{\Omega}} v_\varepsilon = v_\varepsilon(t_\varepsilon, x_\varepsilon)$. Since Γ_T is compact, there exists a sequence $\varepsilon_n \to 0$ such that $(t_{\varepsilon_n}, x_{\varepsilon_n})$ converges to $(t^*, x^*) \in \Gamma_T$ when $n \to \infty$. By passing to the limit $n \to \infty$ in the inequality $v_{\varepsilon_n}(t_{\varepsilon_n}, x_{\varepsilon_n}) \leq u(t_{\varepsilon_n}, x_{\varepsilon_n})$, one thus get $u(t^*,x^*) = \max_{[0,T]\times\overline{\Omega}} u$. ♢

Remark 44. *The proof we provide here of the weak maximum principle for the heat equation is different in nature from the one we used for the Laplace equation. The latter was indeed based on a mean value formula. Let us comment on two aspects. First, it is also possible to use a mean value formula for the heat equation to prove the maximum principle, even though this is more technical than for the Laplace equation (see [6, Sections 2.3.2 and 2.3.3]). Second, one can also prove a weak maximum principle for the Laplace equation using similar arguments as above, and actually get the weak maximum principle of Theorem 38 only assuming* $-\Delta u \leq 0$ *rather that* $-\Delta u = 0$. *(try it!★★★).*

Exercise 64. ★*Show that the weak maximum principle is equivalent to the following statement: let* $u \in \mathcal{C}^{1,2}((0,T] \times \Omega) \cap \mathcal{C}([0,T] \times \overline{\Omega})$ *be such that* $\partial_t u - \Delta u \leq 0$ *in* $(0,T] \times \Omega$. *Then*

$$u \leq 0 \text{ on } \Gamma_T \Longrightarrow u \leq 0 \text{ on } [0,T] \times \overline{\Omega}.$$

Remark 45. *Following Exercise 64, from a physical viewpoint, the weak maximum principle states a very natural fact: if the temperature is non positive initially and on the boundary of the domain (this is the assumption* $u \leq 0$ *on* Γ_T*) and if the source terms remove energy from the system (this is the assumption* $\partial_t u - \Delta u \leq 0$*), then the temperature is non positive everywhere and for all time.*

This is related to the fact that the heat flows from higher to lower temperature regions.

---

[4] if $t_0 \in (0,T)$, one actually has $\partial_t u(t_0,x_0) = 0$, but if $t_0 = T$, we only get $\partial_t u(t_0,x_0) \geq 0$.

A consequence of this maximum principle is the following uniqueness result for the Dirichlet problem on a bounded domain:

Proposition 63. *Let Ω be a bounded open subset of $\mathbb{R}^d$. There exists at most one solution $u \in \mathcal{C}^{1,2}(\mathbb{R}_+^* \times \Omega) \cap \mathcal{C}(\mathbb{R}_+ \times \overline{\Omega})$ to the heat equation with Dirichlet boundary conditions:*

$$\begin{cases} \partial_t u - \Delta u = f \text{ on } \mathbb{R}_+^* \times \Omega \\ u(t,x) = g(x) \text{ for } t > 0, x \in \partial\Omega \\ u(0,x) = u_0(x) \text{ for } x \in \Omega \end{cases} \tag{8.7}$$

where f, g and u_0 are continuous functions.

Proof: Let us consider two solutions u and v and let us introduce $w = u - v$. The function w solves (8.7) for $f = g = u_0 = 0$. Thus, by applying Thorem 62 to w, one gets $w \leq 0$. The same argument applied to $-w$ yields $w \geq 0$ and thus $w = 0$. ♢

Remark 46. *Contrary to the Laplace equation studied in Part II, we will not discuss the existence of a solution to the heat equation with Dirichlet boundary conditions. The construction of Green functions are indeed tricky. Better approaches will be introduced in the lectures [4, 13], using in particular Fourier transforms and variational approaches.*

It is actually possible to prove the following stronger statement (admitted here, see [6, Section 2.3.3] for a proof using a mean value formula for the heat equation).

Theorem 64 (Strong maximum principle for the heat equation on a bounded domain)**.** *Let $T > 0$ and $\Omega \subset \mathbb{R}^d$ be an open bounded connected subset of $\mathbb{R}^d$, and let*

$$u \in C^{1,2}((0,T] \times \Omega) \cap C([0,T] \times \overline{\Omega})$$

satisfy

$$\partial_t u - \Delta u = 0 \quad \text{in } (0,T] \times \Omega.$$

If u attains a maximum M at some point $(t_0, x_0) \in (0,T] \times \Omega$, i.e.,

$$u(t_0,x_0) = M, \qquad u \leq M \text{ on } [0,T] \times \overline{\Omega},$$

then $u = M$ on $[0,t_0] \times \overline{\Omega}$.

8.3.2 Weak maximum principle on $\mathbb{R}^d$

To get a uniqueness result for the heat equation (8.1) on $\mathbb{R}^d$, we need a weak maximum principle on $\mathbb{R}^d$ (and not on a bounded domain such as in Theorem 62). This requires some assumption on the growth at infinity.

Theorem 65 (Weak maximum principle for the heat equation on $\mathbb{R}^d$)**.** *Let $T > 0$ and let $u \in C^{1,2}((0,T] \times \mathbb{R}^d) \cap C([0,T] \times \mathbb{R}^d)$ satisfy*

$$\begin{cases} \partial_t u - \Delta u \leq 0 \text{ in } (0,T] \times \mathbb{R}^d \\ u(0,x) = u_0(x) \text{ for } x \in \mathbb{R}^d. \end{cases}$$

Let us moreover assume that there exists positive constants a and A such that, for all $(t,x) \in [0,T] \times \mathbb{R}^d$,

$$u(t,x) \leq A \exp(a|x|^2).$$

Then

$$\sup_{[0,T]\times\mathbb{R}^d} u = \sup_{\mathbb{R}^d} u_0.$$

Proof:

The proof of this result can be skipped on first reading.

By dividing the interval $(0,T)$ into subintervals, one can assume that $4aT < 1$. Let $\varepsilon > 0$ be such that $4a(T+\varepsilon) < 1$, and let us consider for $\mu > 0$ and a fixed $y \in \mathbb{R}^d$.

$$v(t,x) = u(t,x) - \frac{\mu}{(T+\varepsilon-t)^{d/2}} \exp\left(\frac{|x-y|^2}{4(T+\varepsilon-t)}\right).$$

Since the perturbation term satisfies the heat equation on $[0,T]\times\mathbb{R}^d$ (check it★), one gets

$$\partial_t v - \Delta v \leq 0.$$

The objective is to show that $v(t,y) \leq \sup_{\mathbb{R}^d} u_0$, and then to let $\mu \to 0$ to conclude the proof.

By applying the weak maximum principle from Theorem 62, for any $R > 0$, one has

$$\max_{[0,T]\times\overline{B(y,R)}} v \leq \max_{\Gamma(R)} v$$

where $\Gamma(R) = \{0\}\times B(y,R) \cup [0,T]\times\partial B(y,R)$ is the parabolic boundary of the cylinder $[0,T]\times\overline{B(y,R)}$.

Let us estimate v on $\Gamma(R)$. For $t = 0$ and $x \in B(y,r)$, one has $v(0,x) = u(0,x) - \frac{\mu}{(T+\varepsilon)^{d/2}} \exp\left(\frac{|x-y|^2}{4(T+\varepsilon)}\right) \leq u_0(x) \leq \sup_{\mathbb{R}^d} u_0$.

For $t \in [0,T]$ and $x \in \partial B(y,R)$, one has (denoting $\gamma = \frac{1}{4(T+\varepsilon)} - a > 0$)

$$\begin{aligned} v(t,x) &= u(t,x) - \frac{\mu}{(T+\varepsilon-t)^{d/2}} \exp\left(\frac{R^2}{4(T+\varepsilon-t)}\right) \\ &\leq A\exp(a|x|^2) - \frac{\mu}{(T+\varepsilon)^{d/2}} \exp\left(\frac{R^2}{4(T+\varepsilon)}\right) \\ &\leq A\exp(a(|y|+R)^2) - \mu(4(a+\gamma))^{d/2} \exp\left((a+\gamma)R^2\right) \leq \sup_{\mathbb{R}^d} u_0 \end{aligned}$$

by choosing R sufficiently large.

Thus, for any $t \in [0,T]$, $v(t,y) \leq \max_{[0,T]\times\overline{B(y,R)}} v \leq \max_{\Gamma(R)} v \leq \sup_{\mathbb{R}^d} u_0$. As a consequence, for any $t \in [0,T]$

$$\begin{aligned} u(t,y) &= v(t,y) + \frac{\mu}{(T+\varepsilon-t)^{d/2}} \\ &\leq \sup_{\mathbb{R}^d} u_0 + \frac{\mu}{(T+\varepsilon-t)^{d/2}} \end{aligned}$$

and by letting $\mu \to 0$, one gets for any $t \in [0,T]$, $u(t,y) \leq \sup_{\mathbb{R}^d} u_0$ which concludes the proof, since $y \in \mathbb{R}^d$ was arbitrary.

♢

Remark 47. *It is known that without the growth assumption on u, the weak maximum principle does not hold.*

As a corollary, we get the following uniqueness result for the solution to the heat equation.

Proposition 66. *For $u_0 \in \mathcal{C}(\mathbb{R}^d)$, and $f \in \mathcal{C}(\mathbb{R}_+^*\times\mathbb{R}^d)$, there exists at most one solution $u \in \mathcal{C}^{1,2}(\mathbb{R}_+^*\times\mathbb{R}^d) \cap \mathcal{C}(\mathbb{R}_+\times\mathbb{R}^d)$ to the heat equation*

$$\begin{cases} \partial_t u - \Delta u = f \text{ in } \mathbb{R}_+^*\times\mathbb{R}^d \\ u(0,x) = u_0(x) \text{ for } x \in \mathbb{R}^d \end{cases}$$

satisfying the growth estimate:

$$\forall (t,x) \in \mathbb{R}_+ \times \mathbb{R}^d,\ |u(t,x)| \leq A \exp(a|x|^2)$$

for some positive constants a and A.

Proof: The proof consists in applying the weak maximim principle of Theorem 65 to $w = \pm(u-v)$ where u and v are two solutions. ♢

As a consequence of Theorem 61 and Proposition 66, for u_0 bounded and continuous and $f \in \mathcal{C}^{1,2}(\mathbb{R}_+ \times \mathbb{R}^d)$ with compact support, we thus have an existence and uniqueness result to classical solutions to the heat equation (8.1) satisfying the growth estimate mentioned in Proposition 66. Indeed, u defined by (8.5) does satisfy this growth estimate (check it★★).

Remark 48. *Coming back to Remark 22, the growth estimate can be seen as a mathematical requirement to select the "right solution" to the heat equation on $\mathbb{R}^d$, namely the one defined by convolution (see Equation* (8.4) *and* (8.5)*). See Remark 22 for additional comments.*

8.3.3 Regularity

In addition to the infinite speed of propagation of perturbations already discussed in Remark 42, solutions to the heat equation also satisfy an instantaneous smoothing property.

Theorem 67. *Let us assume that $u \in \mathcal{C}^{1,2}((0,T)\times\Omega)$ for some open set $\Omega \subset \mathbb{R}^d$ satisfies the homogeneous heat equation:*

$$\partial_t u - \Delta u = 0 \text{ in } (0,T)\times\Omega.$$

Then $u \in \mathcal{C}^\infty((0,T)\times\Omega)$.

Notice that we do not assume anything about the regularity of u on the boundary of the cylinder $(0,T)\times\Omega$.

Proof:

This proof can be skipped on first reading.
Step 1: Localization of the argument. Let us fix an arbitrary point $(t_0,x_0) \in (0,T)\times\Omega$. Let us introduce the parabolic cylinders: for $t \geq 0$, $x \in \mathbb{R}^d$ and $r > 0$,

$$C(t,x;r) = \{(s,y) \in \mathbb{R}\times\mathbb{R}^d, |x-y| \leq r,\ t-r^2 \leq s \leq t\}.$$

Choose $r > 0$ sufficiently small that

$$C = C(t_0,x_0;r) \subset (0,T)\times\Omega.$$

Define also the smaller cylinders

$$C' = C\Big(t_0,x_0;\tfrac{3r}{4}\Big), \quad C'' = C\Big(t_0,x_0;\tfrac{r}{2}\Big),$$

which have the same top center point (t_0,x_0). Notice that $C'' \subset C' \subset C$ (see Figure 8.1). Our objective is to show that u is $\mathcal{C}^\infty(C'')$, which will conclude the proof since $(t_0,x_0) \in (0,T)\times\Omega$ is arbitrary.

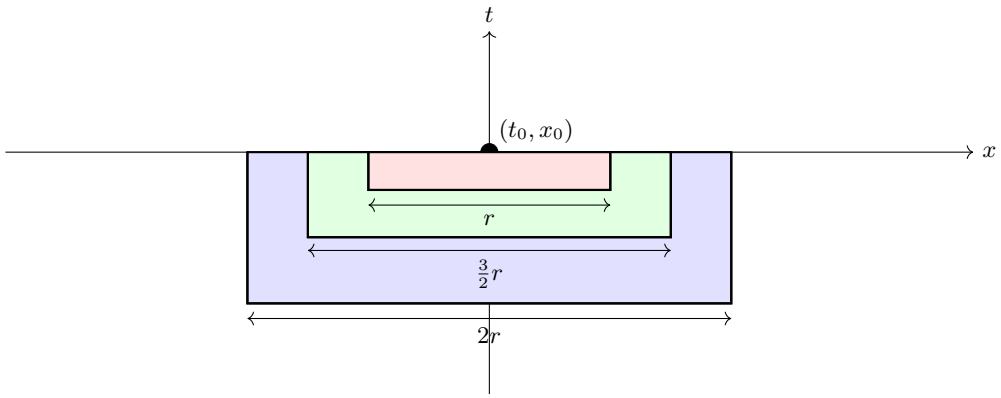


Figure 8.1: One-dimensional representation of the three cylinders C'' (red) $\subset C'$ (green) $\subset C$ (blue), with same top center (t_0, x_0).

Choose a smooth cutoff function $\zeta : C \to [0,1]$ such that

$$\zeta = 1 \text{ on } C', \quad \zeta = 0 \text{ near the parabolic boundary of } C.$$

Let us recall that the parabolic boundary of C is $\{t - r^2\} \times B(x,r) \cup [t - r^2, t] \times \partial B(x,r)$. Extend ζ by 0 in $[0,t_0] \times \mathbb{R}^d \setminus C$.

Let us introduce

$$v(t,x) := \zeta(t,x)u(t,x) \text{ for } t \in [0,t_0],\ x \in \mathbb{R}^d.$$

Notice that

$$v(0,\cdot) = 0 \text{ on } \mathbb{R}^d.$$

Moreover, $\partial_t v = \zeta \partial_t u + \partial_t \zeta u$ and $\Delta v = \zeta \Delta u + 2\nabla\zeta \cdot \nabla u + u\Delta\zeta$ so that

$$\partial_t v - \Delta v = f \text{ where } f = \partial_t \zeta u - 2\nabla\zeta \cdot \nabla u - u\Delta\zeta \text{ in } (0,t_0) \times \mathbb{R}^d. \tag{8.8}$$

Step 2: An integral representation of u, if $u \in C^\infty((0,T) \times \Omega)$. Let us assume temporarily that $u \in C^\infty((0,T) \times \Omega)$ (which is what we actually would like to prove... we will come back to that in the next step), so that f is regular with compact support. Let us introduce (see (8.5))

$$\tilde{v}(t,x) := \int_0^t \int_{\mathbb{R}^d} \Phi(x - y, t - s)\, f(s,y)\, dy\, ds.$$

According to Theorem 61, $\tilde{v}$ satisfies the same partial differential equation as v:

$$\begin{cases} \partial_t \tilde{v} - \Delta \tilde{v} = f & \text{in } (0,t_0) \times \mathbb{R}^d, \\ \tilde{v}(0,\cdot) = 0 & \text{on } \mathbb{R}^d. \end{cases}$$

Since $|v|$ and $|\tilde{v}|$ are in $L^\infty(\mathbb{R}_+ \times \mathbb{R}^d)$, Proposition 66 implies $v = \tilde{v}$. Therefore, since ζ

and thus f are zero outside of the cylinder C,

$$v(t,x) = \int_0^t \int_{\mathbb{R}^n} \Phi(x-y,t-s)\, f(s,y)\, dy\, ds$$
$$= \iint_C \Phi(x-y,t-s)\, f(s,y)\, dy\, ds. \tag{8.9}$$

Using (8.8) and the fact that $\zeta = 1$ in C'', one gets, with an integration by parts: for all $(t,x) \in C''$,

$$\begin{aligned} u(t,x) &= \iint_C \Phi(x-y,t-s)\big[\partial_s\zeta(s,y) - \Delta\zeta(s,y)\big]u(s,y) - 2\nabla\zeta(s,y)\cdot\nabla u(s,y)\, dy\, ds \\ &= \iint_C \left[\Phi(x-y,t-s)(\partial_s\zeta(s,y) + \Delta\zeta(s,y)) + 2\nabla_y\Phi(x-y,t-s)\cdot\nabla\zeta(s,y)\right] u(s,y)\, dyds \\ &= \iint_C K(t,x,s,y)u(s,y)\, dyds \end{aligned} \tag{8.10}$$

where $K(t,x,s,y) = \Phi(x-y,t-s)(\partial_s\zeta(s,y) + \Delta\zeta(s,y)) + 2\nabla_y\Phi(x-y,t-s)\cdot\nabla\zeta(s,y)$. Notice that the supports of $\partial_s\zeta$ and $\nabla\zeta$ are included in $C \setminus C'$. Thus, for fixed $(t,x) \in C''$, $K(t,x,s,y) = 0$ for all $(s,y) \in C'$ and $(s,y) \mapsto K(t,x,s,y)$ is smooth on $C \setminus C'$.

Step 3: Mollification to get (8.10) **for** $u \in C^{1,2}((0,T)\times\Omega)$**.** We have proven (8.10) assuming $u \in C^\infty((0,T)\times\Omega)$. However, u still satisfies (8.10) if it is only in $\mathcal{C}^{1,2}((0,T)\times\Omega)$. Indeed, one can introduce the time-space mollification of u (in the spirit of the proof of Theorem 39): for $\varepsilon > 0$,

$$u_\varepsilon(t,x) = \int_{\mathbb{R}_+\times\mathbb{R}^d} \eta_\varepsilon(t-s,x-y)u(s,y)\, dy = \int_{\mathbb{R}_+\times\mathbb{R}^d} \eta_\varepsilon(s,y)u(t-s,x-y)\, dy$$

where $\eta_\varepsilon(t,x) = \varepsilon^{-d+1}\eta(t,x)$ and $\eta(t,x) = C\exp\left(\frac{1}{t^2+|x|^2-1}\right)1_{|x|^2+t^2<1}$. Notice that $(\partial_t - \Delta)u_\varepsilon = \int_{\mathbb{R}_+\times\mathbb{R}^d}\eta_\varepsilon(s,y)(\partial_t-\Delta)u(t-s,x-y)\, dy = 0$. Moreover, u_ε is smooth. One can thus derive (8.10) with u^ε replacing u, and then pass to the limit $\varepsilon \to 0$ to obtain (8.10) with $u \in \mathcal{C}^{1,2}((0,T)\times\Omega)$.

Step 4: Conclusion. The conclusion of the proof now consists in contemplating formula (8.10) and realizing that the right-hand side indeed defines a C^∞ function on C'', by standard arguments on the regularity of functions defined by integrals (dominated convergence), using the regularity of the function K.

♢

Remark 49. *For the non-homogeneous problem $\partial_t u - \Delta u = f$, regularity of u is determined by the regularity of f: u is not $\mathcal{C}^\infty$ at times t and positions x where f is not $\mathcal{C}^\infty$.*

8.4 Towards variational methods

As already mentioned, very powerful variational methods to solve partial differential equations will be introduced in forthcoming lectures [4, 11], in order in particular to prove existence and uniqueness of solutions to the heat equation with Dirichlet (or Neumann) boundary conditions. Let us provide a result in that direction, which gives an alternative proof to a variant of the uniqueness result of Proposition 63.

Proposition 68 (Uniqueness by a variational method)**.** *Let Ω be a bounded open subset of $\mathbb{R}^d$. There exists at most one solution $u \in \mathcal{C}^{1,2}(\mathbb{R}_+ \times \overline{\Omega})$ to the heat equation with Dirichlet boundary conditions:*

$$\begin{cases} \partial_t u - \Delta u = f \text{ for } t > 0, x \in \Omega \\ u(t,x) = g(x) \text{ for } t > 0, x \in \partial\Omega \\ u(0,x) = u_0(x) \text{ for } x \in \Omega \end{cases} \tag{8.11}$$

where f, g and u_0 are continuous functions.

Proof: As usual, since the problem is linear, this amounts to proving that if $f = g = u_0 = 0$, then $u = 0$. Let us multiply the partial differential equation satisfied by u by u itself, and integrate over $x \in \Omega$. One gets:

$$\int_\Omega u \partial_t u = \int_\Omega u \Delta u.$$

This yields, using the regularity of u to justify the calculation (notice that $u = 0$ on $\partial\Omega$ so that there are no boundary terms after integration by parts)

$$\frac{d}{dt}\frac{1}{2}\int_\Omega u^2 = -\int_\Omega |\nabla u|^2 \leq 0 \tag{8.12}$$

and thus, for all $t \geq 0$,

$$\int_\Omega u^2(t,x)\,dx \leq \int_\Omega u^2(0,x)\,dx = 0$$

which concludes the proof. ♢

A consequence of the previous calculation is that for a classical solution to (8.11) with $f = g = 0$, for all $t \geq 0$,

$$\|u(t,\cdot)\|_{L^2(\Omega)} \leq \|u_0\|_{L^2(\Omega)}.$$

The solution at time t continuously depends on the initial condition, in the L^2 norm.

Exercise 65 (The backward heat equation)**.** ★*The sign of the diffusion term is essential. Consider the 1d heat equation with the opposite sign on the diffusion term*

$$\begin{cases} \partial_t u + \partial_{x,x} u = 0 \quad \text{for } t > 0,\ 0 < x < \pi, \\ u(t,0) = u(t,\pi) = 0 \quad \text{for } t \geq 0. \end{cases} \tag{8.13}$$

For $k \geq 1$, check that

$$u_k(t,x) = \frac{1}{k} e^{k^2 t} \sin(kx)$$

is a classical solution. Using

$$\int_0^\pi \sin^2(kx)\,dx = \frac{\pi}{2},$$

show that

$$\|u_k(0,\cdot)\|_{L^2(0,\pi)} = \frac{1}{k}\sqrt{\frac{\pi}{2}} \underset{k\to\infty}{\longrightarrow} 0, \qquad \|u_k(t,\cdot)\|_{L^2(0,\pi)} = \frac{e^{k^2 t}}{k}\sqrt{\frac{\pi}{2}} \underset{k\to\infty}{\longrightarrow} +\infty$$

for every fixed $t > 0$. What does this say about the continuous dependence of the solution on its initial condition? Compare with the solution $k^{-1}e^{-k^2 t}\sin(kx)$ of the heat equation with the usual sign.

Chapter 9

Discretization with the finite difference method

The objective of this chapter is to discuss a numerical method to discretize the heat equation in dimension $d = 1$ on the domain

$$\Omega = (0, L)$$

for $L > 0$ with Dirichlet boundary conditions[1]:

$$\begin{cases} \partial_t u - \partial_{x,x} u = f \text{ on } (0,T) \times \Omega, \\ u(t,0) = a \text{ and } u(t,L) = b \text{ for } t \in (0,T), \\ u(0,x) = u_0(x) \text{ for } x \in \Omega. \end{cases} \tag{9.1}$$

The functions $f : (0,T) \times \Omega \to \mathbb{R}$ and $u_0 : \Omega \to \mathbb{R}$ are continuous[2], and we assume that there exists a solution[3] $u \in \mathcal{C}^{1,2}((0,T) \times \Omega) \cap \mathcal{C}^0([0,T] \times \overline{\Omega})$.

The idea is to first apply the finite difference method to the Laplace operator $\partial_{x,x}$ to obtain an ordinary differential equation in time (see Chapter 6), and then standard numerical schemes for ordinary differential equation such as the Euler method to discretize the problem in time (see Chapter 3).

9.1 The numerical scheme

Let us introduce, for two integers N and I, a time-step size $\Delta t = \frac{T}{N} > 0$ and a mesh size $h = \frac{L}{I} > 0$ together with the associated grids in time and space: $(t_n)_{0 \le n \le N}$ and $(x_j)_{0 \le j \le I}$ where $t_n = n\Delta t$ and $x_j = jh$.

Let us first discretize the problem in space, using the three point approximation operator D_h^2 already introduced in Section 6.1.1, and evaluating the partial differential equation (9.1) at

[1] Generalization to higher dimensions is straightforward, see e.g. Chapter 6 for a presentation of the finite difference method in dimension 2.

[2] We assume that the boundary conditions a and b are constant in time. Generalization to time-dependent boundary condition is straightforward.

[3] The continuity of u on $[0,T] \times \overline{\Omega}$ implies in particular the so-called compatibility conditions: u_0 is continuous on $\overline{\Omega}$ with $u_0(0) = a$ and $u_0(L) = b$.

the interior points $(x_j)_{1\leq j\leq I-1}$. We thus obtain, an ordinary differential equation on a vector $U(t)\in\mathbb{R}^{I-1}$ (the so-called space discretized problem)[4]:

$$\begin{cases}\dot{U}(t) = -AU(t) + C + F(t) \text{ for } t\in(0,T)\\ U(0) = U^0\end{cases} \tag{9.2}$$

where $A\in\mathbb{R}^{(I-1)\times(I-1)}$, $C\in\mathbb{R}^{I-1}$, $F(t)\in\mathbb{R}^{I-1}$ and $U^0\in\mathbb{R}^{I-1}$ are defined by:

$$A = \frac{1}{h^2}\begin{bmatrix} 2 & -1 & 0 & \cdots & 0\\ -1 & 2 & -1 & \cdots & 0\\ \vdots & \ddots & \ddots & \ddots & \vdots\\ \vdots & \ddots & -1 & 2 & -1\\ 0 & \cdots & 0 & -1 & 2\end{bmatrix} \tag{9.3}$$

$$C = \frac{1}{h^2}\begin{pmatrix} a\\ 0\\ \vdots\\ 0\\ b\end{pmatrix}\quad F(t) = \begin{pmatrix} f(t,x_1)\\ f(t,x_2)\\ \vdots\\ f(t,x_{I-2})\\ f(t,x_{I-1})\end{pmatrix} \text{ and } U^0 = \begin{pmatrix} u_0(x_1)\\ u_0(x_2)\\ \vdots\\ u_0(x_{I-2})\\ u_0(x_{I-1})\end{pmatrix}. \tag{9.4}$$

The i-th component $U_i(t)$ of the vector $U(t)$ is supposed to be an approximation of $u(t,x_i)$, for $i\in\{1,\dots,I-1\}$ (notice that the values of u at $x_0=0$ and $x_I=L$ are known from (9.1)). Notice that the ordinary differential equation (9.2) admits a unique solution in $\mathcal{C}^1([0,T],\mathbb{R}^{I-1})$ since the right-hand side is Lipschitz in U.

The problem (9.2) is an ordinary differential equation which can be discretized using standard numerical schemes. For example, for the explicit Euler scheme, one has

$$\frac{U(t_{n+1})-U(t_n)}{\Delta t}\approx\dot{U}(t_n) = -AU(t_n)+C+F(t_n)$$

which yields: $\forall n\in\{0,\dots,N-1\}$,

$$U^{n+1} = (\mathrm{Id}-\Delta tA)U^n + \Delta tC + \Delta tF(t_n). \tag{9.5}$$

The implicit Euler scheme consists in approximating the ordinary differential equation by:

$$\frac{U(t_{n+1})-U(t_n)}{\Delta t}\approx\dot{U}(t_{n+1}) = -AU(t_{n+1})+C+F(t_{n+1})$$

which yields: $\forall n\in\{0,\dots,N-1\}$,

$$(\mathrm{Id}+\Delta tA)\,U^{n+1} = U^n + \Delta tC + \Delta tF(t_{n+1}). \tag{9.6}$$

The Crank-Nicolson scheme is based on the following approximation:

$$\frac{U(t_{n+1})-U(t_n)}{\Delta t}\approx\dot{U}(t_{n+1/2}) = -A\frac{U(t_n)+U(t_{n+1})}{2}+C+\frac{F(t_n)+F(t_{n+1})}{2}$$

which yields an average of the two previous schemes: $\forall n\in\{0,\dots,N-1\}$,

$$\left(\mathrm{Id}+\frac{\Delta t}{2}A\right)U^{n+1} = \left(\mathrm{Id}-\frac{\Delta t}{2}A\right)U^n + \Delta tC + \frac{\Delta t}{2}\left(F(t_n)+F(t_{n+1})\right). \tag{9.7}$$

[4] We adopt a similar notation as in Section 6.2 and introduce a matrix A which is associated with the discretization of $-\partial_{x,x}$ and is thus a positive definite matrix (see Proposition 47).

In those schemes, U_i^n is supposed to be an approximation of $u(t_n, x_i)$. In particular, recall that (see (9.4))

$$\forall i \in \{1, \dots I-1\},\ U_i^0 = u_0(x_i). \tag{9.8}$$

Notice that the implicit Euler and Crank-Nicolson schemes require to solve a linear system at each time-step. Since A is positive definite (see Proposition 47), these linear systems always admit a unique solution, and the numerical schemes are thus well-posed.

9.2 Convergence analysis

We will follow the general principle already used previously to prove convergence (Lax principle): consistency analysis, stability analysis, to finally get convergence.

9.2.1 Consistency

Let us adapt the general definition of consistency already used in Section 6.2.2 for the Poisson equation. A finite difference discretization of a partial differential equation (involving time and space differential operators) is said to be consistent if, when plugging the solution to the original (continuous in space and time) problem in the finite difference equation, the remaining error ε goes to zero when $h \to 0$. Typically, we will use the l^∞ norm (in time and space) on the vector ε to define and quantify the consistency (or truncation) error. Let us illustrate this on our example.

Let us recall a useful notation already introduced in Section 6.2.2. The linear operator $\pi : \mathcal{C}(\Omega) \to \mathbb{R}^{I-1}$ is defined by, for any continuous fonction $v \in \mathcal{C}(\Omega)$,

$$\forall j \in \{1, \dots, I-1\},\ (\pi(v))_j = v(x_j).$$

Let us first consider the explicit Euler scheme. One has on the one hand

$$\frac{U^{n+1} - U^n}{\Delta t} = -AU^n + C + F(t_n). \tag{9.9}$$

Let us define on the other hand the family of error vectors $(\varepsilon^n)_{0 \le n \le N-1}$ by: $\forall n \in \{0, \dots, N-1\}$,

$$\frac{\pi(u(t_{n+1}, \cdot)) - \pi(u(t_n, \cdot))}{\Delta t} = -A\pi(u(t_n, \cdot)) + C + F(t_n) + \varepsilon^n. \tag{9.10}$$

For each $n \in \{0, \dots, N-1\}$, $\varepsilon^n \in \mathbb{R}^{I-1}$ is the error done on the time interval (t_n, t_{n+1}) by plugging the solution to the original problem in the numerical scheme. Assuming that the solution u is sufficiently regular ($u \in \mathcal{C}^{2,4}([0,T] \times \overline{\Omega})$), one can check that (do it★):

$$\sup_{0 \le n \le N-1} \|\varepsilon^n\|_\infty \le C(\Delta t + h^2) \tag{9.11}$$

where C depends on u, but not on Δt and h. The numerical scheme is said to be consistent with order 1 in time and 2 in space. Notice that we use l^∞ norms both in time and space (see (6.8) for the definition of the vector norm $\|\cdot\|_\infty$).

Remark 50. *Unfortunately, the convention used in textbooks to define the consistency error for ordinary differential equations (see Definition 22) and for partial differential equation (see Definition 46) is slightly different. Let us explain this on a simple example, without differential operators in space. Let us consider the ordinary differential equation*

$$\dot{x} = f(x)$$

discretized by the explicit Euler scheme on the time interval $(0, \Delta t)$*:*

$$\frac{x^1 - x(0)}{\Delta t} = f(x(0)).$$

Following Definition 22, the consistency error made on the first time step is defined by

$$\varepsilon_{\Delta t} = x(\Delta t) - x^1.$$

Following Definition 46, the consistency error made on the first time step is defined by

$$\frac{x(\Delta t) - x(0)}{\Delta t} = f(x^n) + \varepsilon.$$

Therefore one has:

$$\frac{\varepsilon_{\Delta t}}{\Delta t} = \varepsilon.$$

This explains why in Definition 22 the scheme is said to be of order p_t *(in time) if* $\varepsilon_{\Delta t} = O(\Delta t^{p_t+1})$*, while in* (9.11) *(like in* (6.7)*) the scheme is said to be of order* p_t *if* $\varepsilon = O(\Delta t^{p_t})$*: this indeed corresponds to the same scaling of the error.*

By using the same reasoning, one can check that the implicit Euler scheme is also consistent of order 1 in time and order 2 in space. Indeed, the implicit Euler scheme is written as:

$$\frac{U^{n+1} - U^n}{\Delta t} = -AU^{n+1} + C + F(t_{n+1}). \tag{9.12}$$

The error vectors $(\varepsilon^n)_{0 \le n \le N-1}$ are thus defined by: $\forall n \in \{0, \dots, N-1\}$,

$$\frac{\pi(u(t_{n+1}, \cdot)) - \pi(u(t_n, \cdot))}{\Delta t} = -A\pi(u(t_{n+1}, \cdot)) + C + F(t_{n+1}) + \varepsilon^n. \tag{9.13}$$

Assuming that the solution u is sufficiently regular ($u \in \mathcal{C}^{2,4}([0,T] \times \overline{\Omega})$), one can check that (do it★):

$$\sup_{0 \le n \le N-1} \|\varepsilon^n\|_\infty \le C(\Delta t + h^2) \tag{9.14}$$

Finally the Crank-Nicolson scheme is of order 2 in time and space. Indeed, the Crank-Nicolson scheme is written as:

$$\frac{U^{n+1} - U^n}{\Delta t} = -A\frac{U^n + U^{n+1}}{2} + C + \frac{F(t_n) + F(t_{n+1})}{2}. \tag{9.15}$$

The error vectors $(\varepsilon^n)_{0 \le n \le N-1}$ are thus defined by: $\forall n \in \{0, \dots, N-1\}$,

$$\frac{\pi(u(t_{n+1}, \cdot)) - \pi(u(t_n, \cdot))}{\Delta t} = -A\frac{\pi(u(t_n, \cdot)) + \pi(u(t_{n+1}, \cdot))}{2} + C + \frac{F(t_n) + F(t_{n+1})}{2} + \varepsilon^n. \tag{9.16}$$

Assuming that the solution u is sufficiently regular ($u \in \mathcal{C}^{4,4}([0,T] \times \overline{\Omega})$), one can check that (do it★):

$$\sup_{0 \le n \le N-1} \|\varepsilon^n\|_\infty \le C((\Delta t)^2 + h^2) \tag{9.17}$$

9.2.2 Stability and convergence

A stability criterium to get convergence. Our objective is to analyze the error: for all $n \in \{0, \dots, N\}$,

$$E^n = \pi(u(t_n, \cdot)) - U^n \tag{9.18}$$

where, we recall, u is the solution to the original problem (9.1). These are vectors in $\mathbb{R}^{I-1}$. By definition of U^0 (see (9.8)), $E^0 = 0$.

By linearity, one can easily check that the error vectors satisfy the following equations.

- For the explicit Euler scheme, by subtracting (9.9) from (9.10), one gets $\frac{E^{n+1}-E^n}{\Delta t} = -AE^n + \varepsilon^n$ and thus
$$E^{n+1} = (\mathrm{Id} - \Delta t A)E^n + \Delta t \varepsilon^n. \tag{9.19}$$
- For the implicit Euler scheme, by subtracting (9.12) from (9.13), one gets $\frac{E^{n+1}-E^n}{\Delta t} = -AE^{n+1} + \varepsilon^n$ and thus
$$E^{n+1} = (\mathrm{Id} + \Delta t A)^{-1} \left(E^n + \Delta t \varepsilon^n\right). \tag{9.20}$$
- For the Crank-Nicolson scheme, by subtracting (9.15) from (9.16), one gets $\frac{E^{n+1}-E^n}{\Delta t} = -A\frac{E^n+E^{n+1}}{2} + \varepsilon^n$ and thus
$$E^{n+1} = \left(\mathrm{Id} + \frac{\Delta t}{2}A\right)^{-1} \left[\left(\mathrm{Id} - \frac{\Delta t}{2}A\right) E^n + \Delta t \varepsilon^n\right]. \tag{9.21}$$

We have already studied in the consistency analysis at which rate $\|\varepsilon^n\|_\infty$ goes to zero when Δt and h goes to 0. We need to understand how the errors made at each time-step propagate over time, in order to obtain an error estimate on the errors $(\|E^n\|)_{1\leq n\leq N}$ for some vector norm $\|\cdot\|$. As in Section 6.2.2, we will either use the l^∞-norm defined by (6.8) or the l^2-norm defined by (6.10).

Let us consider for example the explicit Euler scheme. From (9.19), one gets by induction: for all $n \in \{1, \dots, N\}$,
$$E^n = \Delta t \sum_{k=0}^{n-1} (\mathrm{Id} - \Delta t A)^{n-1-k} \varepsilon^k.$$
Therefore, using a vectorial norm $\|\cdot\|$ and the associated operator norm $\|\cdot\|$ on matrices, one gets:
$$\begin{aligned}
\|E^n\| &\leq \Delta t \sum_{k=0}^{n-1} \|\mathrm{Id} - \Delta t A\|^{n-1-k} \|\varepsilon^k\| \\
&\leq C\Delta t \left(\sum_{k=0}^{n-1} \|\mathrm{Id} - \Delta t A\|^{n-1-k}\right) (\Delta t + h^2) \\
&= C\Delta t \left(\sum_{k=0}^{n-1} \|\mathrm{Id} - \Delta t A\|^{k}\right) (\Delta t + h^2)
\end{aligned}$$
In particular, if
$$\|\mathrm{Id} - \Delta t A\| \leq 1 \tag{9.22}$$
one gets, for all $n \in \{1, \dots, N\}$,
$$\|E^n\| \leq CT(\Delta t + h^2) \tag{9.23}$$
which yields the expected convergence result: up to a multiplicative factor, the error is the consistency error. The condition (9.22) is a stability condition, which remains to be checked.

Likewise, for the implicit Euler scheme, the stability condition is:
$$\|(\mathrm{Id} + \Delta t A)^{-1}\| \leq 1. \tag{9.24}$$

And for the Crank-Nicolson scheme, the stability condition is:
$$\left\|\left(\mathrm{Id} + \frac{\Delta t}{2}A\right)^{-1} \left(\mathrm{Id} - \frac{\Delta t}{2}A\right)\right\| \leq 1 \text{ and } \left\|\left(\mathrm{Id} + \frac{\Delta t}{2}A\right)^{-1}\right\| \leq 1. \tag{9.25}$$
Notice that if (9.22) and (9.24) hold with Δt replaced by $\Delta t/2$, then (9.25) also holds (check it!★).

In the following, we will focus on proving (9.22) and (9.24), for the l^2 and l^∞ norms, respectively defined by (6.10) and (6.8).

Remark 51. *In the convergence analysis, we are actually focussing on the asymptotic regime $\Delta t \to 0$ and $h \to 0$. Thus, it may be enough to check the stability conditions* (9.22), (9.24) *and* (9.25) *only for sufficiently small time-steps and/or mesh sizes. If the stability conditions hold whatever Δt and h, the schemes are said to be unconditionally stable. If the stability conditions hold under some conditions on Δt and h, the schemes are said to be conditionally stable.*

Remark 52. *The reader may have noticed that the conditions* (9.22), (9.24) *and* (9.25) *are only sufficient conditions to get convergence. Let us comment on that. First, Exercise 66 explore an essentially necessary and sufficient condition to get convergence. Second, these conditions are equivalent to the fact that the numerical scheme is dissipative in the l^2-norm, in the sense that, for any E^n, $\|E^{n+1}\|_2 \leq \|E^n\|_2$ where E^{n+1} is defined from E^n by the underlying numerical scheme: $\frac{E^{n+1}-E^n}{\Delta t} = -AE^n$ for explicit Euler, $\frac{E^{n+1}-E^n}{\Delta t} = -AE^{n+1}$ for implicit Euler, and $\frac{E^{n+1}-E^n}{\Delta t} = -A\frac{E^n+E^{n+1}}{2}$ for Crank-Nicolson. It is natural to expect this since at the continuous level, if $\partial_t e - \Delta e = 0$ on $(0,T)\times\Omega$, and e satisfies homogeneous Dirichlet boundary conditions on $\partial\Omega$, one indeed has $\frac{d}{dt}\int_\Omega e^2 \leq 0$ (see* (8.12)*).*

Exercise 66. ★★★*Show that one can get an error estimate of the form*

$$\|E^n\| \leq C\frac{\exp(\alpha T)-1}{\alpha}(\Delta t^p + h^2) \tag{9.26}$$

with $p = 1$ for the Euler schemes and $p = 2$ for Crank-Nicolson, if one replaces the conditions (9.22), (9.24), *and* (9.25) *by, respectively,*

$$\|\mathrm{Id} - \Delta t A\| \leq 1 + \alpha\Delta t \tag{9.27}$$

$$\|(\mathrm{Id} + \Delta t A)^{-1}\| \leq 1 + \alpha\Delta t \tag{9.28}$$

and

$$\left\|\left(\mathrm{Id} + \frac{\Delta t}{2}A\right)^{-1}\left(\mathrm{Id} - \frac{\Delta t}{2}A\right)\right\| \leq 1 + \alpha\Delta t \text{ and } \left\|\left(\mathrm{Id} + \frac{\Delta t}{2}A\right)^{-1}\right\| \leq 1 + \alpha\Delta t \tag{9.29}$$

for some constant $\alpha > 0$ independent of Δt and h

Stability in l^2-norm. As already explained in Section 6.2.2, for any symmetric matrix S, the operator norm associated with the l^2 norm is $\|S\|_2 = \sqrt{(\max \mathrm{spec}(S^2))} = \rho(S)$ where the spectral radius $\rho(S)$ denotes the maximum of the absolute values of S. We will thus analyze the l^2 stability of the explicit Euler and implicit Euler schemes by computing the spectra of the matrices $(\mathrm{Id} - \Delta t A)$ and $(\mathrm{Id} + \Delta t A)^{-1}$, relying on Lemma 49 which gives the spectrum of $h^2 A$.

For the explicit Euler scheme, the eigenvalues of $\mathrm{Id} - \Delta t A$ are

$$\left(1 - \Delta t\frac{\lambda_i}{h^2}\right)_{1\leq i\leq I-1}$$

where, we recall, $\lambda_i = 4\sin^2\left(\frac{i\pi}{2I}\right)$ (see Lemma 49). Thus, one has

$$\rho(\mathrm{Id} - \Delta t A) = \max\left(\left|1 - \Delta t\frac{\lambda_1}{h^2}\right|, \left|1 - \Delta t\frac{\lambda_{I-1}}{h^2}\right|\right).$$

Since $(1 - \Delta t\mu)^2 \leq 1$ is equivalent (for positive μ and Δt) to $\Delta t \leq 2/\mu$, one gets the stability conditions

$$\Delta t \leq \frac{2h^2}{\lambda_{I-1}}.$$

Notice that $\lambda_{I-1} \leq 4$ and that $\lim_{I\to\infty} \lambda_{I-1} = 4$. It is thus natural to enforce the stronger condition:

$$\Delta t \leq \frac{h^2}{2} \tag{9.30}$$

which is called a CFL condition, named after Richard Courant, Kurt Friedrichs, and Hans Lewy. The Euler explicit scheme is thus conditionally stable in l^2-norm: it is stable if (9.30) is satisfied[5]. In practice, this condition may be restrictive, since it imposes to take a very small timestep is the mesh size is small. This is why the implicit Euler scheme is very often used in this context, since it is unconditionally stable in l^2-norm, as we now explain.

For the implicit Euler scheme, the eigenvalues of $\mathrm{Id} + \Delta t A$ are $\left(1 + \Delta t \frac{\lambda_i}{h^2}\right)_{1\leq i\leq I-1}$. These eigenvalues are positive whatever Δt and h, and thus

$$\rho((\mathrm{Id} + \Delta t A)^{-1}) = \left(1 + \Delta t \frac{\lambda_1}{h^2}\right)^{-1} < 1$$

for any Δt and h. The Implicit Euler scheme is thus unconditionally stable in l^2-norm.

Let us summarize our findings in a lemma.

Lemma 69.

- *Under the CFL condition* (9.30)*, the explicit Euler scheme is stable in l^2-norm:*

$$\|(\mathrm{Id} - \Delta t A)\|_2 \leq 1.$$

- *The implicit Euler scheme is unconditionally stable in l^2-norm: whatever Δt and h,*

$$\|(\mathrm{Id} + \Delta t A)^{-1}\|_2 \leq 1.$$

Exercise 67 (Discretization of the backward heat equation)**.** ★*Consider the backward 1d heat Equation* (8.13) *with homogeneous Dirichlet boundary conditions. Using the matrix A defined in* (9.3)*, show that its space discretization is*

$$\dot{U}(t) = AU(t)$$

and that the explicit Euler scheme is

$$U^{n+1} = (\mathrm{Id} + \Delta t A)U^n. \tag{9.31}$$

Let V be an eigenvector of A associated with an eigenvalue $\mu > 0$. Show that, if $U^0 = V$, then

$$U^n = (1 + \Delta t\, \mu)^n V.$$

Deduce that every spatial mode is amplified for every $\Delta t > 0$, and that the highest-frequency modes are amplified the most. In particular, show that

$$\|\mathrm{Id} + \Delta t A\|_2 = 1 + \Delta t \frac{\lambda_{I-1}}{h^2} > 1.$$

Why can no CFL condition make this scheme dissipative? Relate this growth to the continuous instability observed in Exercise 65.

Exercise 68. ★★*Compute the spectrum of $\left(\mathrm{Id} + \frac{\Delta t}{2} A\right)^{-1} \left(\mathrm{Id} - \frac{\Delta t}{2} A\right)$. Deduce that, for the Crank-Nicolson scheme,* (9.25) *holds for the l^2-norm whatever Δt and h: the Crank-Nicolson scheme is unconditionally stable in l^2-norm.*

[5] We essentially argued through sufficient conditions. In practice, it is indeed observed that if (9.30), the results are plagued by instabilities.

Stability in l^∞-norm. Let us first consider the explicit Euler scheme. We would like to show that $\|\mathrm{Id} - \Delta t A\|_\infty \leq 1$. To prove this result, we rely on a discrete maximum principle.

Lemma 70. *Under the CFL condition* (9.30)*, the coefficients of the matrix* $\mathrm{Id} - \Delta t A$ *are non negative. In particular, the explicit Euler scheme* (9.5) *(with initial condition* (9.8)*) satisfies the following maximum principle: if* $a \geq 0$*,* $b \geq 0$*,* $f \geq 0$ *and* $u_0 \geq 0$*, then for all* $n \in \{1, \ldots, N\}$*,* $U^n \geq 0$*.*

Proof: This is straightforward by computing the coefficients of the matrix $\mathrm{Id} - \Delta t A$: the off diagonal coefficients $\frac{\Delta t}{h^2}$ are positive, and the diagonal coefficients $1 - 2\frac{\Delta t}{h^2}$ are non negative if and only if the CFL condition (9.30) is satisfied. ♢

Remark 53. *In the spirit of Remark 52, it is natural to exhibit conditions under which a numerical scheme for the heat equation satisfies a maximum principle, since the heat equation also satisfies a maximum principle (see Theorem 62 and Exercise 64). Trying to devise numerical schemes which satisfy qualitative properties similar to the original continuous problem is a very fruitful principle of numerical analysis.*

Lemma 71. *Under the CFL condition* (9.30)*, the explicit Euler scheme is stable in* l^∞*-norm:* $\|(\mathrm{Id} - \Delta t A)\|_\infty \leq 1$*.*

Proof: Let us consider, for a vector $E \in \mathbb{R}^{I-1}$, $F = (I - \Delta t A)E$. Our objective is to prove that

$$\|F\|_\infty \leq \|E\|_\infty.$$

Let us set $E_0 = 0$ and $E_I = 0$ and introduce (see (9.30))

$$\nu = \frac{\Delta t}{h^2} \in [0, 1/2]$$

Then for any $i \in \{1, \ldots, I-1\}$,

$$\begin{aligned} |F_i| &= |\nu U_{i-1} + (1 - 2\nu)U_i + \nu U_{i+1}| \\ &\leq \nu \|U\|_\infty + (1 - 2\nu)\|U\|_\infty + \nu \|U\|_\infty \\ &= \|U\|_\infty. \end{aligned}$$

This concludes the proof. ♢

Let us finally consider the implicit Euler scheme. We will show that it is unconditionally stable in l^∞-norm. This is again very much related to a discrete maximum principle.

Lemma 72. *The coefficients of the matrix* $(\mathrm{Id} + \Delta t A)^{-1}$ *are non negative whatever* Δt *and* h*. In particular, the implicit Euler scheme* (9.6) *(with initial condition* (9.8)*) satisfies the following maximum principle: if* $a \geq 0$*,* $b \geq 0$*,* $f \geq 0$ *and* $u_0 \geq 0$*, then for all* $n \in \{1, \ldots, N\}$*,* $U^n \geq 0$*.*

Proof: Let us assume that $(I + \Delta t A)V = F$ with $F \geq 0$ and let us consider $m = \min_{1 \leq j \leq l-1} V_j$. Let us argue by contradiction and assume that $m < 0$. Let us moreover extend the vector V by $V_0 = V_I = 0$.

Let us consider $k = \max\{j, V_j = m\}$. We have $k \in \{1, \ldots, I-1\}$, $V_k = m$, $V_{k+1} > V_k$ and $V_{k-1} \geq V_k$ which yields

$$V_{k+1} - 2V_k + V_{k-1} > 0.$$

Besides, one has $V_k - \frac{\Delta t}{h^2}(V_{k+1} - 2V_k + V_{k-1}) = F_k$ and thus

$$V_{k+1} - 2V_k + V_{k-1} = \frac{h^2}{\Delta t}(V_k - F_k) \leq 0$$

since $V_k = m < 0$ and $F_k \geq 0$.

This leads to a contradiction, and thus $V \geq 0$. ♢

Lemma 73. *The implicit Euler scheme is unconditionally stable in l^∞-norm: whatever Δt and h, $\|(\mathrm{Id} + \Delta t A)^{-1}\|_\infty \leq 1$.*

Proof: Let us assume that $(I + \Delta t A)V = F$. We would like to prove that $\|V\|_\infty \leq \|F\|_\infty$. Let us set $V_0 = 0$ and $V_I = 0$ and introduce $\nu = \frac{\Delta t}{h^2}$. Let $k \in \{1, \dots, I-1\}$ be such that

$$|V_k| = \max_{1 \leq j \leq I-1} |V_j| = \|V\|_\infty.$$

If $V_k = \|V\|_\infty$, then $V_{k-1} \leq V_k$ and $V_{k+1} \leq V_k$ and thus

$$F_k = -\nu V_{k-1} + (1+2\nu)V_k - \nu V_{k+1} \geq -\nu\|V\|_\infty + (1+2\nu)\|V\|_\infty - \nu\|V\|_\infty = \|V\|_\infty.$$

Therefore $\|F\|_\infty \geq F_k \geq \|V\|_\infty$.

If $V_k = -\|V\|_\infty$, then $V_{k-1} \geq -V_k$ and $V_{k+1} \geq -V_k$ and thus

$$F_k = -\nu V_{k-1} + (1+2\nu)V_k - \nu V_{k+1} \leq \nu\|V\|_\infty - (1+2\nu)\|V\|_\infty \nu\|V\|_\infty = -\|V\|_\infty.$$

Therefore $-\|F\|_\infty \leq F_k \leq -\|V\|_\infty$.

In both cases, we obtain $\|V\|_\infty \leq \|F\|_\infty$. This concludes the proof. ♢

Remark 54. *For the Crank-Nicolson scheme, one can check that* (9.25) *is satisfied for the l^∞ norm under a CFL condition (do it!★). However,* (9.25) *is not satisfied for any choice of the parameters $(\Delta t, h)$.*

Conclusion Let us gather in the following theorem the convergence results we have obtained. Let us recall that a numerical scheme is said to be convergent with order p_t in time and p_x in space, for a vector norm $\|\cdot\|$, if the error vectors (9.18) satisfy:

$$\sup_{0 \leq n \leq N-1} \|E^n\| \leq C\left(\Delta t^{p_t} + h^{p_x}\right).$$

Theorem 74. *Let us consider the three schemes: explicit Euler* (9.5)*, implicit Euler* (9.6) *and Crank-Nicolson* (9.7)*.*

- *The explicit Euler scheme is convergent under the CFL condition* (9.7)*, with order* 1 *in time and* 2 *in space, in l^2 and l^∞ norm.*
- *The implicit Euler scheme is convergent with order* 1 *in time and* 2 *in space, in l^2 and l^∞ norm.*
- *The Crank-Nicolson scheme is convergent with order* 2 *in time and* 2 *in space, unconditionally in l^2-norm, and under a CFL condition for the l^∞ norm.*

9.3 Heat equation and random walk

This section can be skipped on first reading.

In this last section, we would like to draw a connection between the discretization of the heat equation and the random walk. In this section, we consider the explicit Euler scheme for $(t, x) \in \mathbb{R}_+ \times \mathbb{R}$ (there are thus no boundary conditions): for all $n \geq 0$,

$$U^{n+1} = (\mathrm{Id} - \Delta A)U^n.$$

We assume that Δt and h satisfies the CFL condition (see (9.30)):

$$\Delta t \leq \frac{h^2}{2}.$$

Let us introduce $\nu = \frac{\Delta t}{h^2} \in \left[0, \frac{1}{2}\right]$. We thus have, for any $j \in \mathbb{Z}$,

$$U_j^{n+1} = \nu U_{j-1}^n + (1-2\nu)U_j^n + \nu U_{j+1}^n.$$

Let us now introduce a Markov chain on $\mathbb{Z}$ defined by:

$$\forall n \geq 0, \quad X_n = X_0 + \sum_{k=1}^{n} Z_k$$

where the random variables $(Z_k)_{k\geq 1}$ are independent and identically distributed with law:

$$\mathbb{P}(Z_1 = \pm 1) = \nu \text{ and } \mathbb{P}(Z_1 = 0) = 1 - 2\nu.$$

The Markov chain $(X_n)_{n\geq 0}$ is thus a random walk on $\mathbb{Z}$. Let us denote

$$V_j^n = \mathbb{E}(U^0(X_n)|X_0 = j).$$

Proposition 75. *For any $j \in \mathbb{Z}$, one has: $V_j^0 = U_j^0$ and for $n \geq 0$,*

$$V_j^{n+1} = \nu V_{j-1}^n + (1-2\nu)V_j^n + \nu V_{j+1}^n.$$

As a consequence, $U_j^n = V_j^n$ for all $n \geq 0$ and $j \in \mathbb{Z}$.

Proof: One has, using the Markov property:

$$\begin{aligned}
V_j^{n+1} &= \mathbb{E}(U^0(X_{n+1})|X_0 = j) \\
&= \sum_{k\in\mathbb{Z}} \mathbb{E}(U^0(X_{n+1})1_{X_1=k}|X_0 = j) \\
&= \sum_{k\in\mathbb{Z}} \mathbb{E}(U^0(X_{n+1})|X_1 = k \text{ and } X_0 = j)\, \mathbb{P}(X_1 = k|X_0 = j) \\
&= \sum_{k\in\mathbb{Z}} \mathbb{E}(U^0(X_n)|X_0 = k)\, \mathbb{P}(X_1 = k|X_0 = j) \\
&= \nu V_{j-1}^n + (1-2\nu)V_j^n + \nu V_{j+1}^n
\end{aligned}$$

where we used in the last line that $\mathbb{E}(U^0(X_n)|X_0 = k) = V_k^n$ and

$$\mathbb{P}(X_1 = k|X_0 = j) = \mathbb{P}(Z_1 = k - j) = \begin{cases} \nu \text{ if } k = j \pm 1 \\ (1-2\nu) \text{ if } k = j \\ 0 \text{ otherwise.} \end{cases}$$

♢

We thus conclude that the solution to the explicit Euler discretization admits a probabilistic representation: for all $n \geq 0$ and $j \in \mathbb{Z}$,

$$U_j^n = \mathbb{E}(U^0(X_n)|X_0 = j) = \mathbb{E}\left(U^0\left(j + \sum_{k=1}^{n} Z_k\right)\right).$$

Notice that on this representation, the weak maximum principle is obvious: if $U^0 \geq 0$, then $U^n \geq 0$ for all $n \geq 0$.

There is actually a way to pass to the limit $h \to 0$ and $\Delta t \to 0$ this representation formula to get a representation of the solution of the heat equation:

$$\begin{cases} \partial_t u = \partial_{x,x} u \text{ for } t > 0,\ x \in \mathbb{R} \\ u(0,x) = u_0(x) \text{ for } x \in \mathbb{R} \end{cases}$$

as

$$u(t,x) = \mathbb{E}\left(u_0\left(x + \sqrt{2}W_t\right)\right)$$

where $(W_t)_{t\geq 0}$ is the so-called Brownian motion, obtained from the random walk $(Z_k)_{k\geq 0}$ by passing to the limit $\Delta t \to 0$, with $\Delta t = \frac{h^2}{2}$ (i.e. $h = \sqrt{2\Delta t}$): $\lim_{\Delta t \to 0} \sqrt{\Delta t} \sum_{k=1}^{\lfloor t/\Delta t \rfloor} Z_k = W_t$.

These deep connections between partial differential equations and stochastic processes will be explored in the lecture [3].

Part IV

Transport phenomena

Chapter 10

Models based on advection and wave equations

The heat equation studied in Part III describes diffusion: a localized profile spreads out and becomes smoother as time evolves. In this part, we consider two other evolution equations, which describe the propagation of information at a finite speed. The first one is the advection (or transport) equation. Depending on the physical meaning of the unknown, it appears either in conservative form

$$\partial_t U + \operatorname{div}(bU) = S, \tag{10.1}$$

or in non-conservative form

$$\partial_t q + b \cdot \nabla q = s, \tag{10.2}$$

where b is a velocity field with values in $\mathbb{R}^d$. The second equation is the wave equation

$$\partial_{t,t} u - c^2 \Delta u = f, \tag{10.3}$$

where $c > 0$ is the wave speed.

These equations are closely related. In dimension one, the advection equation with constant velocity translates a profile without changing its shape. The solution to the wave equation is, roughly speaking, the superposition of two such profiles, one moving to the right and the other one to the left. This relation will be made precise in Chapter 11.

As in the previous parts, we first derive the equations from elementary modeling principles, then study classical solutions, and finally introduce finite difference approximations.

10.1 Conservation laws and the advection equation

10.1.1 A quantity per unit volume: the conservative form

Let $U(t,x)$ denote the amount of a scalar quantity per unit volume at time t and point $x \in \mathbb{R}^d$. One could for example think of the mass density $U = \rho$. Thus, for a fixed region $A \subset \mathbb{R}^d$, the total amount contained in A is

$$\int_A U(t,x)\, dx.$$

Let $J(t,x) \in \mathbb{R}^d$ denote the flux of this quantity, and let $S(t,x)$ be its production rate per unit time and unit volume. Recall from the first chapter of Part III, and in particular Equation (7.2), that the balance law is

$$\partial_t U + \operatorname{div}(J) = S. \tag{10.4}$$

A positive value of S corresponds to production of the quantity, and a negative value to consumption.

Assume now that the quantity is carried by a prescribed velocity field

$$b : x \in \mathbb{R}^d \longmapsto b(x) \in \mathbb{R}^d.$$

The amount crossing an infinitesimal surface with unit normal n, per unit time and unit area, is $Ub \cdot n$. The advective flux is therefore

$$J = bU. \tag{10.5}$$

Inserting (10.5) into (10.4) yields the conservative advection equation

$$\partial_t U + \text{div } (bU) = S. \tag{10.6}$$

Using the product rule,

$$\text{div } (bU) = b \cdot \nabla U + U \text{div } b, \tag{10.7}$$

the same equation can be rewritten as

$$\partial_t U + b \cdot \nabla U + U \text{div } b = S. \tag{10.8}$$

The term $U \text{div } b$ accounts for dilution when the flow expands and concentration when it contracts (see Exercise 18).

Remark 55 (Time-dependent velocity fields)**.** *Throughout this part, we assume for simplicity that the velocity field b does not depend explicitly on time. All the arguments extend directly to a time-dependent velocity field $b(t, x)$ (see also Remark 1).*

Exercise 69. ★*Let $b : \mathbb{R}^d \to \mathbb{R}^d$ and $U : \mathbb{R}^d \to \mathbb{R}$ be continuously differentiable. Prove formula* (10.7)*.*

Exercise 70. ★★*Assume that U and b are smooth and decay sufficiently fast at infinity, and that U solves* (10.6) *with $S = 0$. Prove formally that*

$$\frac{d}{dt} \int_{\mathbb{R}^d} U(t, x)\, dx = 0.$$

What boundary term appears if the integral is over a bounded domain Ω?

10.1.2 A quantity per unit mass: the non-conservative form

Let us now consider $q(t, x)$ a scalar quantity per unit mass, sometimes called a specific quantity (e.g. the specific internal energy).

Let $\rho(t, x) > 0$ be the mass density of the fluid which carries the quantity. Conservation of the mass of the fluid gives

$$\partial_t \rho + \text{div } (\rho b) = 0. \tag{10.9}$$

The amount per unit volume of the quantity of interest is

$$U = \rho q. \tag{10.10}$$

If q is produced at a rate $s(t, x)$ per unit time and unit mass, then the volumetric production rate is $S = \rho s$. Applying the conservative balance law to $U = \rho q$ gives (see Equation (10.6))

$$\partial_t(\rho q) + \text{div } (\rho q b) = \rho s. \tag{10.11}$$

The product rule yields

$$\begin{aligned}\partial_t(\rho q) + \operatorname{div}(\rho q b) &= q\left(\partial_t \rho + \operatorname{div}(\rho b)\right) + \rho\left(\partial_t q + b \cdot \nabla q\right) \\ &= \rho\left(\partial_t q + b \cdot \nabla q\right),\end{aligned}$$

where we used mass conservation (10.9) in the last equality. Since $\rho > 0$, Equation (10.11) is equivalent to

$$\partial_t q + b \cdot \nabla q = s. \tag{10.12}$$

The operator

$$\frac{Dq}{Dt} := \partial_t q + b \cdot \nabla q$$

is called the material derivative: it measures the rate of change of q seen by an observer moving with the fluid.

Definition 76 (Characteristic curve for the advection equation)**.** *Let $b : \mathbb{R}^d \to \mathbb{R}^d$ be a continuous velocity field. A* characteristic curve *is a differentiable curve $\tau \mapsto X(\tau)$ satisfying*

$$\frac{dX}{d\tau}(\tau) = b(X(\tau)). \tag{10.13}$$

The associated curve in space-time is $\tau \mapsto (\tau, X(\tau))$.

Indeed, if q satisfies

$$\partial_t q + b \cdot \nabla q = s,$$

then the chain rule gives, along every characteristic,

$$\frac{d}{d\tau} q(\tau, X(\tau)) = s(\tau, X(\tau)). \tag{10.14}$$

In particular, when $s = 0$, the function q is constant along the characteristics.

For an incompressible flow, $\operatorname{div} b = 0$, and therefore

$$\operatorname{div}(bw) = b \cdot \nabla w \tag{10.15}$$

for every differentiable scalar field w. The conservative and non-conservative differential operators then coincide. In physical terms, if ρ is constant, $U = \rho q$ differs from q only by a constant factor, and $S = \rho s$. Notice that Equation (10.9) indeed implies $\operatorname{div} b = 0$ when ρ is constant.

Exercise 71 (The flow of an ordinary differential equation)**.** ★★*Let $b : \mathbb{R}^d \to \mathbb{R}^d$ be a $\mathcal{C}^1$ vector field and let $\phi(x_0, t)$ be the flow of the ordinary differential equation $\dot{x} = b(x)$ introduced in Definition 10. Assume for simplicity that the flow is defined for all positive and negative times. Define the time-reversed flow map*

$$a(t, x) = \phi(x, -t).$$

Using the flow property, prove that

$$a(t, \phi(x_0, t)) = x_0.$$

Deduce that every component of the vector-valued function a satisfies

$$\partial_t a + b \cdot \nabla a = 0, \qquad a(0, x) = x.$$

More generally, prove that for every $q_0 \in \mathcal{C}^1(\mathbb{R}^d)$,

$$q(t, x) = q_0(\phi(x, -t))$$

satisfies $\partial_t q + b \cdot \nabla q = 0$ and $q(0, x) = q_0(x)$.

10.1.3 Advection with a constant velocity

The simplest advection equation is obtained in dimension one with a constant velocity $a \in \mathbb{R}$. Since a constant velocity field has zero divergence, the conservative and non-conservative forms coincide. We use the generic notation u for the unknown and write

$$\partial_t u + a\partial_x u = f. \tag{10.16}$$

When $f = 0$, the equation states that u is constant along each straight line $x - at =$ constant. Indeed, for any differentiable function $u_0 : \mathbb{R} \to \mathbb{R}$, the function

$$u(t, x) = u_0(x - at) \tag{10.17}$$

satisfies (10.16). If $a > 0$, the initial profile moves to the right at speed a; if $a < 0$, it moves to the left. Contrary to the heat equation, its shape and its amplitude do not change.

Exercise 72. ★*Check directly that* (10.17) *satisfies* $\partial_t u + a\partial_x u = 0$ *and* $u(0, x) = u_0(x)$.

Exercise 73. ★*Let* u *be given by* (10.17). *Assume that* u_0 *is integrable. Prove that, for all* $t \geq 0$,

$$\int_{\mathbb{R}} u(t, x)\, dx = \int_{\mathbb{R}} u_0(x)\, dx.$$

Prove also that $\sup_{x\in\mathbb{R}} |u(t, x)| = \sup_{x\in\mathbb{R}} |u_0(x)|$.

10.1.4 Initial and boundary conditions

On the whole space, the advection equation is naturally supplemented with an initial condition

$$u(0, x) = u_0(x). \tag{10.18}$$

On a bounded domain, boundary data are needed only where the velocity points into the domain. This is a major difference with the Poisson and heat equations.

Consider for example a positive constant velocity $a > 0$ on the interval $\Omega = (0, L)$. The characteristics enter the interval at $x = 0$ and leave it at $x = L$. The natural problem is therefore

$$\begin{cases} \partial_t u + a\partial_x u = f & \text{for } t > 0,\ x \in (0, L), \\ u(0, x) = u_0(x) & \text{for } x \in (0, L), \\ u(t, 0) = g(t) & \text{for } t > 0. \end{cases} \tag{10.19}$$

No boundary value should be prescribed at $x = L$: the value there is determined by what has previously entered the domain or was present at time 0. If $a < 0$, the roles of $x = 0$ and $x = L$ are reversed.

More generally, if $\Omega \subset \mathbb{R}^d$ is a smooth bounded domain and $n(x)$ denotes its outward unit normal, the incoming part of the boundary is

$$\Gamma_- = \{(t, x) : t > 0,\ x \in \partial\Omega,\ b(x) \cdot n(x) < 0\}. \tag{10.20}$$

Boundary data are prescribed on Γ_-.

Exercise 74. ★*For* $a > 0$, *draw in the* (t, x) *plane the characteristics associated with problem* (10.19). *Identify those which meet the initial line* $t = 0$ *and those which meet the incoming boundary* $x = 0$. *Repeat the exercise when* $a < 0$.

10.2 Some models involving advection

10.2.1 A volumetric quantity: transport of a pollutant

Let $U(t,x)$ denote the mass of a pollutant per unit volume of fluid, and let $b(x)$ be the prescribed fluid velocity. Since U is a volumetric density, its advective flux is bU. If $S(t,x)$ denotes production or removal of pollutant per unit time and unit volume, conservation gives

$$\partial_t U + \operatorname{div}(bU) = S. \tag{10.21}$$

If molecular diffusion cannot be neglected, Fick's law adds the diffusive flux $-D\nabla U$, where $D > 0$ is the diffusion coefficient. The total flux is then

$$J = bU - D\nabla U,$$

and the advection–diffusion equation reads

$$\partial_t U + \operatorname{div}(bU) - D\Delta U = S. \tag{10.22}$$

Equation (10.22) combines the propagation studied in this part and the diffusion studied in Part III.

If the flow is incompressible, $\operatorname{div} b = 0$, the same equation can be written as

$$\partial_t U + b \cdot \nabla U - D\Delta U = S.$$

10.2.2 A specific quantity: transport of a passive scalar

Let $b(x) \in \mathbb{R}^d$ be the prescribed velocity of a fluid. A fluid particle initially located at x_0 follows a trajectory $X(t;x_0)$ which solves the ordinary differential equation

$$\frac{dX}{dt}(t;x_0) = b(X(t;x_0)), \qquad X(0;x_0) = x_0. \tag{10.23}$$

Let $q(t,x)$ denote a scalar property per unit mass of fluid, for example its salinity or the mass fraction of a chemical species in a simplified model without diffusion or reaction. A particle label is another example. We assume that this quantity is passive: it is transported by the fluid but does not modify the velocity field b.

If there is no source or sink, the value carried by each particle remains constant along its trajectory:

$$\frac{d}{dt} q(t, X(t;x_0)) = 0. \tag{10.24}$$

Using the chain rule and (10.23), we obtain

$$\frac{d}{dt} q(t, X(t;x_0)) = \partial_t q(t, X(t;x_0)) + b(X(t;x_0)) \cdot \nabla q(t, X(t;x_0)).$$

Since every fluid point lies on one such trajectory, the passive scalar satisfies the advection equation

$$\partial_t q + b \cdot \nabla q = 0. \tag{10.25}$$

Remark 56 (Temperature)**.** *Temperature is an intensive quantity rather than an amount per unit mass. However, when the heat capacity c_p is constant, it is proportional to the specific thermal energy $c_p T$. In the simplified case of an incompressible fluid, its evolution is*

$$\rho c_p (\partial_t T + b \cdot \nabla T) - \operatorname{div}(k \nabla T) = Q,$$

where k is the thermal conductivity and Q the heat production per unit volume. Temperature is therefore a passive scalar only when thermal diffusion and heat production are neglected. In a compressible fluid, pressure work produces additional terms which are outside the scope of these lectures.

10.3 The wave equation in physics

The wave equation (10.3) describes many physical phenomena. The reader has probably heard of gravitational waves or seismic waves for example. Let us describe a few examples where this equation is used. We first consider in Sections 10.3.1 and 10.3.2 examples in mechanics, where the wave equation naturally appears as a consequence of Newton's laws of motion. We will then show how the wave equation models electromagnetic phenomena in Section 10.3.3.

For simplicity, the derivations below will mostly be done in dimension $d = 1$, and actually, we will also then restrict the mathematical analysis and the numerical discretization to $d = 1$ (except in Section 11.4).

10.3.1 Vibrations and acoustics

The wave equation models vibrating strings (guitars, pianos), membranes (drumheads), and sound waves traveling through air or fluids.

Derivation of the wave equation for a vibrating string. Let us derive the one-dimensional wave equation from Newton's second law. Consider a flexible string at rest on the interval $(0, L)$. We denote by $u(t, x)$ its transverse displacement at time t and horizontal position x. We make the following modeling assumptions.

- The string has a constant mass per unit length $\rho > 0$.
- The tension has a constant magnitude $T > 0$ and is tangent to the string.
- The displacements and slopes are small, so that horizontal motion and nonlinear geometric effects can be neglected.
- A transverse force $f(t, x)$ acts per unit length of the string.

Let us apply Newton's law to the portion of string between x and $x + \Delta x$. Its mass is approximately $\rho\Delta x$, so its transverse acceleration gives the force

$$\rho\Delta x\, \partial_{t,t} u(t, x).$$

If $\theta(t, x)$ is the angle between the tangent to the string and the horizontal axis, the vertical component of the tension is $T \sin\theta$. The resultant vertical tension on the small segment is therefore

$$T \sin\theta(t, x + \Delta x) - T \sin\theta(t, x).$$

For small slopes,

$$\sin\theta \simeq \tan\theta = \partial_x u.$$

Adding the external force $f(t, x)\Delta x$, Newton's law gives

$$\rho\Delta x\, \partial_{t,t} u(t, x) \simeq T\left(\partial_x u(t, x + \Delta x) - \partial_x u(t, x)\right) + f(t, x)\Delta x.$$

Dividing by Δx and sending Δx to zero yields

$$\rho\partial_{t,t} u - T\partial_{x,x} u = f. \tag{10.26}$$

Introducing the propagation speed

$$c = \sqrt{\frac{T}{\rho}}, \tag{10.27}$$

and replacing f/ρ by f, we obtain the wave equation

$$\partial_{t,t}u - c^2\partial_{x,x}u = f. \tag{10.28}$$

In dimension $d \geq 2$, the same equation

$$\partial_{t,t}u - c^2\Delta u = f$$

models, for example, the small transverse vibrations of a membrane when $d = 2$. The wave equation thus naturally appears to model acoustics and vibrations.

Exercise 75. ★*Check the physical dimensions in* (10.26) *and show that* c *defined by* (10.27) *has the dimension of a velocity.*

Exercise 76. ★*Let* $\phi : \mathbb{R} \to \mathbb{R}$ *be twice differentiable. Check that* $u(t,x) = \phi(x - ct)$ *and* $v(t,x) = \phi(x + ct)$ *both solve the homogeneous wave equation. In which directions do these two profiles move?*

Acoustic waves. Small pressure perturbations in a compressible fluid also satisfy a wave equation. In a simplified one-dimensional linear model, let $v(t,x)$ denote the fluid velocity and $p(t,x)$ the pressure perturbation. Conservation of mass and Newton's law lead to a first-order system of the form

$$\partial_t p + K\partial_x v = 0, \qquad \rho\partial_t v + \partial_x p = 0,$$

where $K > 0$ is the bulk modulus and $\rho > 0$ the density. Differentiating the first equation in time and the second one in space gives

$$\partial_{t,t}p - \frac{K}{\rho}\partial_{x,x}p = 0.$$

Thus the acoustic speed is $c = \sqrt{K/\rho}$. The same reasoning shows that v satisfies a wave equation.

Exercise 77. ★★*Starting from the first-order system above, prove that* v *satisfies* $\partial_{t,t}v - c^2\partial_{x,x}v = 0$.

10.3.2 Strain waves in elastic materials

A one-dimensional case: longitudinal strain waves in an elastic bar. Consider a straight homogeneous bar with constant cross-sectional area $S > 0$. The material point located at x in the reference configuration has longitudinal displacement $u(t,x)$ at time t. We make the small-deformation approximation: the longitudinal strain is

$$\varepsilon(t,x) = \partial_x u(t,x). \tag{10.29}$$

For a linearly elastic material, Hooke's law relates the axial stress σ to the strain:

$$\sigma(t,x) = E\varepsilon(t,x) = E\partial_x u(t,x), \tag{10.30}$$

where $E > 0$ is Young's modulus. The internal force across a section of the bar is $S\sigma$.

Let $\rho > 0$ be the mass density of the material and let $f(t,x)$ be an external longitudinal force per unit volume. Applying Newton's law to the portion between x and $x + \Delta x$ gives

$$\rho S \Delta x \, \partial_{t,t} u(t,x) \simeq S\big(\sigma(t, x+\Delta x) - \sigma(t,x)\big) + S\Delta x \, f(t,x).$$

After division by $S\Delta x$ and passage to the limit, we obtain the balance of linear momentum

$$\rho \partial_{t,t} u - \partial_x \sigma = f.$$

Using Hooke's law (10.30), this becomes

$$\partial_{t,t} u - c_{\text{el}}^2 \partial_{x,x} u = \frac{f}{\rho}, \qquad c_{\text{el}} = \sqrt{\frac{E}{\rho}}. \tag{10.31}$$

The displacement is longitudinal, unlike the transverse displacement of the vibrating string. In the absence of external forces, differentiating the equation with respect to x shows that the strain $\varepsilon = \partial_x u$ and hence the stress $\sigma = E\varepsilon$ also propagate according to the wave equation. This is why these disturbances are called strain waves.

Exercise 78. ★*Check the physical dimensions in* (10.31)*. In the homogeneous case $f = 0$, prove that both the strain ε and the stress σ satisfy a wave equation with speed c_{el}.*

Generalization to three-dimensional elastic materials. For a three-dimensional elastic solid, the displacement is a vector field $u(t,x) \in \mathbb{R}^3$. In the small-deformation regime, the strain is described by the symmetric matrix

$$\varepsilon(u) = \frac{1}{2}\left(\nabla u + (\nabla u)^{\mathsf{T}}\right), \qquad \varepsilon_{ij}(u) = \frac{1}{2}\left(\partial_{x_j} u_i + \partial_{x_i} u_j\right). \tag{10.32}$$

The stress is also a symmetric matrix, denoted by Σ: the force per unit area exerted across a surface with unit normal n is the vector Σn. For a homogeneous isotropic linearly elastic material, Hooke's law takes the form

$$\Sigma = \lambda_{\text{L}} \operatorname{tr}(\varepsilon(u)) \operatorname{Id} + 2\mu_{\text{L}} \varepsilon(u), \tag{10.33}$$

where $\lambda_{\text{L}} > 0$ and $\mu_{\text{L}} > 0$ are the Lamé coefficients and Id is the identity matrix.

Newton's law now becomes the balance of linear momentum

$$\rho \partial_{t,t} u - \operatorname{div} \Sigma = f, \qquad (\operatorname{div} \Sigma)_i = \sum_{j=1}^{3} \partial_{x_j} \Sigma_{ij}, \tag{10.34}$$

where $f(t,x) \in \mathbb{R}^3$ is an external force per unit volume. Since $\operatorname{tr}(\varepsilon(u)) = \operatorname{div} u$, substituting Hooke's law (10.33) into Equation (10.34) yields

$$\rho \partial_{t,t} u - \mu_{\text{L}} \Delta u - (\lambda_{\text{L}} + \mu_{\text{L}}) \nabla(\operatorname{div} u) = f. \tag{10.35}$$

This is a system of three coupled wave equations, often called the Navier equation of linear elasticity.

In the absence of external forces, the dilatation $p = \operatorname{div} u$ and the rotation $w = \operatorname{curl} u$ satisfy

$$\partial_{t,t} p - c_P^2 \Delta p = 0, \qquad c_P = \sqrt{\frac{\lambda_{\text{L}} + 2\mu_{\text{L}}}{\rho}}, \qquad \partial_{t,t} w - c_S^2 \Delta w = 0, \qquad c_S = \sqrt{\frac{\mu_{\text{L}}}{\rho}}. \tag{10.36}$$

The first equation describes longitudinal compression waves, or P-waves, whereas the second describes transverse shear waves, or S-waves. These are, for example, the two principal types of elastic waves observed in seismology. Thus, in several space dimensions, linear elasticity naturally leads to coupled wave equations and to several propagation speeds.

Exercise 79. ★★*Assume that* $f = 0$. *Apply respectively* div *and* curl *to Equation* (10.35) *and prove Equations* (10.36). *Use in particular* $\operatorname{curl}(\nabla q) = 0$ *for a scalar function* q.

10.3.3 Electromagnetic waves

Let $E(t,x) \in \mathbb{R}^3$ and $B(t,x) \in \mathbb{R}^3$ denote respectively the electric and magnetic fields. In a homogeneous medium with constant electric permittivity $\epsilon > 0$ and magnetic permeability $\mu > 0$, and in the absence of charges and electric currents, Maxwell's equations read

$$\operatorname{div} E = 0, \qquad \operatorname{div} B = 0, \qquad \partial_t B + \operatorname{curl} E = 0, \qquad \operatorname{curl} B = \mu\epsilon\, \partial_t E. \tag{10.37}$$

We use the vector identity (★check it!)

$$\operatorname{curl}(\operatorname{curl} V) = \nabla(\operatorname{div} V) - \Delta V, \tag{10.38}$$

where the Laplacian acts on each component of V.

Taking the curl of Faraday's equation $\partial_t B + \operatorname{curl} E = 0$ and using the last Maxwell equation gives

$$\mu\epsilon\, \partial_{t,t} E + \operatorname{curl}(\operatorname{curl} E) = 0.$$

Since $\operatorname{div} E = 0$, identity (10.38) yields

$$\partial_{t,t} E - c_{\mathrm{em}}^2 \Delta E = 0, \qquad c_{\mathrm{em}} = \frac{1}{\sqrt{\mu\epsilon}}, \tag{10.39}$$

where the differential operators act on each component of the vector field E. The same calculation shows that

$$\partial_{t,t} B - c_{\mathrm{em}}^2 \Delta B = 0. \tag{10.40}$$

Thus every component of the electric and magnetic fields satisfies a wave equation, while the divergence equations in (10.37) impose additional constraints on the fields. In vacuum, $\epsilon = \epsilon_0$ and $\mu = \mu_0$, and

$$c_{\mathrm{em}} = c_0 = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \simeq 3.0 \times 10^8 \ \mathrm{m\,s^{-1}},$$

the speed of light.

Exercise 80. ★★*Starting from* (10.37), *derive Equation* (10.40). *Which of Maxwell's equations is used to replace* $\operatorname{curl}(\operatorname{curl} B)$ *by* $-\Delta B$*?*

10.3.4 Initial and boundary conditions for the wave equation

Let us conclude this section with a discussion on the initial and boundary conditions for the wave equation, focussing on the 1d case (10.28) for simplicity.

Since the wave equation is of second order in time, two initial conditions are required:

$$u(0,x) = u_0(x), \qquad \partial_t u(0,x) = v_0(x). \tag{10.41}$$

For the vibrating string, u_0 is its initial vertical displacement and v_0 its initial vertical velocity.

If the string is fixed at both endpoints, homogeneous Dirichlet boundary conditions are used:

$$u(t,0) = u(t,L) = 0. \tag{10.42}$$

For a free endpoint, the transverse component of the tension vanishes, which leads to a homogeneous Neumann condition

$$\partial_x u(t,0) = 0 \quad \text{or} \quad \partial_x u(t,L) = 0. \tag{10.43}$$

As usual, initial and boundary data must be compatible to get classical solutions. For example, conditions (10.41)–(10.42) require $u_0(0) = u_0(L) = 0$.

Remark 57 (Absorbing boundary conditions)**.** *For the one-dimensional wave equation, the conditions*

$$\partial_t u - c\partial_x u = 0 \quad \text{at } x = 0, \qquad \partial_t u + c\partial_x u = 0 \quad \text{at } x = L$$

allow respectively a left-going wave and a right-going wave to leave the interval without reflection. They are called absorbing boundary conditions. Their meaning will become clear from the decomposition into traveling waves in Equation (11.7)*.*

Chapter 11

Mathematical analysis of advection and wave equations

In this chapter, we obtain representation formulas for classical solutions of the advection and wave equations, and present essential qualitative properties of the solutions. For the advection equation, the main tool is the method of characteristics, which reduces the partial differential equation to an ordinary differential equation along suitable curves. For the wave equation in dimension one, we express classical solutions using the so-called d'Alembert's formula. We then use integration by parts to obtain an energy identity and uniqueness results for classical solutions in any dimension.

As will become clear below, the heat and wave equations both involve the Laplace operator, but their solutions behave very differently. The heat equation has one time derivative and dissipates spatial variations. Its solutions become smoother and information propagates throughout space immediately (see Section 8.3). The wave equation has two time derivatives and conserves an energy in the absence of forcing. Disturbances propagate at the finite speed c (see Section 11.3). Similarly, the advection equation propagates a profile in the direction of the prescribed velocity field; it does not smooth discontinuities (see Section 11.1.2).

11.1 Characteristics for the advection equation

11.1.1 A representation formula for classical solutions on $\mathbb{R}^d$

Let $b \in \mathcal{C}^1(\mathbb{R}^d, \mathbb{R}^d)$ and assume for simplicity that the flow ϕ of the ordinary differential equation

$$\dot{X}(t) = b(X(t)), \qquad X(0) = x_0, \tag{11.1}$$

is defined for all positive and negative times. We consider the non-conservative advection problem

$$\begin{cases} \partial_t u + b(x) \cdot \nabla u = f(t,x) & \text{for } t > 0,\ x \in \mathbb{R}^d, \\ u(0,x) = u_0(x) & \text{for } x \in \mathbb{R}^d. \end{cases} \tag{11.2}$$

The homogeneous equation corresponds to $f = 0$.

For a fixed point (t, x), the backward characteristic ending at the point x and time $s = t$ is

$$X(s; t, x) = \phi(x, s - t), \qquad 0 \le s \le t. \tag{11.3}$$

Indeed, $X(t; t, x) = x$ and

$$\frac{dX}{ds}(s; t, x) = b(X(s; t, x)).$$

Along this characteristic, the chain rule gives, u being a $\mathcal{C}^1$ solution to (11.2),

$$\frac{d}{ds}u(s, X(s;t,x)) = f(s, X(s;t,x)).$$

Integrating between 0 and t, we obtain

$$u(t,x) = u_0(\phi(x,-t)) + \int_0^t f(s, \phi(x, s-t))\, ds. \tag{11.4}$$

This yields a representation of $\mathcal{C}^1$ solutions to (11.2) as stated in the next theorem.

Theorem 77 (Representation formula of solutions to the advection equation)**.** *Let $b \in \mathcal{C}^1(\mathbb{R}^d, \mathbb{R}^d)$ have a well-defined global-in-time flow. Let $u_0 \in \mathcal{C}^1(\mathbb{R}^d)$ and $f \in \mathcal{C}^1([0,T]\times\mathbb{R}^d)$. Then the function defined by* (11.4) *is the unique classical solution to* (11.2) *on $[0,T]\times\mathbb{R}^d$.*

Proof: The computation performed above along the backward characteristics proves uniqueness: for $u_0 = f = 0$, then $u = 0$ thanks to (11.4). For existence, let us consider v defined by the right-hand side of (11.4):

$$v(t,x) = u_0(\phi(x,-t)) + \int_0^t f(s, \phi(x, s-t))\, ds.$$

Let $x_0 \in \mathbb{R}^d$. The flow property (see Equation (2.7)) gives

$$\phi(\phi(x_0,t),-t) = x_0, \qquad \phi(\phi(x_0,t), s-t) = \phi(x_0,s).$$

Consequently, one gets

$$v(t, \phi(x_0,t)) = u_0(x_0) + \int_0^t f(s, \phi(x_0,s))\, ds.$$

Differentiating with respect to t shows that v satisfies (11.2) at every point $(t,x) = (t, \phi(x_0,t))$. Since, for every $t \in \mathbb{R}$, $x_0 \mapsto \phi(x_0,t)$ is a diffeomorphism of $\mathbb{R}^d$, the equation is satisfied at every point (t,x). ♢

Exercise 81. ★★*In dimension one, let $b(x) = x$ and $f = 0$. Compute the flow and use formula* (11.4) *to solve the advection equation with initial condition u_0.*

Exercise 82 (A source term depending on the unknown)**.** ★★*Let u be a classical solution to*

$$\partial_t u + b(x)\cdot\nabla u = F(t,x,u), \qquad u(0,x) = u_0(x)$$

for a function F. For $x_0 \in \mathbb{R}^d$, let X be the characteristic associated with the velocity field b, s.t. $X(0) = x_0$, and set $v(t) = u(t, X(t))$. Using the chain rule, prove that (X, v) satisfies the coupled system of ordinary differential equations

$$\begin{cases} \dot{X}(t) = b(X(t)), \\ \dot{v}(t) = F(t, X(t), v(t)), \end{cases} \qquad X(0) = x_0, \quad v(0) = u_0(x_0).$$

Explain how solving this system for every x_0 can be used to reconstruct the solution u.

11.1.2 Qualitative properties

The representation formula immediately gives several properties which are the counterparts on the advection equation of the maximum principles studied for the Poisson and heat equations.

Proposition 78. *Let $b \in \mathcal{C}^1(\mathbb{R}^d, \mathbb{R}^d)$ have a well defined global-in-time flow ϕ, and let u solve the problem (11.2) with $f = 0$. Then, for all $t \geq 0$:*

1. *if $u_0 \geq 0$, then $u(t, \cdot) \geq 0$;*
2. *$\inf u_0 \leq u(t, x) \leq \sup u_0$ for all $x \in \mathbb{R}^d$;*
3. *if $u_0 = 0$ outside a subset $K \subset \mathbb{R}^d$, then $u(t, x) = 0$ outside $\phi(K, t) = \{\phi(y, t) : y \in K\}$;*
4. *if $|u_0|^2$ is integrable, then*

$$\int_{\mathbb{R}^d} |u(t,x)|^2 \, dx = \int_{\mathbb{R}^d} |u_0(x_0)|^2 J(x_0, t) \, dx_0,$$

   where $J(x_0, t) = \det \nabla_{x_0} \phi(x_0, t)$ is the Jacobian matrix associated with ϕ, namely $\nabla_{x_0} \phi(x_0, t)$ is the $d \times d$ matrix with (i, j) component $\frac{\partial \phi_i}{\partial x_{0,j}}(x_0, t)$;

5. *under the assumptions of the previous item, if $|\operatorname{div} b| \leq L$, then*

$$e^{-Lt} \int_{\mathbb{R}^d} |u_0(x)|^2 \, dx \leq \int_{\mathbb{R}^d} |u(t,x)|^2 \, dx \leq e^{Lt} \int_{\mathbb{R}^d} |u_0(x)|^2 \, dx.$$

   In particular, if $\operatorname{div} b = 0$, the L^2 norm is conserved.

Proof: The first two properties follow directly from $u(t, x) = u_0(\phi(x, -t))$. The support property follows from the fact that $x_0 \mapsto \phi(x_0, t)$ is a diffeomorphism. The change of variables $x = \phi(x_0, t)$ gives the fourth property. Finally, formula (2.10) yields

$$e^{-Lt} \leq J(x_0, t) \leq e^{Lt},$$

which proves the last property. ♢

Remark 58. *The advection equation preserves regularity, but does not create any. For example, if u_0 has a jump discontinuity, this discontinuity is simply transported by the flow. This is again very different from the regularizing effect of the heat equation studied in Section 8.3.*

Exercise 83. ★*This exercise is about a generalization of items 4 and 5 in Proposition 78. Let $p \in [1, \infty)$ be such that $|u_0|^p$ is integrable. Show that*

$$\int_{\mathbb{R}^d} |u(t,x)|^p \, dx = \int_{\mathbb{R}^d} |u_0(x_0)|^p J(x_0, t) \, dx_0.$$

Show that if $|\operatorname{div} b| \leq L$, then

$$e^{-Lt} \int_{\mathbb{R}^d} |u_0(x)|^p \, dx \leq \int_{\mathbb{R}^d} |u(t,x)|^p \, dx \leq e^{Lt} \int_{\mathbb{R}^d} |u_0(x)|^p \, dx.$$

In particular, if $\operatorname{div} b = 0$, all the L^p norms are conserved.

Exercise 84. ★★*Assume that u solves (11.2) and that $|f| \leq M$ on $[0, T] \times \mathbb{R}^d$. Prove that*

$$\sup_{x \in \mathbb{R}^d} |u(t,x)| \leq \sup_{x \in \mathbb{R}^d} |u_0(x)| + tM.$$

11.1.3 Advection equation on an interval with an incoming boundary condition

In order to discuss boudary conditions on advection equations, let us consider a simple one-dimensional case, namely problem (10.19), with $a > 0$. The backward characteristic starting at (t, x) meets either the initial line $t = 0$ if $x - at \geq 0$, or the incoming boundary $x = 0$ if $x - at < 0$. Consequently, a classical solution must satisfy

$$u(t,x) = \begin{cases} u_0(x-at) + \displaystyle\int_0^t f(s, x-a(t-s))\,ds, & x \geq at, \\ g(t-x/a) + \displaystyle\int_{t-x/a}^t f(s, x-a(t-s))\,ds, & x < at. \end{cases} \tag{11.5}$$

The two expressions coincide on $x = at$ if the compatibility condition $u_0(0) = g(0)$ holds.

Proposition 79. *Assume that u_0, g, and f are continuously differentiable and satisfy the compatibility conditions required on the line $x = at$. Then formula* (11.5) *defines the unique classical solution to problem* (10.19).

Proof: The formula follows by integrating the source term along the backward characteristic until it reaches either $t = 0$ or $x = 0$. It gives uniqueness. Existence is checked by differentiating the two expressions in their respective regions. The compatibility assumptions ensure the required regularity across $x = at$. ♢

Exercise 85. ★*Derive the analogue of* (11.5) *when $a < 0$. Where should the boundary value be prescribed?*

Exercise 86. ★★*Consider $a > 0$, $f = 0$, and $g = 0$. Prove that the solution to problem* (10.19) *vanishes identically for $t \geq L/a$. Interpret this result.*

11.2 Explicit solutions of the wave equation, $d = 1$

11.2.1 D'Alembert's formula for the wave equation

The case with no source term. We now consider the wave equation in dimension one (10.28) and with zero right-hand side, on the whole line:

$$\begin{cases} \partial_{t,t} u - c^2 \partial_{x,x} u = 0 & \text{for } t > 0,\ x \in \mathbb{R}, \\ u(0,x) = u_0(x) & \text{for } x \in \mathbb{R}, \\ \partial_t u(0,x) = v_0(x) & \text{for } x \in \mathbb{R}. \end{cases} \tag{11.6}$$

Introduce the characteristic variables

$$\xi(x,t) = x + ct, \qquad \eta(x,t) = x - ct,$$

and write $u(t,x) = U(\xi(x,t), \eta(x,t))$. By the chain rule,

$$\partial_t u(t,x) = c(\partial_\xi U - \partial_\eta U)(\xi(x,t), \eta(x,t)), \qquad \partial_x u(t,x) = (\partial_\xi U + \partial_\eta U)(\xi(x,t), \eta(x,t)),$$

and therefore

$$\partial_{t,t} u(t,x) - c^2 \partial_{x,x} u(t,x) = -4c^2 \partial_{\xi,\eta} U(\xi(x,t), \eta(x,t)).$$

The wave equation is thus equivalent to $\partial_{\xi,\eta} U = 0$. Integrating successively with respect to ξ and η gives

$$U(\xi, \eta) = F(\xi) + G(\eta),$$

and hence

$$u(t, x) = F(x + ct) + G(x - ct). \tag{11.7}$$

The solution is thus the superposition of a left-going and a right-going wave.

The initial conditions imply

$$F(x) + G(x) = u_0(x), \qquad cF'(x) - cG'(x) = v_0(x).$$

Adding and subtracting these relations leads to the following formula[1].

Theorem 80 (D'Alembert's formula)**.** *Let $u_0 \in \mathcal{C}^2(\mathbb{R})$ and $v_0 \in \mathcal{C}^1(\mathbb{R})$. The unique classical solution to* (11.6) *is*

$$u(t, x) = \frac{u_0(x - ct) + u_0(x + ct)}{2} + \frac{1}{2c} \int_{x-ct}^{x+ct} v_0(y)\, dy. \tag{11.8}$$

Proof: The preceding change of variables shows that every classical solution has the form $F(x + ct) + G(x - ct)$. The initial conditions determine F and G and give the stated formula, which proves uniqueness. Conversely, direct differentiation proves existence. ♢

Exercise 87. ★*Carry out the direct verification at the end of the proof of Theorem 80.*

Exercise 88. ★*Assume that $v_0 = 0$. Describe the two waves generated by the initial displacement u_0. What are their amplitudes and directions of motion?*

Exercise 89. ★★*Assume that $u_0 = 0$ and that v_0 is nonnegative and compactly supported. Use d'Alembert's formula to describe qualitatively the solution.*

Generalization with a source term. Let us now consider the non-homogeneous problem

$$\begin{cases} \partial_{t,t} u - c^2 \partial_{x,x} u = f(t, x), \\ u(0, x) = u_0(x), \\ \partial_t u(0, x) = v_0(x). \end{cases} \tag{11.9}$$

Proposition 81 (Generalization of d'Alembert's formula)**.** *Let $u_0 \in \mathcal{C}^2(\mathbb{R})$, $v_0 \in \mathcal{C}^1(\mathbb{R})$ and assume that f and $\partial_x f$ belong to $\mathcal{C}^0([0, T] \times \mathbb{R})$. Then the unique classical solution to* (11.9) *is*

$$u(t, x) = \frac{u_0(x - ct) + u_0(x + ct)}{2} + \frac{1}{2c} \int_{x-ct}^{x+ct} v_0(y)\, dy + \frac{1}{2c} \int_0^t \int_{x-c(t-s)}^{x+c(t-s)} f(s, y)\, dy\, ds. \tag{11.10}$$

The last term in (11.10) is the wave-equation analogue of the variation of constants formula for ordinary differential equations and of formula (11.4) for the advection equation. This is simply Duhamel's principle (see Exercise 16): each instantaneous forcing acts as a new initial-velocity contribution, and all these contributions are superposed.

Exercise 90. ★*By differentiating under the integral sign, prove that the last term in* (11.10) *has zero initial displacement and velocity and satisfies the wave equation with right-hand side f. Then, prove Proposition 81.*

[1] Jean le Rond d'Alembert derived this formula in 1747 as a solution to the problem of a vibrating string.

11.2.2 The wave equation as two advection equations

The one-dimensional wave operator factorizes as

$$\partial_{t,t} - c^2\partial_{x,x} = (\partial_t - c\partial_x)(\partial_t + c\partial_x) = (\partial_t + c\partial_x)(\partial_t - c\partial_x). \tag{11.11}$$

For a solution to the homogeneous wave equation (11.6), introduce

$$w_+ = \partial_t u + c\partial_x u, \qquad w_- = \partial_t u - c\partial_x u. \tag{11.12}$$

By (11.11), these functions satisfy

$$\partial_t w_+ - c\partial_x w_+ = 0, \qquad \partial_t w_- + c\partial_x w_- = 0. \tag{11.13}$$

Thus w_+ is transported to the left and w_- to the right. Conversely, the identities

$$\partial_t u = \frac{w_+ + w_-}{2}, \qquad \partial_x u = \frac{w_+ - w_-}{2c}$$

allow one to reconstruct the derivatives of u.

Exercise 91. ★*Check equations* (11.13) *directly. Compute the initial values* $w_+(0,x)$ *and* $w_-(0,x)$ *in terms of* u_0 *and* v_0*.*

Exercise 92. ★★*Use the representation formula for the two advection equations* (11.13) *to recover d'Alembert's formula.*

11.3 Mathematical properties of the wave equation

11.3.1 Finite speed of propagation

D'Alembert's formula shows that, for $t > 0$ and $x \in \mathbb{R}$, $u(t,x)$ depends only on the initial data (u_0, v_0) in the interval

$$[x - ct, x + ct]. \tag{11.14}$$

This interval is the section at time 0 of the backward characteristic cone with vertex (t,x). This remark yields the following Proposition.

Proposition 82 (Finite speed of propagation)**.** *Let* $t_0 > 0$ *and* $x_0 \in \mathbb{R}$*, and introduce the backward cone*

$$\mathcal{C}_c = \{(t,x) \in [0,t_0] \times \mathbb{R} : |x - x_0| \leq c(t_0 - t)\}.$$

Let u *and* $\widetilde{u}$ *be two* $\mathcal{C}^2$ *solutions to the same wave equation* (10.28)*, with the same right-hand side* f*. If*

$$u(0,x) = \widetilde{u}(0,x) \quad \textit{and} \quad \partial_t u(0,x) = \partial_t \widetilde{u}(0,x) \qquad \textit{for every } x \in [x_0 - ct_0, x_0 + ct_0],$$

then

$$u(t,x) = \widetilde{u}(t,x) \qquad \textit{for every } (t,x) \in \mathcal{C}_c.$$

In particular, in the homogeneous case $f = 0$*, if*

$$u(0,x) = 0 \quad \textit{and} \quad \partial_t u(0,x) = 0 \qquad \textit{for every } x \in [x_0 - ct_0, x_0 + ct_0],$$

then u *vanishes throughout* $\mathcal{C}_c$*.*

Proof: The difference $w = u - \widetilde{u}$ satisfies the homogeneous wave equation. For every $(t, x) \in \mathcal{C}_c$, one has

$$[x - ct, x + ct] \subset [x_0 - ct_0, x_0 + ct_0].$$

D'Alembert's formula applied to w therefore gives $w(t, x) = 0$. The last statement follows by taking $\widetilde{u} = 0$. ♢

As a consequence of Proposition 82, for the homogeneous wave equation, if the initial data $u(0, \cdot)$ and $\partial_t u(0, \cdot)$ are supported in $[-R, R]$ for some $R > 0$, then

$$\operatorname{supp} u(t, \cdot) \subset [-R - ct, R + ct] \qquad \text{for every } t \geq 0.$$

The situation sharply contrasts with the heat equation: since the heat kernel is strictly positive, a nonzero, nonnegative localized initial condition produces a solution which is immediately strictly positive throughout $\mathbb{R}^d$; see Remark 42.

Exercise 93. ★*Draw the backward characteristic cone associated with* (t_0, x_0)*. Use it to determine which part of a source term* f *can influence the value of the solution to* (11.9) *at* (t_0, x_0)*.*

11.3.2 Non-regularization

We saw in Theorem 67 that the heat equation has an instantaneous smoothing effect: a classical solution is infinitely differentiable at every positive time, even when the initial data are much less regular. The wave equation behaves differently. D'Alembert's formula only translates the initial profiles and does not create new derivatives.

Proposition 83 (Preservation of regularity)**.** *Let* $k \geq 2$*,* $u_0 \in \mathcal{C}^k(\mathbb{R})$*, and* $v_0 \in \mathcal{C}^{k-1}(\mathbb{R})$*. Then the classical solution to the homogeneous wave equation* (11.6) *belongs to* $\mathcal{C}^k(\mathbb{R}_+ \times \mathbb{R})$*.*

Proof: This follows from d'Alembert's formula. Let V be an antiderivative of v_0. Since $v_0 \in \mathcal{C}^{k-1}(\mathbb{R})$, one has $V \in \mathcal{C}^k(\mathbb{R})$. Formula (11.8) can be rewritten as

$$u(t, x) = \frac{u_0(x - ct) + u_0(x + ct)}{2} + \frac{V(x + ct) - V(x - ct)}{2c}. \tag{11.15}$$

Each term on the right-hand side is the composition of a $\mathcal{C}^k$ function with a linear function of (t, x). Therefore $u \in \mathcal{C}^k(\mathbb{R}_+ \times \mathbb{R})$. ♢

The preceding result gives the regularity that is naturally expected from the initial conditions. In general, the solution is not smoother. To see this, let $\phi \in \mathcal{C}^k(\mathbb{R})$ be a function which does not belong to $\mathcal{C}^{k+1}(\mathbb{R})$, and choose

$$u_0(x) = \phi(x), \qquad v_0(x) = -c\phi'(x). \tag{11.16}$$

The corresponding solution is the right-going wave

$$u(t, x) = \phi(x - ct). \tag{11.17}$$

For every $t \geq 0$, the function $u(t, \cdot)$ belongs to $\mathcal{C}^k(\mathbb{R})$ but not to $\mathcal{C}^{k+1}(\mathbb{R})$. Any lack of regularity of ϕ is simply transported to the right at speed c.

Exercise 94. ★*Let* $\phi \in \mathcal{C}^k(\mathbb{R})$ *and set* $u_0 = \phi$*,* $v_0 = c\phi'$*. Use d'Alembert's formula to identify the solution and the direction in which its possible lack of regularity propagates.*

11.4 Towards variational methods for the wave equation

In this section, we consider the wave equation in a general bounded $\mathcal{C}^1$ domain $\Omega \subset \mathbb{R}^d$. Let $T > 0$, and let $c > 0$. We consider the wave equation on $(0,T) \times \Omega$, supplemented with initial conditions and either homogeneous Dirichlet or homogeneous Neumann boundary conditions.

The homogeneous Dirichlet problem is

$$\begin{cases} \partial_{t,t} u - c^2 \Delta u = f & \text{in } (0,T) \times \Omega, \\ u = 0 & \text{on } [0,T] \times \partial\Omega, \\ u(0,x) = u_0(x) & \text{for } x \in \Omega, \\ \partial_t u(0,x) = v_0(x) & \text{for } x \in \Omega. \end{cases} \tag{11.18}$$

The homogeneous Neumann problem is

$$\begin{cases} \partial_{t,t} u - c^2 \Delta u = f & \text{in } (0,T) \times \Omega, \\ \partial_n u = 0 & \text{on } [0,T] \times \partial\Omega, \\ u(0,x) = u_0(x) & \text{for } x \in \Omega, \\ \partial_t u(0,x) = v_0(x) & \text{for } x \in \Omega. \end{cases} \tag{11.19}$$

where $\partial_n u = \nabla u \cdot n$ and n denotes the outward unit normal to Ω. Throughout this subsection, the wave speed is a fixed constant $c > 0$.

We again only consider classical solutions to (11.18) or (11.19), namely functions

$$u \in \mathcal{C}^2([0,T] \times \overline{\Omega}) \tag{11.20}$$

which satisfy all the equations, initial conditions, and boundary conditions pointwise. This requires that the data are sufficiently regular and satisfy the necessary compatibility conditions. The Dirichlet problem requires $u_0 = v_0 = 0$ on $\partial\Omega$, whereas the Neumann problem requires $\partial_n u_0 = \partial_n v_0 = 0$ on $\partial\Omega$.

11.4.1 Conservation of energy

For a classical solution to either problem, we define the energy at time t by

$$E(t) = \frac{1}{2} \int_\Omega \left((\partial_t u(t,x))^2 + c^2 |\nabla u(t,x)|^2 \right) \, dx. \tag{11.21}$$

The two terms can be interpreted respectively as kinetic and elastic potential energy, up to multiplicative physical constants.

Multiplying the wave equation by $\partial_t u$ and integrating over Ω gives

$$\int_\Omega \partial_{t,t} u \, \partial_t u - c^2 \int_\Omega \Delta u \, \partial_t u = \int_\Omega f \partial_t u.$$

Using Green's identity from Exercise 37, we have

$$-\int_\Omega \Delta u \, \partial_t u = \int_\Omega \nabla u \cdot \nabla \partial_t u - \int_{\partial\Omega} \partial_n u \, \partial_t u \, d\sigma.$$

Differentiation under the integral sign therefore yields the energy identity

$$E'(t) = \int_\Omega f(t,x) \partial_t u(t,x) \, dx + c^2 \int_{\partial\Omega} \partial_n u(t,x) \partial_t u(t,x) \, \sigma(dx). \tag{11.22}$$

The last term represents the energy flux through the boundary of the domain. These computations lead to the following result.

Proposition 84 (Conservation of energy)**.** *Let u be a classical solution to either the Dirichlet problem* (11.18) *or the Neumann problem* (11.19)*. If $f = 0$, then*

$$E(t) = E(0) \qquad \text{for all } t \in [0, T].$$

Proof: For the Neumann problem, the boundary integral in (11.22) vanishes because $\partial_n u = 0$ on $\partial\Omega$. For the Dirichlet problem, for any $x \in \partial\Omega$, differentiating $u(t, x) = 0$ with respect to time gives $\partial_t u(t, x) = 0$, so the boundary integral again vanishes. Since $f = 0$, Equation (11.22) gives $E'(t) = 0$. ♢

Remark 59. *For the heat equation with homogeneous Dirichlet boundary conditions, the quantity $\int u^2$ decreases in time; see Equation* (8.12)*. For the wave equation, energy is not dissipated but continually exchanged between its kinetic and potential parts.*

Exercise 95. ★*Derive identity* (11.22) *in detail, justifying why the boundary integral vanishes for each of the two boundary conditions.*

Exercise 96. ★★*Assume homogeneous Dirichlet or Neumann boundary conditions and let $f \neq 0$. Using $2ab \leq a^2 + b^2$, derive an estimate of the energy in terms of its initial value and the source term.*

Remark 60 (A local conservation law for energy)**.** *Define the energy density and energy flux by*

$$e = \frac{1}{2}\left((\partial_t u)^2 + c^2|\nabla u|^2\right), \qquad q = -c^2 \partial_t u \nabla u. \tag{11.23}$$

A direct computation using the wave equation gives

$$\partial_t e + \operatorname{div}(q) = f \partial_t u. \tag{11.24}$$

This has exactly the form of the conservation law (10.4)*: energy is itself a quantity which is transported by waves.*

Exercise 97. ★*Prove* (11.24)*. Integrate it over Ω and recover* (11.22)*.*

11.4.2 Uniqueness from the energy method

The energy identity provides a proof of uniqueness in any space dimension which does not rely on an explicit representation formula.

Proposition 85. *There is at most one classical solution to the Dirichlet problem* (11.18)*. The same statement holds for the Neumann problem* (11.19)*.*

Proof: Let u_1 and u_2 be two solutions with the same data. Their difference $w = u_1 - u_2$ satisfies the homogeneous wave equation, homogeneous initial conditions, and homogeneous boundary conditions. By Proposition 84, its energy is constant and equal to its value at time 0, namely 0. Thus $\partial_t w = 0$ and $\nabla w = 0$. In particular, $w(t, x) = w(0, x) = 0$ for every (t, x). ♢

11.4.3 Finite speed of propagation from the energy method

We now give another proof of finite speed of propagation. Contrary to the proof of Proposition 82, the following argument does not use an explicit representation formula and works in any space dimension.

Let $x_0 \in \mathbb{R}^d$ and $t_0 > 0$, and introduce the backward cone

$$\mathcal{C}_c = \left\{(t, x) \in [0, t_0] \times \mathbb{R}^d : |x - x_0| \leq c(t_0 - t)\right\}. \tag{11.25}$$

Theorem 86 (Finite speed of propagation)**.** *Let u be a $\mathcal{C}^2$ function satisfying*

$$\partial_{t,t}u - c^2\Delta u = 0 \tag{11.26}$$

in a neighborhood of $\mathcal{C}_c$. Assume that

$$u(0,x) = 0 \quad \textit{and} \quad \partial_t u(0,x) = 0 \qquad \textit{for } x \in B(x_0, ct_0). \tag{11.27}$$

Then

$$u(t,x) = 0 \qquad \textit{for every } (t,x) \in \mathcal{C}_c. \tag{11.28}$$

Proof: For $0 \le t < t_0$, define the energy contained in the shrinking ball $B(x_0, c(t_0-t))$ by

$$E(t) = \frac{1}{2}\int_{B(x_0,c(t_0-t))} \left((\partial_t u(t,x))^2 + c^2|\nabla u(t,x)|^2\right)\, dx. \tag{11.29}$$

The radius of this ball decreases at speed c. Thus, if e is a smooth function, differentiation of an integral over the shrinking ball gives

$$\frac{d}{dt}\int_{B(x_0,c(t_0-t))} e(t,x)\,dx = \int_{B(x_0,c(t_0-t))} \partial_t e(t,x)\,dx - c\int_{\partial B(x_0,c(t_0-t))} e(t,x)\,d\sigma. \tag{11.30}$$

This formula follows, for example, by using polar coordinates centered at x_0 and differentiating with respect to the upper bound $c(t_0-t)$.

Applying (11.30) to the energy density and integrating by parts in space, we obtain

$$\begin{aligned} E'(t) = &\int_{B(x_0,c(t_0-t))} \left(\partial_t u\,\partial_{t,t}u + c^2\nabla u\cdot\nabla\partial_t u\right)\, dx \\ &- \frac{c}{2}\int_{\partial B(x_0,c(t_0-t))} \left((\partial_t u)^2 + c^2|\nabla u|^2\right)\, d\sigma \\ = &\, c^2\int_{\partial B(x_0,c(t_0-t))} \partial_t u\,\partial_n u\,d\sigma - \frac{c}{2}\int_{\partial B(x_0,c(t_0-t))} \left((\partial_t u)^2 + c^2|\nabla u|^2\right)\, d\sigma, \end{aligned}$$

where we used (11.26) in the last equality and $\partial_n u = \nabla u\cdot n$ denotes the outward normal derivative. On the boundary of the ball, decompose the gradient into its normal and tangential components:

$$|\nabla u|^2 = (\partial_n u)^2 + |\nabla_\tau u|^2.$$

It follows that

$$E'(t) = -\frac{c}{2}\int_{\partial B(x_0,c(t_0-t))} \left((\partial_t u - c\partial_n u)^2 + c^2|\nabla_\tau u|^2\right)\, d\sigma \le 0. \tag{11.31}$$

By (11.27), one has $E(0) = 0$; notice that $u(0,\cdot) = 0$ on the ball also implies $\nabla u(0,\cdot) = 0$ there. Since E is nonnegative and nonincreasing, $E(t) = 0$ for every $t \in [0, t_0)$. Therefore

$$\partial_t u(t,x) = 0 \quad \text{and} \quad \nabla u(t,x) = 0 \qquad \text{for } |x - x_0| < c(t_0 - t).$$

For every point (t,x) in the interior of $\mathcal{C}_c$, the vertical segment (s,x), $0 \le s \le t$, remains in $\mathcal{C}_c$. Hence

$$u(t,x) = u(0,x) + \int_0^t \partial_t u(s,x)\,ds = 0.$$

The result on the boundary of the cone follows by continuity. ♢

Exercise 98. ★★*Prove formula* (11.30) *using polar coordinates. Deduce from the theorem that if two classical solutions have initial data which coincide on $B(x_0, ct_0)$, then the solutions coincide throughout $\mathcal{C}_c$.*

Chapter 12

Discretization with the finite difference method

The objective of this chapter is to discretize the advection and wave equations in dimension one. As in Chapters 3, 6 and 9, we follow the principle

$$\text{consistency} + \text{stability} \Longrightarrow \text{convergence}.$$

As will be explained below, for both the advection and wave equations, stability of the explicit schemes requires a relation of the form $\Delta t \leq Ch$. This is a Courant–Friedrichs–Lewy (CFL) condition (already encountered in Chapter 9): during one time step, information should not travel across more than approximately one mesh cell.

12.1 The upwind scheme for the advection equation

Let $a > 0$ and consider

$$\begin{cases} \partial_t u + a\partial_x u = f & \text{on } (0,T)\times(0,L), \\ u(t,0) = g(t) & \text{for } t \in (0,T), \\ u(0,x) = u_0(x) & \text{for } x \in (0,L). \end{cases} \tag{12.1}$$

Let $N, I \geq 1$, $\Delta t = T/N$, $h = L/I$, $t_n = n\Delta t$, and $x_j = jh$. We denote by U_j^n an approximation of $u(t_n, x_j)$.

Since information travels from left to right, it is natural to approximate the space derivative using the backward difference (see Section 12.1.4 for an example where another consistent discretization yields to an unstable scheme)

$$\partial_x u(t_n, x_j) \simeq \frac{U_j^n - U_{j-1}^n}{h}.$$

Together with explicit Euler in time, this gives the upwind scheme

$$\frac{U_j^{n+1} - U_j^n}{\Delta t} + a\frac{U_j^n - U_{j-1}^n}{h} = f(t_n, x_j), \tag{12.2}$$

for $1 \leq j \leq I$ and $0 \leq n \leq N-1$, with

$$U_j^0 = u_0(x_j), \qquad U_0^n = g(t_n). \tag{12.3}$$

Introducing the Courant number

$$\nu = \frac{a\Delta t}{h}, \tag{12.4}$$

the scheme can be written

$$U_j^{n+1} = (1-\nu)U_j^n + \nu U_{j-1}^n + \Delta t f(t_n, x_j). \tag{12.5}$$

Remark 61. *If $a < 0$, information travels from right to left and one must use the forward difference $(U_{j+1}^n - U_j^n)/h$. The word "upwind" means that the numerical derivative uses values located on the side from which the information arrives.*

Exercise 99. ★*Write the upwind scheme for $a < 0$, including the appropriate boundary condition. Rewrite it as a convex combination under a suitable CFL condition.*

12.1.1 Consistency

Let us insert the exact solution into the numerical scheme. We define the consistency error ε_j^n by

$$\frac{u(t_{n+1}, x_j) - u(t_n, x_j)}{\Delta t} + a\frac{u(t_n, x_j) - u(t_n, x_{j-1})}{h} = f(t_n, x_j) + \varepsilon_j^n. \tag{12.6}$$

Assuming that u is twice continuously differentiable, Taylor expansions give

$$\frac{u(t_{n+1}, x_j) - u(t_n, x_j)}{\Delta t} = \partial_t u(t_n, x_j) + O(\Delta t)$$

and

$$\frac{u(t_n, x_j) - u(t_n, x_{j-1})}{h} = \partial_x u(t_n, x_j) + O(h).$$

Since u satisfies the advection equation, we obtain

$$\max_{0\le n\le N-1} \max_{1\le j\le I} |\varepsilon_j^n| \le C(\Delta t + h), \tag{12.7}$$

where C depends on second derivatives of u, but not on Δt and h. The scheme is consistent of order one in time and space.

Exercise 100. ★*Using Taylor expansions with integral or mean-value remainders, prove estimate* (12.7) *and give an explicit possible value for C.*

12.1.2 A discrete maximum principle and stability

Assume the CFL condition

$$0 \le \nu = \frac{a\Delta t}{h} \le 1. \tag{12.8}$$

The coefficients $1-\nu$ and ν in formula (12.5) are then nonnegative and sum to one.

Proposition 87 (Discrete maximum principle)**.** *Assume* (12.8) *and $f = 0$. Then, for every n,*

$$\min\left(\min_{0\le j\le I} U_j^0, \min_{0\le k\le n} g(t_k)\right) \le U_j^n \le \max\left(\max_{0\le j\le I} U_j^0, \max_{0\le k\le n} g(t_k)\right).$$

In particular, nonnegative initial and boundary data produce a nonnegative numerical solution.

Proof: Formula (12.5) shows that U_j^{n+1} lies between U_j^n and U_{j-1}^n. The result follows by induction on n. ♢

Let $V^n = (V_j^n)_{1\le j\le I}$ satisfy the homogeneous scheme with zero incoming boundary value. The same convexity argument gives

$$\|V^{n+1}\|_\infty \le \|V^n\|_\infty. \tag{12.9}$$

Thus the upwind scheme is stable in l^∞ norm under the CFL condition (12.8).

Exercise 101. ★★*Assume an infinite or periodic grid and $f = 0$. Prove that the upwind scheme conserves the discrete mass $h\sum_j U_j^n$. Under which assumptions is the discrete l^1 norm also conserved?*

12.1.3 Convergence

Define the error

$$E_j^n = u(t_n, x_j) - U_j^n. \tag{12.10}$$

We assume for simplicity that the initial and boundary values are inserted exactly, so $E_j^0 = 0$ and $E_0^n = 0$. Subtracting the numerical scheme from Equation (12.6) gives

$$E_j^{n+1} = (1-\nu)E_j^n + \nu E_{j-1}^n + \Delta t \varepsilon_j^n. \tag{12.11}$$

Under the CFL condition (12.8), we deduce

$$\|E^{n+1}\|_\infty \leq \|E^n\|_\infty + \Delta t \|\varepsilon^n\|_\infty.$$

Iterating and using $n\Delta t \leq T$, one obtains (this is exactly the stability result from Proposition 87)

$$\max_{0\leq n\leq N} \|E^n\|_\infty \leq T \max_{0\leq n\leq N-1} \|\varepsilon^n\|_\infty. \tag{12.12}$$

Theorem 88 (Convergence of the upwind scheme)**.** *Assume that the solution to* (12.1) *is twice continuously differentiable. Under the CFL condition* (12.8)*, the upwind scheme satisfies*

$$\max_{0\leq n\leq N} \max_{1\leq j\leq I} |u(t_n, x_j) - U_j^n| \leq CT(\Delta t + h).$$

It is convergent of order one in time and space.

Proof: Combine the consistency estimate (12.7) and the stability estimate (12.12). ♢

12.1.4 Why a centered explicit scheme is unstable?

One may be tempted to use the centered approximation

$$\partial_x u(t_n, x_j) \simeq \frac{U_{j+1}^n - U_{j-1}^n}{2h},$$

leading, when $f = 0$, to

$$U_j^{n+1} = U_j^n - \frac{\nu}{2}(U_{j+1}^n - U_{j-1}^n). \tag{12.13}$$

This scheme is consistent, but unstable. To see this, on an infinite or periodic grid, consider a discrete oscillation

$$U_j^n = G^n \exp(\mathrm{i}j\theta).$$

Inserting it into (12.13) gives the amplification factor

$$G = 1 - \mathrm{i}\nu \sin\theta, \qquad |G|^2 = 1 + \nu^2 \sin^2\theta.$$

For any nonzero ν, some modes satisfy $|G| > 1$ and are amplified at each time step. Consistency alone is therefore not sufficient for convergence.

Exercise 102. ★★*Carry out the calculation above. Which modes are not amplified? Observe the instability numerically starting from a smooth initial condition with a small random perturbation.*

Exercise 103. ★*Assume that $a > 0$ and consider, on an infinite grid, or on a periodic grid with an even number of points, the scheme obtained by using the forward difference, namely*

$$\frac{U_j^{n+1} - U_j^n}{\Delta t} + a\frac{U_{j+1}^n - U_j^n}{h} = 0. \tag{12.14}$$

This is sometimes called the downwind scheme*, since the space derivative is discretized in the direction opposite to that from which the information arrives. Write the scheme in terms of the Courant number $\nu = a\Delta t/h$. Starting from $U_j^0 = (-1)^j$, prove that*

$$U_j^n = (1+2\nu)^n (-1)^j.$$

Deduce that the scheme is unstable for every $\nu > 0$.

12.1.5 Numerical diffusion

The exact advection equation preserves the shape of the initial profile, but the upwind approximation generally smooths it. This can already be seen from (12.5): unless $\nu = 0$ or 1, the new value is an average of two neighboring values.

There is also an exact algebraic way to see this diffusion. The backward difference can be decomposed as

$$\frac{U_j^n - U_{j-1}^n}{h} = \frac{U_{j+1}^n - U_{j-1}^n}{2h} - \frac{h}{2}\frac{U_{j+1}^n - 2U_j^n + U_{j-1}^n}{h^2}. \tag{12.15}$$

Consequently, when $f = 0$, the upwind scheme can be written exactly as

$$\frac{U_j^{n+1} - U_j^n}{\Delta t} + a\frac{U_{j+1}^n - U_{j-1}^n}{2h} = \frac{ah}{2}\frac{U_{j+1}^n - 2U_j^n + U_{j-1}^n}{h^2}. \tag{12.16}$$

The left-hand side is the centered explicit discretization of the advection equation, while the right-hand side is a discrete diffusion term with positive coefficient $ah/2$. Thus the upwind scheme is exactly a centered scheme to which numerical diffusion has been added.

A formal Taylor expansion gives a more precise interpretation. Up to higher-order terms, the upwind scheme is equivalent to

$$\partial_t u + a\partial_x u = \frac{ah}{2}(1-\nu)\partial_{x,x} u. \tag{12.17}$$

The coefficient in the modified equation differs from the coefficient $ah/2$ in (12.16), because the forward time difference also contributes at the same order. Using $\partial_{t,t} u = a^2 \partial_{x,x} u$ subtracts $a^2\Delta t/2 = ah\nu/2$ from the diffusion coefficient.

Thus the leading numerical error behaves like an artificial diffusion. It vanishes when $\nu = 1$: in that case

$$U_j^{n+1} = U_{j-1}^n,$$

which translates the grid values by exactly one cell per time step.

Exercise 104. ★★ *Use the identity*

$$\frac{U_{j+1}^n - U_j^n}{h} = \frac{U_{j+1}^n - U_{j-1}^n}{2h} + \frac{h}{2}\frac{U_{j+1}^n - 2U_j^n + U_{j-1}^n}{h^2}$$

to interpret the downwind scheme (12.14) *as a centered scheme with a negative diffusion term. Discuss the instability of this scheme using this rewriting, in light of the exercises* (65) *and* (67).

Exercise 105. ★★★ *Derive formally the modified equation* (12.17) *by expanding a smooth solution up to second order in Δt and h. Use the advection equation to replace $\partial_{t,t} u$ by $a^2 \partial_{x,x} u$.*

12.2 A centered scheme for the wave equation

We now consider the one-dimensional wave equation with homogeneous Dirichlet boundary conditions:

$$\begin{cases} \partial_{t,t} u - c^2 \partial_{x,x} u = f & \text{on } (0,T)\times(0,L), \\ u(t,0) = u(t,L) = 0 & \text{for } t \in (0,T), \\ u(0,x) = u_0(x) & \text{for } x \in (0,L), \\ \partial_t u(0,x) = v_0(x) & \text{for } x \in (0,L). \end{cases} \tag{12.18}$$

We use the same grids $t_n = n\Delta t$ and $x_j = jh$ as above. Centered finite differences give

$$\partial_{t,t} u(t_n, x_j) \simeq \frac{U_j^{n+1} - 2U_j^n + U_j^{n-1}}{\Delta t^2}$$

and

$$\partial_{x,x}u(t_n,x_j) \simeq \frac{U^n_{j+1}-2U^n_j+U^n_{j-1}}{h^2}.$$

A natural numerical scheme is therefore

$$\frac{U^{n+1}_j-2U^n_j+U^{n-1}_j}{\Delta t^2} - c^2\frac{U^n_{j+1}-2U^n_j+U^n_{j-1}}{h^2} = f(t_n,x_j) \tag{12.19}$$

for $1 \le j \le I-1$ and $1 \le n \le N-1$, with

$$U^n_0 = U^n_I = 0, \qquad U^0_j = u_0(x_j). \tag{12.20}$$

Because the recurrence (12.19) uses two time levels, the initial velocity is needed to compute U^1. Taylor expansion at $t=0$ and the differential equation give a natural way to define U^1 as:

$$U^1_j = u_0(x_j) + \Delta t\, v_0(x_j) + \frac{c^2\Delta t^2}{2h^2}\left(u_0(x_{j+1})-2u_0(x_j)+u_0(x_{j-1})\right) + \frac{\Delta t^2}{2}f(0,x_j). \tag{12.21}$$

The scheme (12.19)–(12.20)–(12.21) is explicit. Let A be the $(I-1)\times(I-1)$ matrix associated with the discretization of $-\partial_{x,x}$ with homogeneous Dirichlet boundary conditions, as in Equation (9.3), and introduce

$$U^n = (U^n_j)_{1\le j\le I-1} \in \mathbb{R}^{I-1}, \qquad F^n = \big(f(t_n,x_j)\big)_{1\le j\le I-1} \in \mathbb{R}^{I-1}.$$

The recurrence can then be written in matrix form as

$$U^{n+1}-2U^n+U^{n-1} = -c^2\Delta t^2 AU^n + \Delta t^2 F^n, \qquad 1 \le n \le N-1. \tag{12.22}$$

Exercise 106. ★*Derive Equation* (12.21)*. Which regularity assumptions on* u_0*,* v_0*, and* f *are natural for this approximation?*

12.2.1 Consistency

For $0 \le n \le N$, introduce the vector of exact nodal values

$$u^n_h = \big(u(t_n,x_j)\big)_{1\le j\le I-1} \in \mathbb{R}^{I-1}. \tag{12.23}$$

The scheme involves two distinct error vectors. First, for $1 \le n \le N-1$, the consistency error in the recurrence is $\varepsilon^n = (\varepsilon^n_j)_{1\le j\le I-1} \in \mathbb{R}^{I-1}$, where

$$\begin{aligned}\varepsilon^n_j = {} & \frac{u(t_{n+1},x_j)-2u(t_n,x_j)+u(t_{n-1},x_j)}{\Delta t^2} \\ & - c^2\frac{u(t_n,x_{j+1})-2u(t_n,x_j)+u(t_n,x_{j-1})}{h^2} - f(t_n,x_j).\end{aligned} \tag{12.24}$$

Equivalently, in vector form,

$$u^{n+1}_h - 2u^n_h + u^{n-1}_h = -c^2\Delta t^2 Au^n_h + \Delta t^2 F^n + \Delta t^2\varepsilon^n. \tag{12.25}$$

Second, the error made at the first time step is the vector $\eta^1 = (\eta^1_j)_{1\le j\le I-1} \in \mathbb{R}^{I-1}$ defined by

$$\begin{aligned}\eta^1_j = {} & u(\Delta t,x_j) - u_0(x_j) - \Delta t\, v_0(x_j) \\ & - \frac{c^2\Delta t^2}{2h^2}\left(u_0(x_{j+1})-2u_0(x_j)+u_0(x_{j-1})\right) - \frac{\Delta t^2}{2}f(0,x_j).\end{aligned} \tag{12.26}$$

Thus, when the initial data are inserted exactly in the scheme, $\eta^1 = u^1_h - U^1$.

The centered approximation of a second derivative is of order two. If the solution u is sufficiently regular, Taylor expansions yield

$$\max_{1\leq n\leq N-1} \|\varepsilon^n\|_\infty \leq C(\Delta t^2 + h^2) \quad \text{and} \quad \|\eta^1\|_\infty \leq C(\Delta t^3 + \Delta t^2 h^2). \tag{12.27}$$

The second estimate is the accuracy at the first time step required for a globally second-order approximation.

Exercise 107. ★*Using Taylor expansions in time and space, prove both estimates in* (12.27).

12.2.2 Stability and the CFL condition

For the homogeneous problem $F^n = 0$, recurrence (12.22) can be studied mode by mode. The matrix A is symmetric positive definite. Its eigenvalues are

$$\mu_k = \frac{4}{h^2}\sin^2\left(\frac{k\pi}{2I}\right), \qquad 1 \leq k \leq I-1, \tag{12.28}$$

see Lemma 49.

Since A admits an orthonormal basis of eigenvectors, the recurrence can be studied independently on each eigenmode. If y^n is the coefficient of U^n on an eigenvector with eigenvalue μ, it satisfies

$$y^{n+1} - (2 - c^2\Delta t^2\mu)y^n + y^{n-1} = 0. \tag{12.29}$$

The roots of its characteristic polynomial are distinct and have modulus one when

$$0 < c^2\Delta t^2\mu < 4. \tag{12.30}$$

At the two endpoints, the characteristic polynomial has a double root and the corresponding scalar recurrence may contain a linearly growing term. Since every eigenvalue satisfies $\mu_k < 4/h^2$, the condition

$$\frac{c\Delta t}{h} \leq 1 \tag{12.31}$$

ensures stability of every mode. If this CFL condition is violated and the mesh is sufficiently fine, the highest-frequency modes grow exponentially.

Lemma 89 (Stability estimate in l^2)**.** *Let $(U^n)_{0\leq n\leq N}$ satisfy*

$$U^{n+1} - 2U^n + U^{n-1} = -c^2\Delta t^2 AU^n + \Delta t^2 F^n, \qquad 1 \leq n \leq N-1, \tag{12.32}$$

and define its discrete initial velocity by

$$V^0 = \frac{U^1 - U^0}{\Delta t} + \frac{c^2\Delta t}{2}AU^0. \tag{12.33}$$

Under the CFL condition (12.31)*, for every $0 \leq n \leq N$,*

$$\max_{0\leq m\leq n} \|U^m\|_2 \leq \|U^0\|_2 + \frac{L}{2c}\left(\|V^0\|_2 + \Delta t\sum_{m=1}^{n-1}\|F^m\|_2\right), \tag{12.34}$$

where an empty sum is understood to be zero. The scheme is stable in l^2 norm, under the CFL condition (12.31).

Proof: Let p_k be an eigenvector of A associated with $\mu_k = 4h^{-2}\sin^2(\alpha_k)$, where $\alpha_k = k\pi/(2I)$. For the corresponding coefficient y_k^n of U^n, introduce $\theta_k \in (0,\pi)$ by

$$\cos(\theta_k) = 1 - \frac{c^2\Delta t^2 \mu_k}{2}.$$

If z_k is the coefficient of V^0 on p_k, the homogeneous scalar recurrence has the explicit solution

$$y_k^n = \cos(n\theta_k) y_k^0 + \Delta t \, \frac{\sin(n\theta_k)}{\sin(\theta_k)} z_k. \tag{12.35}$$

Writing $\lambda = c\Delta t/h$, one has

$$\sin(\theta_k) = 2\lambda \sin(\alpha_k)\sqrt{1-\lambda^2\sin^2(\alpha_k)}.$$

Under $\lambda \le 1$,

$$\frac{\Delta t}{\sin(\theta_k)} \le \frac{h}{c\sin(2\alpha_k)} \le \frac{L}{2c},$$

where we used $\sin(k\pi/I) \ge \sin(\pi/I) \ge 2/I$. Formula (12.35) and orthonormality of the eigenvectors therefore give the homogeneous version of (12.34). Finally, by linearity, each F^m acts as a new discrete initial velocity at time t_m. Superposing these contributions gives (12.34). ♢

Exercise 108. ★★*Study the roots of the characteristic polynomial associated with* (12.29) *and prove condition* (12.30).

Exercise 109. ★★★*For $f = 0$, multiply* (12.19) *by $U_j^{n+1} - U_j^{n-1}$, sum over j, and perform a discrete integration by parts. Identify a quantity involving the time differences and space differences which is conserved by the scheme. Investigate its positivity under* (12.31).

Lemma 90 (Stability estimate in l^∞)**.** *Let $(U^n)_{0\le n\le N}$ satisfy* (12.32)*, and let V^0 be defined by* (12.33)*. For $W = (W_j)_{1\le j\le I-1}$, completed by $W_0 = W_I = 0$, set*

$$|W|_{1,h} = \left(\frac{1}{I-1} \sum_{j=0}^{I-1} \left(\frac{W_{j+1} - W_j}{h} \right)^2 \right)^{1/2}. \tag{12.36}$$

Assume the strict CFL condition

$$\frac{c\Delta t}{h} \le 1 - \delta, \qquad \delta \in (0,1). \tag{12.37}$$

Then, for every $0 \le n \le N$,

$$\max_{0\le m\le n} \|U^m\|_\infty \le L \left[|U^0|_{1,h} + \frac{1}{c\sqrt{\delta(2-\delta)}} \left(\|V^0\|_2 + \Delta t \sum_{m=1}^{n-1} \|F^m\|_2 \right) \right]. \tag{12.38}$$

In particular, this gives uniform l^∞ control for the homogeneous scheme when $|U^0|_{1,h}$ and $\|V^0\|_2$ are bounded independently of h and Δt.

Proof: We use the notation and the modal representation introduced in the proof of Lemma 89. Under condition (12.37),

$$\frac{\sqrt{\mu_k}\,\Delta t}{\sin(\theta_k)} = \frac{1}{c\sqrt{1-\lambda^2\sin^2(\alpha_k)}} \le \frac{1}{c\sqrt{\delta(2-\delta)}}.$$

Multiplying the modal estimates by $\sqrt{\mu_k}$, summing over k, and using (12.36) consequently yield, for the homogeneous scheme,

$$|U^n|_{1,h} \leq |U^0|_{1,h} + \frac{1}{c\sqrt{\delta(2-\delta)}}\|V^0\|_2. \tag{12.39}$$

A telescopic sum and the Cauchy–Schwarz inequality give the discrete one-dimensional estimate

$$\|W\|_\infty \leq L|W|_{1,h}. \tag{12.40}$$

Combining (12.39) and (12.40) proves the homogeneous version of (12.38). The same superposition argument as in the proof of Lemma 89 gives the estimate with a source term. ♢

Remark 62. *Estimate* (12.38) *is not a discrete maximum principle of the type obtained for the upwind scheme in Proposition 87. Indeed, the wave recurrence contains the coefficient* $-U^{n-1}$. *Uniform control in* l^∞ *is obtained here from the discrete gradient quantity* $|U^0|_{1,h}$ *and the one-dimensional inequality* (12.40).

12.2.3 Convergence

Define the global error vector at time t_n by

$$E^n = u_h^n - U^n = \left(u(t_n, x_j) - U_j^n\right)_{1\leq j\leq I-1} \in \mathbb{R}^{I-1}. \tag{12.41}$$

Subtracting the numerical scheme (12.19) from the defining identity (12.24) gives

$$E^{n+1} - 2E^n + E^{n-1} = -c^2\Delta t^2 A E^n + \Delta t^2 \varepsilon^n, \qquad 1 \leq n \leq N-1, \tag{12.42}$$

with $E^0 = 0$ and $E^1 = \eta^1$. Thus the global error satisfies the same recurrence as the numerical solution, with the consistency error as a source term. Stability controls the accumulation of these local errors, exactly as for the heat and advection equations.

Theorem 91 (Convergence of the centered wave scheme)**.** *Assume that the solution to* (12.18) *is sufficiently regular and that the first time step is given by* (12.21). *The following two results hold.*

- *Under the CFL condition*
$$\frac{c\Delta t}{h} \leq 1,$$
there exists a constant C, *independent of* h *and* Δt, *such that*
$$\max_{0\leq n\leq N} \|E^n\|_2 \leq C(\Delta t^2 + h^2).$$

- *Under the strict CFL condition*
$$\frac{c\Delta t}{h} \leq 1-\delta, \qquad \delta \in (0,1),$$
where δ *is independent of* h *and* Δt, *there exists a constant* C_δ, *also independent of* h *and* Δt, *such that*
$$\max_{0\leq n\leq N} \|E^n\|_\infty \leq C_\delta(\Delta t^2 + h^2).$$

Thus the scheme is convergent of order two in time and space in l^2 *under the CFL condition, and in* l^∞ *under the strict CFL condition.*

Proof: For the first result, apply Lemma 89, with $F^n = \varepsilon^n$, to the error recurrence (12.42). Since $E^0 = 0$ and $E^1 = \eta^1$, its discrete initial velocity is $\eta^1/\Delta t$. Therefore,

$$\|E^n\|_2 \leq \frac{L}{2c}\left(\frac{\|\eta^1\|_2}{\Delta t} + \Delta t \sum_{m=1}^{n-1} \|\varepsilon^m\|_2\right).$$

For the second result, apply Lemma 90, again with $F^n = \varepsilon^n$, to the same recurrence. Since $|E^0|_{1,h} = 0$, one obtains

$$\|E^n\|_\infty \leq \frac{L}{c\sqrt{\delta(2-\delta)}}\left(\frac{\|\eta^1\|_2}{\Delta t} + \Delta t \sum_{m=1}^{n-1} \|\varepsilon^m\|_2\right).$$

Finally, $\|W\|_2 \leq \|W\|_\infty$, $n\Delta t \leq T$, and the two consistency estimates (12.27) give the announced orders of convergence. ♢

Remark 63. *The proof above deliberately emphasizes the same "consistency–stability–convergence" mechanism as in the preceding chapters. A fully detailed estimate can alternatively be obtained by the discrete energy method suggested in Exercise 109.*

Remark 64 (Comparison of the CFL conditions)**.** *For the explicit schemes studied in these lectures, the stability conditions have different scalings:*

- *heat equation:* $\Delta t \leq Ch^2$*;*
- *advection equation:* $|a|\Delta t \leq h$*;*
- *wave equation:* $c\Delta t \leq h$*.*

For transport and waves, the condition has a geometric interpretation: the numerical domain of dependence must contain the domain of dependence of the differential equation. For diffusion, the more restrictive h^2 *scaling reflects the large eigenvalues of the discrete Laplacian and the fact that the explicit Euler method must damp every spatial mode.*

Exercise 110. ★★*For the transport scheme with* $\nu > 1$*, show that the exact characteristic starting from* (t_{n+1}, x_j) *meets the preceding time level to the left of* x_{j-1}*. Explain geometrically why a scheme using only* U_j^n *and* U_{j-1}^n *cannot then recover the correct domain of dependence.*

Appendix A

Glossary of mathematical terms

This bilingual glossary collects a selection of mathematical and numerical-analysis terms used throughout these lecture notes. English terms are listed in alphabetical order. Depending on the context, other French translations may also be appropriate.

English	French
almost everywhere (a.e.)	presque partout (p.p.)
antiderivative	primitive
assumption	hypothèse
backward Euler scheme	schéma d'Euler implicite
backward heat equation	équation de la chaleur rétrograde
ball	boule
boundary	bord, frontière
boundary condition	condition au bord, condition aux limites
boundary value problem	problème aux limites
bounded	borné
bounded above	majoré
bounded below	minoré
centered difference	différence centrée
chain rule	règle de dérivation des fonctions composées
characteristic curve	courbe caractéristique
characteristic function	fonction indicatrice
closed set	ensemble fermé
closure	adhérence
compactly supported	à support compact
to compute	calculer
consistency	consistance
convex combination	combinaison convexe
curl	rotationnel
data	données
derivative	dérivée
derive the formula	démontrez, obtenez la formule
to differentiate = to take the derivative	dériver (au sens de : prendre la dérivée)
differentiable	dérivable

English	French
downwind scheme	schéma décentré aval
eigenmode	mode propre
eigenvalue	valeur propre
eigenvector	vecteur propre
finite difference	différence finie
flow	flot
forcing term	terme de forçage
forward Euler scheme	schéma d'Euler explicite
Green's kernel	noyau de Green
grid	grille
grid point	point de grille, nœud
heat equation	équation de la chaleur
homogeneous boundary condition	condition au bord homogène
ill-posed problem	problème mal posé
incoming boundary	frontière entrante
initial condition	condition initiale
initial value problem	problème avec une condition initiale
inner product	produit scalaire
integration by parts	intégration par parties
interval $(a, b]$	l'intervalle $]a, b]$
inverse function	fonction inverse
kernel	noyau
Lipschitz continuous	lipschitzien
local truncation error	erreur locale de troncature
maximum principle	principe du maximum
mean value formula	formule de la moyenne
mesh	maillage
mesh size	pas d'espace, taille de maille
modeling	modélisation
mollification	régularisation par convolution
mollifier	fonction régularisante
negative	strictement négatif
non-negative	positif ou nul
non-positive	négatif ou nul
normal derivative	dérivée normale
normal mode	mode propre
open set	ensemble ouvert
ordinary differential equation (ODE)	équation différentielle ordinaire (EDO)
outward unit normal	vecteur normal unitaire sortant
partial differential equation (PDE)	équation aux dérivées partielles (EDP)
periodic boundary condition	condition au bord périodique
phase portrait	portrait de phase
piecewise constant	constant par morceaux
positive	strictement positif
random walk	marche aléatoire
remainder	reste
residual	résidu

English	**French**
right-hand side	second membre (d'une égalité/inégalité)
smooth function/domain	fonction/domaine régulier, lisse
Span$(v_1, \dots v_n)$	Vect$(v_1, \dots, v_n)$
= vector space spanned by $(v_1, \dots v_n)$	= espace vectoriel engendré par $(v_1, \dots v_n)$
spectral radius	rayon spectral
strain tensor	tenseur des déformations
stress tensor	tenseur des contraintes
Taylor expansion	développement de Taylor
upwind scheme	schéma décentré amont
variational formulation	formulation variationnelle
vector field	champ de vecteurs, champ vectoriel
wave equation	équation des ondes
well-posed problem	problème bien posé

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